

RBSE · BSER AJMER · CLASS 12 · BOARD EXAM 2027

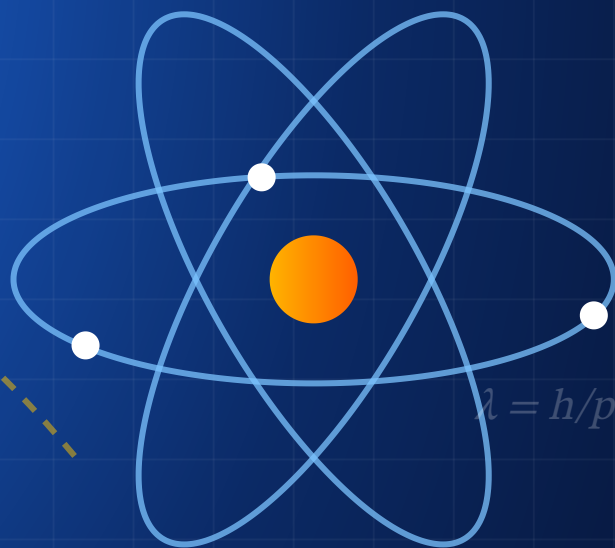
$$1/f = 1/v - 1/u$$

PHYSICS

BOARD EXAM COMPLETE GUIDE

Notes · Formulas · Diagrams · Most-repeated board questions · Answer-writing tips · 5 Model Papers in the new 56-mark pattern

ENGLISH MEDIUM



$$E = h\nu$$

All 14 chapters · notes + diagrams

$$V = IR$$

Chapter-wise weightage (56 marks)

PYQ analysis of official papers

Step-marked derivations

5 Model Papers + answers

30–45 day revision map

$$\varepsilon = -d$$



VS SIR

Vijay Kumar Sharma
20+ years · Physics expert · Kota

RBSE Class 12 Physics: Board Exam Complete Guide

Rajasthan Board (BSER, Ajmer) · Senior Secondary · Theory 56 marks

by VS Sir (Vijay Kumar Sharma)

20+ years teaching Physics, Kota

First edition: 2026 · Language: English

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About the questions: Questions marked [RBSE year] are previous-year questions from the official question papers published by the Board of Secondary Education, Rajasthan (rajeduboard.rajasthan.gov.in), given here in summarised form. All other questions, notes and the 5 model papers are original material written by the author in the board pattern. This book is an independent study aid and is not published or endorsed by BSER.

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ABOUT THE AUTHOR

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- 20+ years of teaching Physics in Kota – Class 12 boards, NEET and JEE
- Knows exactly how board examiners mark: keywords, diagrams and step marks
- Author of the NEET + JEE Physics 30-Day Rapid Revision series (Class 11 & 12, English & Hindi)

A message from VS Sir

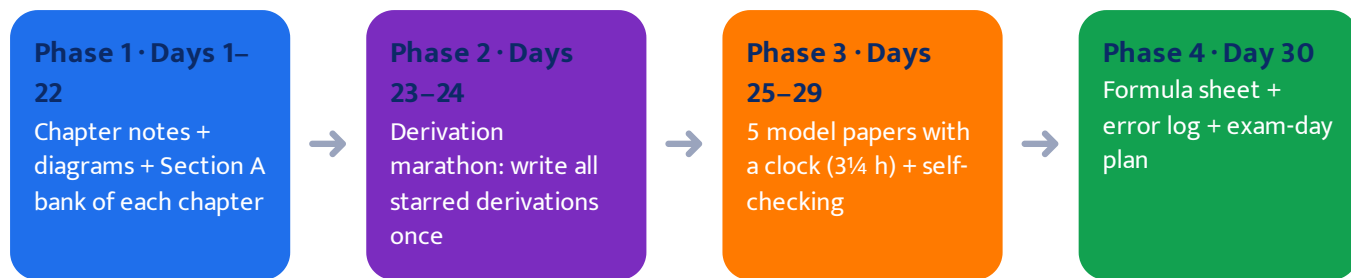
Dear student,

The RBSE Physics paper is very predictable if you know where to look. More than half of the marks come from Optics, AC, EMI, Magnetism and Electrostatics, and the same derivations return year after year. This book is built on the official BSER question papers: it shows you what is asked, how much it is worth, and exactly how to write the answer to get full marks.

Follow the revision map, write every derivation once with your own hand, and solve the five model papers with a clock. 56 out of 56 is possible.

— VS Sir

Your 30–45 day revision map



Have 45 days? Add a second round

- Days 31–35: second round of all derivations (Sections C, D)
- Days 36–40: re-solve every wrong question from the 5 model papers
- Days 41–44: official BSER papers of 2023, 2024, 2025 with a clock
- Day 45: formula sheet + diagrams only

How to study each chapter (2–3 hours)

1. **Read** the chapter snapshot: marks, what was asked, what is likely.
2. **Learn** definitions word-for-word – 1-mark answers need exact keywords.
3. **Write** each derivation once on paper with the diagram – reading is not enough.
4. **Practise** the question bank: MCQ and blanks in 1 min each; short answers in 5 min.
5. **Check** with the answers; mark every mistake in an error log and revisit on Day 30.

Golden rule of board Physics

Diagram + formula + final result in a box = full marks. Examiners look for the labelled diagram and the boxed answer first.

30-day plan (tick each day)

Daily target: 2–3 hours. Heavy chapters (Optics, AC, Magnetism) get extra days.

Day 1 Ch 1 Electric charges & fields – I	Day 2 Ch 1 Dipole, flux, Gauss law	Day 3 Ch 2 Potential & capacitance	Day 4 Ch 3 Current, Ohm, cells	Day 5 Ch 3 Kirchhoff, Wheatstone bridge
Day 6 Ch 4 Biot–Savart, Ampere	Day 7 Ch 4 Force, galvanometer	Day 8 Ch 5 Magnetism & matter	Day 9 Ch 6 EMI: Faraday, Lenz, inductance	Day 10 Ch 7 AC: rms, R/L/C circuits
Day 11 Ch 7 LCR, resonance, power, transformer	Day 12 Ch 8 EM waves	Day 13 Ch 9 Mirrors, refraction, TIR	Day 14 Ch 9 Lenses, lens maker, prism	Day 15 Ch 9 Microscope, telescope
Day 16 Ch 10 Huygens, YDSE	Day 17 Ch 10 Diffraction, polarisation	Day 18 Ch 11 Photoelectric effect, de Broglie	Day 19 Ch 12 Atoms: Bohr model	Day 20 Ch 13 Nuclei: BE, fission, fusion
Day 21 Ch 14 Semiconductors, p–n junction	Day 22 Ch 14 Rectifiers, special diodes	Day 23 Final Derivation marathon: Electricity & Magnetism	Day 24 Final Derivation marathon: AC, Optics, Modern	Day 25 Final Model Paper 1 (3¼ h)
Day 26 Final Model Paper 2	Day 27 Final Model Paper 3	Day 28 Final Model Paper 4	Day 29 Final Model Paper 5	Day 30 Final Formula sheet + exam-day plan

Paper pattern (as in the 2025 board paper)

Subject code SS-40 · Time **3 hours 15 minutes** · Maximum marks **56** · **18** numbered questions · all compulsory
· internal choice only in Q14–18 · if Hindi and English differ, the Hindi version is valid.

Section	Q. no.	Type	Items	Marks each	Total
अ / A	1	Multiple choice (MCQ)	18	$\frac{1}{2}$	9
अ / A	2	Fill in the blanks	10	$\frac{1}{2}$	5
अ / A	3	Very short answer	10	1	10
ब / B	4–13	Short answer	10	$1\frac{1}{2}$	15
स / C	14–16	Long answer (with OR)	3	3	9
द / D	17–18	Essay type (with OR)	2	4	8
Total			53		56

How the 100 marks are made

Theory paper

56

Practical

30

Sessional
(school)

14

🕒 Time plan (195 min)

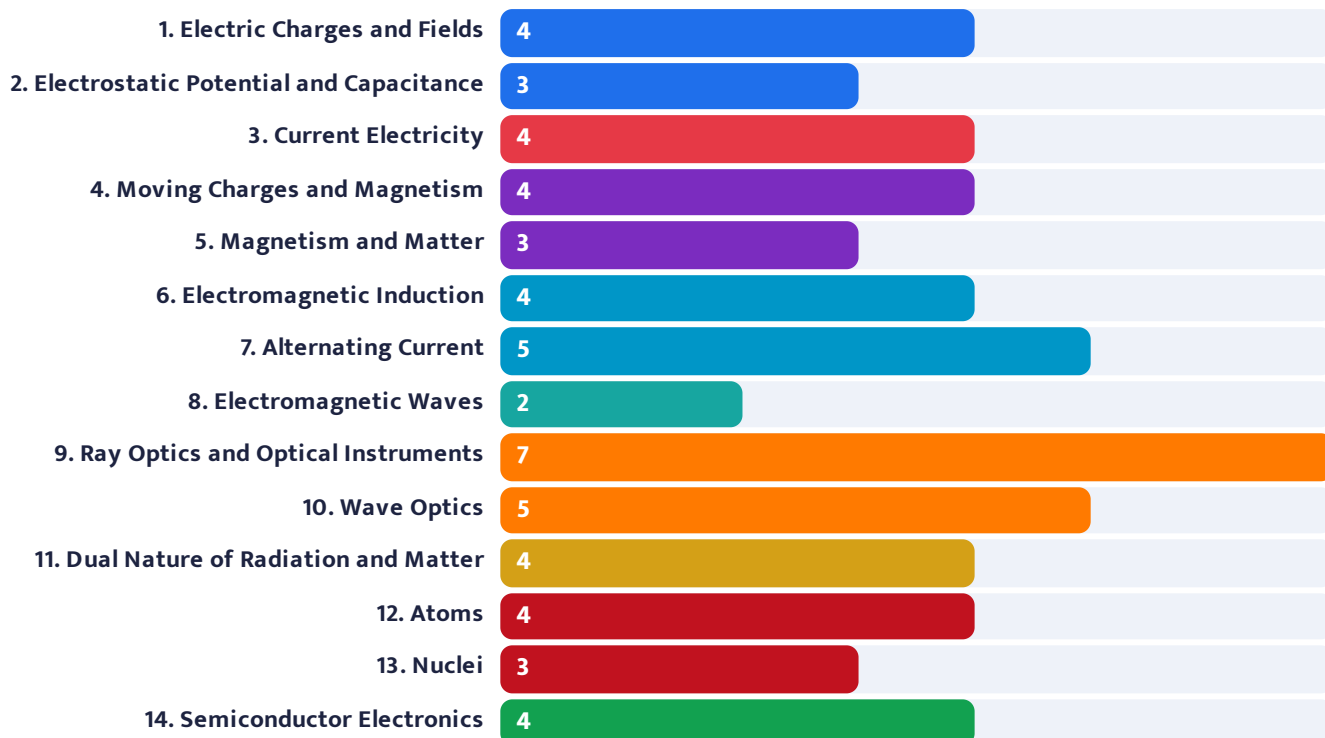
- Section A (28 items): 50 min
- Section B ($10 \times 1\frac{1}{2}$): 60 min
- Section C (3×3): 35 min
- Section D (2×4): 40 min
- Revision: 10 min

⚠️ Pattern has changed recently

2023 and 2024 papers had 20 questions; 2025 had 18 questions with $\frac{1}{2}$ -mark MCQs and choice in Q14–18. The 2026 CWSN paper also used 18 questions. Always check the latest BSER model paper for your exam year.

Chapter-wise weightage (56 marks)

Marks per chapter as per the RBSE syllabus blueprint. Spend your time in this ratio.



Unit view: top 5 units = 42 of 56 marks (75%)



Note

Chapter marks are from the RBSE blueprint for 2025-26 (sum = 56). Websites quote slightly different splits for Current Electricity and Semiconductors ($\pm 1-2$ marks); the board can change the blueprint each year, so confirm with the latest syllabus on rajeduboard.rajasthan.gov.in.

What the board keeps asking

Topic-wise analysis of the official BSER Physics papers (SS-40). ✓ = asked in that year (any section). *2026 = CWSN paper of the same pattern.

Repeated topic	Ch	2023	2024	2025	2026*
Electric field of dipole (axis / equatorial)	1	✓	✓	✓	✓
Gauss's law: infinite plane / line charge	1	–	✓	✓	✓
Potential of point charge, PE of charge system	2	✓	–	✓	–
Wheatstone bridge (Kirchhoff)	3	✓	✓	✓	–
EMF, internal resistance, cells in series	3	–	✓	✓	–
Biot–Savart: field on axis of loop	4	–	✓	✓	✓
Ampere's law: wire / solenoid	4	✓	–	✓	✓
Faraday, Lenz, self / mutual inductance	6	✓	✓	✓	✓
AC: rms, reactance, LCR, resonance, power	7	✓	✓	✓	✓
Transformer / AC generator	7	✓	✓	–	✓
EM waves: properties, spectrum	8	✓	✓	✓	–
Mirror formula, prism, lens maker, microscope	9	✓	✓	✓	✓
Huygens' principle (Snell's law), YDSE	10	✓	✓	✓	✓
Photoelectric effect, Einstein equation	11	✓	✓	✓	✓
Bohr model, energy of H-atom	12	✓	✓	✓	✓
BE/nucleon graph, fission, fusion, mass defect	13	✓	✓	✓	–
Energy bands, p-n junction, rectifier	14	✓	✓	✓	✓

🎯 Conclusion

Every topic in this table came at least twice in the last three regular papers. Learn these derivations and numericals first – they cover almost all of Sections B, C and D.

Contents

1	Electric Charges and Fields Charge & its properties · Coulomb's law · Electric field & field lines · Electric dipole	4 marks	10
2	Electrostatic Potential and Capacitance Potential & potential energy · Equipotential surfaces · Potential of a dipole · Conductors & dielectrics	3 marks	22
3	Current Electricity Drift velocity & mobility · Ohm's law & resistivity · emf & internal resistance · Cells in series/parallel	4 marks	34
4	Moving Charges and Magnetism Lorentz force · Circular motion in B · Biot–Savart law · Ampère's law & solenoid	4 marks	46
5	Magnetism and Matter Bar magnet & field lines · Torque & PE of dipole · Gauss's law · M, H, χ , μ_r	3 marks	60
6	Electromagnetic Induction Magnetic flux · Faraday's & Lenz's laws · Motional emf · Eddy currents	4 marks	71
7	Alternating Current AC generator · rms value · R, L, C circuits · Series LCR & resonance	5 marks	82
8	Electromagnetic Waves Displacement current · Maxwell's idea · Nature of EM waves · $c = 1/\sqrt{(\mu_0\epsilon_0)}$	2 marks	94
9	Ray Optics and Optical Instruments Mirror formula · Refraction & TIR · Lens maker's formula · Power & combination	7 marks	103
10	Wave Optics Wavefront & Huygens · Coherent sources · Young's double slit · Single-slit diffraction	5 marks	118
11	Dual Nature of Radiation and Matter Work function · Photoelectric effect · Stopping potential · Einstein's equation	4 marks	128
12	Atoms α -scattering & nuclear model · Closest approach · Hydrogen spectrum · Bohr's postulates	4 marks	138
13	Nuclei Size & density · Isotopes, isobars · Mass defect · B.E. per nucleon	3 marks	148
14	Semiconductor Electronics Energy bands · n- and p-type · p-n junction · Diode & rectifiers	4 marks	158
①	Model Paper 1 Full paper · 18 questions · 56 marks · 3 h 15 min · with marking scheme	56 marks	169
②	Model Paper 2 Full paper · 18 questions · 56 marks · 3 h 15 min · with marking scheme	56 marks	184
③	Model Paper 3 Full paper · 18 questions · 56 marks · 3 h 15 min · with marking scheme	56 marks	198
	Model Paper 4 Full paper in the 2025 pattern · SS-40 · 3¼ h · marking scheme + analysis	56 marks	214
	Model Paper 5 Full paper in the 2025 pattern · SS-40 · 3¼ h · marking scheme + analysis	56 marks	228
★	Final Toolkit: Formulas, Derivations, Errors and Exam-Day Plan 14-chapter formula sheet with units · derivation checklist · errors log · exam-day plan	Day 30	241

CHAPTER

1

Electric Charges and Fields

17

Day 1-2

Board marks: 4

PYQ frequency ★★★

RBSE Class 12

Charge & its properties

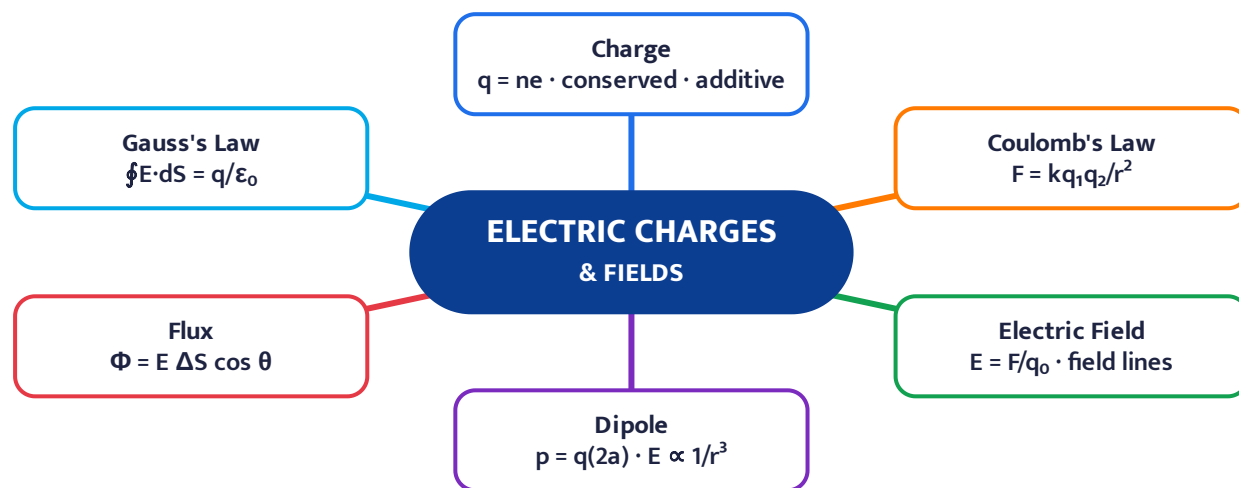
Coulomb's law

Electric field & field lines

Electric dipole

Electric flux

Gauss's law & applications



Chapter at a glance

Board snapshot

- **Marks: 4** (of 56). Usually one MCQ / fill-in (½ each) + one 1½-mark numerical **or** a 3-mark derivation (Section C, with OR).
- **2025:** MCQ – SI unit of electric flux; MCQ – force when $r \rightarrow 2r$; fill-in – dipole field $\propto 1/r^3$; Section B – λ of an infinite line charge ($E = 4.5 \times 10^4$ N/C at 4 cm).
- **2024:** MCQ – flux through a Gaussian sphere when its radius is tripled; fill-in – field lines of $+q$ are radially outward; Section C – dipole field on the equatorial line **OR** field of an infinite plane sheet (Gauss).
- **2023:** MCQ – permittivity of free space ϵ_0 ; fill-in – field inside a charged spherical shell; VSA – field lines, dipole field. **2026 (CWSN):** Section C – dipole field **OR** infinite plane sheet. **2020:** electron orbiting an infinite line charge.
- **Most likely in 2027:** (1) dipole field – axial / equatorial (3 marks); (2) Gauss's law with infinite line charge or plane sheet; (3) ½-mark items on flux unit, Coulomb's law ratio, field inside a shell.

1.1 Electric Charge and its Basic Properties

Electric charge is an intrinsic property of matter due to which it produces and experiences electric and magnetic effects. Charges are of two kinds – **positive** and **negative**; **like charges repel and unlike charges attract**. When two bodies are rubbed, **electrons are transferred** from one to the other: the body that loses electrons becomes positive. SI unit of charge: **coulomb (C)**.

**Three basic properties**

- **Additivity:** total charge = algebraic sum, $q = q_1 + q_2 + \dots$ (charge is a **scalar**).
- **Conservation:** the total charge of an isolated system remains constant; charge can be neither created nor destroyed.
- **Quantisation:** $q = \pm ne$, $n = 1, 2, 3, \dots$;
 $e = 1.6 \times 10^{-19} \text{ C}$ is the smallest free charge.

**Solved Example**

How many electrons must be removed from a body to give it a charge of $+3.2 \times 10^{-7} \text{ C}$?

Solution: $n = \frac{q}{e} = \frac{3.2 \times 10^{-7}}{1.6 \times 10^{-19}} = 2 \times 10^{12} \text{ electrons.}$

**Note**

For macroscopic charges (μC) quantisation is not noticeable – charge is treated as **continuous**. $1 \text{ C} \approx 6.25 \times 10^{18} \text{ electrons.}$

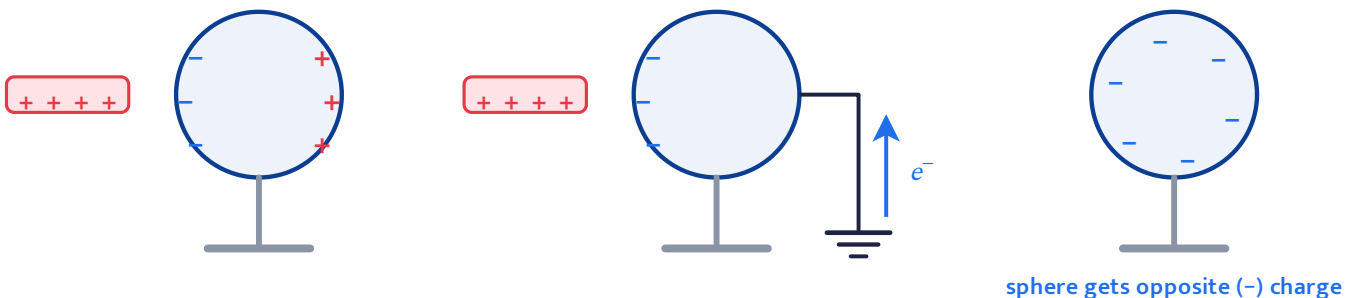
1.2 Conductors, Insulators and Charging by Induction

Conductors (metals, human body, earth) allow electric charge to move freely through them because they contain free electrons. **Insulators** (glass, plastic, nylon, porcelain) do not. Charge given to a conductor spreads over its **outer surface**. **Earthing (grounding)** connects a body to the earth, which acts as an infinite sink of charge. In **charging by induction**, a charged body redistributes the charges of a nearby conductor **without touching it**; the conductor finally gets a charge **opposite** to that of the inducing body.

1. Bring a +ve rod near

2. Earth the far side

3. Remove earth, then rod



Charging a metal sphere by induction – the charged rod never touches the sphere

**Common Trap**

- A charged body also attracts a **neutral** body (due to induction). So **attraction is not a sure test** of charge – **repulsion is**.
- In induction, remove the earth connection **first** and the rod **afterwards**; otherwise the sphere becomes neutral again.

1.3 Coulomb's Law and Superposition

Coulomb's law: the force of attraction or repulsion between two stationary point charges is **directly proportional to the product of their magnitudes** and **inversely proportional to the square of the distance** between them; it acts along the line joining the two charges.

Coulomb's law

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

Permittivity of free space (vacuum)

$$\vec{F}_{21} = k \frac{q_1 q_2}{r^2} \hat{r}_{21} = -\vec{F}_{12}$$

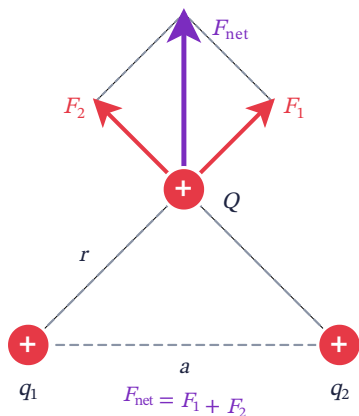
Vector form: force on q_2 due to q_1 ; $\hat{r}_{21}$ points from 1 to 2

$$F_m = \frac{F_0}{K}, \quad K = \frac{\epsilon}{\epsilon_0}$$

In a medium: $K =$ **dielectric constant** (relative permittivity)

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n}$$

Superposition: net force = vector sum of forces due to each charge separately



Superposition: forces add as vectors

Solved Example

Charges $+2 \mu\text{C}$ and $-3 \mu\text{C}$ are 30 cm apart in air. Find the force between them.

Solution: $F = \frac{kq_1 q_2}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 3 \times 10^{-6}}{(0.3)^2} = 0.6 \text{ N}$, attractive.

VS Sir's Exam Tip

Ratio questions: $F \propto \frac{q_1 q_2}{r^2}$. Distance doubled $\Rightarrow F/4$; distance halved $\Rightarrow 4F$; medium $K \Rightarrow F/K$.

1.4 Electric Field

The **electric field** at a point is the force experienced by a **unit positive test charge** placed at that point:

$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$. The test charge is taken very small so that it does not disturb the source charges. It is a **vector**; SI

unit N C^{-1} ($= \text{V m}^{-1}$). A charge q in a field experiences $\vec{F} = q\vec{E}$ – along $\vec{E}$ for $+q$ and **opposite** to $\vec{E}$ for $-q$.

Electric field

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

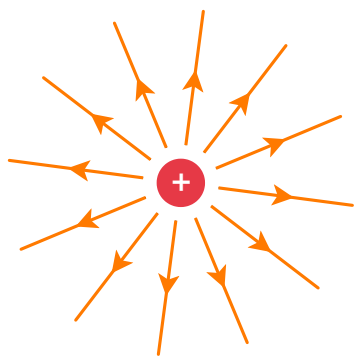
Point charge: radially outward for $+q$, inward for $-q$; $E \propto 1/r^2$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots = \sum_i \frac{kq_i}{r_i^2} \hat{r}_i$$

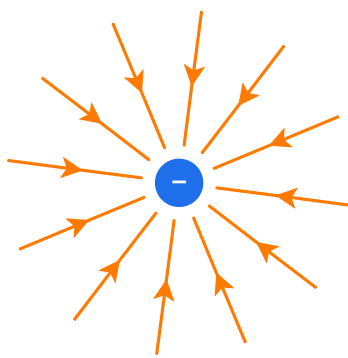
System of charges – superposition (vector sum)

1.5 Electric Field Lines

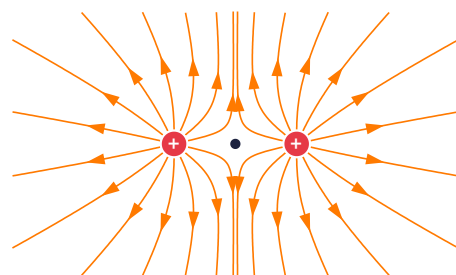
An **electric field line** is a curve drawn in such a way that the **tangent at any point gives the direction of the electric field** at that point. The relative closeness of the lines shows the strength of the field.



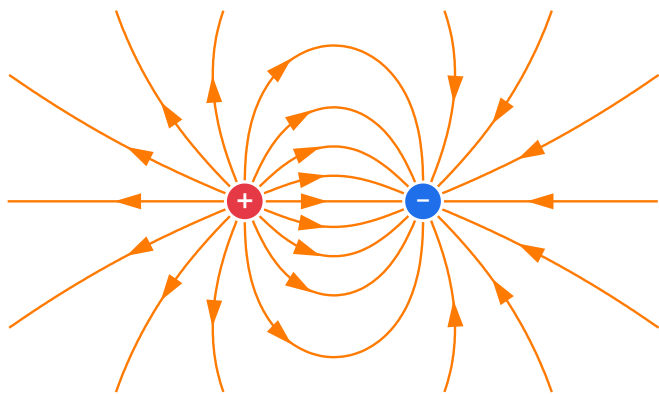
Isolated +q: radially outward



Isolated -q: radially inward



Two equal +q: neutral point midway



Dipole (+q, -q): lines start on +q, end on -q

Properties of field lines (write any three)

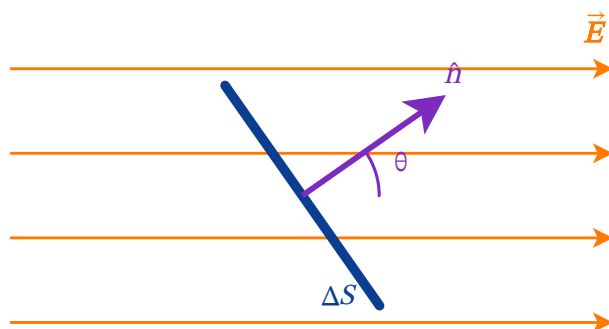
- They **start from positive** charge and **end on negative** charge (or at infinity).
- They are continuous curves in a charge-free region and **do not form closed loops**.
- **Two field lines never intersect** (else E would have two directions at one point).
- Crowded lines $\Rightarrow$ strong field; parallel, equally spaced lines $\Rightarrow$ uniform field.

1.6 Electric Flux

Electric flux through a surface is the number of electric field lines crossing it normally. For a small area element: $\Delta\phi = \vec{E} \cdot \vec{\Delta S} = E \Delta S \cos\theta$, where θ is the angle between $\vec{E}$ and the **area vector** (outward normal). Flux is a **scalar**.

Memory Trick

- SI unit: $\text{N m}^2 \text{C}^{-1} = \text{V m}$.
- $\theta = 0^\circ$: maximum; $\theta = 90^\circ$: zero.
- Outward flux +ve, inward flux -ve.



$$\Phi = E \Delta S \cos \theta \quad (\theta: \text{between } \vec{E} \text{ and normal})$$

Flux through a tilted area

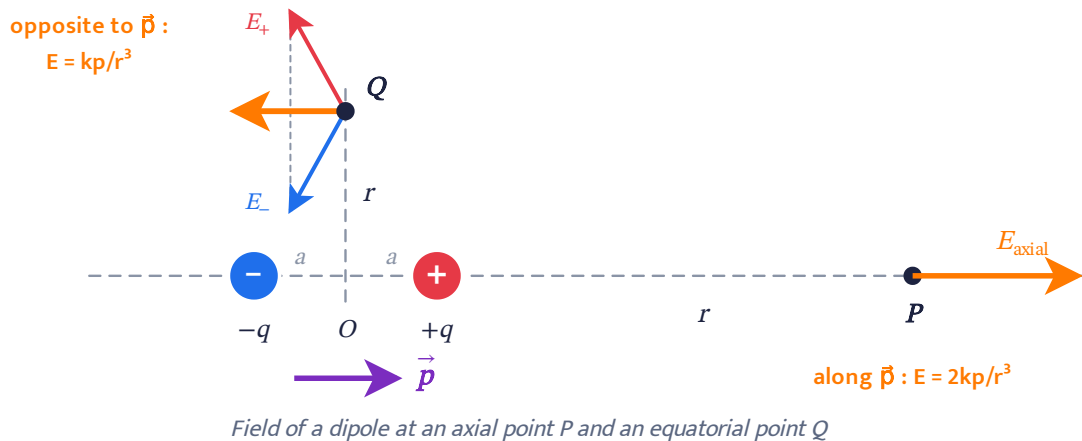
Solved Example

A uniform field $E = 3 \times 10^3 \text{ N/C}$ passes through a square of side 10 cm whose normal makes 60° with the field. Find the flux.

Solution: $\phi = E S \cos\theta = 3 \times 10^3 \times (0.1)^2 \times \cos 60^\circ = 15 \text{ N m}^2 \text{C}^{-1}$.

1.7 Electric Dipole

A pair of **equal and opposite point charges** (+q and -q) separated by a small distance 2a is called an **electric dipole**. **Dipole moment** $\vec{p} = q \times 2\vec{a}$; it is a **vector directed from -q to +q**; SI unit **C m**. The total charge of a dipole is zero, but its field is not zero; at large distances it falls as $1/r^3$ (faster than the $1/r^2$ of a point charge). Molecules like H₂O and HCl are **polar** – they have a permanent dipole moment.



Dipole formulas

$$E_{axial} = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2} \approx \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3} \quad \text{On the axis; direction along } \vec{p}$$

$$E_{eq} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{3/2}} \approx \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \quad \text{On the equatorial line; direction opposite to } \vec{p}$$

$$E_{axial} : E_{eq} = 2:1 \quad \text{Short dipole } (r \gg a), \text{ same distance}$$

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin \theta \quad \text{Torque in a uniform field; net force} = 0$$

Solved Example

A dipole has charges $\pm 10 \mu\text{C}$ separated by 5 mm. Find E at 15 cm from its centre (i) on the axis, (ii) on the equatorial line.

Solution: $p = 10^{-5} \times 5 \times 10^{-3} = 5 \times 10^{-8} \text{ C m}$.

$$(i) E = \frac{2kp}{r^3} = \frac{2 \times 9 \times 10^9 \times 5 \times 10^{-8}}{(0.15)^3} = 2.67 \times 10^5 \text{ N/C along } \vec{p}.$$

$$(ii) E = \frac{kp}{r^3} = 1.33 \times 10^5 \text{ N/C, opposite to } \vec{p}.$$

VS Sir's Exam Tip

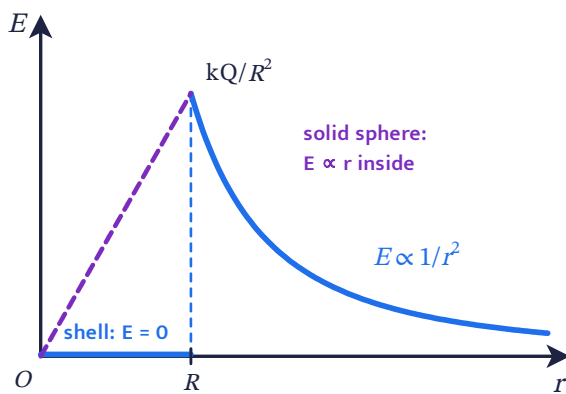
- Uniform field:** net force = 0 but torque $\neq 0 \Rightarrow$ only rotation. **Non-uniform field:** a net force also acts.
- τ is maximum ($= pE$) at $\theta = 90^\circ$; $\tau = 0$ at $\theta = 0^\circ$ (**stable** equilibrium) and $\theta = 180^\circ$ (**unstable** equilibrium).

1.8 Continuous Charge Distribution and Gauss's Law

Distribution	Charge density	SI unit	Example
Linear	$\lambda = \frac{dq}{dl}$	C m^{-1}	long wire, ring
Surface	$\sigma = \frac{dq}{dS}$	C m^{-2}	plane sheet, shell
Volume	$\rho = \frac{dq}{dV}$	C m^{-3}	solid charged sphere

Gauss's law: the total electric flux through **any closed surface** is equal to $\frac{1}{\epsilon_0}$ times the **net charge enclosed** by the surface: $\oint \vec{E} \cdot d\vec{S} = \frac{q_{enc}}{\epsilon_0}$. The closed surface chosen is called a **Gaussian surface**; choose it to match the symmetry so that E is constant on it.

Charge system	Field E	Variation
Infinite line charge (λ)	$\frac{\lambda}{2\pi\epsilon_0 r}$	$E \propto \frac{1}{r}$
Infinite plane sheet (σ)	$\frac{\sigma}{2\epsilon_0}$	independent of distance
Thin spherical shell: outside ($r > R$)	$\frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$	as if q were at the centre
Shell: on surface ($r = R$)	$\frac{\sigma}{\epsilon_0}$	maximum
Shell: inside ($r < R$)	0	no charge enclosed



E-r graph: shell (solid line), solid sphere (dashed)

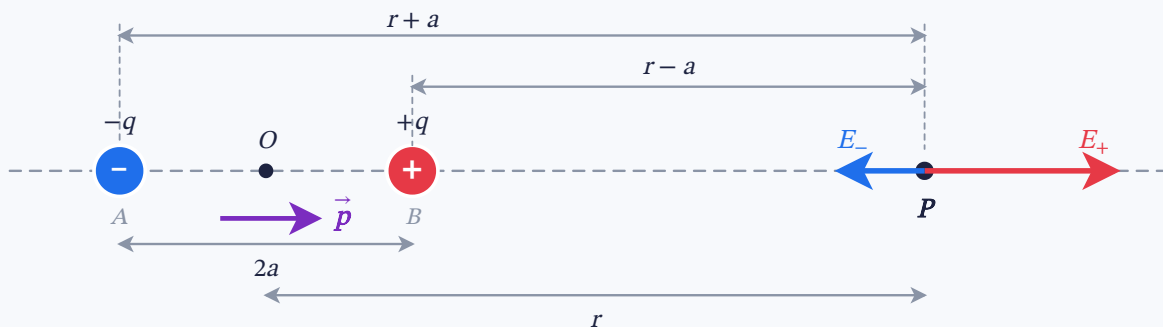
! Common Trap

- Flux through a closed surface depends **only on the charge inside** – not on its shape, size or the position of the charge inside.
- But $\vec{E}$ on the surface is due to **all** charges (inside and outside).
- Charge q at the centre of a cube: flux through each face = $q / 6\epsilon_0$.

Important derivations ★

🔧 Derivation ★ [3 marks]

Derive an expression for the electric field at a point on the **axial line** of an electric dipole.



Solution: 1. Dipole: $-q$ at A, $+q$ at B, $AB = 2a$, centre O. Point P on the axis, $OP = r$ (figure). (½)

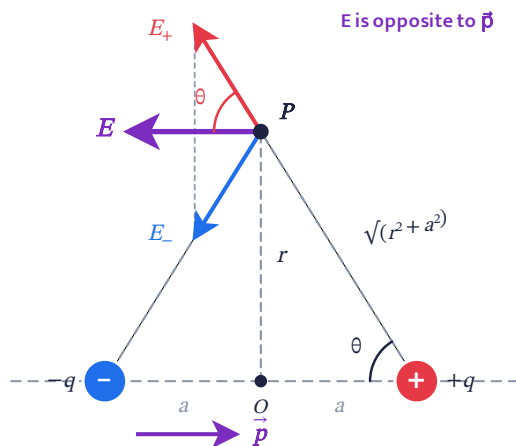
2. Field at P due to $+q$ (distance $r - a$), along BP (along $\vec{p}$): $E_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2}$ (½)

3. Field at P due to $-q$ (distance $r + a$), along PA (opposite to $\vec{p}$): $E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$ (½)

4. $E_+ > E_-$, so net field $E = E_+ - E_- = \frac{q}{4\pi\epsilon_0} \left[\frac{(r+a)^2 - (r-a)^2}{(r^2 - a^2)^2} \right] = \frac{q}{4\pi\epsilon_0} \frac{4ar}{(r^2 - a^2)^2}$ (½)

5. With $p = q \cdot 2a$: $E_{axial} = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2}$, directed **along $\vec{p}$** . (½)

6. For a short dipole ($r \gg a$): $E_{axial} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$ (½)



How to write for full marks

- Draw the figure first: label $-q, +q, 2a, r, P, \theta$, the two fields and the resultant.
- Write the **direction** of the final field (along $\vec{p}$ on the axis, opposite on the equator) – it carries marks.
- Finish with the short-dipole result in a box and the ratio $E_{ax}:E_{eq} = 2:1$.

Derivation ★ [3 marks]

[RBSE 2024, 2026] Derive an expression for the electric field at a point on the **equatorial line** of an electric dipole.

Solution: 1. P is on the perpendicular bisector, $OP = r$; distance of P from each charge $= \sqrt{r^2 + a^2}$ (figure). (½)

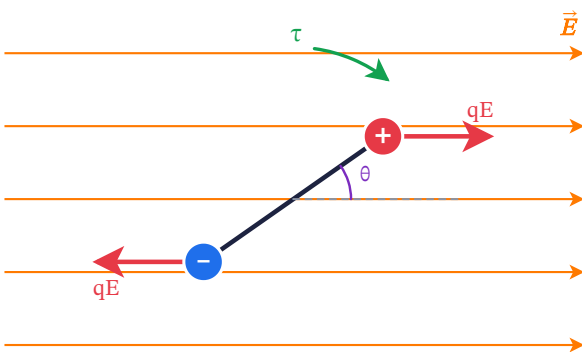
2. Magnitudes are equal: $E_+ = E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2 + a^2}$; $\vec{E}_+$ away from $+q$, $\vec{E}_-$ towards $-q$. (½)

3. Resolve: components **perpendicular** to the axis ($E\sin\theta$) are equal and opposite – they **cancel**. Components **parallel** to the axis ($E\cos\theta$) add, both opposite to $\vec{p}$. (½)

4. $E = 2E_+\cos\theta$, with $\cos\theta = \frac{a}{\sqrt{r^2 + a^2}} \Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{2qa}{(r^2 + a^2)^{3/2}}$ (½)

5. With $p = q \cdot 2a$: $E_{eq} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{3/2}}$, i.e. $\vec{E} = -\frac{\vec{p}}{4\pi\epsilon_0(r^2 + a^2)^{3/2}}$ (**opposite to $\vec{p}$**). (½)

6. Short dipole ($r \gg a$): $E_{eq} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$; hence $E_{axial} = 2E_{eq}$ at the same distance. (½)



Net force = 0; $\tau = pE \sin\theta$ turns $\vec{p}$ towards $\vec{E}$

Couple on a dipole in a uniform field

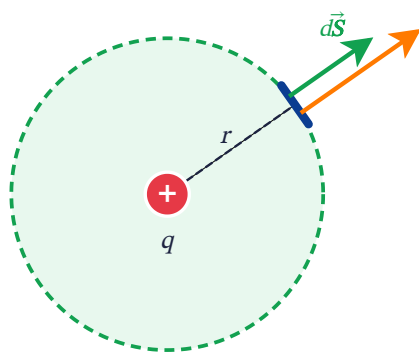
Derivation ★ [1½ marks]

Derive the torque on an electric dipole placed in a uniform electric field.

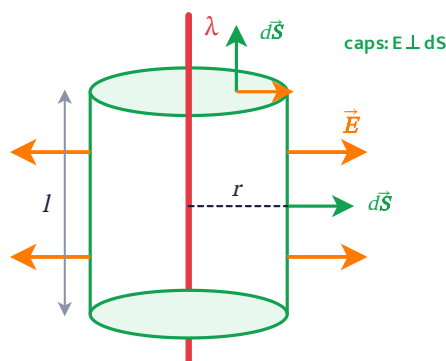
Solution: 1. Dipole at angle θ with $\vec{E}$: force $+q\vec{E}$ on $+q$ and $-q\vec{E}$ on $-q$. Net force = 0. (½)

2. The two equal, opposite, non-collinear forces form a **couple**; perpendicular distance between them $= 2a\sin\theta$. (½)

3. $\tau = qE \times 2a \sin\theta \Rightarrow \tau = pE \sin\theta$, in vector form $\vec{\tau} = \vec{p} \times \vec{E}$. It tends to align $\vec{p}$ along $\vec{E}$. (½)

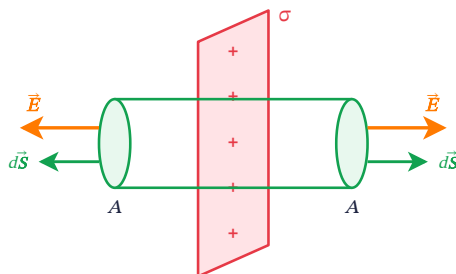


Gaussian sphere

Sphere of radius r around a point charge q 

cylindrical Gaussian surface

Infinite line charge: coaxial cylinder



Gaussian cylinder (pill-box) through the sheet

Infinite plane sheet: cylinder of cross-section A through the sheet**Derivation ★ [3 marks]**

State Gauss's law and verify it for a point charge.

Solution: Statement: total flux through a closed surface $= \frac{1}{\epsilon_0} \times$ net charge enclosed: $\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$. (1)

1. Take a sphere of radius r with q at its centre. At every point

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}, \text{ along } d\vec{S} (\theta = 0). \text{ (}\frac{1}{2}\text{)}$$

$$2. \phi = \oint E dS = E \times 4\pi r^2 \text{ (}\frac{1}{2}\text{)}$$

$$3. \phi = \frac{q}{4\pi\epsilon_0 r^2} \times 4\pi r^2 \Rightarrow \phi = \frac{q}{\epsilon_0} - \text{independent of } r. \text{ Hence proved. (1)}$$

Derivation ★ [3 marks]

Using Gauss's law, find the electric field due to an infinitely long straight uniformly charged wire.

Solution: 1. Linear charge density λ . By symmetry $\vec{E}$ is radial and has the same magnitude at equal distance r . Take a coaxial cylinder of radius r , length l . (1/2)

2. Flat ends: $\vec{E} \perp d\vec{S} \Rightarrow \text{flux} = 0$. Curved surface: $\vec{E} \parallel d\vec{S} \Rightarrow \text{flux} = E \times 2\pi r l$. (1)

$$3. \text{Charge enclosed} = \lambda l. \text{ Gauss: } E \cdot 2\pi r l = \frac{\lambda l}{\epsilon_0} \text{ (}\frac{1}{2}\text{)}$$

$$4. E = \frac{\lambda}{2\pi\epsilon_0 r}, \text{ radially outward for } \lambda > 0; E \propto \frac{1}{r}. \text{ (1)}$$

Derivation ★ [3 marks]

[RBSE 2024, 2026] Using Gauss's law, derive the electric field due to a uniformly charged infinite plane sheet.

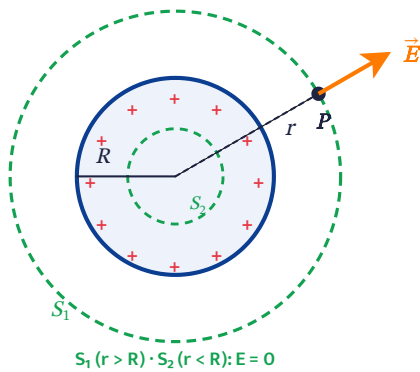
Solution: 1. Surface charge density σ . By symmetry $\vec{E}$ is normal to the sheet, outward on both sides ($\sigma > 0$), same magnitude at equal distances. (1/2)

2. Gaussian surface: a cylinder of cross-section A , cutting the sheet perpendicularly, ends equidistant from the sheet. (1/2)

3. Curved surface: $\vec{E} \perp d\vec{S} \Rightarrow \text{flux} = 0$. Two flat ends: $\vec{E} \parallel d\vec{S} \Rightarrow \text{flux} = EA + EA = 2EA$. (1)

$$4. \text{Charge enclosed} = \sigma A. \text{ Gauss: } 2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0} \text{ (}\frac{1}{2}\text{)}$$

5. E is **independent of the distance** from the sheet (uniform field); in vector form $\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$. (1/2)



Thin shell: Gaussian spheres S_1 (outside) and S_2 (inside)

🔧 Derivation ★ [3 marks]

Using Gauss's law, find the electric field due to a uniformly charged thin spherical shell at a point (i) outside, (ii) inside it.

Solution: 1. Shell of radius R , charge q ($\sigma = q/4\pi R^2$). By symmetry $\vec{E}$ is radial. (½)

2. **Outside ($r > R$):** Gaussian sphere S_1 of radius r : $E \times 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \text{ — as if all charge were at the centre. (1)}$$

3. **On the surface ($r = R$):** $E = \frac{q}{4\pi\epsilon_0 R^2} = \frac{\sigma}{\epsilon_0}$. (½)

4. **Inside ($r < R$):** Gaussian sphere S_2 encloses no charge:
 $E \times 4\pi r^2 = 0 \Rightarrow E = 0$. (1)

Board question bank



Section A – Multiple choice (½ mark each)

Questions marked [RBSE year] are from official BSER papers (reworded); others are board-pattern questions. Answers at the end of the chapter.

- 1 [RBSE 2025] The SI unit of electric flux is
 (a) C m (b) N C⁻¹ (c) N m² C⁻¹ (d) N m⁻² C
- 2 [RBSE 2025] The force between two point charges is F . If the distance between them is doubled, the force becomes
 (a) $4F$ (b) $F/2$ (c) $F/4$ (d) $2F$
- 3 [RBSE 2024] A point charge is at the centre of a Gaussian sphere of radius 15 cm. If the radius is made three times, the flux through the sphere
 (a) remains the same (b) becomes 3 times
 (c) becomes 9 times (d) becomes 1/9 times
- 4 [RBSE 2023] The value of the permittivity of free space ϵ_0 is
 (a) $8.854 \times 10^{12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ (b) $8.854 \times 10^{-12} \text{ N m}^2 \text{ C}^{-2}$
 (c) $8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ (d) $9 \times 10^9 \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
- 5 Which of the following can NOT be the charge on a body?
 (a) $2.4 \times 10^{-19} \text{ C}$ (b) $1.6 \times 10^{-18} \text{ C}$ (c) $3.2 \times 10^{-19} \text{ C}$ (d) $4.8 \times 10^{-19} \text{ C}$
- 6 The direction of the electric dipole moment is
 (a) perpendicular to the axis (b) from $+q$ to $-q$
 (c) from $-q$ to $+q$ (d) along the equatorial line
- 7 An electric dipole is placed at an angle θ ($0 < \theta < 180^\circ$) in a uniform electric field. It experiences
 (a) zero net force but a torque (b) neither force nor torque
 (c) both a net force and a torque (d) a net force but no torque

- 8 A charge q is placed at the centre of a cube. The electric flux through one face is
 (a) q / ϵ_0 (b) $q / 6\epsilon_0$ (c) $q / 8\epsilon_0$ (d) $6q / \epsilon_0$
- 9 The electric field near a uniformly charged infinite plane sheet (σ) is
 (a) σ/ϵ_0 (uniform) (b) $\sigma/2\epsilon_0 r$ ($\propto 1/r$)
 (c) $\sigma/2\epsilon_0$ (uniform) (d) zero
- 10 The electric field at distance r from an infinitely long uniformly charged wire varies as
 (a) $1/r$ (b) $1/r^3$
 (c) $1/r^2$ (d) independent of r

Fill in the blanks (½ mark)

Question	Answer
[RBSE 2025] At a large distance r , the electric field of a dipole varies as _____.	$1/r^3$
[RBSE 2024] The field lines of an isolated positive charge are radially _____.	outward
[RBSE 2023] The electric field inside a uniformly charged thin spherical shell is _____.	zero
The SI unit of electric dipole moment is _____.	C m
When a medium of dielectric constant K fills the space between two charges, the force becomes _____ times.	$1/K$
The torque on a dipole in a uniform field is maximum when $\theta =$ _____.	90°

Very short answer (1 mark)

Question	Model answer
[RBSE 2023] What are electric field lines?	A curve whose tangent at any point gives the direction of the electric field at that point.
Why can two field lines never intersect?	At the point of intersection E would have two directions, which is impossible.
What is meant by quantisation of charge?	Charge on a body is always an integral multiple of e : $q = \pm ne$.
Define electric dipole moment.	Product of either charge and the separation: $p = q \times 2a$; directed from $-q$ to $+q$.
What is the direction of the field at a point on the equatorial line of a dipole?	Opposite to the dipole moment $\vec{p}$ (parallel to the axis).
State Gauss's law.	Flux through any closed surface $= (1/\epsilon_0) \times$ net charge enclosed: $\oint \vec{E} \cdot d\vec{S} = q / \epsilon_0$.

Short answer – Section B (1½ marks)

 Q1 (1½ marks)

[RBSE 2020] An electron revolves in a circular orbit of diameter 0.30 m around an infinitely long line charge of density 10^{-6} C/m. Find its speed. ($e = 1.6 \times 10^{-19}$ C, $m = 9.1 \times 10^{-31}$ kg)

Solution: Electric force provides the centripetal force: $eE = \frac{mv^2}{r}$ with $E = \frac{2k\lambda}{r}$ (½) $\Rightarrow v = \sqrt{\frac{2k\lambda e}{m}}$ (independent of r) (½)

$$v = \sqrt{\frac{2 \times 9 \times 10^9 \times 10^{-6} \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = 5.6 \times 10^7 \text{ m/s (½)}$$

 Q2 (1½ marks)

[RBSE 2025] An infinite line charge produces a field of 4.5×10^4 N/C at a distance of 4 cm. Find the linear charge density.

Solution: $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$ (½) $\Rightarrow$

$$\lambda = \frac{Er}{2k} = \frac{4.5 \times 10^4 \times 0.04}{2 \times 9 \times 10^9} \text{ (½)} = 1 \times 10^{-7} \text{ C/m (½)}$$

 Q3 (1½ marks)

A dipole of moment 4×10^{-9} C m is aligned at 30° with a uniform field of 5×10^4 N/C. Find the torque on it.

Solution: $\tau = pE \sin \theta$ (½) $= 4 \times 10^{-9} \times 5 \times 10^4 \times \frac{1}{2}$ (½) $= 10^{-4} \text{ N m (½)}$

 Q4 (1½ marks)

Two small charged spheres of 2×10^{-7} C and 3×10^{-7} C are 30 cm apart in air. Find the force between them.

Solution: $F = \frac{kq_1q_2}{r^2}$ (½) $= \frac{9 \times 10^9 \times 2 \times 10^{-7} \times 3 \times 10^{-7}}{(0.3)^2}$ (½) $= 6 \times 10^{-3} \text{ N, repulsive (½)}$

 Q5 (1½ marks)

A point charge of $2.0 \mu\text{C}$ is at the centre of a cubic Gaussian surface of side 9.0 cm. What is the net flux through the surface?

Solution: By Gauss's law $\phi = \frac{q}{\epsilon_0}$ – the side of the cube does not matter (½); $\phi = \frac{2 \times 10^{-6}}{8.854 \times 10^{-12}}$ (½) $= 2.26 \times 10^5 \text{ N m}^2 \text{ C}^{-1} \text{ (½)}$

 Last-minute revision

- $q = ne$; $e = 1.6 \times 10^{-19}$ C; charge is additive, conserved and quantised.
- $F = kq_1q_2/r^2$, $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$, $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$; in a medium F/K .
- $E = F/q_0$ (N C^{-1}); field lines: from + to –, never cross, no closed loops.
- Flux $\phi = ES \cos \theta$, unit $\text{N m}^2 \text{ C}^{-1}$; Gauss: $\oint \mathbf{E} \cdot d\mathbf{S} = q/\epsilon_0$.
- Dipole: $p = q \cdot 2a$ (– to +); $E(\text{axial}) = 2kp/r^3$ (along p), $E(\text{eq}) = kp/r^3$ (opposite to p); $\tau = pE \sin \theta$.
- Line charge $\lambda/2\pi\epsilon_0 r$ ($\propto 1/r$); plane sheet $\sigma/2\epsilon_0$ (constant); shell: kq/r^2 outside, 0 inside.

 Answer Key & Quick Solutions

1(c) 2(c) 3(a) 4(c) 5(a) 6(c) 7(a) 8(b) 9(c) 10(a)

1. (c) $\phi = ES \Rightarrow (\text{N C}^{-1})(\text{m}^2) = \text{N m}^2 \text{ C}^{-1} (= \text{V m})$.

2. (c) $F \propto 1/r^2 \Rightarrow F' = F/2^2 = F/4$.

3. (a) $\phi = q/\epsilon_0$ depends only on the enclosed charge, not on the radius.

4. (c) $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$; $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ is the value of $k = 1/4\pi\epsilon_0$.

5. (a) $q = ne$; $2.4 \times 10^{-19} / 1.6 \times 10^{-19} = 1.5$ is not an integer.

6. (c) By convention $\vec{p}$ points from the negative to the positive charge.
7. (a) $+qE$ and $-qE$ cancel but form a couple: $\tau = pE \sin \theta$.
8. (b) Total flux q / ϵ_0 is shared equally by 6 faces.
9. (c) Gauss: $2EA = \sigma A / \epsilon_0 \Rightarrow E = \sigma / 2\epsilon_0$, no r in the result.
10. (a) $E = \lambda / 2\pi\epsilon_0 r \Rightarrow E \propto 1 / r$.

CHAPTER

2

Electrostatic Potential and Capacitance

Day 3

17

Board marks: 3

PYQ frequency ★★★

RBSE Class 12

Potential & potential energy

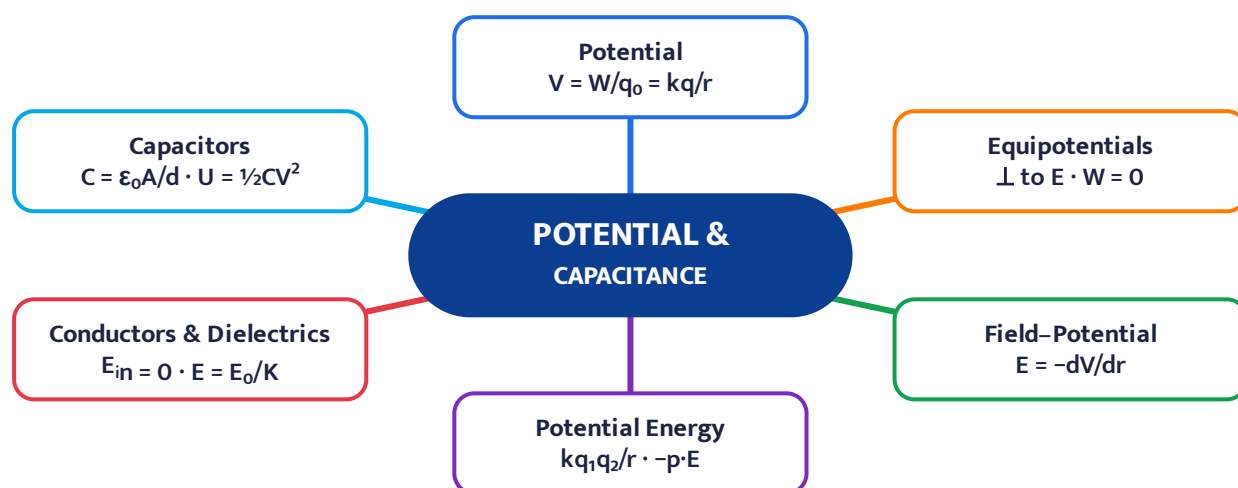
Equipotential surfaces

Potential of a dipole

Conductors & dielectrics

Capacitors & combinations

Energy stored



Chapter at a glance

Board snapshot

- **Weightage: 3 marks** – usually one MCQ / fill-in / VSA (½–1 mark) plus one 1½-mark short answer, or one 3-mark derivation.
- **2025:** MCQ – form of energy stored in a capacitor; 1½ marks – derive the potential at distance r from a charge $+Q$. **2024:** MCQ – dielectric strength of air (3×10^6 V/m); VSA – equipotential surfaces of a positive charge; 1½ marks – three 6 F capacitors in parallel.
- **2023:** 1½ marks – potential of 2×10^{-9} C at 9×10^{-4} m; 1½ marks – potential energy of a system of three point charges. **2026 (CWSN):** fill-in – dielectric strength; 1½ marks – capacitor circuit. **2020:** definition of dielectric constant; potential of a dipole at (r, θ) **OR** spherical capacitor.
- **Most likely in 2027:** (1) potential due to a point charge / dipole; (2) combination of capacitors and parallel-plate capacitance; (3) energy stored in a capacitor and potential energy of point charges.

2.1 Electrostatic Potential and Potential Difference

The electrostatic force is **conservative** – the work done in moving a charge between two points does not depend on the path. So we can define **potential energy** and, per unit charge, a **potential**. The **electrostatic potential at a point** is the work done by an external agent in bringing a unit positive test charge **slowly** (without acceleration) from infinity to that point. It is a **scalar**; SI unit **volt** ($1 \text{ V} = 1 \text{ J C}^{-1}$).

Potential difference $V_B - V_A$ is the work done by an external agent in taking a unit positive charge from A to B. Only potential **differences** have physical meaning; the zero of potential (infinity or the Earth) is a matter of choice.



Potential formulas

$$V = \frac{kq}{r} = \frac{q}{4\pi\epsilon_0 r}$$

Point charge – put the sign of q ;
 $V \propto 1/r$

$$V = k \sum_i \frac{q_i}{r_i}$$

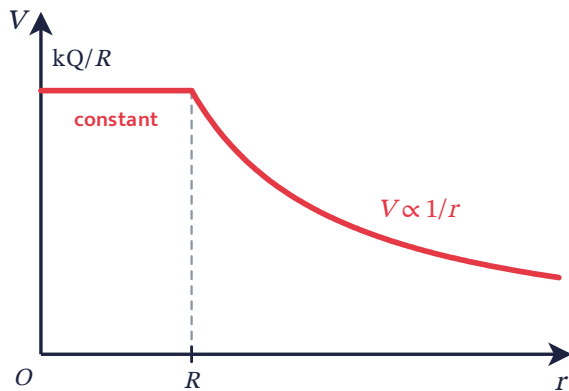
System of charges – plain algebraic (scalar) sum

$$V = \frac{kp \cos \theta}{r^2}$$

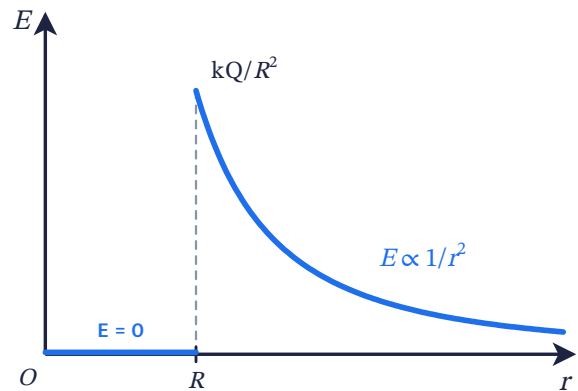
Short dipole: axial kp/r^2 ,
equatorial **zero**; $V \propto 1/r^2$

$$V_{out} = \frac{kQ}{r}, V_{in} = V_s = \frac{kQ}{R}$$

Charged spherical shell /
conducting sphere –
constant inside



V - r graph of a charged shell: constant inside, $1/r$ outside



E - r graph: zero inside, $1/r^2$ outside



Solved Example

Charges 3×10^{-8} C and -2×10^{-8} C are 15 cm apart. Find the points on the line joining them where the potential is zero.

Solution: Let x be the distance from the positive charge.

Between them: $\frac{3}{x} = \frac{2}{15-x} \Rightarrow x = 9$ cm. **Beyond** the negative

charge: $\frac{3}{x} = \frac{2}{x-15} \Rightarrow x = 45$ cm.



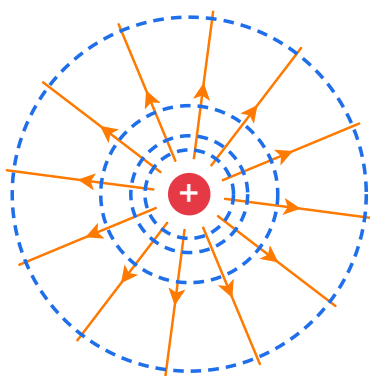
Solved Example

[RBSE 2023] Find the potential at 9×10^{-4} m from a charge of 2×10^{-9} C.

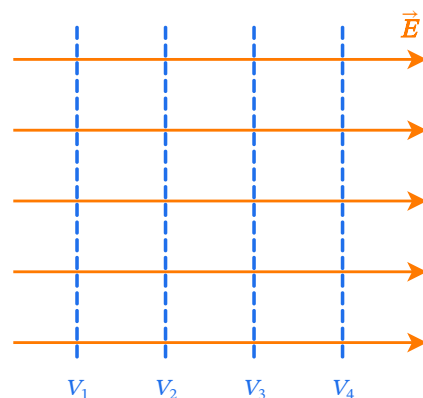
Solution: $V = \frac{kq}{r} = \frac{9 \times 10^9 \times 2 \times 10^{-9}}{9 \times 10^{-4}} = 2 \times 10^4$ V.

2.2 Equipotential Surfaces and $E = -dV/dr$

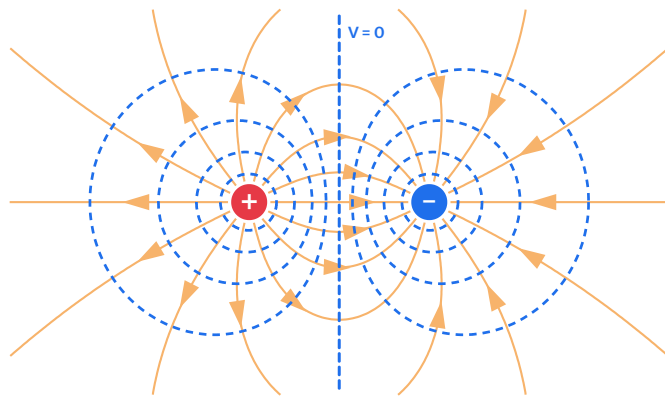
An **equipotential surface** is a surface on which the potential is the same at every point. A single point charge gives **concentric spheres**; a uniform field gives **parallel planes** perpendicular to $\vec{E}$; a dipole gives the curves shown in the figure.



Positive point charge: concentric spheres, closer near q



Uniform field: planes $\perp \vec{E}$, $V_1 > V_2 > V_3 > V_4$



Dipole: equipotentials (dashed blue) cut the field lines at 90°

Properties of equipotential surfaces

- The electric field is always **normal** to an equipotential surface.
- No work** is done in moving a charge over it ($W = q\Delta V = 0$).
- Two equipotential surfaces **never intersect**.
- Where surfaces are closer (equal ΔV), the field is **stronger**.

Field-potential relation

$$E = -\frac{dV}{dr}$$

Field = negative potential gradient; $\vec{E}$ points towards **falling** V

$$V_A - V_B = \int_A^B \vec{E} \cdot d\vec{l}, \quad E = \frac{V}{d} \quad \text{Potential from field; uniform field between plates}$$

Solved Example

$V = (5x^2 + 3)$ volt, x in metre. Find $\vec{E}$ at $x = 2$ m.

Solution: $E_x = -\frac{dV}{dx} = -10x = -20 \text{ V/m}$
 $\Rightarrow 20 \text{ V/m along } -x$.

Common Trap

- $V = 0$ does **not** imply $E = 0$ (equatorial line of a dipole), and $E = 0$ does **not** imply $V = 0$ (inside a charged shell).
- A +ve charge moves from high to low V ; an electron from low to high V .

2.3 Potential Energy of Charges and of a Dipole

The **potential energy of a system** of charges is the work done by an external agent in assembling it from infinity. For a charge q at a point of potential V , $U = qV$. For a dipole in a uniform field, a torque $\tau = pE \sin\theta$ tries to align $\vec{p}$ with $\vec{E}$ (shown in Derivation 7).

Potential energy

$$U = \frac{kq_1q_2}{r_{12}}$$

Two charges (with signs): + for like, - for unlike charges

$$U = k \left(\frac{q_1q_2}{r_{12}} + \frac{q_2q_3}{r_{23}} + \frac{q_1q_3}{r_{13}} \right) \quad \text{Three charges – sum over every pair}$$

$$U = -\vec{p} \cdot \vec{E} = -pE \cos\theta$$

Dipole in uniform field:
 min $-pE$ at $\theta = 0^\circ$ (stable), max $+pE$ at 180° (unstable)

$$\frac{1}{2}mv^2 = q\Delta V$$

Charged particle accelerated through ΔV (in eV: KE = $q\Delta V$)

Solved Example

Charges $+2 \mu\text{C}$, $+2 \mu\text{C}$ and $-4 \mu\text{C}$ sit at the corners of an equilateral triangle of side 10 cm. Find the potential energy of the system.

Solution:

$$U = \frac{k}{a} (q_1q_2 + q_2q_3 + q_1q_3) = \frac{9 \times 10^9}{0.1} (4 - 8 - 8) \times 10^{-12} = -1.08 \text{ J (negative: the system is bound)}$$

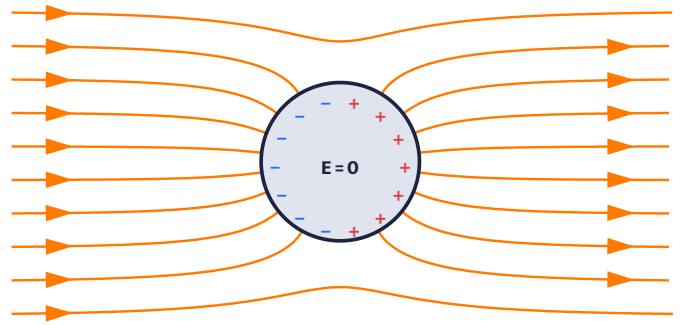
VS Sir's Exam Tip

- Work done by the **field** = $-\Delta U$; by an **external agent** (slowly) = $+\Delta U = q\Delta V$.
- Write q with its **sign** in every pair term; add the three pair-terms – never take magnitudes.

2.4 Conductors and Dielectrics

Key results for a conductor in electrostatics

- $E = 0$ inside the conductor (free charges rearrange until the field vanishes).
- Just outside, $\vec{E}$ is **normal** to the surface, $E = \sigma / \epsilon_0$.
- Excess charge resides **only on the surface**.
- The whole conductor is at **one potential** (an equipotential volume).
- Field inside a hollow cavity is zero – **electrostatic shielding**.

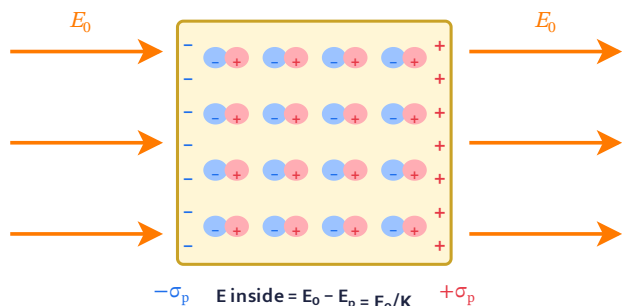


Neutral conductor in a uniform field: induced charges, $E = 0$ inside

NCERT Line to Remember

In a thunderstorm, stay **inside a car** (closed metal body) – it shields you, since the field inside a conducting shell is zero.

Dielectrics are insulators with no free charges. In an external field, **non-polar** molecules (O_2 , N_2 , H_2) acquire induced dipole moments, and **polar** molecules (H_2O , HCl) tend to align their permanent dipoles. Either way the dielectric becomes **polarised**: bound charges $\pm\sigma_p$ appear on its faces and set up a field E_p opposite to E_0 , so the net field inside is reduced: $E = E_0 - E_p = E_0 / K$.



Polarised dielectric slab: bound charges on the faces reduce the field

Dielectric constant and strength

- [RBSE 2020] **Dielectric constant** $K = \frac{\epsilon}{\epsilon_0} = \frac{C}{C_0} = \frac{E_0}{E}$: the ratio of the capacitance with the dielectric completely filling the space between the plates to that in vacuum. $K \geq 1$ (vacuum 1, water ≈ 80 , metal $\rightarrow \infty$).
- **Dielectric strength**: the **maximum electric field** a dielectric can withstand without breakdown. For air it is about 3×10^6 V/m [RBSE 2024, 2026].

2.5 Capacitors and Capacitance

A **capacitor** is two conductors carrying charges $+Q$ and $-Q$, separated by an insulator. As $Q \propto V$, the ratio $C = \frac{Q}{V}$ is a constant – the **capacitance**. It depends only on the **geometry** and the medium, not on Q or V . SI unit **farad** ($1 \text{ F} = 1 \text{ C V}^{-1}$); $1 \mu\text{F} = 10^{-6} \text{ F}$, $1 \text{ pF} = 10^{-12} \text{ F}$.

Capacitance

$$C = \frac{\epsilon_0 A}{d}$$

Parallel plates, air (A = plate area, d = separation)

$$C = \frac{K\epsilon_0 A}{d} = KC_0$$

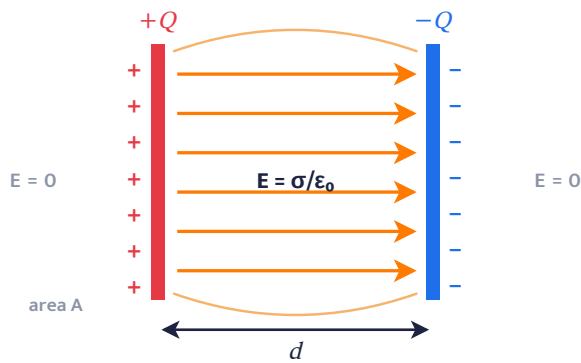
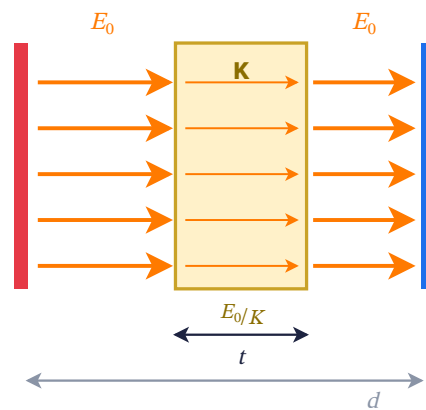
Completely filled with dielectric K

$$C = \frac{\epsilon_0 A}{d - t + t/K}$$

Dielectric slab $t < d$ (metal slab: $K \rightarrow \infty$, $C = \frac{\epsilon_0 A}{d - t}$)

$$C = 4\pi\epsilon_0 R$$

Isolated conducting sphere of radius R

Parallel-plate capacitor: uniform field $E = \sigma/\epsilon_0$ between the platesDielectric slab of thickness $t (< d)$ between the plates**Effect of inserting a dielectric K (filling completely)**Battery **disconnected** (Q constant)

C

Q

V

E

U

 $\times K$

same

 $\div K$ $\div K$ $\div K$ Battery **connected** (V constant) $\times K$ $\times K$

same

same

 $\times K$ **Solved Example**

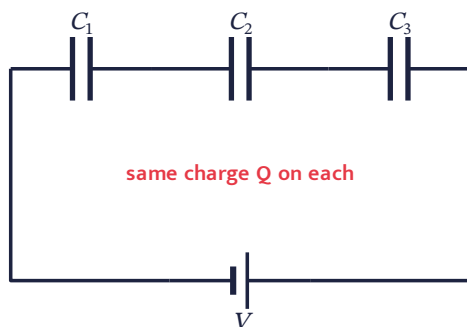
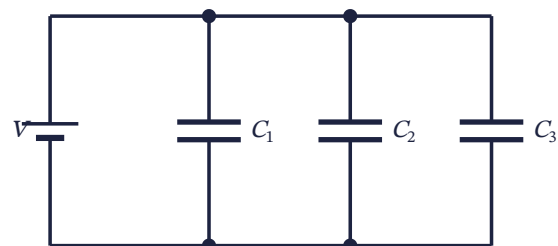
A parallel-plate capacitor has $C_0 = 8 \text{ pF}$ in air. A slab of $K = 6$ fills the gap completely. Find the new capacitance, and the charge when connected to 100 V.

Solution: $C = KC_0 = 6 \times 8 = 48 \text{ pF}$;

$Q = CV = 48 \times 10^{-12} \times 100 = 4.8 \times 10^{-9} \text{ C}$.

Common Trap

- Capacitance depends on A, d and K only – **not** on the charge or the battery voltage.
- "Plates are charged to Q" means +Q on one and -Q on the other (net charge zero).

2.6 Combination of Capacitors and Energy StoredSeries: same charge Q on each, $V = V_1 + V_2 + V_3$ Parallel: same potential difference V, $Q = Q_1 + Q_2 + Q_3$ **Combinations, energy and sharing of charge**

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Series – C_s is **less** than the smallest; $V_1:V_2 = C_2:C_1$

$$C_p = C_1 + C_2 + C_3$$

Parallel – C_p is **more** than the largest

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

Energy stored in a capacitor (J)

$$u = \frac{1}{2}\epsilon_0 E^2$$

Energy density in free space (J m^{-3}); in a dielectric $\frac{1}{2}K\epsilon_0 E^2$

$$V_c = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

Common potential after joining two capacitors (+ plate to + plate)

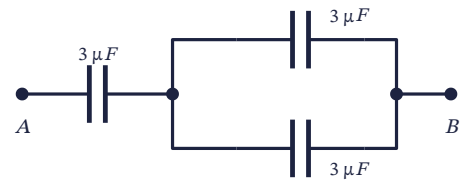
$$\Delta U = \frac{C_1C_2(V_1 - V_2)^2}{2(C_1 + C_2)}$$

Energy lost as heat/radiation on sharing charge

Solved Example

Find the equivalent capacitance of the network between A and B (a $3\ \mu\text{F}$ capacitor in series with two $3\ \mu\text{F}$ capacitors in parallel).

Solution: Parallel pair: $3 + 3 = 6\ \mu\text{F}$. Series: $\frac{3 \times 6}{3 + 6} = 2\ \mu\text{F}$.



Capacitor network

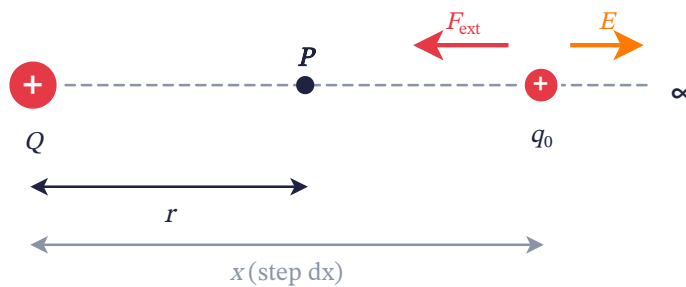
Solved Example

A $600\ \text{pF}$ capacitor is charged to $200\ \text{V}$ and then joined to an uncharged $600\ \text{pF}$ capacitor. Find the common potential and the energy lost.

Solution: $V_c = \frac{600 \times 200}{1200} = 100\ \text{V}$. $U_i = \frac{1}{2}(600\ \text{pF})(200)^2 = 1.2 \times 10^{-5}\ \text{J}$; $U_f = \frac{1}{2}(1200\ \text{pF})(100)^2 = 6 \times 10^{-6}\ \text{J}$. Loss = $6 \times 10^{-6}\ \text{J}$ (half).

VS Sir's Exam Tip

- Series $\rightarrow$ charge same, **voltage divides**; parallel $\rightarrow$ voltage same, **charge divides**. (Opposite to resistors.)
- The energy is stored **in the electric field** between the plates – the form of energy stored in a capacitor is electrostatic potential energy [RBSE 2025].

2.7 Important Derivations ★

Unit positive charge brought slowly from infinity to P against the repulsion of $+Q$

Derivation ★ [3 marks]

Derivation 1 – [RBSE 2025]. Derive an expression for the electrostatic potential at a distance r from a point charge $+Q$.

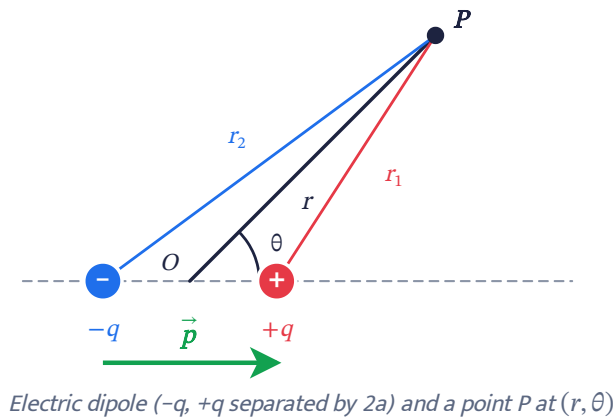
Solution: (i) Potential at P = work done by an external force in bringing a **unit positive charge** slowly from infinity to P (figure). (½)

(ii) At distance x from Q the electric field is $E = \frac{kQ}{x^2}$, directed away from Q . The external force on the unit charge must balance the repulsion, so $F_{\text{ext}} = -E$ (towards Q). (½)

(iii) Work for a small step dx : $dW = F_{\text{ext}} dx = -\frac{kQ}{x^2} dx$. (½)

(iv) Total work: $V = \int_{\infty}^r \left(-\frac{kQ}{x^2} \right) dx = -kQ \left[-\frac{1}{x} \right]_{\infty}^r = \frac{kQ}{r}$. (1)

(v) $V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ – scalar, positive for $+Q$ (negative for $-Q$), falls as $1/r$. (½)



🔧 Derivation ★ [3 marks]

Derivation 2 – [RBSE 2020]. Derive an expression for the potential at a point P at distance r from the centre of a short electric dipole, the line OP making angle θ with the dipole axis.

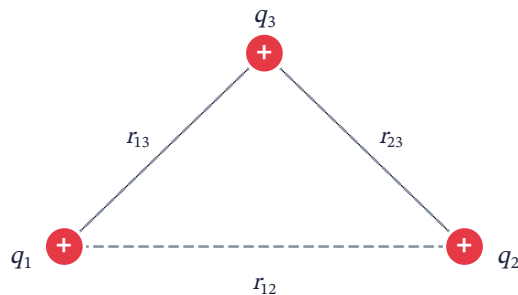
Solution: (i) Let the dipole consist of $-q$ and $+q$ at distance $2a$; $p = q \cdot 2a$. Let $r_1 = BP$ (from $+q$) and $r_2 = AP$ (from $-q$). By the superposition principle $V = kq \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$. (1)

(ii) From the geometry (cosine rule): $r_1^2 = r^2 + a^2 - 2ar \cos \theta$ and $r_2^2 = r^2 + a^2 + 2ar \cos \theta$. For $r \gg a$, neglecting a^2 : $r_1 \approx r \left(1 - \frac{a \cos \theta}{r} \right)$, $r_2 \approx r \left(1 + \frac{a \cos \theta}{r} \right)$. (1)

(iii) Hence $\frac{1}{r_1} - \frac{1}{r_2} \approx \frac{1}{r} \left(\frac{2a \cos \theta}{r} \right) = \frac{2a \cos \theta}{r^2}$. (1/2)

(iv) $V = \frac{kq 2a \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$. (1/2)

(v) Special cases: **axial** ($\theta = 0^\circ$) $V = \frac{kp}{r^2}$; **equatorial** ($\theta = 90^\circ$) $V = 0$. **Dipole potential varies as $1/r^2$.**



Three point charges placed at the corners of a triangle

🔧 Derivation ★ [3 marks]

Derivation 3 – [RBSE 2023]. Obtain an expression for the potential energy of a system of three point charges q_1 , q_2 , q_3 .

Solution: Potential energy = work done by an external agent in bringing the charges **one by one from infinity** to their positions (figure). (1/2)

(i) Bringing q_1 : no other charge is present, so $W_1 = 0$. (1/2)

(ii) Bringing q_2 to distance r_{12} from q_1 : potential due to q_1 there is $\frac{kq_1}{r_{12}}$, so $W_2 = \frac{kq_1 q_2}{r_{12}}$. (1/2)

(iii) Bringing q_3 : potential due to q_1 and q_2 at its position is $k \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right)$, so $W_3 = kq_3 \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right)$. (1/2)

(iv) Adding: $U = W_1 + W_2 + W_3 = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right)$. (1)

The result is independent of the order; the sign of each q must be used.

🔧 Derivation ★ [3 marks]

Derivation 4 (Part 1). Obtain the capacitance of a parallel-plate capacitor (figure in 2.5).

Solution: (i) Two plates of area A , separation d (small), carry $+Q$ and $-Q$; surface charge density $\sigma = \frac{Q}{A}$. (½)

(ii) Field of each plate is $\frac{\sigma}{2\epsilon_0}$; the fields add between the plates and cancel outside: $E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$ (uniform, from + to -). (1)

(iii) Potential difference: $V = Ed = \frac{Qd}{\epsilon_0 A}$. (½)

(iv) $C = \frac{Q}{V} \Rightarrow C = \frac{\epsilon_0 A}{d}$ - depends only on A and d (and the medium). (1)

🔧 Derivation ★ [3 marks]

Derivation 4 (Part 2). A dielectric slab of thickness $t < d$ and dielectric constant K is placed between the plates. Find the capacitance.

Solution: (i) Field in air gap: $E_0 = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$; field inside the slab: $E = \frac{E_0}{K}$ (figure). (1)

(ii) $V = E_0(d - t) + \frac{E_0}{K}t = \frac{Q}{\epsilon_0 A} \left(d - t + \frac{t}{K} \right)$. (1)

(iii) $C = \frac{Q}{V} = \frac{\epsilon_0 A}{d - t + t/K}$. For $t = d$: $C = \frac{K\epsilon_0 A}{d} = KC_0$; for a metal slab ($K \rightarrow \infty$): $C = \frac{\epsilon_0 A}{d - t}$. (1)

🔧 Derivation ★ [3 marks]

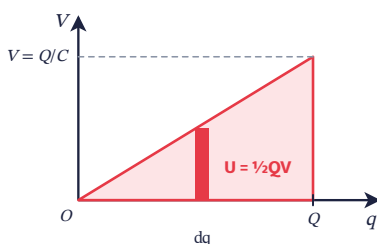
Derivation 5 – [RBSE 2024]. Derive the equivalent capacitance of capacitors joined (a) in series and (b) in parallel (figures in 2.6). Three 6 F capacitors are joined in parallel – find C_{eq} .

Solution: (a) Series: the same charge Q is on every capacitor (inner plates are charged by induction). (½)

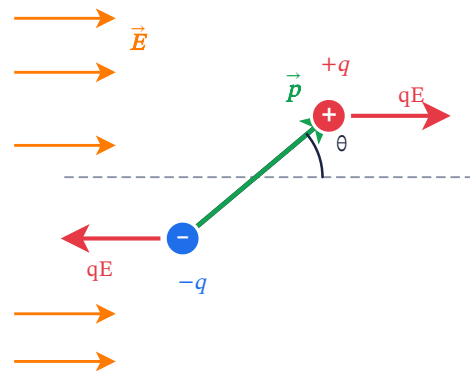
$V = V_1 + V_2 + V_3 = Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$. (½) With $V = \frac{Q}{C_s}$: $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$. (½)

(b) Parallel: the same potential difference V is across each. (½) $Q = Q_1 + Q_2 + Q_3 = (C_1 + C_2 + C_3)V$. (½) With $Q = C_p V$: $C_p = C_1 + C_2 + C_3$. (½)

Numerical: $C_{eq} = 6 + 6 + 6 = 18 \text{ F}$. (In series they would give 2 F.)



(a) q - V graph of a capacitor: shaded area = stored energy (Derivation 6)



(b) Dipole in a uniform field: equal and opposite forces form a couple (Derivation 7)

 Derivation ★ [3 marks]

Derivation 6 – [RBSE 2025]. Derive an expression for the energy stored in a capacitor and hence the energy density of the electric field.

Solution: (i) Charge the capacitor in small steps (figure (a) above). When charge q is already on it, the potential difference is $V = \frac{q}{C}$. (½)

(ii) Work done to add a further charge dq : $dW = V dq = \frac{q}{C} dq$. (½)

(iii) Total work to charge from 0 to Q : $W = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}$. This is stored as potential energy in the field:

$$U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV. \quad (1)$$

(iv) For a parallel-plate capacitor $C = \frac{\epsilon_0 A}{d}$, $V = Ed$: $U = \frac{1}{2} \frac{\epsilon_0 A}{d} E^2 d^2 = \frac{1}{2} \epsilon_0 E^2 (Ad)$. (½)

(v) Volume between plates = Ad , so $u = \frac{U}{Ad} = \frac{1}{2} \epsilon_0 E^2$ (J m^{-3}). (½)

 Derivation ★ [3 marks]

Derivation 7. Find the torque on an electric dipole in a uniform electric field and the potential energy of the dipole.

Solution: (i) Forces on the two charges (figure (b) above): $+q\vec{E}$ and $-q\vec{E}$ – equal and opposite, so net force is zero but they form a **couple**. (½)

(ii) Perpendicular distance between the forces = $2a \sin \theta$. Torque $\tau = qE \times 2a \sin \theta \Rightarrow \tau = pE \sin \theta$, i.e. $\vec{\tau} = \vec{p} \times \vec{E}$. (1)

(iii) Work done by an external torque in rotating the dipole from θ_0 to θ :

$$W = \int_{\theta_0}^{\theta} pE \sin \theta d\theta = pE(\cos \theta_0 - \cos \theta). \quad (1)$$

(iv) Taking $\theta_0 = 90^\circ$ as the zero of energy ($\cos \theta_0 = 0$): $U = -pE \cos \theta = -\vec{p} \cdot \vec{E}$. (½)

 VS Sir's Exam Tip

- Examiner's checklist: labelled figure $\rightarrow$ each step with its formula $\rightarrow$ final result boxed. In integrals, write the **limits** (∞ to r ; 0 to Q) – a missing limit costs ½ mark.
- Scalar potential: add potentials **algebraically** with signs; never use vector addition for V .

 Asked earlier by RBSE

The **spherical capacitor** (2020, alternative to the dipole potential) is outside the current rationalised NCERT book. Learn only the result: for inner sphere radius a and outer radius b , $C = 4\pi\epsilon_0 \frac{ab}{b-a}$; for an isolated sphere ($b \rightarrow \infty$) $C = 4\pi\epsilon_0 a$.

Board question bank



Section A – Multiple choice (½ mark each)

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- 1 [RBSE 2025] The energy stored in a charged capacitor is in the form of
 - (a) magnetic energy
 - (b) heat energy
 - (c) electrostatic potential energy of the electric field
 - (d) kinetic energy of the charges
- 2 [RBSE 2024] The dielectric strength of air is approximately
 - (a) 3×10^8 V/m
 - (b) 3×10^6 V/m
 - (c) 3×10^3 V/m
 - (d) 9×10^9 V/m
- 3 The potential at a distance r from a point charge is 60 V. At a distance $3r$ it will be
 - (a) 180 V
 - (b) 20 V
 - (c) 6.7 V
 - (d) 60 V
- 4 Three $6 \mu\text{F}$ capacitors are joined in series. The equivalent capacitance is
 - (a) $18 \mu\text{F}$
 - (b) $0.5 \mu\text{F}$
 - (c) $2 \mu\text{F}$
 - (d) $6 \mu\text{F}$
- 5 If the plate separation of a parallel-plate capacitor is halved and the plate area is doubled, its capacitance becomes
 - (a) half
 - (b) unchanged
 - (c) 2 times
 - (d) 4 times
- 6 The work done in moving a charge between two points of the same equipotential surface is
 - (a) depends on the path
 - (b) negative
 - (c) positive
 - (d) zero
- 7 A conducting sphere of radius 10 cm has a potential of 100 V on its surface. The potential at 5 cm from its centre is
 - (a) 200 V
 - (b) 50 V
 - (c) zero
 - (d) 100 V
- 8 The energy stored in a $10 \mu\text{F}$ capacitor charged to 100 V is
 - (a) 5 J
 - (b) 0.005 J
 - (c) 0.5 J
 - (d) 0.05 J
- 9 The electric potential at any point on the equatorial line of an electric dipole is
 - (a) $\frac{kp}{r}$
 - (b) $\frac{kp}{r^2}$
 - (c) zero
 - (d) $\frac{2kp}{r^3}$
- 10 If $V = 4x^2$ volt (x in metre), the electric field at $x = 1$ m is
 - (a) 16 V/m along $+x$
 - (b) 8 V/m along $-x$
 - (c) 4 V/m along $-x$
 - (d) 8 V/m along $+x$

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2026] The maximum electric field a dielectric can withstand without breakdown is its ____.	dielectric strength
The SI unit of capacitance is ____.	farad
The work done in moving a charge on an equipotential surface is ____.	zero
Capacitors joined in series carry the same ____.	charge
The dielectric constant of a conductor (metal) is ____.	infinite
The potential of a point charge varies with distance as ____.	1/r

Very short answer (1 mark)

Question	Model answer
[RBSE 2024] Draw the equipotential surfaces of a positive point charge.	Concentric spheres centred on the charge (figure in 2.2); they are crowded near the charge and normal to the radial field.
[RBSE 2020] Define the dielectric constant.	Ratio of the capacitance with the dielectric filling the space between the plates to that in vacuum: $K = C / C_0 = \epsilon / \epsilon_0$.

Question	Model answer
Define 1 volt.	Potential difference between two points if 1 joule of work is done in moving 1 coulomb between them: $1 \text{ V} = 1 \text{ J C}^{-1}$.
Why is the electric field always normal to an equipotential surface?	If it had a component along the surface, work would be done in moving a charge on it, contradicting constant V.
Write the expression for energy density of an electric field.	$u = \frac{1}{2} \epsilon_0 E^2 \text{ (J m}^{-3}\text{)}.$
Why does a dielectric increase the capacitance?	Polarisation reduces the field to E_0 / K , so V falls to V/K for the same Q: $C = Q / V$ rises K times.

Short answer (Section B, 1½ marks)

 Short answer (1½ marks)

[RBSE 2024] Three capacitors of 6 F each are connected in parallel. Find the equivalent capacitance. What would it be in series?

Solution: Parallel: $C_p = C_1 + C_2 + C_3$ (½) $= 6 + 6 + 6 = 18 \text{ F. (½)}$ In series: $\frac{1}{C_s} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2} \Rightarrow 2 \text{ F. (½)}$

 Short answer (1½ marks)

[RBSE 2023] Find the electric potential at a distance of $9 \times 10^{-4} \text{ m}$ from a point charge of $2 \times 10^{-9} \text{ C}$.

Solution: $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ (½) $= \frac{9 \times 10^9 \times 2 \times 10^{-9}}{9 \times 10^{-4}}$ (½) $= 2 \times 10^4 \text{ V. (½)}$

 Short answer (1½ marks)

[RBSE 2023] Write the expression for the potential energy of a system of three point charges q_1, q_2, q_3 at mutual distances r_{12}, r_{23}, r_{13} . What does a negative value mean?

Solution: $U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right)$ (1) (derivation 3). A negative U means the charges form a **bound system** – work must be done on them to separate them to infinity. (½)

 Short answer (1½ marks)

Find the equivalent capacitance between A and B of the network in section 2.6 (3 μF – then two 3 μF in parallel).

Solution: The two upper 3 μF capacitors are in **parallel**: $3 + 3 = 6 \mu\text{F}$. (½) This 6 μF is in **series** with the first 3 μF :
 $C_{AB} = \frac{3 \times 6}{3 + 6}$ (½) = **2 μF** . (½)

 Short answer (1½ marks)

Write any three properties of equipotential surfaces.

Solution: (i) The field is normal to an equipotential surface at every point. (ii) No work is done in moving a charge over it. (iii) Two equipotential surfaces never intersect (otherwise V would have two values at one point). (½ each)

 Answer Key & Quick Solutions

1(c) 2(b) 3(b) 4(c) 5(d) 6(d) 7(d) 8(d) 9(c) 10(b)

1. (c) Work done in charging is stored as electrostatic potential energy in the electric field between the plates, $U = \frac{1}{2} CV^2$.
2. (b) Air breaks down in a field of about $3 \times 10^6 \text{ V/m}$ (NCERT value).
3. (b) $V \propto 1/r \Rightarrow 60/3 = 20 \text{ V}$.
4. (c) $\frac{1}{C_s} = \frac{1}{6} \times 3 = \frac{1}{2} \Rightarrow C_s = 2 \mu\text{F}$.
5. (d) $C = \epsilon_0 A / d \Rightarrow (2A) / (d/2) = 4$ times.
6. (d) $W = q(V_B - V_A) = 0$ since $V_B = V_A$.
7. (d) Inside a conductor the potential is constant and equals the surface value.
8. (d) $U = \frac{1}{2} CV^2 = \frac{1}{2} \times 10 \times 10^{-6} \times 10^4 = 0.05 \text{ J}$.
9. (c) $V = \frac{kp \cos \theta}{r^2}$ with $\theta = 90^\circ$ gives 0.
10. (b) $E = -\frac{dV}{dx} = -8x = -8 \text{ V/m}$, i.e. 8 V/m along $-x$.

 Last-minute revision

- $V = \frac{kq}{r}$ (scalar, add with signs); dipole: $V = \frac{kp \cos \theta}{r^2}$ – equatorial $V = 0$.
- $E = -\frac{dV}{dr}$; $E \perp$ equipotential; $W = 0$ on an equipotential; no two ever meet.
- Three charges: $U = k \sum_{\text{pairs}} \frac{q_i q_j}{r_{ij}}$; dipole: $U = -pE \cos \theta, \tau = pE \sin \theta$.
- Conductor: $E = 0$ inside, charge on surface, one potential; shielding. Dielectric: $E = E_0 / K$.
- $C = \frac{Q}{V}$; $C = \frac{\epsilon_0 A}{d}$; with slab $\frac{\epsilon_0 A}{d - t + t/K}$; filled: KC_0 .
- Series: $\frac{1}{C_s} = \sum \frac{1}{C}$ (same Q); parallel: $C_p = \sum C$ (same V).
- $U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$; $u = \frac{1}{2} \epsilon_0 E^2$; dielectric strength of air = $3 \times 10^6 \text{ V/m}$.

CHAPTER

3

Current Electricity

17

Day 4–5

Board marks: 4

PYQ frequency ★★★

RBSE Class 12

Drift velocity & mobility

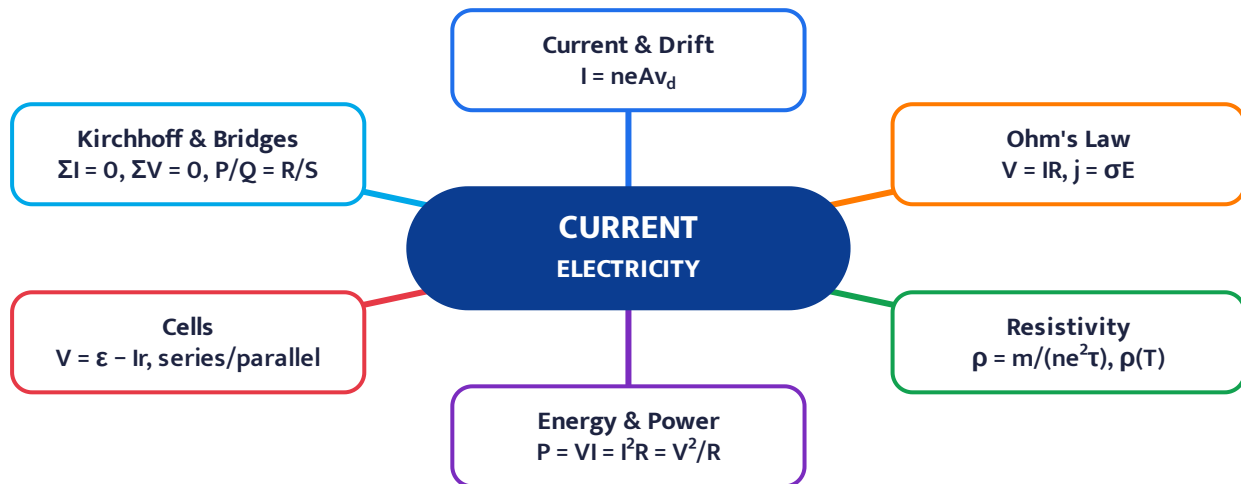
Ohm's law & resistivity

emf & internal resistance

Cells in series/parallel

Kirchhoff's rules

Wheatstone bridge



Chapter at a glance



Board snapshot

- **Weightage: 4 marks** – usually one MCQ / fill-in (½–1 mark) plus one 3-mark derivation or a 1½-mark question.
- **2025:** MCQ – ammeter reading in a circuit; fill-in – mobility; 3 marks – balance condition of Wheatstone bridge by Kirchhoff's laws **OR** internal resistance + two cells in series (ε_{eq}, r_{eq}).
- **2024:** MCQ – SI unit of resistivity; MCQ – balanced Wheatstone bridge (30 Ω, 18 Ω, 10 Ω, S = 6 Ω); fill-in – mobility; VSA – V–I graph (Ohm's law); 1½ marks – define emf and internal resistance.
- **2023:** MCQ – tolerance (colour code); 3 marks – Wheatstone bridge diagram + balance condition. **2020:** Kirchhoff's second law; three resistors in parallel (find the third).
- **Most likely in 2027:** (1) Wheatstone bridge balance by Kirchhoff's laws (asked 2023 and 2025); (2) cells in series / parallel – ε_{eq} and r_{eq} ; (3) drift velocity, mobility and $I = neAv_d$.

3.1 Electric Current and Current Density

Electric current is the rate of flow of charge through any cross-section of a conductor: $I = \frac{dq}{dt}$ (steady current) or $I = \frac{dq}{dt}$ (instantaneous). SI unit **ampere** ($1 \text{ A} = 1 \text{ C s}^{-1}$). The conventional direction of current is the direction of flow of **positive** charge – opposite to the flow of electrons. Current has a direction but is a **scalar** (it does not follow the laws of vector addition).



Current and current density

$$I = \frac{dq}{dt}$$

Current (A); $1 \text{ A} = 6.25 \times 10^{18}$ electrons per second

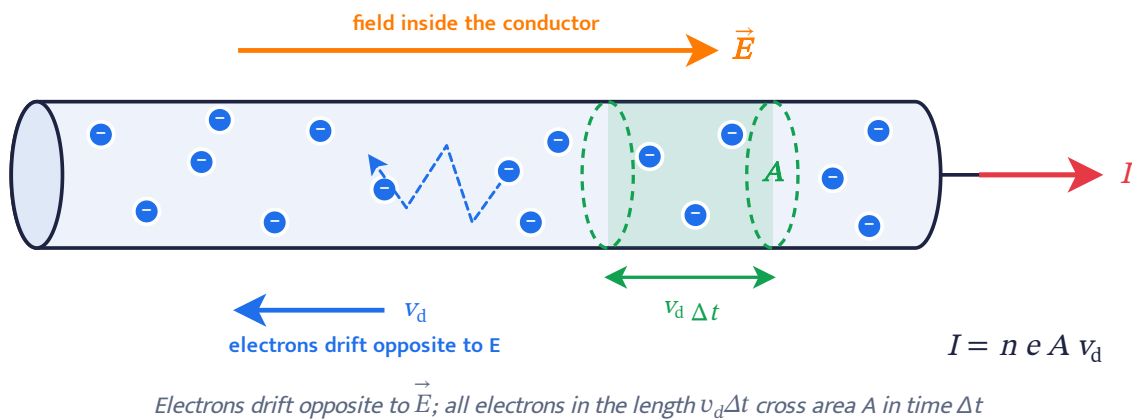
$$j = \frac{I}{A}$$

Current density – current per unit area normal to flow; vector, A m^{-2}

3.2 Drift Velocity, Relaxation Time and Mobility

Free electrons in a metal move randomly with thermal speeds ($\sim 10^5$ m/s) and keep colliding with the positive ions; with no field their **average velocity is zero**. When a field $\vec{E}$ is applied, every electron is accelerated opposite to $\vec{E}$ between collisions and acquires a small net velocity – the **drift velocity**.

- **Drift velocity** (v_d): the average velocity with which free electrons drift opposite to the applied electric field ($\sim 10^{-4}$ m/s).
- **Relaxation time** (τ): the average time between two successive collisions of a free electron ($\sim 10^{-14}$ s). It **decreases** as temperature rises.
- **Mobility** (μ): the magnitude of drift velocity per unit electric field, $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$. SI unit $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$; always positive.



Drift and mobility

$$v_d = \frac{eE}{m}\tau$$

Drift speed (derived in 3.10)

$$I = neAv_d$$

n = number of free electrons per m^3

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

Mobility

$$\sigma = \frac{ne^2\tau}{m} = ne\mu$$

Conductivity (S m^{-1})

Solved Example

A copper wire of cross-section 1.0 mm^2 carries 2.0 A . Taking $n = 8.5 \times 10^{28} \text{ m}^{-3}$, find the drift speed of the electrons.

Solution:
$$v_d = \frac{I}{neA} = \frac{2.0}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 1.0 \times 10^{-6}} = 1.47 \times 10^{-4} \text{ m/s (about 0.15 mm/s).}$$

Solved Example

Electrons drift at $2.5 \times 10^{-4} \text{ m/s}$ in a field of 0.5 V/m . Find their mobility.

Solution:
$$\mu = \frac{v_d}{E} = \frac{2.5 \times 10^{-4}}{0.5} = 5 \times 10^{-4} \text{ m}^2 \text{V}^{-1} \text{s}^{-1}.$$

Common Trap

- If drift speed is so small, why does a bulb glow at once? Because the **electric field** is set up along the whole wire almost at the speed of light, and all free electrons start drifting together.
- In a wire of non-uniform cross-section, **I is the same** everywhere, but j and v_d are larger where the wire is thinner ($v_d \propto 1/A$).

3.3 Ohm's Law, Resistance and Resistivity

Ohm's law: if the temperature and other physical conditions remain unchanged, the current through a conductor is directly proportional to the potential difference across its ends: $V \propto I$ or $V = IR$. The constant **R** is the **resistance** (SI unit **ohm**, $\Omega = \text{V A}^{-1}$). For a uniform wire $R = \rho \frac{l}{A}$, where **ρ is the resistivity** – it depends only on the material and temperature, not on the shape. SI unit of resistivity: **$\Omega \text{ m}$** . Its reciprocal $\sigma = 1 / \rho$ is the **conductivity** (S m^{-1}).



Ohm's law and resistivity

$$V = IR, \quad R = \rho \frac{l}{A} \quad \text{Ohm's law; resistance of a uniform wire}$$

$$\vec{j} = \sigma \vec{E}$$

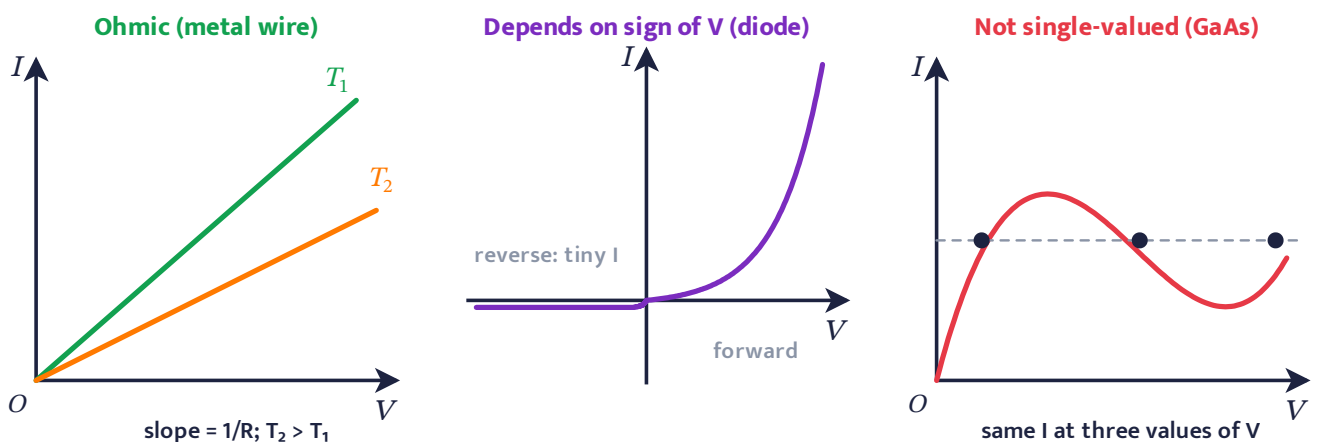
Ohm's law – **vector (microscopic) form**

$$\rho = \frac{m}{ne^2\tau}$$

Resistivity from electron theory

$$R' = n^2 R$$

Wire stretched to n times its length (volume constant)



V - I characteristics: the **straight line through the origin** is the graph of an ohmic conductor; the other two are non-ohmic



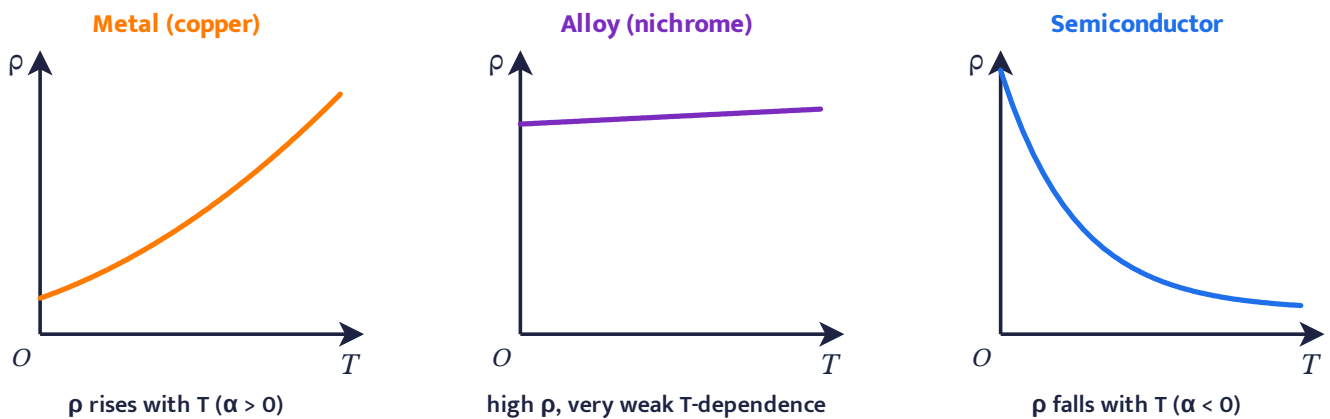
Limitations of Ohm's law (non-ohmic devices)

- V stops being proportional to I (graph not a straight line).
- The relation depends on the **sign** of V – e.g. a diode.
- The relation is **not unique** – more than one V for the same I , e.g. GaAs.

Material	ρ ($\Omega \text{ m}$)
Metals (Ag, Cu, Al)	$10^{-8} - 10^{-6}$
Alloys (nichrome, manganin)	$\sim 10^{-6}$
Semiconductors (Si, Ge)	$\sim 10^{-5} - 10^6$
Insulators (glass, rubber)	$10^{14} - 10^{16}$

3.4 Temperature Dependence of Resistivity

Over a limited range, $\rho_T = \rho_0 [1 + \alpha(T - T_0)]$, where α is the **temperature coefficient of resistivity** (unit $^{\circ}\text{C}^{-1}$ or K^{-1}). From $\rho = \frac{m}{ne^2\tau}$: in a **metal** n is constant but τ decreases with T , so ρ **increases** (α positive). In a **semiconductor** n increases rapidly with T , so ρ **decreases** (α negative). **Alloys** (nichrome, manganin, constantan) have high ρ and very small α – used for standard resistors.



Resistivity versus temperature for a metal, an alloy and a semiconductor

Solved Example

A silver wire has resistance $2.1 \, \Omega$ at 27.5°C and $2.7 \, \Omega$ at 100°C . Find the temperature coefficient of resistivity of silver.

Solution: $\alpha = \frac{R_2 - R_1}{R_1(T_2 - T_1)} = \frac{2.7 - 2.1}{2.1 \times 72.5} = 3.9 \times 10^{-3} \, ^\circ\text{C}^{-1}$.

3.5 Electrical Energy, Power and Combination of Resistors

When a current I flows through a resistor R for time t , the energy $W = VIt$ is converted into heat (**Joule heating**).

Electric power is the rate at which electrical energy is consumed: $P = VI = I^2R = \frac{V^2}{R}$ (unit watt). Power is transmitted at high voltage because the line loss $I^2R_c = \frac{P^2R_c}{V^2}$ falls as V rises.

Power and combinations

$$P = VI = I^2R = \frac{V^2}{R} \quad \text{Electric power (W)}$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J} \quad \text{Commercial unit ("unit")}$$

$$R_s = R_1 + R_2 + R_3 \quad \text{Series}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad \text{Parallel}$$

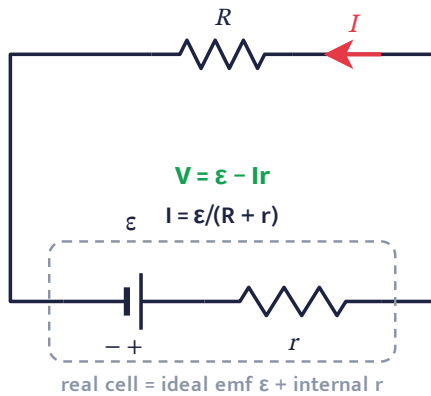
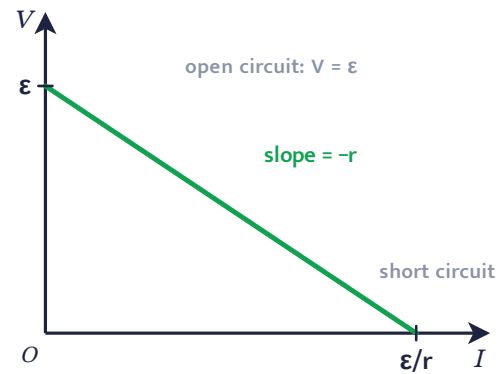
Solved Example

Find the resistance of a 100 W, 220 V bulb and the current it draws.

Solution: $R = \frac{V^2}{P} = \frac{220^2}{100} = 484 \, \Omega$; $I = \frac{P}{V} = \frac{100}{220} \approx 0.45 \text{ A}$.

3.6 Cells: emf, Internal Resistance and Terminal Voltage

- emf (ϵ)** of a cell: the potential difference between its terminals when **no current** is drawn from it (open circuit). Equivalently, the work done by the cell in taking unit positive charge once round the complete circuit. Unit: volt. (emf is not a force.)
- Internal resistance (r)**: the resistance offered by the electrolyte (and electrodes) of the cell to the flow of current inside the cell. It increases with distance between electrodes and concentration of electrolyte, and decreases with area of the electrodes.
- Terminal voltage (V)**: the potential difference between the terminals when current I is drawn: $V = \epsilon - Ir$ (discharging), $V = \epsilon + Ir$ (charging).

Cell of emf ε and internal resistance r driving an external resistance R Terminal voltage falls linearly with current: $V = \varepsilon - Ir$ **Cell formulas**

$$I = \frac{\varepsilon}{R + r} \quad \text{Current in the circuit}$$

$$V = IR = \varepsilon - Ir \quad \text{Terminal voltage}$$

$$r = R \left(\frac{\varepsilon}{V} - 1 \right) \quad \text{Internal resistance from } \varepsilon, V \text{ and } R$$

Solved Example

A cell of emf 1.5 V gives a terminal voltage of 1.2 V across a 4 Ω resistor. Find its internal resistance.

Solution: $I = V / R = 1.2 / 4 = 0.3 \text{ A}$. $r = \frac{\varepsilon - V}{I} = \frac{0.3}{0.3} = 1 \Omega$.
(Check: $r = 4(1.5 / 1.2 - 1) = 1 \Omega$.)

Common Trap

- $V = \varepsilon$ only on **open circuit** ($I = 0$). A voltmeter of finite resistance reads slightly **less** than ε .
- On short circuit ($R = 0$): $V = 0$ and the current is maximum, $I = \varepsilon / r$.

3.7 Cells in Series and in Parallel

Combination	Equivalent emf	Equivalent internal resistance
Two cells in series (+ of one to - of other)	$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$	$r_{eq} = r_1 + r_2$
Series, one cell reversed	$\varepsilon_{eq} = \varepsilon_1 - \varepsilon_2$	$r_{eq} = r_1 + r_2$
Two cells in parallel	$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$	$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$
n identical cells in series	$n\varepsilon$	nr
m identical cells in parallel	ε	r / m

Solved Example

Cells of emf 2 V ($r = 0.5 \Omega$) and 1.5 V ($r = 0.5 \Omega$) are joined in series with a 4 Ω resistor. Find the current.

Solution: $\varepsilon_{eq} = 3.5 \text{ V}$, $r_{eq} = 1 \Omega \Rightarrow I = \frac{3.5}{4 + 1} = 0.7 \text{ A}$.

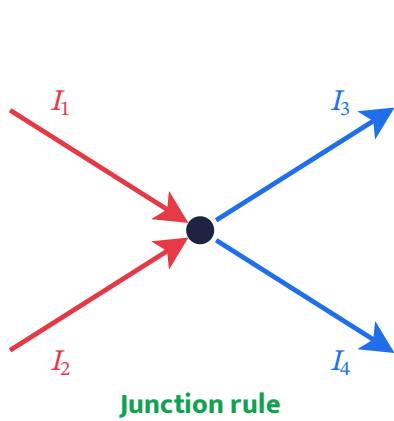
Solved Example

Cells of 2 V (1 Ω) and 1 V (1 Ω) are joined in parallel (like terminals together). Find ε_{eq} and r_{eq} .

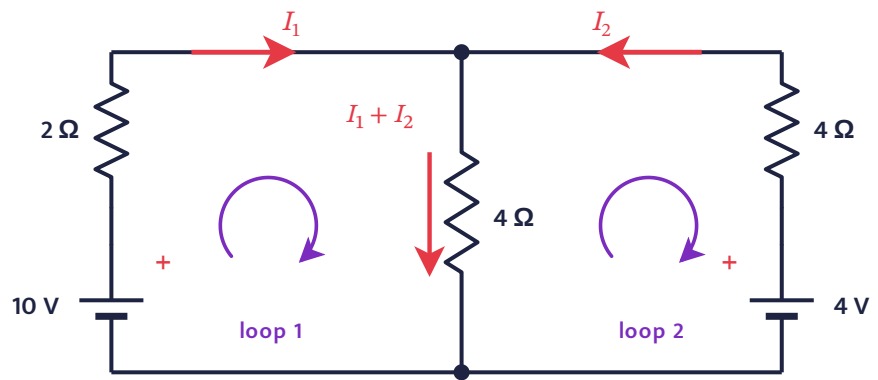
Solution: $\varepsilon_{eq} = \frac{2 \times 1 + 1 \times 1}{1 + 1} = 1.5 \text{ V}$; $r_{eq} = \frac{1 \times 1}{2} = 0.5 \Omega$.

3.8 Kirchhoff's Rules

- **First rule (junction rule):** at any junction, the sum of currents entering the junction is equal to the sum of currents leaving it ($\sum I = 0$). It follows from **conservation of charge**.
- **Second rule (loop rule):** the algebraic sum of changes in potential around any closed loop of resistors and cells is zero ($\sum \varepsilon = \sum IR$). It follows from **conservation of energy**.



$$I_1 + I_2 = I_3 + I_4$$



Loop rule: $\sum(\text{emf}) = \sum(IR)$ round any closed loop

Junction rule (left) and a two-loop circuit (right) solved below

💡 Sign convention (loop rule)

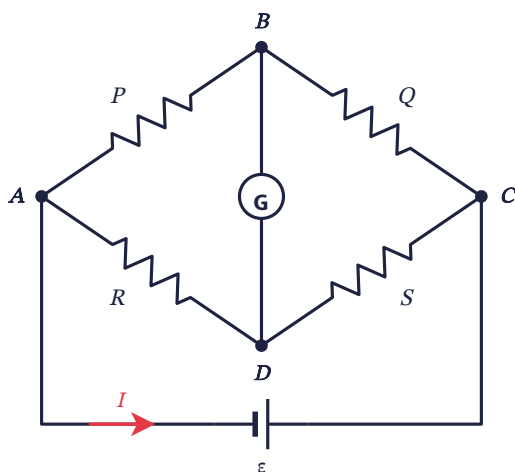
- Resistor traversed **along** the current: $-IR$; against it: $+IR$.
- Cell traversed from $-$ to $+$: $+\varepsilon$; from $+$ to $-$: $-\varepsilon$.
- A negative answer only means the real current is opposite to the assumed one.

🔑 Solved Example

Find I_1 , I_2 and the current in the middle 4Ω in the circuit above.

Solution: Loop 1: $10 - 2I_1 - 4(I_1 + I_2) = 0 \Rightarrow 6I_1 + 4I_2 = 10$. Loop 2: $4 - 4I_2 - 4(I_1 + I_2) = 0 \Rightarrow 4I_1 + 8I_2 = 4$. Solving: $I_1 = 2\text{ A}$, $I_2 = -0.5\text{ A}$ (actual direction reversed), middle $= I_1 + I_2 = 1.5\text{ A}$.

3.9 Wheatstone Bridge



Wheatstone bridge ABCD

Four resistances P, Q, R, S are joined to form a quadrilateral ABCD. A cell is connected between A and C and a galvanometer between B and D. When the galvanometer shows **no deflection** ($I_g = 0$), points B and D are at the same potential and the bridge is said to be **balanced**:

$$\frac{P}{Q} = \frac{R}{S}$$

🎯 VS Sir's Exam Tip

Balance condition is proved by Kirchhoff's laws in Derivation 2 (section 3.10) – asked in 2023 and 2025. Draw the bridge with **all six currents marked**.

! Common Trap

- In a balanced bridge the galvanometer arm carries no current – it may be removed or replaced by any resistor without changing anything.
- Balance does **not** depend on the emf of the cell or on G.

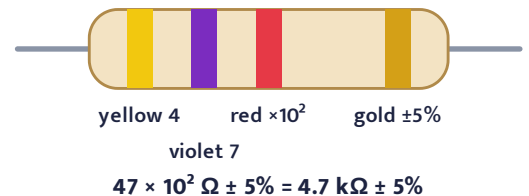
🔧 Solved Example

[RBSE 2024] In a balanced Wheatstone bridge $P = 30 \Omega$, $Q = 18 \Omega$ and $R = 10 \Omega$. Find S.

Solution: $S = \frac{QR}{P} = \frac{18 \times 10}{30} = 6 \Omega$.

📝 Asked earlier by RBSE

- Resistor colour code** (2023 MCQ on tolerance) is **outside the current rationalised NCERT book**, so learn only this much: digits Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9 ("BB ROY of Great Britain has a Very Good Wife"). The 3rd band is the multiplier (power of 10).
Tolerance (4th band): gold $\pm 5\%$, silver $\pm 10\%$, no band $\pm 20\%$.



Example: yellow–violet–red–gold resistor

3.10 Important Derivations ★**🔧 Derivation ★ [3 marks]**

Derivation 1 (Part 1). Obtain an expression for the drift velocity of free electrons in a conductor and hence show that $I = neAv_d$.

Solution: (i) In field $\vec{E}$ each electron feels force $-e\vec{E}$, so its acceleration is $\vec{a} = -\frac{e\vec{E}}{m}$. (½)

(ii) If electron i had velocity $\vec{u}_i$ just after its last collision, then after time t_i : $\vec{v}_i = \vec{u}_i - \frac{e\vec{E}}{m}t_i$. (½)

(iii) Averaging over all electrons: average of $\vec{u}_i = 0$ (random directions) and average of $t_i = \tau$ (relaxation time). Hence

$$\vec{v}_d = -\frac{e\vec{E}}{m}\tau, \text{ i.e. } v_d = \frac{eE\tau}{m}. \quad (1)$$

(iv) Consider a conductor of area A with n free electrons per unit volume (figure in 3.2). In time Δt , all electrons within length $v_d \Delta t$ cross the section: number = $nAv_d \Delta t$, charge $\Delta Q = neAv_d \Delta t$. (½)

$$(v) I = \frac{\Delta Q}{\Delta t} \Rightarrow I = neAv_d. \quad (½)$$

🔧 Derivation ★ [3 marks]

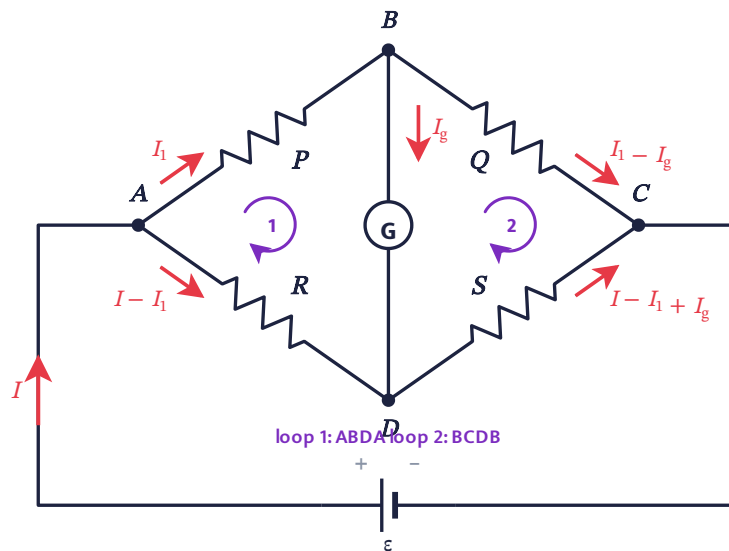
Derivation 1 (Part 2). Using the drift velocity, deduce Ohm's law and obtain the expressions for resistivity and mobility.

Solution: (i) Current density $j = \frac{I}{A} = nev_d = ne \cdot \frac{eE\tau}{m} \Rightarrow j = \frac{ne^2\tau}{m}E$, i.e. $\vec{j} = \sigma \vec{E}$ with $\sigma = \frac{ne^2\tau}{m}$. (1)

(ii) For a wire of length l: $E = \frac{V}{l}$ and $j = \frac{I}{A} \Rightarrow \frac{I}{A} = \sigma \frac{V}{l} \Rightarrow V = \frac{l}{\sigma A} I = IR$ – this is **Ohm's law**, since R is constant at constant temperature. (1)

(iii) $R = \rho \frac{l}{A}$ with $\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$. (½)

(iv) Mobility $\mu = \frac{v_d}{E} = \frac{\sigma e\tau}{m}$ (so $\sigma = ne\mu$). (½)



Wheatstone bridge with branch currents (from the junction rule) and the two loops used

🔧 Derivation ★ [3 marks]

Derivation 2 – [RBSE 2025, 2023]. Draw a labelled diagram of a Wheatstone bridge and, using Kirchhoff's laws, obtain the condition for its balance.

Solution: (i) **Diagram** as above: P, Q, R, S in arms AB, BC, AD, DC; galvanometer G between B and D; cell between A and C. (½)

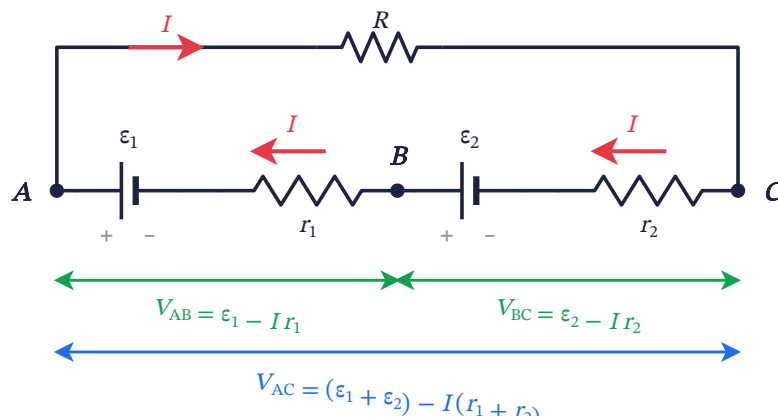
(ii) **Junction rule:** let I enter at A; I_1 flows in P and $(I - I_1)$ in R; at B, I_g goes through G, so $(I_1 - I_g)$ flows in Q and $(I - I_1 + I_g)$ in S. (½)

(iii) **Loop ABDA:** $-I_1P - I_gG + (I - I_1)R = 0 \dots(1)$ (½)

(iv) **Loop BCDB:** $-(I_1 - I_g)Q + (I - I_1 + I_g)S + I_gG = 0 \dots(2)$ (½)

(v) **Balance:** $I_g = 0$. Then (1) gives $I_1P = (I - I_1)R$ and (2) gives $I_1Q = (I - I_1)S$. (½)

(vi) Dividing: $\frac{P}{Q} = \frac{R}{S}$ – the balance condition. (½)



Two cells in series between A and C, driving an external resistor R

🔧 Derivation ★ [3 marks]

Derivation 3 – [RBSE 2025]. Define internal resistance of a cell. Two cells of emf $\varepsilon_1, \varepsilon_2$ and internal resistances r_1, r_2 are joined in series. Find the equivalent emf and equivalent internal resistance.

Solution: (i) **Internal resistance:** the resistance offered by the electrolyte and electrodes of a cell to the flow of current inside it. (½)

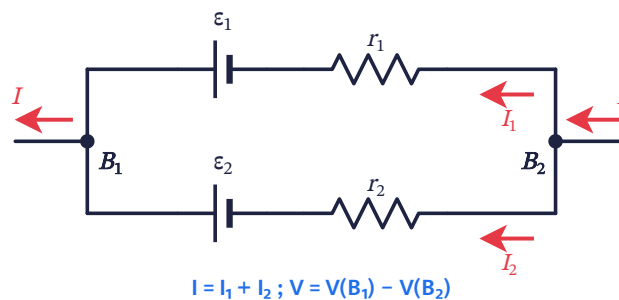
(ii) Let current I flow (figure). For cell 1 (A is its + terminal): $V_{AB} = V_A - V_B = \varepsilon_1 - Ir_1$. (½)

(iii) For cell 2: $V_{BC} = V_B - V_C = \varepsilon_2 - Ir_2$. (½)

(iv) Adding: $V_{AC} = V_{AB} + V_{BC} = (\varepsilon_1 + \varepsilon_2) - I(r_1 + r_2)$. (½)

(v) If the combination is replaced by a single cell (ε_{eq}, r_{eq}): $V_{AC} = \varepsilon_{eq} - Ir_{eq}$. Comparing: $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$ and $r_{eq} = r_1 + r_2$. (½)

(vi) If cell 2 is reversed, $\varepsilon_{eq} = \varepsilon_1 - \varepsilon_2$ ($\varepsilon_1 > \varepsilon_2$) but still $r_{eq} = r_1 + r_2$. (½)



Two cells in parallel between B_1 and B_2

🔧 Derivation ★ [3 marks]

Derivation 4. Two cells (ε_1, r_1) and (ε_2, r_2) are connected in parallel. Find the equivalent emf and internal resistance.

Solution: (i) Both cells have the same terminal voltage $V = V(B_1) - V(B_2)$: $V = \varepsilon_1 - I_1 r_1$ and $V = \varepsilon_2 - I_2 r_2$. (½)

(ii) So $I_1 = \frac{\varepsilon_1 - V}{r_1}$, $I_2 = \frac{\varepsilon_2 - V}{r_2}$. By the junction rule $I = I_1 + I_2 = \frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$. (1)

(iii) Rearranging: $V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$. (½)

(iv) Comparing with $V = \varepsilon_{eq} - Ir_{eq}$: $\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$, $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$. (1)

🎯 VS Sir's Exam Tip

- Examiner's checklist: labelled diagram with current directions → equations numbered (1), (2) → condition / result boxed. A derivation without a diagram rarely gets full marks.
- In the Wheatstone derivation write the **loop names** (ABDA, BCDB) – they show that you applied Kirchhoff's second law.

Board question bank

📝 Section A – Multiple choice (½ mark each)

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

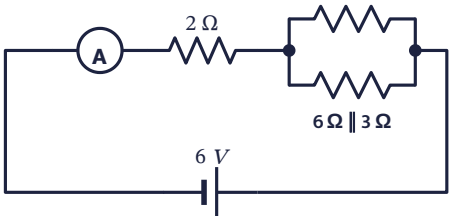
1 [RBSE 2024] The SI unit of resistivity is

(a) $\Omega \text{ m}$

(b) $\Omega \text{ m}^{-1}$

(c) S m^{-1}

(d) Ω

- 2 [RBSE 2024] In a balanced Wheatstone bridge the arms are $P = 30\ \Omega$, $Q = 18\ \Omega$, $R = 10\ \Omega$ and S . The value of S is
 (a) $3\ \Omega$ (b) $54\ \Omega$ (c) $16.7\ \Omega$ (d) $6\ \Omega$
- 3 [RBSE 2023] In the colour code of a carbon resistor, a gold fourth band means a tolerance of
 (a) $\pm 10\%$ (b) $\pm 5\%$ (c) $\pm 20\%$ (d) $\pm 1\%$
- 4 In the circuit shown (ideal 6 V cell and ideal ammeter), the ammeter reads

 (a) 3 A (b) 1.5 A (c) 0.67 A (d) 0.55 A
- 5 The SI unit of mobility of charge carriers is
 (a) $\text{V m}^{-1} \text{ s}$ (b) $\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$
 (c) m s^{-1} (d) $\text{m V}^{-1} \text{ s}^{-1}$
- 6 A wire is connected to a fixed potential difference. If its length is doubled, the drift velocity of electrons
 (a) remains the same (b) becomes four times
 (c) becomes double (d) becomes half
- 7 Kirchhoff's second (loop) rule is based on the conservation of
 (a) momentum (b) mass (c) energy (d) charge
- 8 A cell of emf 2 V and internal resistance $0.5\ \Omega$ is connected to a $3.5\ \Omega$ resistor. The terminal voltage is
 (a) 1.5 V (b) 1.75 V (c) 0.25 V (d) 2 V
- 9 When the temperature of a semiconductor is increased, its resistivity
 (a) decreases
 (b) remains unchanged
 (c) increases
 (d) first increases then becomes constant
- 10 The resistance of a 100 W , 220 V electric bulb is
 (a) $2.2\ \Omega$ (b) $0.45\ \Omega$ (c) $484\ \Omega$ (d) $22\text{ k}\Omega$

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2025, 2024] The drift velocity per unit electric field is called _____.	mobility
The average time between two successive collisions of a free electron is called _____.	relaxation time
Kirchhoff's junction rule is based on the conservation of _____.	charge
The resistivity of a metal _____ with increase in temperature.	increases
In a balanced Wheatstone bridge the current through the galvanometer is _____.	zero
In an open circuit the terminal voltage of a cell is equal to its _____.	emf

Very short answer (1 mark)

Question	Model answer
[RBSE 2024] Draw the V–I graph of a conductor obeying Ohm's law.	A straight line through the origin (figure in 3.3); slope of I–V graph = $1/R$.
[RBSE 2020] State Kirchhoff's second law.	In any closed loop, the algebraic sum of emfs = algebraic sum of IR products: $\sum \varepsilon = \sum IR$ (energy conservation).
Is current density a scalar or a vector? Give its SI unit.	Vector ; $A\ m^{-2}$.
How does the relaxation time of electrons in a metal change with temperature?	It decreases (more frequent collisions), so resistivity increases.
Why is manganin used for making standard resistors?	High resistivity and very small temperature coefficient α .
Write one relation between emf ε , terminal voltage V and internal resistance r .	$V = \varepsilon - Ir$ (or $r = R(\varepsilon / V - 1)$).

Short answer (Section B, 1½ marks)

 Short answer (1½ marks)

[RBSE 2024] Define the emf and the internal resistance of a cell.

Solution: **emf:** the potential difference between the terminals of a cell when no current is drawn from it (open circuit); unit volt. (¾) **Internal resistance:** the resistance offered by the electrolyte and electrodes of the cell to the current flowing inside it; unit ohm. (¾)

 Short answer (1½ marks)

[RBSE 2020] Three resistors are connected in parallel to a 30 V source and draw a total current of 7.5 A. Two of them are $10\ \Omega$ and $12\ \Omega$. Find the third.

Solution: $R_p = \frac{V}{I} = \frac{30}{7.5} = 4\ \Omega$. (½) $\frac{1}{R_3} = \frac{1}{4} - \frac{1}{10} - \frac{1}{12} = \frac{15-6-5}{60} = \frac{4}{60}$ (½) $\Rightarrow R_3 = 15\ \Omega$. (½) (Check: currents $3\ A + 2.5\ A + 2\ A = 7.5\ A$ ✓)

 Short answer (1½ marks)

Distinguish between the emf of a cell and its terminal voltage (three points).

Solution: (i) emf is the p.d. in **open** circuit; terminal voltage is the p.d. when current flows. (ii) emf is independent of R ; $V = \varepsilon - Ir$ depends on the current drawn. (iii) emf is the cause (work per unit charge), terminal voltage is the effect; $V < \varepsilon$ while discharging. (½ each)

 Short answer (1½ marks)

Why is the Wheatstone bridge method of measuring resistance said to be a "null method"? Give one advantage.

Solution: Measurement is made when the galvanometer shows **zero deflection** ($I_g = 0$). (1) Advantage: the result does not depend on the emf or internal resistance of the cell or on the galvanometer resistance. (½)

 Answer Key & Quick Solutions

1(a) 2(d) 3(b) 4(b) 5(b) 6(d) 7(c) 8(b) 9(a) 10(c)

1. (a) $\rho = RA / l \Rightarrow \Omega \times \text{m}^2 / \text{m} = \Omega \text{ m}$. (S m^{-1} is the unit of conductivity.)
2. (d) $P / Q = R / S \Rightarrow S = QR / P = 180 / 30 = 6 \Omega$.
3. (b) Gold $\pm 5\%$, silver $\pm 10\%$, no band $\pm 20\%$.
4. (b) $6 \Omega \parallel 3 \Omega = 2 \Omega$; total $2 + 2 = 4 \Omega \Rightarrow I = 6 / 4 = 1.5 \text{ A}$.
5. (b) $\mu = v_d / E = (\text{m s}^{-1}) / (\text{V m}^{-1}) = \text{m}^2 \text{V}^{-1} \text{s}^{-1}$.
6. (d) $v_d = \frac{eE\tau}{m}$ with $E = V / l \Rightarrow v_d \propto 1 / l$; halves.
7. (c) Loop rule $\Rightarrow$ energy; junction rule $\Rightarrow$ charge.
8. (b) $I = 2 / (3.5 + 0.5) = 0.5 \text{ A}$; $V = IR = 0.5 \times 3.5 = 1.75 \text{ V}$.
9. (a) n increases rapidly with $T \Rightarrow \rho$ decreases (α negative).
10. (c) $R = V^2 / P = 48400 / 100 = 484 \Omega$.

 Last-minute revision

- $I = neAv_d$; $v_d = eE\tau / m$; $\mu = v_d / E = e\tau / m (\text{m}^2 \text{V}^{-1} \text{s}^{-1})$.
- $\rho = m / ne^2\tau (\Omega \text{ m})$; $\vec{j} = \sigma \vec{E}$ is the vector form of Ohm's law.
- Metals: ρ rises with T ; semiconductors: ρ falls; alloys: high ρ , tiny α .
- $V = \varepsilon - Ir$; $I = \varepsilon / (R + r)$; $r = R(\varepsilon / V - 1)$.
- Series cells: $\varepsilon_1 + \varepsilon_2, r_1 + r_2$. Parallel: $\frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}, \frac{r_1 r_2}{r_1 + r_2}$.
- Junction rule $\rightarrow$ charge; loop rule $\rightarrow$ energy.
- Balanced bridge: $P / Q = R / S, I_g = 0$ – prove with loops ABDA and BCDB.
- Colour code tolerance: gold $\pm 5\%$, silver $\pm 10\%$, none $\pm 20\%$.

CHAPTER

4

Moving Charges and Magnetism

17

Day 6–7

Board marks: 4

PYQ frequency ★★★

RBSE Class 12

Lorentz force

Circular motion in B

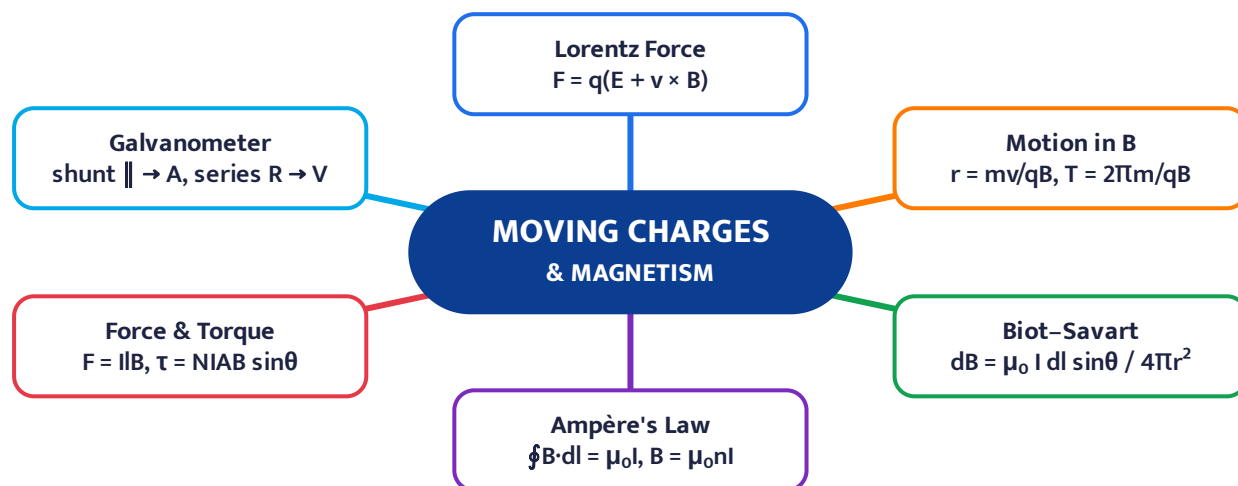
Biot–Savart law

Ampère's law & solenoid

Parallel currents

Torque on a loop

Galvanometer, ammeter, voltmeter



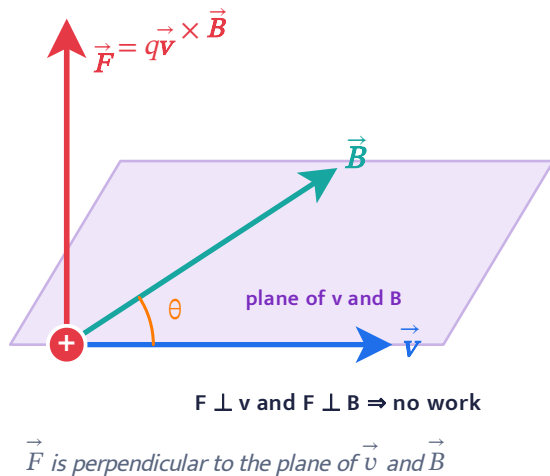
Chapter at a glance

Board snapshot

- **Weightage: 4 marks** – usually one MCQ + one fill-in (½ mark each) and one **3-mark derivation** (Section C) – or a 4-mark Section D question.
- **2025:** MCQ – force on a charge moving parallel to $\vec{B}$; fill-in – shunt (low resistance in parallel) for an ammeter; 3 marks – Ampère's law and field of an infinite wire **OR** Biot–Savart law and field on the axis of a circular loop.
- **2024:** MCQ – path of a charge moving $\perp$ to $\vec{B}$ (circle); fill-in – high resistance in series for a voltmeter; 3 marks – Biot–Savart, field on the axis of a loop **OR** force per unit length between parallel wires. **2023:** fill-in – force between parallel currents; 3 marks – field on the axis of a long solenoid by Ampère's law.
- **2026 (CWSN):** 4 marks (Section D) – Biot–Savart law and field of a loop **OR** Ampère's law and field of an infinite wire.
- **Most likely in 2027:** (1) Biot–Savart law + field on the axis of a circular loop (asked in 2024, 2025, 2026); (2) Ampère's law: infinite wire / solenoid; (3) force between parallel currents and the definition of ampere; (4) galvanometer $\rightarrow$ ammeter / voltmeter.

4.1 Magnetic Force and Lorentz Force

A charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences the **magnetic force** $\vec{F} = q(\vec{v} \times \vec{B})$. In the presence of an electric field also, the total force is the **Lorentz force** $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. SI unit of B is **tesla** ($1\text{ T} = 1\text{ N A}^{-1}\text{ m}^{-1}$); $1\text{ gauss} = 10^{-4}\text{ T}$.



Force on a moving charge

$$F = qvB \sin\theta \quad \theta = \text{angle between } \vec{v} \text{ and } \vec{B}$$

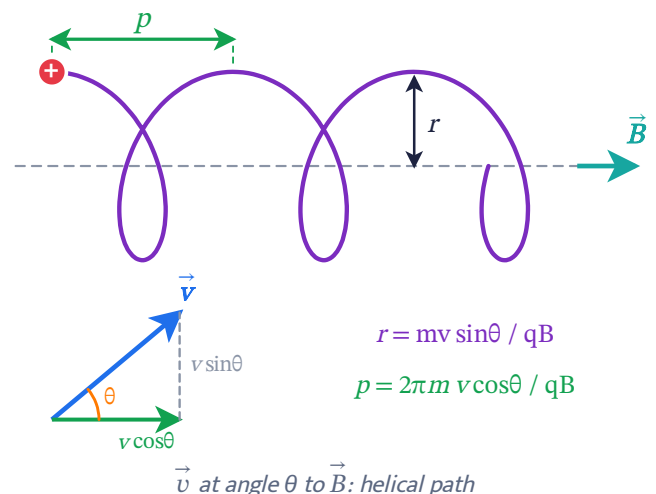
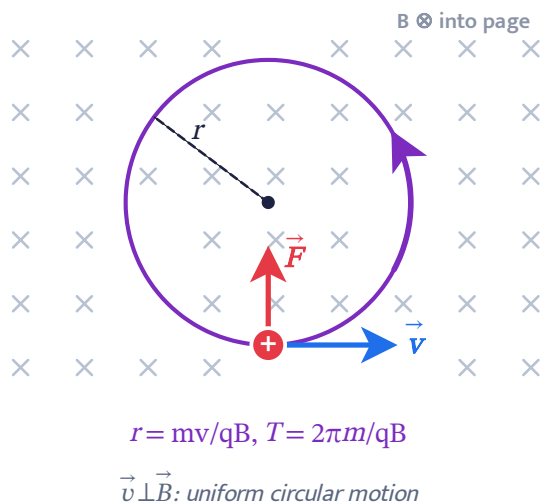
$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) \quad \text{Lorentz force}$$

Key Concept

- $F = 0$ if the charge is at rest or $\vec{v} \parallel \vec{B}$ [RBSE 2025]; F is maximum (qvB) if $\vec{v} \perp \vec{B}$.
- $\vec{F} \perp \vec{v}$ always $\Rightarrow$ the magnetic force does **no work**; speed and kinetic energy are constant.

4.2 Motion of a Charged Particle in a Magnetic Field

If $\vec{v} \perp \vec{B}$, the magnetic force qvB is always perpendicular to the velocity and supplies the **centripetal force**: the particle moves on a **circle** [RBSE 2024]. If $\vec{v}$ makes an angle θ with $\vec{B}$, the component $v \cos\theta$ along $\vec{B}$ is unaffected, so the path is a **helix**.



Circular and helical motion

$$r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

Radius of the circular path (K = kinetic energy)

$$T = \frac{2\pi m}{qB}, \quad \nu = \frac{qB}{2\pi m}$$

Period and (cyclotron) frequency – **independent of v and r**

$$v = \frac{E}{B}$$

Velocity selector – undeflected in crossed $\vec{E}$ and $\vec{B}$

$$p = \frac{2\pi m v \cos\theta}{qB}$$

Pitch of the helix

Solved Example

A proton ($m = 1.67 \times 10^{-27}$ kg) enters a field of 0.1 T perpendicularly with a speed of 2×10^6 m/s. Find the radius and the time period.

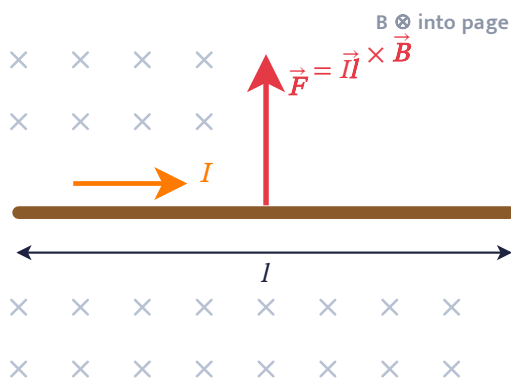
Solution: $r = \frac{mv}{qB} = \frac{1.67 \times 10^{-27} \times 2 \times 10^6}{1.6 \times 10^{-19} \times 0.1} = 0.21$ m; $T = \frac{2\pi m}{qB} = 6.6 \times 10^{-7}$ s.

Common Trap

- Doubling the speed doubles r but leaves T **unchanged**.
- A magnetic field can **never** change the kinetic energy of a charge – only the direction of motion.

Cyclotron (in brief)

A cyclotron accelerates positive ions using a **uniform magnetic field** (which makes them move in semicircles inside two D-shaped hollow electrodes, the "dees") and a high-frequency **alternating electric field** across the gap, whose frequency equals the cyclotron frequency $\nu = \frac{qB}{2\pi m}$ (resonance). The ion gains energy at every crossing of the gap; $K_{\max} = \frac{q^2 B^2 R^2}{2m}$ at the dee radius R .

4.3 Force on a Current-Carrying Conductor

Force on a straight wire in a uniform field

A current is a drift of charges, so a wire of length l in a field $\vec{B}$ feels the sum of the forces on its carriers: $\vec{F} = I(\vec{l} \times \vec{B})$, $F = BIl \sin \theta$. Direction: **Fleming's left-hand rule** (forefinger – B , middle finger – I , thumb – force).

Solved Example

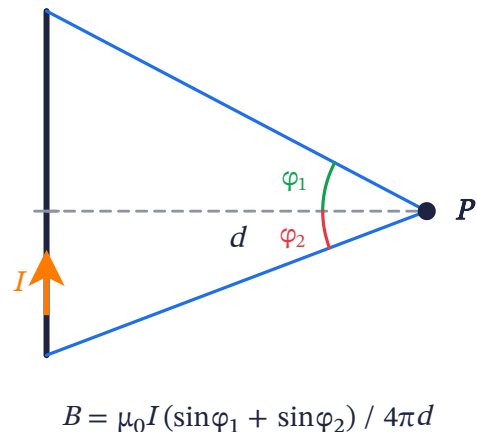
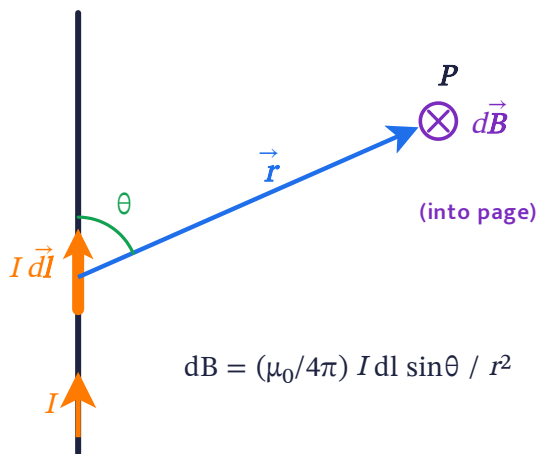
A 3 m wire carrying 5 A lies at 30° to a field of 0.2 T. Find the force.

Solution: $F = BIl \sin \theta = 0.2 \times 5 \times 3 \times \sin 30^\circ = 1.5$ N.

4.4 Biot–Savart Law

The magnetic field $d\vec{B}$ at a point P due to a small current element $I d\vec{l}$ is **directly proportional** to I , dl and $\sin \theta$, and **inversely proportional** to the square of the distance r (θ = angle between $d\vec{l}$ and $\vec{r}$). Its direction is that of $d\vec{l} \times \vec{r}$ – perpendicular to the plane containing them.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}, \quad \frac{\mu_0}{4\pi} = 10^{-7} \text{ T m A}^{-1}$$



Left: current element and $d\vec{B}$ at P. Right: angles for a finite straight wire

Straight conductors

$$B = \frac{\mu_0 I}{2\pi d} \quad \text{Infinitely long straight wire (d = perpendicular distance)}$$

$$B = \frac{\mu_0 I}{4\pi d} (\sin \phi_1 + \sin \phi_2) \quad \text{Finite straight wire}$$

Solved Example

Find B at 10 cm from a long straight wire carrying 5 A.

Solution: $B = \frac{\mu_0 I}{2\pi d} = \frac{2 \times 10^{-7} \times 5}{0.1} = 1.0 \times 10^{-5} \text{ T.}$

4.5 Magnetic Field of a Circular Current Loop

For a loop of radius R with N turns carrying current I, the field at a point on the axis at distance x from the centre is obtained from the Biot-Savart law (Derivation 1). The direction is given by the **right-hand thumb rule**: curl the fingers of the right hand along the current – the thumb gives the direction of $\vec{B}$ on the axis.

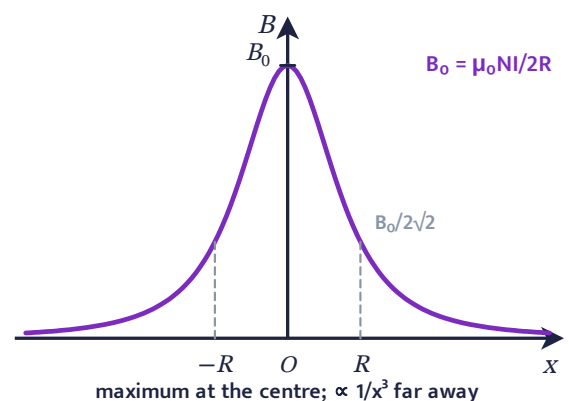
Loop and arc

$$B_{\text{axis}} = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}} \quad \text{On the axis, distance x from the centre}$$

$$B_{\text{centre}} = \frac{\mu_0 N I}{2R} \quad \text{At the centre (x = 0) – maximum}$$

$$B \approx \frac{\mu_0 2m}{4\pi x^3} \quad \text{Far away (x \gg R); } m = NIA$$

$$B_{\text{arc}} = \frac{\mu_0 I \theta}{4\pi R} \quad \text{Centre of an arc of angle } \theta \text{ (radians)}$$



B along the axis falls off with distance from the centre

Solved Example

A 100-turn circular coil of radius 10 cm carries 3.2 A. Find B at its centre.

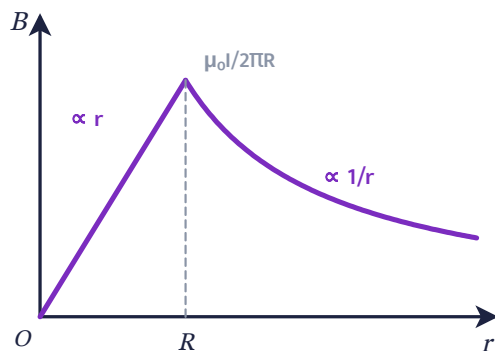
Solution: $B = \frac{\mu_0 N I}{2R} = \frac{4\pi \times 10^{-7} \times 100 \times 3.2}{2 \times 0.10} = 2.0 \times 10^{-3} \text{ T.}$

Memory Trick

- Arc family at the centre: full loop $\frac{\mu_0 I}{2R} \rightarrow$ semicircle $\frac{\mu_0 I}{4R} \rightarrow$ quarter circle $\frac{\mu_0 I}{8R}$.
- Straight parts pointing **towards the centre** give zero.

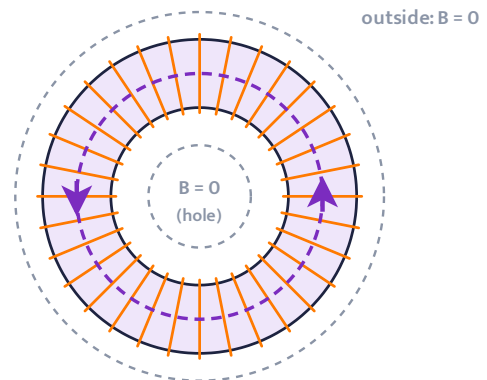
4.6 Ampère's Circuital Law, Solenoid and Toroid

Ampère's circuital law: the line integral of $\vec{B}$ around any closed path equals μ_0 times the total current passing through the surface bounded by it: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$. It is always true, but is used to find B only where the symmetry lets B be taken out of the integral.



Thick wire (radius R), uniform current

B inside and outside a long thick wire



$$B = \mu_0 NI / 2\pi r \text{ (inside the core)}$$

Toroid: dashed circles are Ampèrian loops



Results of Ampère's law

$$B = \frac{\mu_0 I}{2\pi r}$$

Long straight wire, outside ($r \geq R$)

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

Inside a long thick wire ($r < R$)

$$B = \mu_0 n I$$

Inside a long solenoid; n = turns per metre

$$B = \frac{\mu_0 N I}{2\pi r}$$

Toroid (inside the core); zero outside

Solved Example

A solenoid has 500 turns on a length of 0.5 m and carries 2 A. Find B inside it near the middle ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$).

$$\text{Solution: } n = \frac{500}{0.5} = 1000 \text{ m}^{-1};$$

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 1000 \times 2 = 2.5 \times 10^{-3} \text{ T.}$$

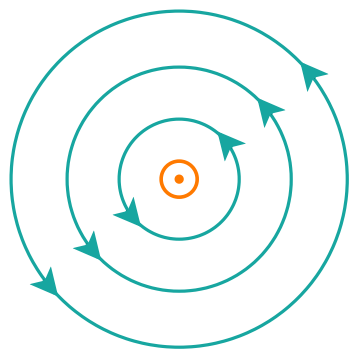


NCERT Line to Remember

- Inside a long solenoid B is **uniform**, along the axis, independent of its radius and length; outside it is nearly **zero**.
- At either end of a long solenoid $B = \frac{\mu_0 n I}{2}$.

4.7 Force Between Two Parallel Currents; the Ampere

Each wire is in the magnetic field of the other. **Parallel currents attract; antiparallel currents repel** – the opposite of the rule for electric charges. The force per unit length is $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$ [RBSE 2023, 2024] (derived in Derivation 2).

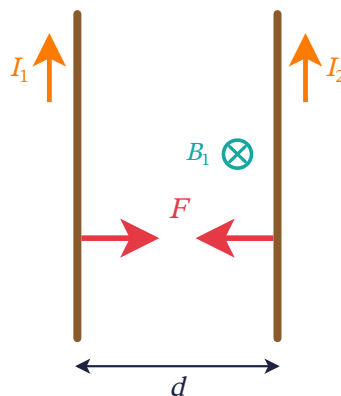


$I \odot$ out of page: B anticlockwise

$$B = \mu_0 I / 2\pi r$$

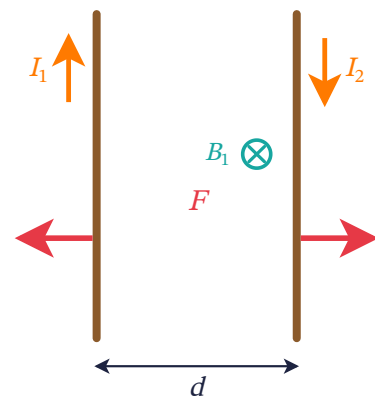
Field lines of a straight wire; like currents attract, unlike currents repel

Like currents attract



$$F/l = \mu_0 I_1 I_2 / 2\pi d$$

Unlike currents repel



Definition of the ampere

One ampere is that steady current which, flowing in each of two infinitely long, straight, parallel conductors of negligible cross-section placed 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre of length between them.

Solved Example

Two long parallel wires 5 cm apart carry 10 A each in the same direction. Find the force per metre.

Solution: $\frac{F}{l} = \frac{2 \times 10^{-7} \times 10 \times 10}{0.05} = 4 \times 10^{-4}$ N/m, attractive.

4.8 Torque on a Current Loop; Magnetic Dipole Moment

In a uniform field the net force on a current loop is **zero**, but the forces on opposite sides form a **couple** (Derivation 6). The loop behaves as a **magnetic dipole** of moment $\vec{m} = NI\vec{A}$ (SI unit $A m^2$), directed along the normal given by the right-hand thumb rule.

Loop in a uniform field

$$\tau = NIAB \sin \theta \quad \theta = \text{angle between normal and } \vec{B}$$

$$\vec{\tau} = \vec{m} \times \vec{B} \quad \text{Vector form; } m = NI A$$

$$U = -\vec{m} \cdot \vec{B} \quad \text{Potential energy of the dipole}$$

Solved Example

A rectangular coil of 100 turns, $10 \text{ cm} \times 5 \text{ cm}$, carries 2 A. Its plane is parallel to a field of 0.1 T. Find the torque.

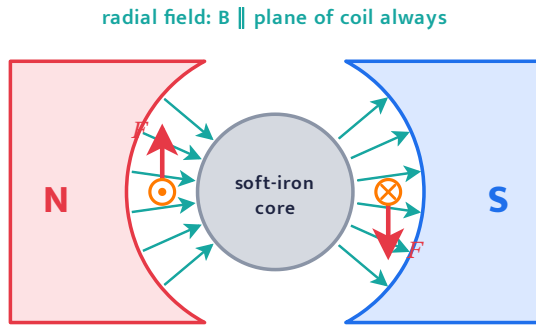
Solution: Plane $\parallel B \Rightarrow$ normal $\perp B$, $\theta = 90^\circ$:
 $\tau = NIAB = 100 \times 2 \times (0.1 \times 0.05) \times 0.1 = 0.1 \text{ N m}$.

Common Trap

θ is the angle between the **normal** and $\vec{B}$, not between the plane and $\vec{B}$: plane $\parallel \vec{B}$ gives the **maximum** torque; plane $\perp \vec{B}$ gives zero.

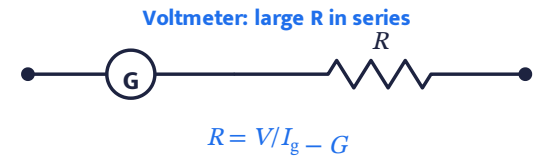
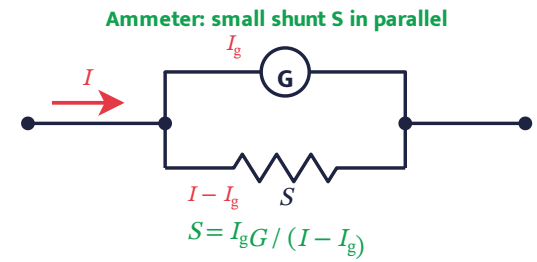
4.9 Moving-Coil Galvanometer, Ammeter and Voltmeter

A **moving-coil galvanometer** detects a current by the torque on a coil in a **radial magnetic field** (concave pole pieces and a soft-iron core), so the plane of the coil is always parallel to $\vec{B}$ and the torque $NIAB$ is balanced by the spring torque $k\phi$: $\phi = \frac{NAB}{k} I$, i.e. deflection $\propto$ current. **Current sensitivity** $= \frac{\phi}{I} = \frac{NAB}{k}$.



$$NIAB = k\varphi \Rightarrow \varphi \propto I$$

Radial field of a moving-coil galvanometer



Galvanometer $\rightarrow$ ammeter (shunt S) and voltmeter (series R)



Conversion formulas

$$S = \frac{I_g G}{I - I_g}$$

Shunt (low resistance, in **parallel**)
– ammeter of range I [RBSE 2025]

$$R = \frac{V}{I_g} - G$$

High resistance in series –
voltmeter of range V [RBSE 2024]



Solved Example

A galvanometer of resistance $50 \, \Omega$ gives full-scale deflection for $5 \, \text{mA}$. How can it be made (a) an ammeter of range $0-5 \, \text{A}$, (b) a voltmeter of range $0-10 \, \text{V}$?

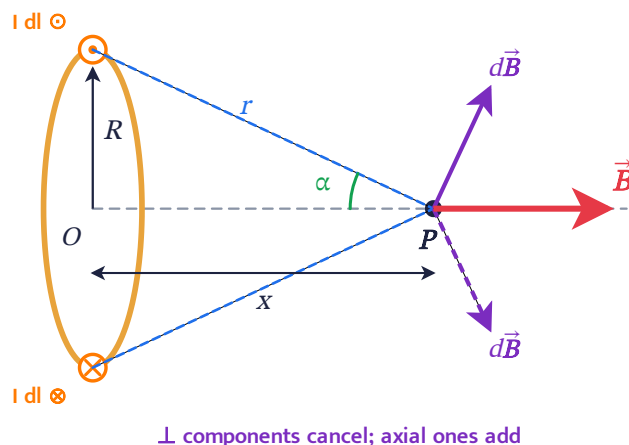
Solution: (a) $S = \frac{I_g G}{I - I_g} = \frac{0.005 \times 50}{5 - 0.005} \approx 0.05 \, \Omega$ in parallel. (b)

$$R = \frac{V}{I_g} - G = \frac{10}{0.005} - 50 = 1950 \, \Omega$$
 in series.


Common Trap

- Ammeter is connected in **series** (its resistance is very small); voltmeter in **parallel** (resistance very large).
- The soft-iron core and radial field make the scale **linear**.

4.10 Important Derivations ★



Circular loop of radius R : field at a point P on the axis, distance x from the centre O

🔧 Derivation ★ [3-4 marks]

Derivation 1 – [RBSE 2025, 2024, 2026]. State the Biot–Savart law and use it to derive the magnetic field at a point on the axis of a circular current loop.

Solution: (i) **Biot–Savart law:** $d\vec{B} = \frac{\mu_0 I d\vec{l} \times \vec{r}}{4\pi r^3}$, i.e. $dB = \frac{\mu_0 I dl \sin\theta}{4\pi r^2}$. (½)

(ii) Loop of radius R , current I , centre O ; P is on the axis at distance x ; $r^2 = R^2 + x^2$ (figure). Every element $d\vec{l}$ is $\perp$ to $\vec{r}$ ($\theta = 90^\circ$), so $dB = \frac{\mu_0 I dl}{4\pi r^2}$, perpendicular to $\vec{r}$. (1)

(iii) Resolve $d\vec{B}$: the components $\perp$ to the axis ($dB \cos\alpha$) cancel for diametrically opposite elements; the **axial** components $dB \sin\alpha$ add, with $\sin\alpha = \frac{R}{r}$. (1)

(iv) $B = \oint dB \sin\alpha = \frac{\mu_0 I}{4\pi r^2} \cdot \frac{R}{r} \oint dl = \frac{\mu_0 IR(2\pi R)}{4\pi r^3} \Rightarrow B = \frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}}$ along the axis (for N turns multiply by N). (1)

(v) **Centre** ($x = 0$): $B = \frac{\mu_0 I}{2R}$. **Far away** ($x \gg R$): $B \approx \frac{\mu_0 IR^2}{2x^3} = \frac{\mu_0}{4\pi} \frac{2m}{x^3}$, $m = I\pi R^2$. Direction: right-hand thumb rule. (½)
(3-mark answer: steps (i)–(iv).)

🔧 Derivation ★ [3-4 marks]

Derivation 2 – [RBSE 2024, 2023]. Derive the expression for the force per unit length between two long parallel current-carrying conductors. Hence define the ampere. (figure in 4.7)

Solution: (i) Let wires a and b , distance d apart, carry currents I_a and I_b in the **same direction**. Field of a at b :

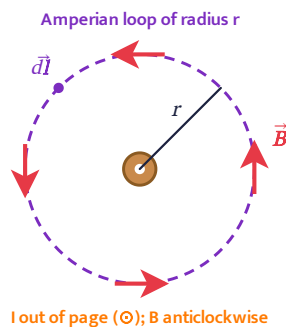
$B_a = \frac{\mu_0 I_a}{2\pi d}$, perpendicular to the plane of the wires. (1)

(ii) Force on a length L of wire b : $F_{ba} = I_b L B_a \sin 90^\circ = \frac{\mu_0 I_a I_b L}{2\pi d}$. By $\vec{F} = I\vec{L} \times \vec{B}$ (or Fleming's rule) it is directed **towards a**. (1)

(iii) Similarly the force on a length L of a due to b is equal in magnitude and **towards b** (Newton's third law) – the wires **attract**; for opposite currents they repel. (½)

(iv) $\frac{F}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$. (½)

(v) **Ampere:** if $I_a = I_b = 1\text{ A}$, $d = 1\text{ m}$, then $F/L = 2 \times 10^{-7}\text{ N m}^{-1}$ – the steady current which produces this force between two very long parallel wires 1 m apart in vacuum is **1 A**. (1)

Circular Amperian loop of radius r around a long straight wire**🔧 Derivation ★ [3-4 marks]**

Derivation 3 – [RBSE 2025, 2026]. State Ampère's circuital law and use it to find the magnetic field due to an infinitely long straight wire carrying current I .

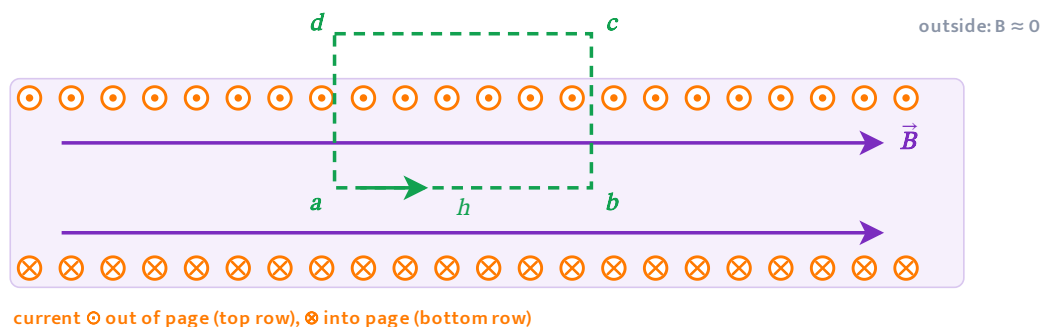
Solution: (i) **Ampère's law:** $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$ – the line integral of $\vec{B}$ round any closed loop = $\mu_0 \times$ net current through the loop. (1)

(ii) By symmetry the field lines are **circles** around the wire (figure). Take a circular Amperian loop of radius r in a plane $\perp$ to the wire; at every point of it B has the **same magnitude** and is **tangential**, so $\vec{B} \parallel d\vec{l}$. (1)

(iii) $\oint \vec{B} \cdot d\vec{l} = B \oint dl = B(2\pi r)$. (½)

(iv) The loop encloses the whole current: $I_{enc} = I \Rightarrow B(2\pi r) = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$. (1)

(v) $B \propto 1/r$; direction by the right-hand thumb rule (or the right-hand grip rule). (½)



$$Bh = \mu_0(nh)I \Rightarrow B = \mu_0 nI$$

Long solenoid: Amperian rectangle $abcd$, side $ab = h$ inside the solenoid**🔧 Derivation ★ [3 marks]**

Derivation 4 – [RBSE 2023]. Using Ampère's law, obtain the magnetic field at a point on the axis of a long solenoid carrying current I with n turns per unit length.

Solution: (i) For a long solenoid the field inside is **uniform and parallel to the axis**; outside it is negligible (figure). (½)

(ii) Take the rectangular Amperian loop $abcd$: ab (length h) lies **inside**, parallel to the axis; cd lies **outside**; bc and da are $\perp$ to the axis. (½)

(iii) On ab : $\int \vec{B} \cdot d\vec{l} = Bh$. On cd : $B \approx 0 \Rightarrow 0$. On bc and da : $B \perp d\vec{l}$ inside and ≈ 0 outside $\Rightarrow 0$. So $\oint \vec{B} \cdot d\vec{l} = Bh$. (1)

(iv) Number of turns threading the loop = nh , so $I_{enc} = nhI$. (½)

(v) $Bh = \mu_0 nhI \Rightarrow B = \mu_0 nI$ (uniform inside; B at an end = $\frac{1}{2}\mu_0 nI$). (½)

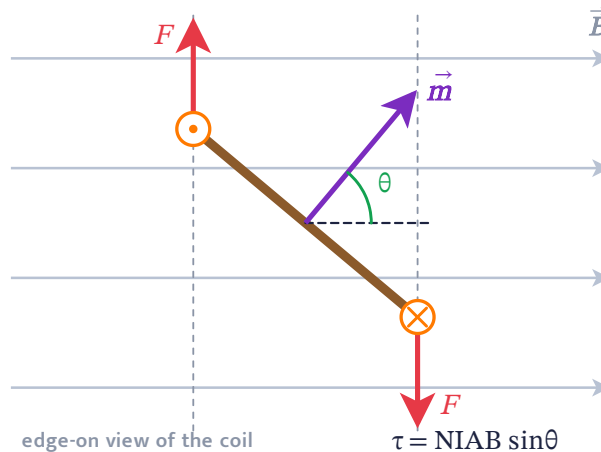
🔧 Derivation ★ [3 marks]

Derivation 5. Show that a charged particle entering a uniform magnetic field perpendicularly moves in a circle. Find the radius and the time period.

Solution: (i) $\vec{v} \perp \vec{B} \Rightarrow F = qvB$, always $\perp$ to $\vec{v}$ (figure in 4.2): it changes only the direction, not the speed, so the path is a **circle** with the force towards the centre. (1)

(ii) Magnetic force = centripetal force: $qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB}$. (1)

(iii) $T = \frac{2\pi r}{v} \Rightarrow T = \frac{2\pi m}{qB}$, $\nu = \frac{qB}{2\pi m}$ – independent of speed and radius. (1)



Edge-on view of a coil in a uniform field: the forces on the two arms form a couple

🔧 Derivation ★ [3 marks]

Derivation 6. Obtain the expression for the torque on a rectangular current loop of N turns in a uniform magnetic field.

Solution: (i) Loop of sides a and b, area $A = ab$, current I; its normal makes angle θ with $\vec{B}$. The a-sides are $\perp$ to $\vec{B}$; the b-sides make angle $(90^\circ - \theta)$ with it. (1/2)

(ii) Forces on the two b-sides: $IbB \cos\theta$, equal, opposite and along the **same line** (the axis of rotation) – they cancel and give **no torque**. (1/2)

(iii) Forces on the two a-sides: $F = IaB$ each, equal and opposite but along **different lines** – a couple, with arm $b \sin\theta$. (1)

(iv) $\tau = F \times b \sin\theta = IaB b \sin\theta \Rightarrow \tau = NIAB \sin\theta$ for N turns, i.e. $\vec{\tau} = \vec{m} \times \vec{B}$ with $m = NIA$. (1)

🔧 Derivation ★ [3 marks]

Derivation 7 – [RBSE 2025, 2024]. How can a galvanometer (resistance G, full-scale current I_g) be converted into (a) an ammeter of range I, (b) a voltmeter of range V? (figure in 4.9)

Solution: (a) Ammeter: connect a low resistance **S (shunt) in parallel** with G. (1/2) Potential difference across both is equal: $I_g G = (I - I_g)S \Rightarrow S = \frac{I_g G}{I - I_g}$. (1) The combined resistance $\frac{GS}{G+S}$ is very small, so the ammeter hardly disturbs the circuit. (1/2)

(b) Voltmeter: connect a high resistance **R in series** with G. (1/2) For range V: $V = I_g(G + R) \Rightarrow R = \frac{V}{I_g} - G$. (1/2) The resistance $G + R$ is very large. (1/2)

VS Sir's Exam Tip

- Examiner's checklist: labelled figure with $d\vec{l}$, $\vec{r}$, $d\vec{B}$ → law written first → symmetry argument (components cancel / B constant on the loop) → integration → result boxed. Name the Amperian loop.
- In Derivation 1 never forget to say that the **perpendicular components cancel** – it carries ½–1 mark.

Board question bank

Section A – Multiple choice (½ mark each)

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- [RBSE 2025]** A charged particle moves in a uniform magnetic field with its velocity parallel to the field. The force on it is
 (a) qvB , along $\vec{v}$ (b) zero
 (c) $qvB / 2$ (d) qvB
- [RBSE 2024]** A charged particle enters a uniform magnetic field perpendicular to it. The path of the particle is
 (a) a straight line (b) a parabola
 (c) a circle (d) a helix
- The magnetic field at the centre of a circular loop of radius R carrying current I is
 (a) $\frac{\mu_0 IR}{2}$ (b) $\frac{\mu_0 I}{4\pi R}$ (c) $\frac{\mu_0 I}{2R}$ (d) $\frac{\mu_0 I}{2\pi R}$
- Two long parallel wires carry currents in the same direction. They
 (a) attract each other (b) repel each other
 (c) rotate about each other (d) exert no force
- A solenoid has 500 turns per metre and carries 2 A. The field inside it is ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$)
 (a) $1.26 \times 10^{-3} \text{ T}$ (b) $1.26 \times 10^{-4} \text{ T}$ (c) $6.3 \times 10^{-4} \text{ T}$ (d) $2.5 \times 10^{-3} \text{ T}$
- A charged particle moves in a circle in a uniform magnetic field. If its speed is doubled, the radius of the path becomes
 (a) double (b) half (c) unchanged (d) four times
- A straight wire of length 2 m carries 3 A perpendicular to a field of 0.5 T. The force on it is
 (a) 0.3 N (b) 3 N (c) 12 N (d) 1.5 N
- To convert a galvanometer of resistance 50Ω and full-scale current 5 mA into an ammeter of range 5 A, the shunt required is about
 (a) 0.5Ω (b) 500Ω (c) 0.05Ω (d) 5Ω
- The magnetic field at 5 cm from a long straight wire carrying 10 A is
 (a) $4 \times 10^{-6} \text{ T}$ (b) $4 \times 10^{-4} \text{ T}$ (c) $2 \times 10^{-5} \text{ T}$ (d) $4 \times 10^{-5} \text{ T}$

10 A magnetic field does no work on a moving charge because the force is

- (a) always along the velocity
- (b) always perpendicular to the velocity
- (c) always zero
- (d) always along the field

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2025] A galvanometer is converted into an ammeter by connecting a low resistance called _____ in parallel with it.	shunt
[RBSE 2024] A galvanometer is converted into a voltmeter by connecting a _____ resistance in series with it.	high
[RBSE 2023] Two parallel conductors carrying currents in the same direction _____ each other.	attract
The SI unit of magnetic field is _____.	tesla
The magnetic dipole moment of a current loop is $m = \text{_____}$ (N turns, area A).	NIA
The magnetic field outside a long ideal solenoid is nearly _____.	zero

Very short answer (1 mark)

Question	Model answer
Write the expression for the Lorentz force.	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$.
Define 1 tesla.	The field in which a charge of 1 C moving at 1 m/s perpendicular to it feels a force of 1 N ($1 \text{ T} = 1 \text{ N A}^{-1} \text{ m}^{-1}$).
State the SI unit of permeability of free space μ_0 .	T m A^{-1} ($= \text{N A}^{-2} = \text{H m}^{-1}$); $\mu_0 = 4\pi \times 10^{-7}$.
Why is the field of a moving-coil galvanometer made radial?	So that the torque $NIAB$ is the same at every deflection and the deflection $\propto$ current (linear scale).
What is the cyclotron frequency of a particle of charge q and mass m ?	$\nu = \frac{qB}{2\pi m}$.
Give the direction rule for the field of a straight current-carrying wire.	Right-hand thumb rule: thumb along the current, the curled fingers give the sense of the circular field lines.

Short answer (Section B, 1½ marks)

Short answer (1½ marks)

State the Biot–Savart law and write the SI unit of μ_0 .

Solution: The field at P due to a current element: $d\vec{B} = \frac{\mu_0 I d\vec{l} \times \vec{r}}{4\pi r^3} \propto I dl \sin\theta, \propto 1/r^2$, direction $\perp$ to $d\vec{l}$ and $\vec{r}$. (1)

Unit of μ_0 : T m A^{-1} . (½)

 Short answer (1½ marks)

Two long parallel wires 5 cm apart carry 10 A each in opposite directions. Find the force per unit length and its nature.

Solution: $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$ (½) = $\frac{2 \times 10^{-7} \times 10 \times 10}{0.05}$ (½) = 4×10^{-4} N/m, **repulsive** (antiparallel currents). (½)

 Short answer (1½ marks)

An electron moves at 3×10^6 m/s perpendicular to a field of 1.0×10^{-3} T. Find the radius of its path ($m = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C).

Solution: $r = \frac{mv}{eB}$ (½) = $\frac{9.1 \times 10^{-31} \times 3 \times 10^6}{1.6 \times 10^{-19} \times 1.0 \times 10^{-3}}$ (½) = 1.7×10^{-2} m (1.7 cm). (½)

 Short answer (1½ marks)

Distinguish between an ammeter and a voltmeter (three points).

Solution: (i) Ammeter: **low** resistance shunt in parallel with G; voltmeter: **high** resistance in series with G. (ii) Ammeter is connected in **series** with the circuit; voltmeter in **parallel** across the component. (iii) Ammeter measures **current**; voltmeter measures **potential difference**. (½ each)

 Answer Key & Quick Solutions

1(b) 2(c) 3(c) 4(a) 5(a) 6(a) 7(b) 8(c) 9(d) 10(b)

1. (b) $F = qvB \sin \theta$ with $\theta = 0^\circ$ gives zero.

2. (c) The force qvB is always $\perp$ to $\vec{v}$ and acts as the centripetal force.

3. (c) From the axis formula with $x = 0$: $B = \frac{\mu_0 I}{2R}$.

4. (a) Parallel currents attract; antiparallel currents repel.

5. (a) $B = \mu_0 nI = 4\pi \times 10^{-7} \times 500 \times 2 = 1.26 \times 10^{-3}$ T.

6. (a) $r = \frac{mv}{qB} \propto v$.

7. (b) $F = BIl = 0.5 \times 3 \times 2 = 3$ N.

8. (c) $S = \frac{I_g G}{I - I_g} = \frac{0.005 \times 50}{4.995} \approx 0.05 \Omega$.

9. (d) $B = \frac{2 \times 10^{-7} \times 10}{0.05} = 4 \times 10^{-5}$ T.

10. (b) $\vec{F} \cdot \vec{v} = 0$ since $\vec{F} = q\vec{v} \times \vec{B}$.



Last-minute revision

- $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$; $F = 0$ if $\vec{v} \parallel \vec{B}$; magnetic force does no work; $F = BIl \sin \theta$ on a wire.
- $\vec{v} \perp \vec{B}$: circle, $r = \frac{mv}{qB}$, $T = \frac{2\pi m}{qB}$ (independent of v); angle θ : helix.
- Biot–Savart: $dB = \frac{\mu_0 I dl \sin \theta}{4\pi r^2}$; loop axis $\frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}}$; centre $\frac{\mu_0 I}{2R}$.
- Ampère: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$; wire $\frac{\mu_0 I}{2\pi r}$; solenoid $\mu_0 nI$; toroid $\frac{\mu_0 NI}{2\pi r}$.
- Parallel currents attract: $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$; $1 \text{ A} \Rightarrow 2 \times 10^{-7} \text{ N/m}$ at 1 m.
- Loop: $\vec{m} = NI\vec{A}$, $\tau = NIAB \sin \theta$, $U = -mB \cos \theta$ (θ from the normal).
- Ammeter: shunt $S = \frac{I_g G}{I - I_g}$ in **parallel**; voltmeter: $R = \frac{V}{I_g} - G$ in **series**.

CHAPTER

5

Magnetism and Matter

17

Day 8

Board marks: 3

PYQ frequency ★★☆☆

RBSE Class 12

Bar magnet & field lines

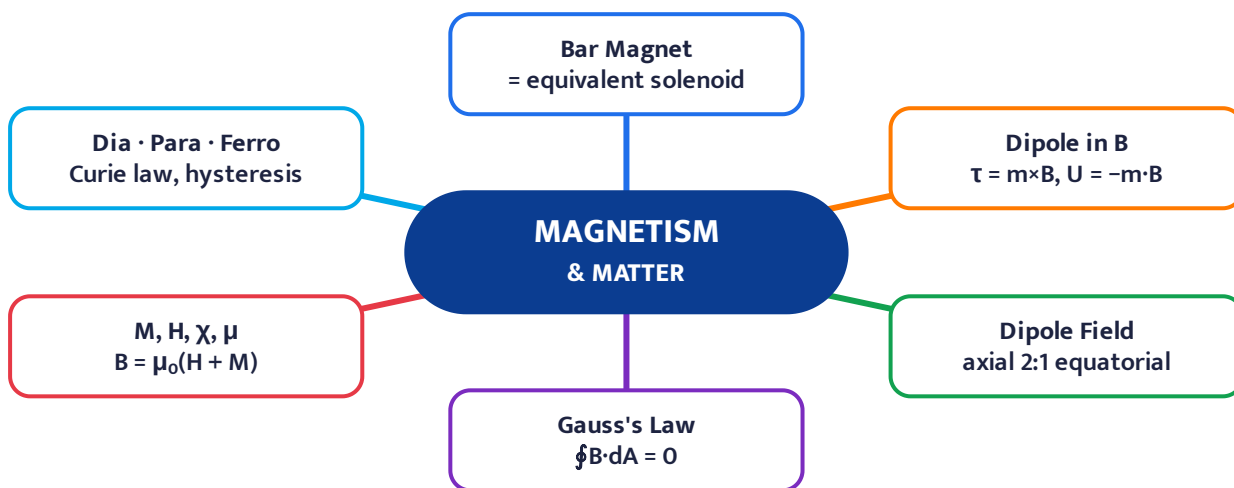
Torque & PE of dipole

Gauss's law

M, H, χ , μ_r

Dia • Para • Ferro

Curie law, hysteresis



Chapter at a glance



Board snapshot

- **Marks: 3** (blueprint). Usually 1 MCQ + 1 fill-in / very-short answer + sometimes a 1½-mark Section-B question.
- **2025:** MCQ – PE of magnetic dipole $U = -mB\cos\theta$; fill-in – susceptibility of a paramagnetic substance (small, positive).
- **2024:** fill-in – magnetisation; VSA – define paramagnetic substance; B – two properties of magnetic field lines.
- **2023:** MCQ – Curie temperature of iron; VSA – characteristics of a magnet, paramagnetic substance. **2020:** diamagnetic substances; soft iron for electromagnets.
- **Most likely in 2027:** (i) torque / PE of a dipole in B, (ii) dia–para–ferro (definition, χ , examples, Curie law), (iii) properties of field lines / Gauss's law.

5.1 Bar Magnet and Magnetic Field Lines

A bar magnet has two poles, **north (N)** and **south (S)**. Like poles repel and unlike poles attract; a freely suspended magnet comes to rest in the **north–south direction**. Poles **always exist in pairs** – if a magnet is broken, each piece is a complete magnet with its own N and S pole. The strength of a magnet is given by its **magnetic dipole moment $\vec{m}$** , directed from S to N; SI unit A m^2 ($= \text{J T}^{-1}$).

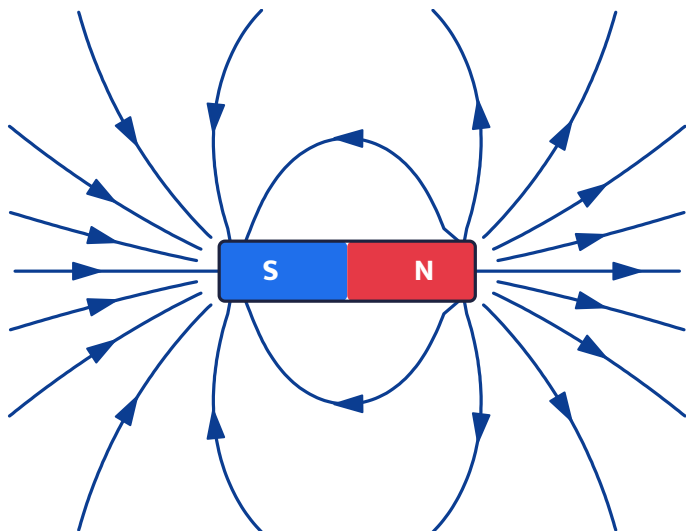
Electric dipole

Magnetic dipole

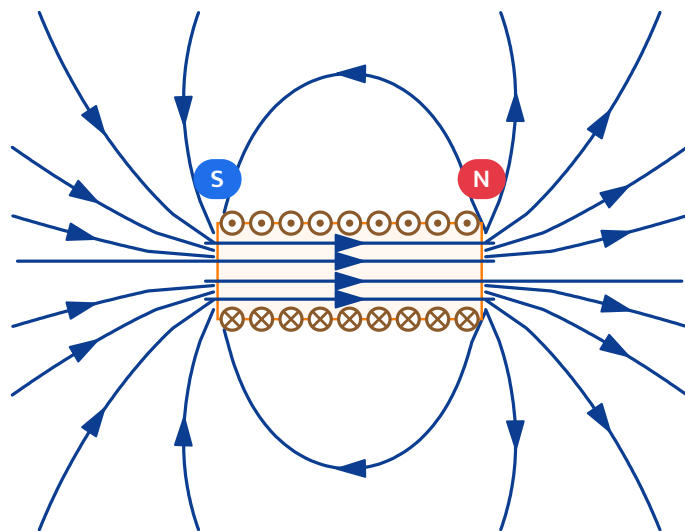
Dipole moment $\vec{p}$ (C m)Dipole moment $\vec{m}$ (A m²)Axial: $E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$; equatorial: $\frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$ Axial: $B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}$; equatorial: $\frac{\mu_0}{4\pi} \frac{m}{r^3}$ Torque $\vec{p} \times \vec{E}$; energy $-\vec{p} \cdot \vec{E}$ Torque $\vec{m} \times \vec{B}$; energy $-\vec{m} \cdot \vec{B}$

Lines start on + and end on – charge

Lines are closed loops – no monopole



Field lines of a bar magnet



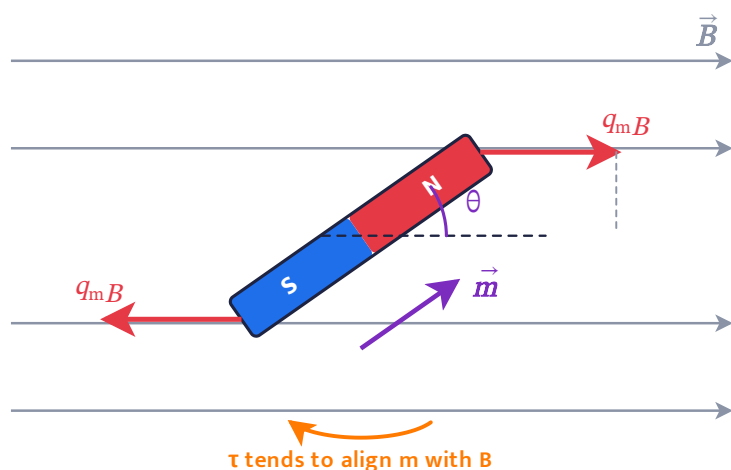
Finite solenoid – the same pattern

**Properties of magnetic field lines**

- They are **continuous closed loops** – outside the magnet from N to S, inside from S to N.
- The **tangent** at any point gives the direction of $\vec{B}$ there.
- Where lines are **closer**, the field is **stronger**.
- Two field lines **never intersect** (B cannot have two directions at a point).

**Bar magnet = equivalent solenoid**

- Field lines of a bar magnet and of a current-carrying finite solenoid are **identical**.
- A solenoid of N turns, area A , current I has $m = NIA$.
- So a bar magnet behaves like a collection of tiny circulating (atomic) currents. (Derivation ★3 below.)

5.2 Magnetic Dipole in a Uniform Magnetic Field

τ tends to align m with B

Equal and opposite forces on the two poles form a couple: torque, but no net force

$$\tau = mB \sin\theta$$

$$U = -mB \cos\theta$$

$$\theta = 0^\circ: U = -mB \text{ (stable)}$$

$$\theta = 180^\circ: U = +mB \text{ (unstable)}$$

$$\text{net force in uniform } B = 0$$

Dipole in uniform B

$$\vec{\tau} = \vec{m} \times \vec{B}, \quad \tau = mB \sin \theta$$

Torque; maximum mB at $\theta = 90^\circ$, zero at 0° and 180°

$$U = -\vec{m} \cdot \vec{B} = -mB \cos \theta$$

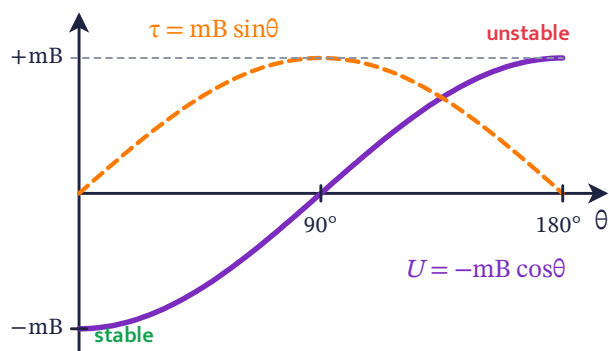
Potential energy (taken zero at $\theta = 90^\circ$)

$$W = mB(\cos \theta_1 - \cos \theta_2)$$

Work done in rotating from θ_1 to θ_2 ; $0^\circ \rightarrow 180^\circ$ gives $2mB$

$$T = 2\pi \sqrt{\frac{I}{mB}}$$

Period of small oscillations (I = moment of inertia)



U and τ versus θ

Solved Example

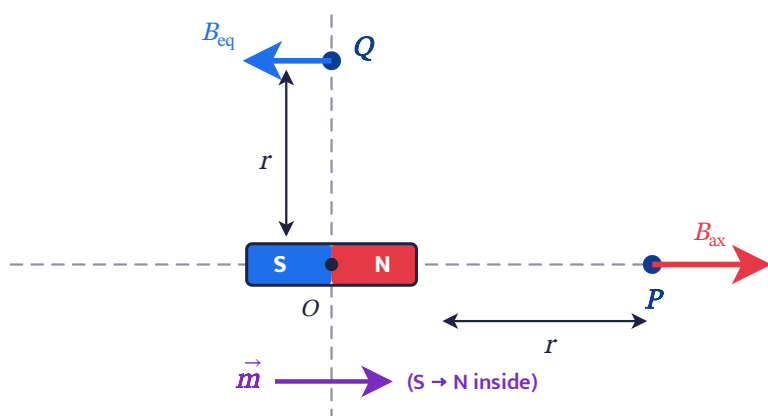
A bar magnet of moment 0.40 J T^{-1} is placed at 60° to a uniform field of 0.25 T . Find (i) the torque, (ii) its potential energy, (iii) the work needed to turn it from stable to unstable equilibrium.

Solution: (i) $\tau = mB \sin 60^\circ = 0.40 \times 0.25 \times 0.866 = \mathbf{0.087 \text{ N m}}$. (ii) $U = -mB \cos 60^\circ = \mathbf{-0.05 \text{ J}}$. (iii) $W = 2mB = 2 \times 0.40 \times 0.25 = \mathbf{0.20 \text{ J}}$.

Common Trap

- Torque is **maximum at 90°** , where PE is **zero**. PE is minimum ($-mB$, stable) at 0° , maximum ($+mB$, unstable) at 180° – at both the torque is zero.
- In a **uniform** field the net force on a dipole is zero (only rotation); in a non-uniform field there is also a net force.

5.3 Field of a Bar Magnet – Axial and Equatorial



Short magnet ($r \gg l$)

$$B_{ax} = (\mu_0/4\pi) \cdot 2m/r^3 \quad \text{along } m$$

$$B_{eq} = (\mu_0/4\pi) \cdot m/r^3 \quad \text{opposite to } m$$

$$B_{ax} : B_{eq} = 2 : 1$$

At equal distance, the axial field is twice the equatorial field; the two point in opposite directions

**Field of a short magnet**

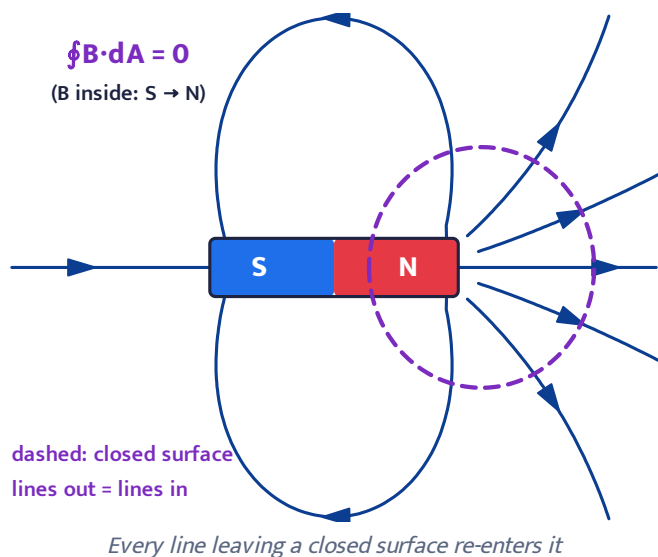
$$B_{axial} = \frac{\mu_0 2m}{4\pi r^3} \quad \text{along } \vec{m} \text{ (short magnet, } r \gg l)$$

$$B_{equatorial} = \frac{\mu_0 m}{4\pi r^3} \quad \text{opposite to } \vec{m}$$

$$\frac{\mu_0}{4\pi} = 10^{-7} \text{ T m A}^{-1} \quad \text{use this value directly}$$

**Memory Trick**

"**Axis 2, Equator 1, both fall as $1/r^3$.**" Axial field is along $\vec{m}$, equatorial field is opposite to $\vec{m}$. A dipole field falls as $1/r^3$ (a single charge as $1/r^2$).

5.4 Gauss's Law in Magnetism

The net magnetic flux through any closed surface is always zero:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

**NCERT Line to Remember**

- The number of lines leaving the surface = number entering it.
- Meaning: **isolated magnetic poles (monopoles) do not exist.** The simplest magnetic element is a dipole (current loop).
- Compare: in electrostatics $\oint \vec{E} \cdot d\vec{A} = q / \epsilon_0$ – not zero, because charges exist.

5.5 Magnetisation and Magnetic Intensity

- **Magnetisation (M):** the net magnetic dipole moment per unit volume of a material, $M = m_{net} / V$. Vector; unit A m^{-1} .
- **Magnetic intensity (H):** $H = \frac{B}{\mu_0} - M$. It depends only on the external (free) current; unit A m^{-1} . For a solenoid $H = nI$.
- **Magnetic susceptibility (χ):** $\chi = M / H$ – how easily a material is magnetised. Dimensionless.
- **Permeability (μ):** $\mu = B / H$; **relative permeability** $\mu_r = \mu / \mu_0 = 1 + \chi$ (dimensionless). Relation: $B = \mu_0(H + M) = \mu_0\mu_r H$ (derivation ★4).

**Solved Example**

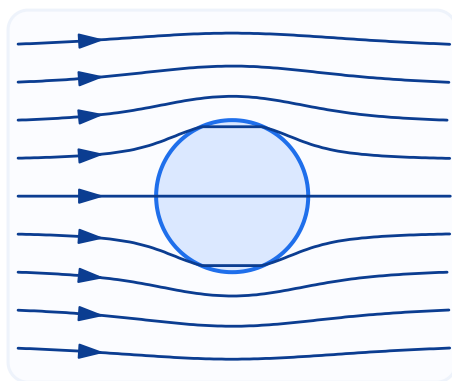
A solenoid with 500 turns per metre carries 1.5 A. Its core has relative permeability 800. Find H, B, M and χ of the core.

Solution: $H = nI = 500 \times 1.5 = 750 \text{ A m}^{-1}$. $B = \mu_0\mu_r H = 4\pi \times 10^{-7} \times 800 \times 750 = 0.75 \text{ T}$. $\chi = \mu_r - 1 = 799$; $M = \chi H = 799 \times 750 \approx 6.0 \times 10^5 \text{ A m}^{-1}$.

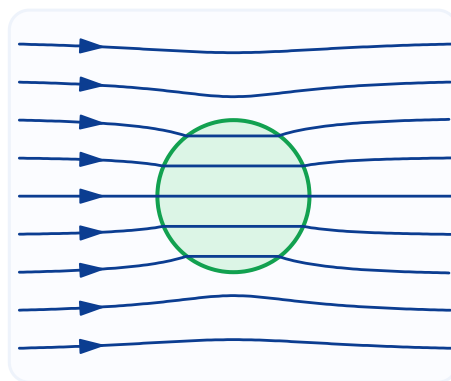
**Common Trap**

- H and M have the **same unit (A m^{-1})**; B is in tesla. χ and μ_r have no unit, but μ has the unit of μ₀ (T m A^{-1}).
- Putting a core in a solenoid does **not** change $H = nI$; it changes M and therefore B.

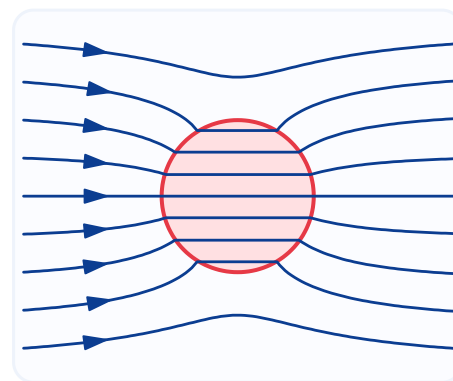
5.6 Magnetic Properties of Materials



Diamagnetic: lines pushed out



Paramagnetic: lines drawn in slightly



Ferromagnetic: lines crowd in strongly

The three classes of materials in a uniform external field

- **Diamagnetic substances:** weakly magnetised **opposite** to the applied field; move from stronger to weaker field (weakly repelled). χ small and **negative**. Examples: bismuth, copper, lead, silicon, water, NaCl, nitrogen (STP).
- **Paramagnetic substances:** weakly magnetised **in the direction** of the applied field; move from weaker to stronger field (weakly attracted). χ small and **positive**. Examples: aluminium, sodium, calcium, oxygen (STP), copper chloride.
- **Ferromagnetic substances:** strongly magnetised in the direction of the field; strongly attracted. χ very **large, positive**. Examples: iron, cobalt, nickel, gadolinium. Reason: **domains** – regions of aligned atomic moments.

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Susceptibility χ	small, negative ($-1 \leq \chi < 0$)	small, positive ($0 < \chi < \epsilon$)	very large, positive ($\chi \gg 1$)
Relative permeability μ_r	$0 \leq \mu_r < 1$	$1 < \mu_r < 1 + \epsilon$	$\mu_r \gg 1$
Effect of temperature	independent of T	Curie law $\chi \propto 1/T$	becomes paramagnetic above T_C

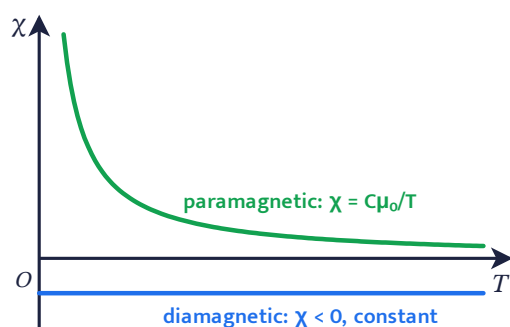
NCERT Line to Remember

- A **superconductor** is a perfect diamagnet: $\chi = -1$, $\mu_r = 0$; it expels the magnetic field completely (**Meissner effect**).
- Diamagnetism is present in **all** substances; in para/ferro materials it is masked.

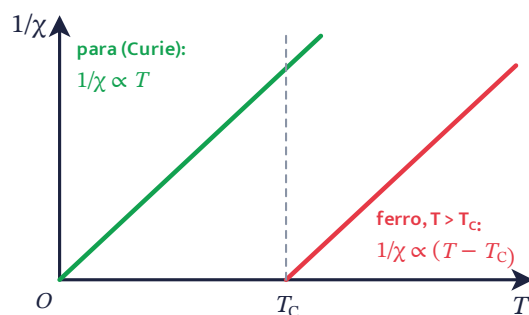
Memory Trick

Sign of χ : **dia (-), para (+ small), ferro (+ huge)**.
 "Dia dislikes the field, Para likes it a little, Ferro loves it."

5.7 Curie Law, Curie Temperature and Hysteresis



Para: $\chi \propto 1/T$; dia: $\chi < 0$, T -independent



Above T_C a ferromagnet behaves as a paramagnet



Temperature dependence

$$\chi = \frac{C\mu_0}{T}$$

Curie law for paramagnets: $\chi \propto 1/T$ (C = Curie constant, T in kelvin)

$$\chi = \frac{C}{T - T_C} \quad (T > T_C)$$

Ferromagnet above its Curie temperature

Curie temperature (T_C): the temperature above which a ferromagnetic substance becomes paramagnetic (domains break up due to thermal agitation). NCERT values: **iron 1043 K**, cobalt 1394 K, nickel 631 K, gadolinium 317 K.

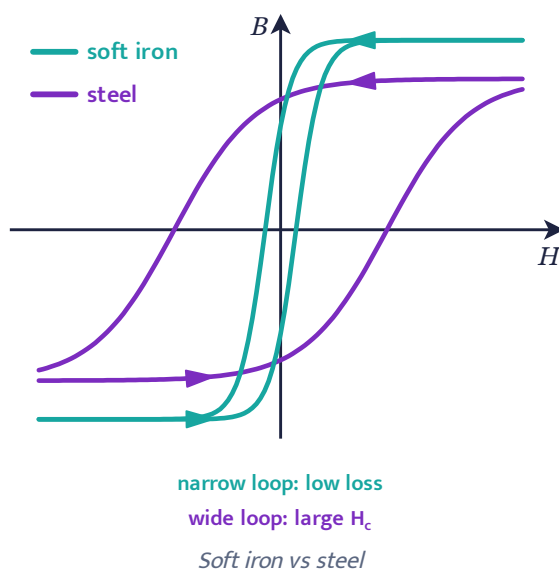
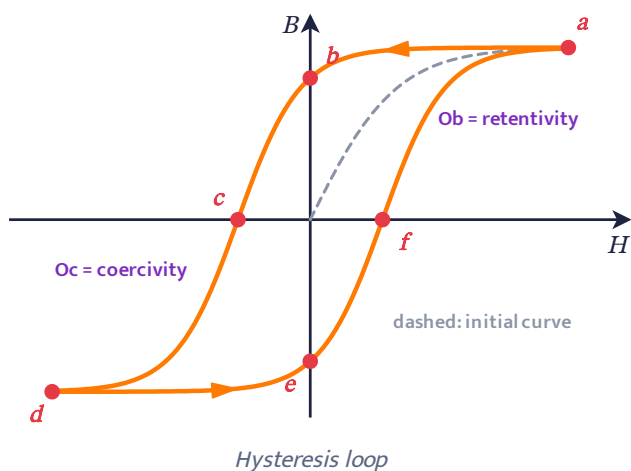


Solved Example

The susceptibility of a paramagnetic salt is 2.8×10^{-4} at 27°C . Find it at -73°C .

Solution: $T_1 = 300\text{ K}$, $T_2 = 200\text{ K}$. Curie law: $\chi_2 = \chi_1 \frac{T_1}{T_2} = 2.8 \times 10^{-4} \times \frac{300}{200} = 4.2 \times 10^{-4}$.

Hysteresis: when a ferromagnet is taken through a cycle of H , B lags behind H and does not retrace its path. **Retentivity (remanence)** = B left when H becomes zero (O_b); **coercivity** = reverse H needed to make B zero (O_c). Loop area = energy lost as heat per unit volume per cycle.



Use	Required properties	Material
Permanent magnet	high retentivity, high coercivity	steel, alnico, cobalt steel
Electromagnet / transformer core	high permeability, low coercivity , small hysteresis loss	soft iron

VS Sir's Exam Tip

Board line (asked 2020): "**Soft iron is used for electromagnets** because it has high permeability and low coercivity – it is magnetised strongly by a small current and loses its magnetism as soon as the current is switched off; its hysteresis loss is small."

Important Derivations ★

🔧 Derivation ★ [1½ marks]

★1 Derive the torque on a bar magnet (magnetic dipole) placed in a uniform magnetic field. Show that the net force on it is zero.

Solution: (See the torque figure in 5.2.) Let the magnet have pole strength q_m , length $2l$ and moment $m = q_m(2l)$, at angle θ with $\vec{B}$. (½)

1. Force on N-pole $= q_m B$ along $\vec{B}$; on S-pole $= q_m B$ opposite to $\vec{B}$. Net force $= q_m B - q_m B = 0$. (½)

2. The two equal, opposite, parallel forces form a couple. Perpendicular distance between their lines of action $= 2l \sin \theta$.

$$\tau = q_m B \times 2l \sin \theta = (q_m 2l) B \sin \theta \Rightarrow \tau = m B \sin \theta, \text{ i.e. } \vec{\tau} = \vec{m} \times \vec{B} \quad (\frac{1}{2})$$

The torque tries to align $\vec{m}$ along $\vec{B}$; it is maximum (mB) at $\theta = 90^\circ$.

🔧 Derivation ★ [1½ marks]

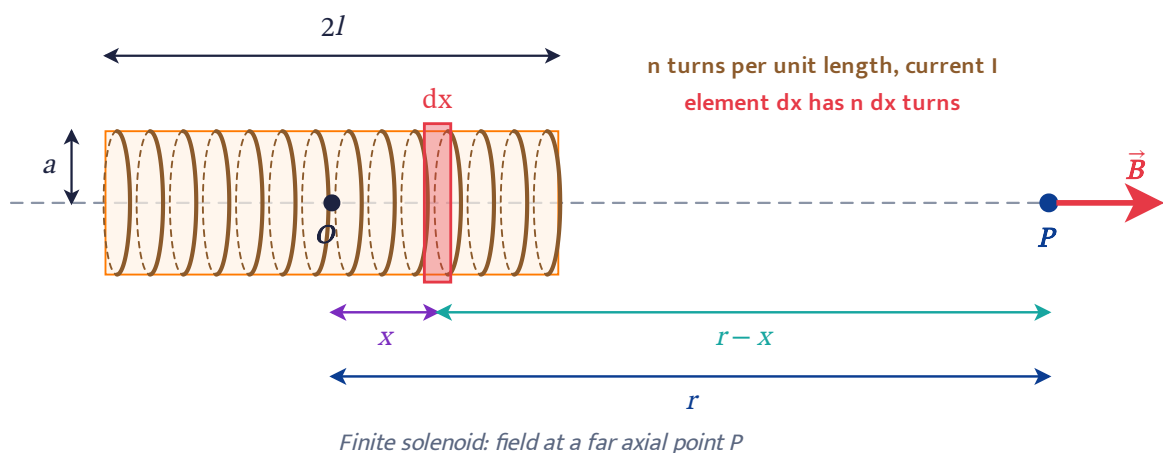
★2 Derive the expression for the potential energy of a magnetic dipole in a uniform magnetic field.

Solution: 1. Torque at angle θ : $\tau = mB \sin \theta$. Work to turn it through $d\theta$: $dW = \tau d\theta = mB \sin \theta d\theta$. (½)

2. Taking $U = 0$ at $\theta = 90^\circ$ (dipole $\perp$ field), PE at angle θ = work done from 90° to θ :

$$U = \int_{90^\circ}^{\theta} mB \sin \theta d\theta = mB [-\cos \theta]_{90^\circ}^{\theta} = -mB(\cos \theta - 0) \quad (\frac{1}{2})$$

3. $U = -mB \cos \theta = -\vec{m} \cdot \vec{B}$ (½) ($\theta = 0^\circ$: $-mB$, stable; $\theta = 180^\circ$: $+mB$, unstable)



🔧 Derivation ★ [3 marks]

★3 Show that a current-carrying solenoid behaves like a bar magnet: find the field at a far point on its axis.

Solution: Solenoid: length $2l$, radius a , n turns per unit length, current I ; P on the axis at distance r from centre O (figure). (½)

1. Take a thin element of thickness dx at distance x from O. It has $n dx$ turns and acts as a circular coil whose centre is at distance $(r - x)$ from P. Field of a loop on its axis gives (½)

$$dB = \frac{\mu_0 (n dx) I a^2}{2 \left[(r - x)^2 + a^2 \right]^{3/2}}$$

2. Total field: $B = \frac{\mu_0 n I a^2}{2} \int_{-l}^{+l} \frac{dx}{\left[(r - x)^2 + a^2 \right]^{3/2}}$ (½)

3. For a far point, $r \gg a$ and $r \gg l$, so $\left[(r - x)^2 + a^2 \right]^{3/2} \approx r^3$; $B = \frac{\mu_0 n I a^2}{2r^3} \int_{-l}^{+l} dx = \frac{\mu_0 n I a^2 (2l)}{2r^3}$ (½)

4. Rewrite: $B = \frac{\mu_0}{4\pi} \frac{2(n \cdot 2l) I (\pi a^2)}{r^3}$. Total turns = $n(2l)$, so magnetic moment $m = n(2l) I \pi a^2$. (½)

5. $B = \frac{\mu_0}{4\pi} \frac{2m}{r^3}$ – exactly the axial field of a bar magnet of moment m . Hence a solenoid is equivalent to a bar magnet. (½)

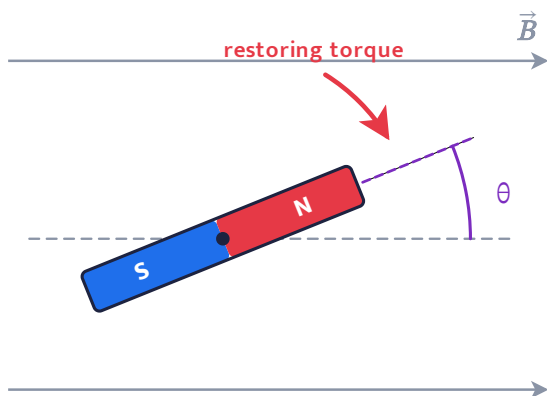
🔧 Derivation ★ [1½ marks]

★4 Starting from $B = \mu_0 (H + M)$, show that $B = \mu_0 \mu_r H$ and $\mu_r = 1 + \chi$.

Solution: 1. The field inside a material: $B = \mu_0 (H + M)$, where H is due to free current and M is the magnetisation. (½)

2. For a linear material $M = \chi H \Rightarrow B = \mu_0 (H + \chi H) = \mu_0 (1 + \chi) H$. (½)

3. Writing $\mu_r = 1 + \chi$: $B = \mu_0 \mu_r H = \mu H$, with $\mu = \mu_0 \mu_r$ = permeability of the material. (½)



pivot: vertical axis through centre

Magnet slightly displaced from B

🔧 Derivation ★ [1½ marks]

★5 A magnet of moment m and moment of inertia I is free to rotate in a uniform field B . Show that its small oscillations are SHM and find the period.

Solution: 1. At angle θ the restoring torque is

$$\tau = -mB \sin \theta \approx -mB \theta \quad (\theta \text{ small}). \quad (\frac{1}{2})$$

$$2. I \frac{d^2 \theta}{dt^2} = -mB \theta \Rightarrow \frac{d^2 \theta}{dt^2} = -\omega^2 \theta \text{ with } \omega^2 = \frac{mB}{I} - \text{SHM}. \quad (\frac{1}{2})$$

$$3. T = 2\pi \sqrt{\frac{I}{mB}} \quad (\frac{1}{2})$$

🎯 How to write for full marks

- Always draw the labelled figure first (poles, θ , $\vec{B}$, $\vec{m}$) – a figure alone often earns ½ mark.
- Write the final result in a box with its vector form ($\vec{\tau} = \vec{m} \times \vec{B}$, $U = -\vec{m} \cdot \vec{B}$) and the SI unit.
- For "define" questions on materials, give **definition + sign of χ + one example** – that is the full 1-mark answer.

Board question bank



Section A – Multiple choice (½ mark each)

[RBSE year] = asked in that board paper. Answers at the end of the chapter.

- 1 [RBSE 2025] The potential energy of a magnetic dipole of moment $\vec{m}$ placed in a uniform field $\vec{B}$ at angle θ is
 (a) $-mB\sin\theta$ (b) $mB\sin\theta$ (c) $mB\cos\theta$ (d) $-mB\cos\theta$
- 2 [RBSE 2023] The Curie temperature of iron is
 (a) 631 K (b) 317 K (c) 1394 K (d) 1043 K
- 3 The SI unit of magnetic dipole moment is
 (a) T (b) Wb (c) $A\ m^2$ (d) $A\ m^{-1}$
- 4 A magnet of moment $0.5\ A\ m^2$ is at 30° to a uniform field of $0.2\ T$. The torque on it is
 (a) $0.05\ N\ m$ (b) $0.025\ N\ m$ (c) $0.10\ N\ m$ (d) $0.087\ N\ m$
- 5 Work done in turning a magnet of moment $1.2\ J\ T^{-1}$ from the direction of a $0.25\ T$ field to the opposite direction is
 (a) $0.6\ J$ (b) $0.15\ J$ (c) $0.3\ J$ (d) zero
- 6 Which of the following is diamagnetic?
 (a) Nickel (b) Oxygen (c) Bismuth (d) Aluminium
- 7 A material has susceptibility 499. Its relative permeability is
 (a) 0.002 (b) 500 (c) 499 (d) 498
- 8 Gauss's law in magnetism, $\oint \vec{B} \cdot d\vec{A} = 0$, shows that
 (a) B is zero inside every closed surface
 (b) isolated magnetic poles do not exist
 (c) magnetic flux is a vector
 (d) magnetic field lines are open curves
- 9 The susceptibility of a paramagnetic substance is χ at $27^\circ C$. At $327^\circ C$ it will be
 (a) χ (b) 2χ (c) $\chi/2$ (d) $\chi/12$
- 10 The core of an electromagnet should have
 (a) negative susceptibility
 (b) high retentivity and high coercivity
 (c) low permeability and high coercivity
 (d) high permeability and low coercivity

Fill in the blanks (½ mark)	Answer
[RBSE 2025] The magnetic susceptibility of a paramagnetic substance is small and _____.	positive
[RBSE 2024] The net magnetic dipole moment per unit volume of a material is called _____.	magnetisation (M)
Above the Curie temperature a ferromagnetic substance becomes _____.	paramagnetic
Inside a bar magnet the field lines go from _____ pole to _____ pole.	S to N
For a superconductor the magnetic susceptibility is _____.	-1
The SI unit of magnetic intensity H is _____.	A m ⁻¹

Very short answer (1 mark)	Model answer
[RBSE 2024] Define paramagnetic substance.	Substances weakly magnetised in the direction of the applied field and weakly attracted; χ small positive. E.g. Al, O ₂ .
[RBSE 2020] What are diamagnetic substances? Give two examples.	Weakly magnetised opposite to the field, move from strong to weak field; χ negative. E.g. Bi, Cu.
[RBSE 2020] Why is soft iron used to make electromagnets?	High permeability and low coercivity (low retentivity): strongly magnetised by small current, demagnetised when current stops.
[RBSE 2023] Write two characteristics of a bar magnet.	(i) Poles always exist in pairs (no isolated pole). (ii) Freely suspended, it rests in the N-S direction; like poles repel.
State Gauss's law for magnetism.	Net magnetic flux through any closed surface is zero: $\oint \vec{B} \cdot d\vec{A} = 0$.
State Curie's law.	Susceptibility of a paramagnet is inversely proportional to absolute temperature: $\chi = C\mu_0 / T$.

Section B – Short answer (1½ marks)

Q1

[RBSE 2024] Write two properties of magnetic field lines.

Solution: (i) They are **continuous closed curves** – outside the magnet from N to S and inside from S to N. (½) (ii) **Two field lines never intersect**, since B at a point has one direction. (½) (iii) The tangent gives the direction of B; crowded lines mean a stronger field. (½)

Q2

Distinguish between diamagnetic, paramagnetic and ferromagnetic substances on the basis of susceptibility and behaviour in a non-uniform field.

Solution: Dia: χ small negative, moves strong $\rightarrow$ weak field (repelled). (½) Para: χ small positive, moves weak $\rightarrow$ strong field (weakly attracted). (½) Ferro: χ very large positive, strongly attracted. (½)

Q3

A short bar magnet has moment 0.4 A m^2 . Find the magnetic field at 10 cm from its centre (i) on its axis, (ii) on its equatorial line.

Solution: (i) $B_{ax} = \frac{\mu_0 2m}{4\pi r^3} = 10^{-7} \times \frac{0.8}{10^{-3}} = 8 \times 10^{-5} \text{ T}$, along $\vec{m}$. (1) (ii) $B_{eq} = B_{ax} / 2 = 4 \times 10^{-5} \text{ T}$, opposite to $\vec{m}$. (½)



Q4

Define retentivity and coercivity. Which of the two decides whether a material is suitable for a permanent magnet?

Solution: Retentivity: B remaining in the material when H is reduced to zero. (½) Coercivity: reverse H needed to reduce B to zero. (½) A permanent magnet needs **high coercivity** (and high retentivity) – e.g. steel. (½)



Answer Key & Quick Solutions

1(d) 2(d) 3(c) 4(a) 5(a) 6(c) 7(b) 8(b) 9(c) 10(d)

- (d) $U = -\vec{m} \cdot \vec{B} = -mB\cos\theta$ (zero at 90°).
- (d) NCERT: Fe 1043 K (Ni 631 K, Co 1394 K, Gd 317 K).
- (c) $m = NIA \Rightarrow A \text{ m}^2 (= \text{J T}^{-1})$.
- (a) $\tau = mB\sin 30^\circ = 0.5 \times 0.2 \times 0.5 = 0.05 \text{ N m}$.
- (a) $W = mB(\cos 0^\circ - \cos 180^\circ) = 2mB = 2 \times 1.2 \times 0.25 = 0.6 \text{ J}$.
- (c) Bi is diamagnetic; Al and O_2 are paramagnetic; Ni is ferromagnetic.
- (b) $\mu_r = 1 + \chi = 500$.
- (b) Zero net flux through every closed surface $\Rightarrow$ no monopole can be enclosed.
- (c) $\chi \propto 1/T$: 300 K $\rightarrow$ 600 K, so χ halves. (Using $^\circ\text{C}$ is the classic mistake.)
- (d) It must magnetise easily and lose magnetism when current stops – soft iron.



Last-minute revision

- m in A m^2 (S $\rightarrow$ N inside); field lines are closed loops, never intersect.
- $\tau = mB\sin\theta$ (max at 90°); $U = -mB\cos\theta$ (min at 0°); $W_{0 \rightarrow 180^\circ} = 2mB$.
- $B_{ax} = \frac{\mu_0 2m}{4\pi r^3}$, $B_{eq} = \frac{\mu_0 m}{4\pi r^3}$ – ratio 2 : 1.
- Gauss's law: $\oint \vec{B} \cdot d\vec{A} = 0 \Rightarrow$ no monopoles.
- $B = \mu_0(H + M)$, $M = \chi H$, $\mu_r = 1 + \chi$; H and M in A m^{-1} .
- χ : dia small –, para small +, ferro huge +. Superconductor $\chi = -1$.
- Curie law $\chi \propto 1/T$ (kelvin!). Fe becomes paramagnetic above 1043 K.
- Permanent magnet: steel (high coercivity). Electromagnet: soft iron (low coercivity, high μ).

CHAPTER

6

Electromagnetic Induction

17

Day 9

Board marks: 4

PYQ frequency ★★★

RBSE Class 12

Magnetic flux

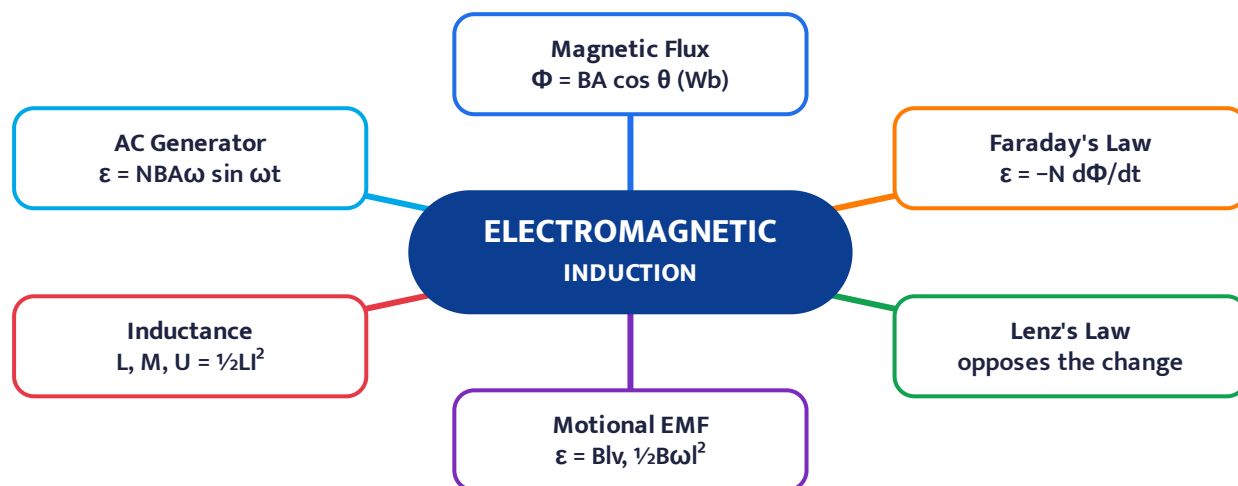
Faraday's & Lenz's laws

Motional emf

Eddy currents

Self & mutual inductance

AC generator



Chapter at a glance

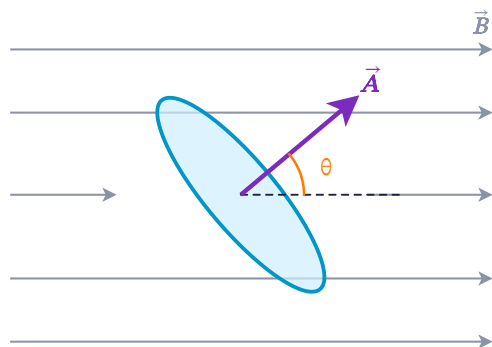
Board snapshot

- **Weightage: 4 marks** – typically one MCQ / fill-in (1/2 mark), one very-short question (1 mark) and a 1½-mark short answer (derivation or numerical).
- **2025:** MCQ – Lenz's law is based on conservation of energy; fill-in – induced emf is proportional to the rate of change of flux; 1½ marks – mutual inductance of two concentric coplanar coils ($r_1 \ll r_2$).
- **2024:** MCQ – device based on mutual induction (transformer); VSA – why self-inductance is called "electrical inertia"; 1½ marks – 5 m E-W wire falling at 2 m/s in $B = 0.3 \times 10^{-4}$ T, find the emf (3×10^{-4} V).
- **2023:** 1½ marks – current falls from 5.0 A to 1.0 A in 0.1 s with average emf 200 V, find L (5 H); 1½ marks – Faraday's and Lenz's laws. **2026 (CWSN):** mutual inductance / flux linkage.
- **Most likely in 2027:** (1) mutual / self-inductance derivation or numerical (asked every year 2023–26); (2) motional emf $\varepsilon = Blv$ numerical; (3) Faraday's & Lenz's laws and "electrical inertia".

6.1 Faraday's Experiments and Magnetic Flux

Faraday and Henry showed that a **changing magnetic flux** through a coil produces an emf (and a current if the circuit is closed) in it – this is **electromagnetic induction**. No battery is needed, and the emf lasts only while the flux is changing.

Experiment	What is done	Observation
1. Magnet and coil	Magnet moved towards / away from a coil with galvanometer	Deflection only during motion; reverses on recession
2. Coil and coil	Magnet replaced by a current-carrying coil	Same results – relative motion matters
3. Key on / off	Coils at rest; current in C_1 switched on / off	Momentary deflection at make and break



$$\Phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

Flux through a plane loop



Magnetic flux

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta$$

Magnetic flux – uniform B; θ = angle between B and the normal to the area

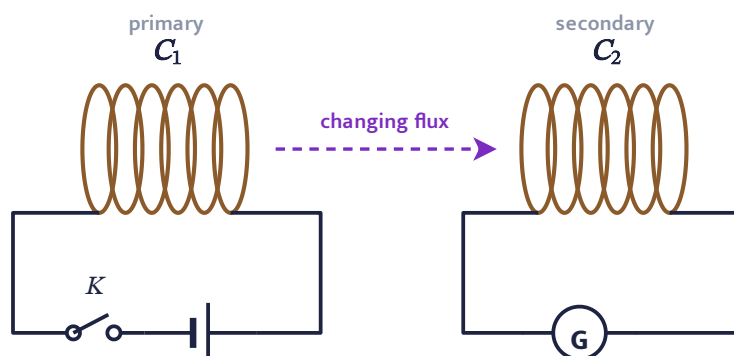
$$1 \text{ Wb} = 1 \text{ T m}^2$$

SI unit **weber**; flux is a **scalar**



Common Trap

θ is measured from the **normal**, not from the plane. Plane $\perp B \Rightarrow \theta = 0, \Phi = BA$ (maximum); plane $\parallel B \Rightarrow \Phi = 0$.



Faraday's coil-coil experiment: G deflects only when the current in C_1 is switched on or off

6.2 Faraday's Laws of Induction

- First law:** whenever the magnetic flux linked with a circuit changes, an emf is induced in it; the emf lasts as long as the change continues.
- Second law:** the magnitude of the induced emf is equal to the **rate of change of magnetic flux** linked with the circuit, $|\varepsilon| = N \left| \frac{d\Phi_B}{dt} \right|$. Flux can be changed by changing B, the area A, or the angle θ .

VS Sir's Exam Tip

If Φ is given as a function of time, **differentiate**: $\Phi = (5t^2 + 3t + 2) \text{ mWb} \Rightarrow \varepsilon = (10t + 3) \text{ mV}$. Never substitute t in Φ – emf depends on the **rate**, not the value, of flux.



Faraday's law

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

N turns; minus sign = Lenz's law

$$I = \frac{\varepsilon}{R} = \frac{N d\Phi}{R dt}$$

Induced current

$$q = \frac{N \Delta \Phi}{R}$$

Induced charge – independent of the time taken



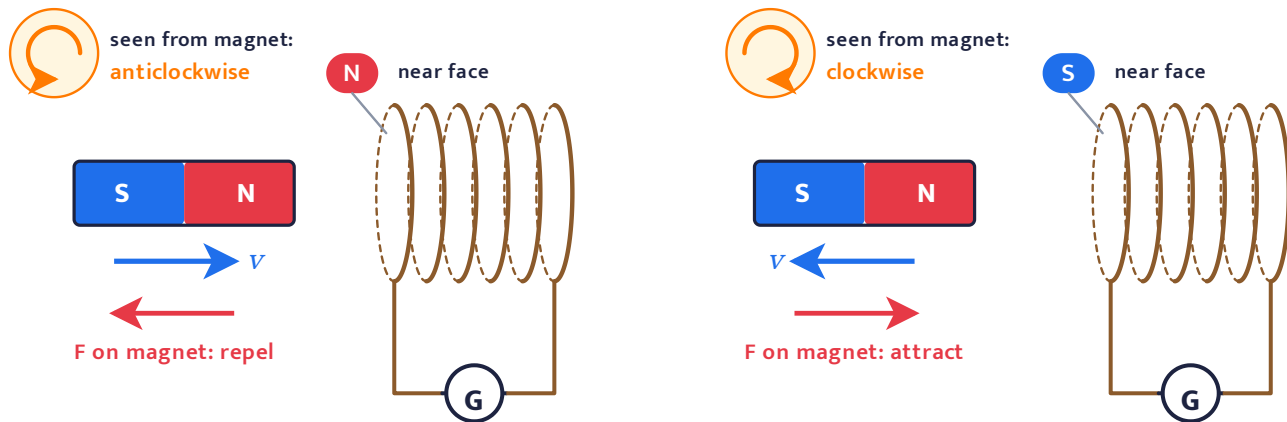
Solved Example

A 200-turn coil of area 0.02 m^2 is perpendicular to a field of 0.4 T . The field falls uniformly to zero in 0.1 s . The coil resistance is 8Ω . Find the emf, current and charge.

Solution: $\varepsilon = \frac{N \Delta(BA)}{\Delta t} = \frac{200 \times 0.4 \times 0.02}{0.1} = \mathbf{16 \text{ V}}$; $I = 16 / 8 = \mathbf{2 \text{ A}}$; $q = I \Delta t = \mathbf{0.2 \text{ C}}$ (also $\frac{N \Delta \Phi}{R} = \frac{200 \times 0.008}{8} = 0.2 \text{ C}$).

6.3 Lenz's Law and Conservation of Energy

Lenz's law: the direction of the induced emf (or current) is such that it **opposes the cause** that produces it – i.e. the change in magnetic flux. This is the **minus sign** in Faraday's law.



N-pole approaching: near face becomes N – the approach is opposed

N-pole receding: near face becomes S – the withdrawal is opposed

Lenz's law = conservation of energy

- If the induced current **helped** the change, the magnet would speed up by itself and produce current for free – energy would be created from nothing.
- Because it **opposes**, work must be done by us; that mechanical work reappears as electrical energy (Joule heat).

Common Trap

- A magnet dropped through a closed metal ring falls with **$a < g$** , while entering and while leaving.
- If the ring has a **cut**, emf is induced but no current flows $\Rightarrow$ no opposition $\Rightarrow a = g$.

6.4 Motional emf

A conductor of length l moving with velocity v perpendicular to a uniform field B develops an emf $\varepsilon = Blv$ across its ends. The derivation (Derivation 1, section 6.8) uses the change of flux of the rectangular circuit PQRS formed with the rails.

Motional emf

$$\varepsilon = Blv$$

Rod, B and v mutually perpendicular

$$F = \frac{B^2 l^2 v}{R}, \quad P = Fv = \frac{B^2 l^2 v^2}{R} \quad \text{Force and power for uniform } v (= \text{Joule heat})$$

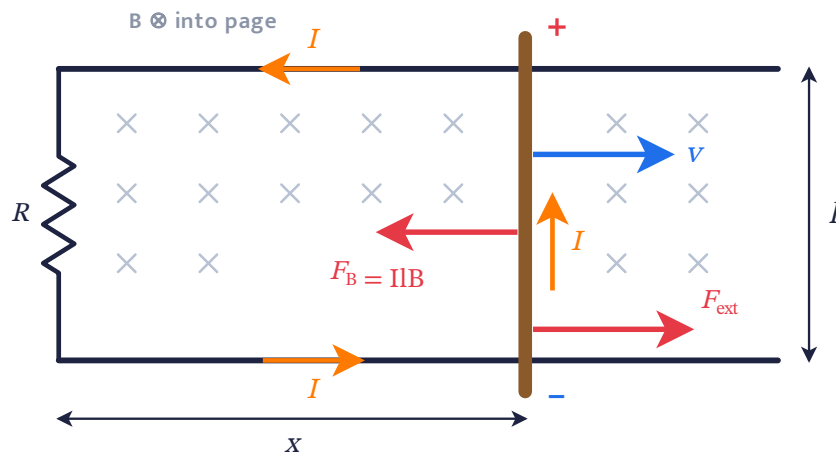
$$\varepsilon = \frac{1}{2} B \omega l^2$$

Rod rotating about one end

Solved Example

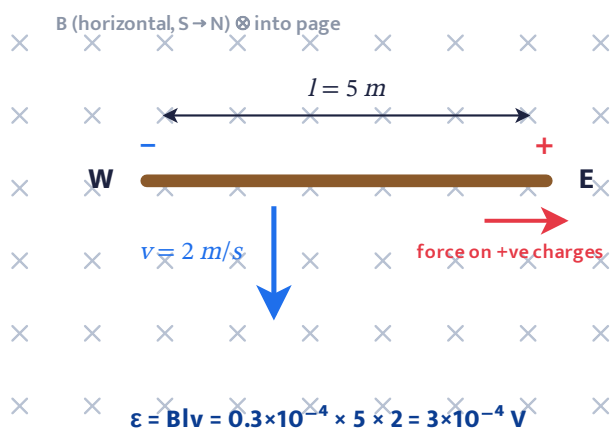
A rod of length 0.4 m slides at 5 m/s on rails ($R = 2 \Omega$) in $B = 0.5$ T. Find ε , I , the force and the power needed.

Solution: $\varepsilon = Blv = 0.5 \times 0.4 \times 5 = 1$ V; $I = 1 / 2 = 0.5$ A; $F = IlB = 0.5 \times 0.4 \times 0.5 = 0.1$ N; $P = Fv = 0.5$ W = $I^2 R$ ✓.



$$\begin{aligned}\Phi &= Blx \\ \varepsilon &= Blv \\ I &= Blv / R \\ F_B &= B^2 l^2 v / R \\ P &= B^2 l^2 v^2 / R \\ &= \text{Joule heat in } R \\ &\quad (\text{anticlockwise current})\end{aligned}$$

Rod sliding on rails in a field into the page – the magnetic force IlB always opposes the motion



[RBSE 2024] Horizontal E–W wire falling in the Earth's horizontal field – the emf is Blv , with the east end positive

Common Trap

- A loop moving **entirely inside** a uniform field has **no** emf (flux constant); emf exists only while entering or leaving.
- Take only the component of v perpendicular to both B and the rod; if $v \parallel B$ or $v \parallel l$, $\varepsilon = 0$.

6.5 Eddy Currents

When a **solid piece of metal** is placed in a changing magnetic flux (or moves in a field), circulating currents are induced in it, like whirlpools – **eddy currents** (Foucault currents). By Lenz's law they oppose the motion and heat the metal.

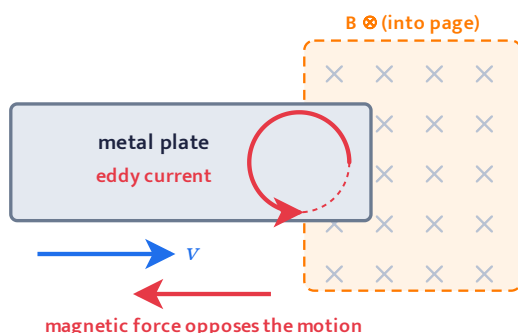


Plate entering a magnetic field: eddy current and the retarding force

Applications

- Magnetic braking** in electric trains.
- Electromagnetic damping** – dead-beat galvanometer.
- Induction furnace** and induction cooking.

Reducing eddy losses

Use a **laminated core** – thin insulated sheets, planes parallel to B – in transformers, motors and generators.

6.6 Self-Inductance and Mutual Inductance

Flux linked with a coil is proportional to the current producing it. The constant of proportionality is the **inductance**; it depends only on the **geometry and the medium** (not on the current). SI unit: **henry** ($1 \text{ H} = 1 \text{ Wb A}^{-1} = 1 \text{ V s A}^{-1}$).

- **Self-inductance (L):** when the current in a coil changes, the emf induced in the **same** coil is $\varepsilon = -L \frac{dI}{dt}$. **1 henry** is the self-inductance of a coil in which an emf of 1 V is induced when the current changes at 1 A/s. L is the **electrical inertia** of the coil – it opposes the change of current.
- **Mutual inductance (M):** a changing current I_1 in one coil induces an emf in a neighbouring coil, $\varepsilon_2 = -M \frac{dI_1}{dt}$; also $N_2 \Phi_2 = M I_1$ and $M_{12} = M_{21}$. M depends on the size, number of turns, separation and relative orientation of the coils and on the medium.



Inductance formulas

$N\Phi = LI, \quad \varepsilon = -L \frac{dI}{dt}$	Self-inductance
$L = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 A l \quad (\times \mu_r \text{ with core})$	Long solenoid (Derivation 3)
$M = \mu_0 n_1 n_2 \pi r_1^2 l$	Two long co-axial solenoids (Derivation 5)
$M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2r_2}$	Concentric coplanar circular coils, $r_1 \ll r_2$ (Derivation 4)
$U = \frac{1}{2} L I^2, \quad u_B = \frac{B^2}{2\mu_0}$	Energy stored; energy per unit volume



Solved Example

A solenoid has 500 turns, length 0.5 m and cross-section 4 cm^2 . Find its self-inductance and the energy stored when it carries 2 A.

Solution: $L = \frac{\mu_0 N^2 A}{l} = \frac{4\pi \times 10^{-7} \times 500^2 \times 4 \times 10^{-4}}{0.5} \approx \mathbf{0.25 \text{ mH}}$
 $U = \frac{1}{2} L I^2 = \frac{1}{2} (2.51 \times 10^{-4}) (2)^2 \approx \mathbf{0.50 \text{ mJ}}$



Solved Example

Two concentric coplanar circular coils have $N_1 = 100, r_1 = 2 \text{ cm}$ (inner) and $N_2 = 50, r_2 = 20 \text{ cm}$ (outer). Find M.

Solution: $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2r_2} = \frac{4\pi \times 10^{-7} \times 100 \times 50 \times \pi \times (0.02)^2}{2 \times 0.2} \approx \mathbf{2.0 \times 10^{-5} \text{ H}}$



Electrical inertia

Mass resists a change in **velocity**; self-inductance resists a change in **current**. $\frac{1}{2} L I^2$ plays the role of $\frac{1}{2} m v^2$ – so L is called the electrical inertia of the circuit. This is why a spark appears at the switch when an inductive circuit is broken suddenly (large $dI/dt \Rightarrow$ large back emf).

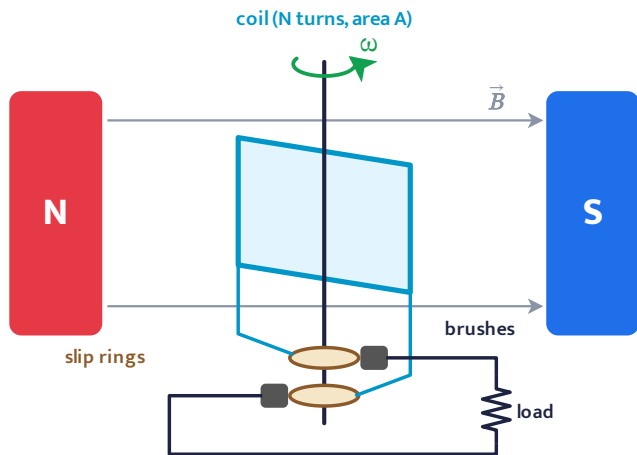


Common Trap

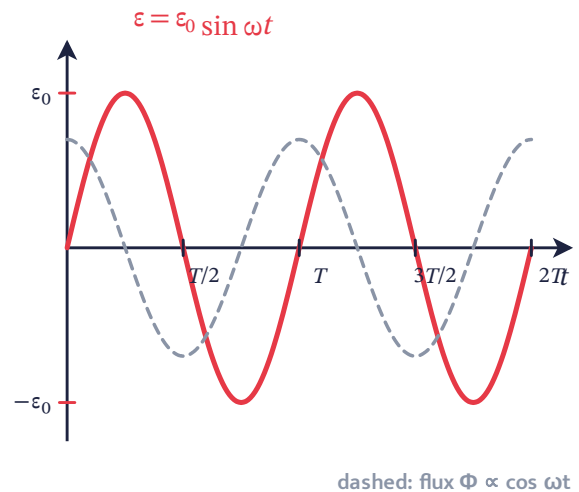
- L and M do **not** depend on the current – only on geometry (N, A, l) and the medium (μ_r).
- $L \propto N^2$ (not N): doubling the turns on the same core gives **4L**.

6.7 AC Generator

An AC generator converts **mechanical energy into electrical energy** by electromagnetic induction. A coil of N turns and area A is rotated with uniform angular speed ω in a uniform field B; **slip rings** and **brushes** take the emf to the external circuit. The emf is sinusoidal: $\varepsilon = NBA\omega \sin \omega t$ (Derivation 7).



Coil rotating between the poles; slip rings and brushes



emf is maximum when the flux is zero

Solved Example

A 50-turn coil of area 0.1 m^2 rotates at 10 revolutions per second in a field of 0.4 T . Find the peak emf.

Solution: $\varepsilon_0 = NBA(2\pi\nu) = 50 \times 0.4 \times 0.1 \times 2\pi \times 10 \approx 126 \text{ V}$. The average emf over a full cycle is zero.

VS Sir's Exam Tip

Coil plane $\perp B \Rightarrow$ flux maximum, **emf = 0**; coil plane $\parallel B \Rightarrow$ flux zero, **emf maximum**. Flux and emf differ in phase by 90° .

6.8 Important Derivations ★**Derivation ★ [3 marks]**

Derivation 1 (Part 1). Derive an expression for the emf induced in a conductor of length l moving with velocity v perpendicular to a uniform magnetic field B (figure in 6.4).

Solution: (i) Let rod PQ of length l slide on two parallel rails with speed v ; B is uniform and perpendicular to the plane of the rails. At distance x from the closed end the area of the circuit is $A = lx$. (½)

(ii) Flux through the circuit: $\Phi_B = BA = Blx$. (½)

(iii) Faraday's law: $\varepsilon = -\frac{d\Phi_B}{dt} = -Bl\frac{dx}{dt}$. Since $\frac{dx}{dt} = v$, $|\varepsilon| = Blv$. (1)

(iv) **Check by the Lorentz force:** a free electron in the rod feels force $F = evB$ along the rod. Charge collects at the ends until $eE = evB$, i.e. $E = vB$; then $\varepsilon = El = Blv$ – the same result. (1)

Derivation ★ [3 marks]

Derivation 1 (Part 2). For the same rod show that the mechanical work done to move it is converted into electrical energy (Joule heat in R).

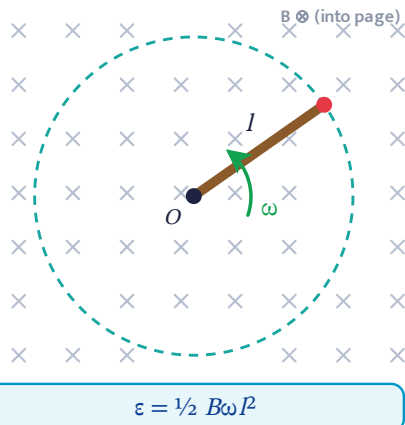
Solution: (i) Current in the circuit of resistance R : $I = \frac{\varepsilon}{R} = \frac{Blv}{R}$. (½)

(ii) By Lenz's law the magnetic force on the rod opposes its motion: $F = IlB = \frac{B^2 l^2 v}{R}$. (1)

(iii) To keep constant speed an external force of the same magnitude must be applied; mechanical power

$$P_{\text{mech}} = Fv = \frac{B^2 l^2 v^2}{R}. \quad (\frac{1}{2})$$

(iv) Electrical (Joule) power: $P_{\text{el}} = I^2 R = \frac{B^2 l^2 v^2}{R}$. Hence $P_{\text{mech}} = P_{\text{el}}$ – energy is conserved. (1)



Rod rotating about end O in a field into the page

Derivation ★ [3 marks]

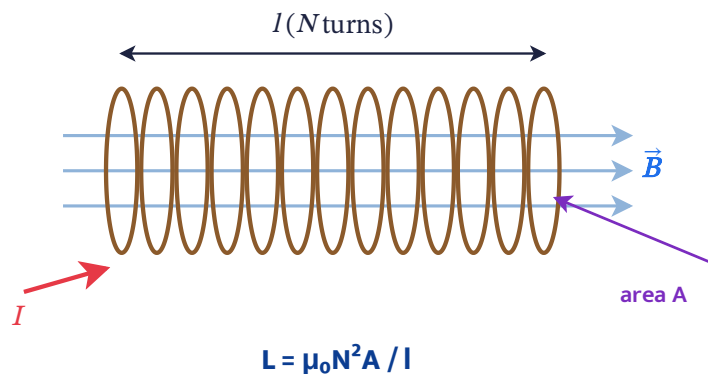
Derivation 2. A rod of length l rotates with angular speed ω about one end, in a uniform field B perpendicular to its plane of rotation. Show that the emf between its ends is $\frac{1}{2}B\omega l^2$.

Solution: (i) Take an element dr at distance r from O; its speed is $v = \omega r$. (½)

(ii) emf across the element: $d\varepsilon = Bv dr = B\omega r dr$. (1)

(iii) Integrating from 0 to l : $\varepsilon = B\omega \int_0^l r dr = \frac{1}{2}B\omega l^2$. (1)

(iv) Check: in time dt the rod sweeps area $\frac{1}{2}l^2\omega dt$, so $\varepsilon = \frac{d\Phi}{dt} = \frac{1}{2}Bl^2\omega$. (½)

Long solenoid of N turns, length l and cross-sectional area A **Derivation ★ [3 marks]**

Derivation 3. Derive an expression for the self-inductance of a long air-cored solenoid of length l , area of cross-section A and N turns.

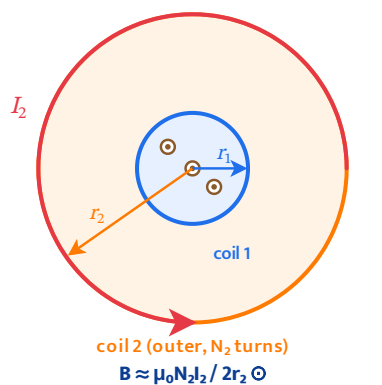
Solution: (i) Self-inductance is defined by flux linkage: $N\Phi_B = LI$. (½)

(ii) For a long solenoid carrying current I , the field inside is uniform: $B = \mu_0 nI = \frac{\mu_0 NI}{l}$. (½)

(iii) Flux through one turn: $\Phi_B = BA = \frac{\mu_0 NIA}{l}$. (½)

(iv) Flux linked with all N turns: $N\Phi_B = \frac{\mu_0 N^2 A}{l} I$. (½)

(v) Comparing with $N\Phi_B = LI$: $L = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 Al$. With a core of relative permeability μ_r , $L = \mu_r \mu_0 n^2 Al$. (1)

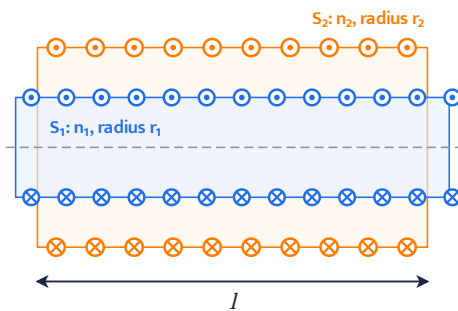
Two concentric coplanar coils, $r_1 \ll r_2$ **Derivation ★ [1½ marks]**

Derivation 4 – [RBSE 2025]. Find the mutual inductance of two concentric, coplanar circular coils of radii r_1 and r_2 ($r_1 \ll r_2$) having N_1 and N_2 turns.

Solution: (i) Let the outer coil 2 carry current I_2 . Field at the common centre: $B_2 = \frac{\mu_0 N_2 I_2}{2r_2}$. (½)

(ii) As $r_1 \ll r_2$, B_2 is almost uniform over the inner coil, so flux through each turn of coil 1 is $\Phi_1 = B_2 \pi r_1^2$. (½)

(iii) Flux linked with coil 1: $N_1 \Phi_1 = MI_2$, so $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2r_2}$. For single turns, $M = \frac{\mu_0 \pi r_1^2}{2r_2}$. (½)



$$M = \mu_0 n_1 n_2 \pi r_1^2 l$$

Two long co-axial solenoids of length l

Derivation ★ [3 marks]

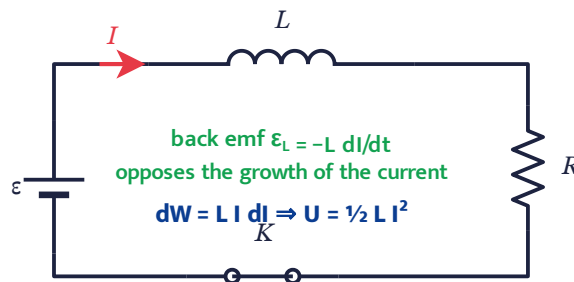
Derivation 5. Obtain the mutual inductance of two long co-axial solenoids S_1 (inner, radius r_1 , n_1 turns per unit length) and S_2 (outer, radius r_2 , n_2 turns per unit length), each of length l .

Solution: (i) Let S_2 carry current I_2 ; the field inside it is $B_2 = \mu_0 n_2 I_2$. (½)

(ii) Flux through each turn of S_1 : $\Phi_1 = B_2 \pi r_1^2$; number of turns of $S_1 = n_1 l$. (½)

(iii) Flux linkage: $N_1 \Phi_1 = (n_1 l)(\mu_0 n_2 I_2) \pi r_1^2 = M I_2$. (1)

(iv) Hence $M = \mu_0 n_1 n_2 \pi r_1^2 l$; the same value results if the current is passed through S_1 ($M_{12} = M_{21}$). (1)



Inductor in a circuit: the source does work against the back emf while the current rises from 0 to I

Derivation ★ [3 marks]

Derivation 6. Derive an expression for the energy stored in an inductor of self-inductance L carrying a current I . Hence obtain the energy density of the magnetic field.

Solution: (i) While the current grows, a back emf $|\varepsilon| = L \frac{dI}{dt}$ opposes it, so the source must do work. (½)

(ii) Work in time dt when the instantaneous current is i : $dW = |\varepsilon| i dt = L i di$. (1)

(iii) Total work from 0 to I : $W = \int_0^I L i di = \frac{1}{2} L I^2$, stored as magnetic energy: $U = \frac{1}{2} L I^2$. (1)

(iv) For a solenoid, $L = \mu_0 n^2 A l$ and $B = \mu_0 n I$: $U = \frac{1}{2} \mu_0 n^2 A l I^2 = \frac{B^2}{2\mu_0} (A l) \Rightarrow$ energy density $u_B = \frac{B^2}{2\mu_0}$. (½)

Derivation ★ [3 marks]

Derivation 7. Describe the principle and working of an AC generator and obtain the expression for the emf induced in its coil (diagram in 6.7).

Solution: (i) **Principle:** when a coil rotates in a magnetic field, the flux linked with it changes and an emf is induced (electromagnetic induction); mechanical energy $\rightarrow$ electrical energy. (½)

(ii) Let the coil (N turns, area A) rotate with angular speed ω in a uniform field B . At $t = 0$ the normal to the coil is along B ; after time t the angle between them is $\theta = \omega t$. (½)

(iii) Flux linked: $\Phi = N B A \cos \omega t$. (½)

(iv) Induced emf: $\varepsilon = -\frac{d\Phi}{dt} = N B A \omega \sin \omega t = \varepsilon_0 \sin \omega t$, where $\varepsilon_0 = N B A \omega$ is the peak emf. (1)

(v) Slip rings and brushes connect the coil to the load; the current $I = \frac{\varepsilon_0}{R} \sin \omega t$ changes direction twice in every rotation – alternating current. (½)

VS Sir's Exam Tip

- Examiner's checklist: labelled figure $\rightarrow$ flux expression $\rightarrow \varepsilon = -d\Phi / dt \rightarrow$ boxed result. Always write the **definition** of the quantity first (e.g. $N\Phi = LI$) – it earns the first $\frac{1}{2}$ mark.
- For numericals write the formula, substitute with units, and give the final answer with unit – each earns separate credit.

Board question bank

Section A – Multiple choice ($\frac{1}{2}$ mark each)

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- [RBSE 2025]** Lenz's law is based on the conservation of
 (a) mass (b) charge (c) momentum (d) energy
- [RBSE 2024]** Which of the following devices works on the principle of mutual induction?
 (a) transformer (b) electric motor
 (c) moving coil galvanometer (d) cyclotron
- The SI unit of magnetic flux is
 (a) tesla (b) weber (c) henry (d) Wb m^{-2}
- The magnetic flux through a coil varies as $\Phi = (4t^2 + 2t + 1)$ mWb. The induced emf at $t = 2$ s is
 (a) 8 mV (b) 36 mV (c) 21 mV (d) 18 mV
- Two coils have mutual inductance 0.5 H. The current in one falls from 6 A to 2 A in 0.5 s. The average emf induced in the other is
 (a) 2 V (b) 1 V (c) 8 V (d) 4 V
- A bar magnet is dropped along the axis of a closed horizontal copper ring. While it passes through the ring its acceleration is
 (a) more than g (b) less than g (c) equal to g (d) zero
- If the number of turns of a solenoid is doubled, keeping its length and area unchanged, its self-inductance becomes
 (a) half (b) unchanged (c) four times (d) double
- The energy stored in an inductor of 40 mH carrying a current of 5 A is
 (a) 0.1 J (b) 0.2 J (c) 0.5 J (d) 1.0 J
- Eddy current loss in the iron core of a transformer is reduced by using
 (a) a laminated core (b) more turns of wire
 (c) a copper core (d) a solid iron core

- 10 The flux through each turn of a 50-turn coil of resistance $5\ \Omega$ changes by 3×10^{-4} Wb. The charge that flows through the coil is
- (a) 3×10^{-4} C (b) 3×10^{-3} C
(c) 3×10^{-2} C (d) depends on the time taken

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2025] The induced emf in a circuit is directly proportional to the _____ of change of magnetic flux linked with it.	rate
The direction of the induced current is given by _____ law.	Lenz's
The SI unit of mutual inductance is _____.	henry
Eddy currents in a transformer are reduced by using a _____ core.	laminated
The emf induced in the coil of an AC generator is maximum when the plane of the coil is _____ to the magnetic field.	parallel
The energy stored in an inductor of inductance L carrying current I is _____.	$\frac{1}{2}LI^2$

Very short answer (1 mark)

Question	Model answer
[RBSE 2024] Why is self-inductance called "electrical inertia"?	Because it opposes any change of current in the coil, just as mass (inertia) opposes any change in velocity.
Define magnetic flux and give its SI unit.	Flux through a surface $= \vec{B} \cdot \vec{A} = BA \cos \theta$ (number of field lines crossing it); unit weber (Wb).
Write the expression for the emf induced in a rod of length l moving with velocity v perpendicular to field B .	$\varepsilon = Blv$.

Question	Model answer
Give one use of eddy currents.	Induction furnace / magnetic braking / dead-beat galvanometer (any one).
What is the emf induced in a coil carrying a steady (constant) current?	Zero , because $dI / dt = 0$ (flux does not change).
Why does a spark appear at the switch when a circuit containing a large inductor is suddenly broken?	Current falls to zero very quickly (large dI / dt) $\Rightarrow$ large back emf $L dI / dt$ across the gap.

Short answer (Section B, 1½ marks)

Short answer (1½ marks)

[RBSE 2024] A horizontal wire 5 m long, placed in the east–west direction, falls with a velocity of 2 m/s in a horizontal magnetic field of 0.3×10^{-4} T (south to north). Find the induced emf across its ends.

Solution: The wire, the velocity (vertical) and the field (horizontal) are mutually perpendicular. (½)

$$\varepsilon = Blv = 0.3 \times 10^{-4} \times 5 \times 2 \text{ (½)} = 3 \times 10^{-4} \text{ V (east end positive). (½)}$$

 Short answer (1½ marks)

[RBSE 2023] The current in a coil changes from 5.0 A to 1.0 A in 0.1 s and an average emf of 200 V is induced. Find the self-inductance of the coil.

Solution: $|\varepsilon| = L \left| \frac{\Delta I}{\Delta t} \right| \quad (1/2) \Rightarrow L = \frac{200 \times 0.1}{5.0 - 1.0} \quad (1/2) = 5 \text{ H. } (1/2)$

 Short answer (1½ marks)

[RBSE 2023] State Faraday's laws of electromagnetic induction and Lenz's law.

Solution: First law: an emf is induced in a circuit whenever the magnetic flux linked with it changes. (½) **Second law:** the magnitude of the induced emf equals the rate of change of flux linked, $\varepsilon = -N d\Phi / dt$. (½) **Lenz's law:** the induced emf (current) acts in such a direction that it opposes the change of flux that produces it. (½)

 Short answer (1½ marks)

Distinguish between self-inductance and mutual inductance (three points).

Solution: (i) Self: emf induced in the **same** coil, $\varepsilon = -L dI / dt$; mutual: emf induced in a **neighbouring** coil, $\varepsilon_2 = -M dI_1 / dt$. (ii) L depends on the coil's own turns, area, length and medium; M also depends on the **separation and orientation** of the two coils. (iii) For a coil $L \propto N^2$; for two coils $M \propto N_1 N_2$. (½ each)

 Last-minute revision

- $\Phi = BA \cos \theta$ (weber, scalar); $\varepsilon = -N d\Phi / dt$; charge $q = N \Delta \Phi / R$ (independent of time).
- Lenz's law (minus sign) = conservation of energy; induced current opposes the **change** of flux.
- Motional emf $\varepsilon = Blv$; force $F = B^2 l^2 v / R$; $P = Fv = I^2 R$. Rotating rod: $\varepsilon = \frac{1}{2} B \omega l^2$.
- Eddy currents: heating + damping; cut by laminated cores.
- $\varepsilon = -L dI / dt$; $L = \mu_0 N^2 A / l = \mu_0 n^2 A l$; $L \propto N^2$; $1 \text{ H} = 1 \text{ Wb/A}$; $U = \frac{1}{2} L I^2$.
- $M = \mu_0 n_1 n_2 \pi r_1^2 l$ (co-axial solenoids); $M = \mu_0 N_1 N_2 \pi r_1^2 / 2r_2$ (concentric coils, $r_1 \ll r_2$).
- AC generator: $\varepsilon = NBA \omega \sin \omega t$; peak when coil plane $\parallel$ B; mechanical $\rightarrow$ electrical energy.

 Answer Key & Quick Solutions

1(d) 2(a) 3(b) 4(d) 5(d) 6(b) 7(c) 8(c) 9(a) 10(b)

1. (d) The induced current opposes the change, so work must be done and is converted to electrical energy – energy is conserved.
2. (a) Changing current in the primary coil induces emf in the secondary coil – mutual induction.
3. (b) $\Phi = BA \Rightarrow T \text{ m}^2 = \text{weber (Wb)}$. Wb m^{-2} is tesla.
4. (d) $\varepsilon = d\Phi / dt = (8t + 2) \text{ mV} = 18 \text{ mV}$ at $t = 2 \text{ s}$.
5. (d) $|\varepsilon| = M \Delta I / \Delta t = 0.5 \times 4 / 0.5 = 4 \text{ V}$.
6. (b) The induced current opposes the relative motion (Lenz) $\Rightarrow$ an upward magnetic force $\Rightarrow a < g$.
7. (c) $L = \mu_0 N^2 A / l \propto N^2 \Rightarrow 4L$.
8. (c) $U = \frac{1}{2} L I^2 = \frac{1}{2} \times 0.04 \times 25 = 0.5 \text{ J}$.
9. (a) Thin insulated laminations break up the large eddy-current loops.
10. (b) $q = N \Delta \Phi / R = 50 \times 3 \times 10^{-4} / 5 = 3 \times 10^{-3} \text{ C}$ – independent of time.

CHAPTER

7

Alternating Current

17

Day 10–11

Board marks: 5

PYQ frequency ★★★

RBSE Class 12

AC generator

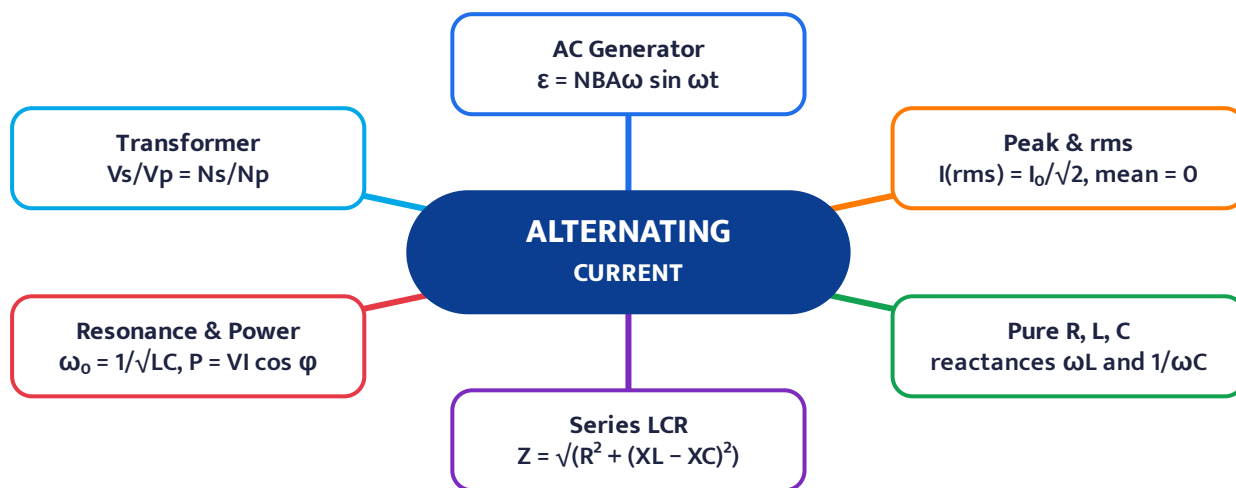
rms value

R, L, C circuits

Series LCR & resonance

Power factor

Transformer



Chapter at a glance



Board snapshot

- **Marks: 5** – usually one Section D (4 marks) or Section C (3 marks) question + one or two ½/1-mark items. The highest-scoring chapter of Unit IV.
- **2025:** MCQ – AC generator converts mechanical → electrical energy; MCQ on voltage in an AC circuit; fill-in – current lags voltage by $\pi/2$ in a pure inductor; **D (4)** – rms value + capacitor in AC (reactance, phase, phasor) **OR** resonance in series LCR + $I-\omega$ graph.
- **2024:** fill-in – mean value of AC over a cycle = 0; **B** – labelled AC generator; **D (4)** – $I_0 = \sqrt{2} I_{rms}$ and average power for $I = 4\sin\omega t$, $V = 200\sin(\omega t + \pi/3)$ **OR** power in a pure inductor = 0 and Z for $R = 24\ \Omega$, $X_L = X_C = 110\ \Omega$.
- **2023:** MCQ – frequency of AC in India (50 Hz); **B** – power factor of pure C and of LCR at resonance; **B** – three energy losses in a transformer + remedies.
- **2026* (CWSN):** wattless current; **C** – LCR impedance OR transformer turns ratio. **2020:** power-factor numericals.
- **Most likely in 2027:** ① AC through a capacitor/inductor with phasor and power (Derivations 2–3) ② series LCR impedance + resonance curve (Derivation 4) ③ transformer theory & losses or AC generator diagram.

7.1 Alternating Current and the AC Generator

A current whose **magnitude changes continuously with time and whose direction reverses periodically** is called **alternating current (AC)**. For sinusoidal AC, $i = I_0\sin\omega t$ and $v = V_0\sin\omega t$, where I_0 , V_0 are the **peak values (amplitudes)** and $\omega = 2\pi\nu$ is the angular frequency. In India the domestic supply is **220 V, 50 Hz**. AC is preferred over DC because its voltage can be easily stepped up or down by a **transformer**, so power can be transmitted over long distances with little loss.

An **AC generator (dynamo)** converts **mechanical energy into electrical energy**. It works on **electromagnetic induction**: a coil rotated in a uniform magnetic field has a continuously changing flux, so an alternating emf is induced in it. (Full answer with labelled diagram: Derivation 6.)

AC generator

$$\varepsilon = NBA\omega \sin \omega t = \varepsilon_0 \sin \omega t \quad \text{emf of AC generator}$$

$$\varepsilon_0 = NBA\omega, \quad \omega = 2\pi\nu \quad \text{Peak emf; } \nu = \text{rotations per second}$$

Solved Example

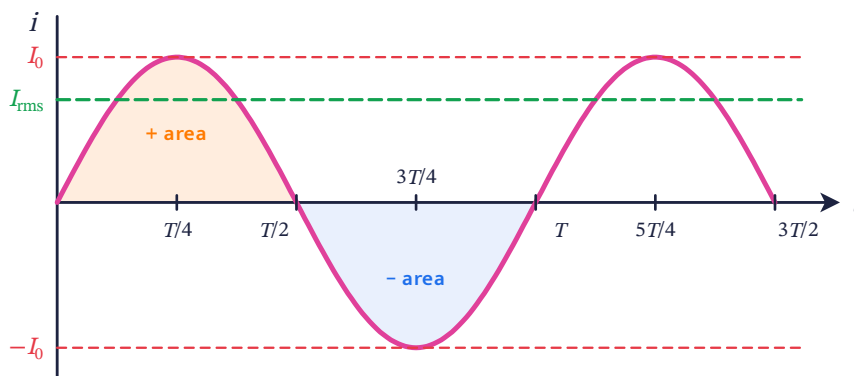
A 100-turn coil of area 0.1 m^2 rotates at 50 rev/s in a field of 0.01 T. Find the peak emf.

Solution:

$$\varepsilon_0 = NBA(2\pi\nu) = 100 \times 0.01 \times 0.1 \times 2\pi \times 50 \approx \mathbf{31.4 \text{ V.}}$$

7.2 Peak, Mean and rms Values

The **mean (average) value of AC over a complete cycle is zero**, because the positive and negative half-cycles are equal. So AC is measured by its **rms (root mean square) or effective value**: the rms value of AC is that **steady (DC) current which produces the same heat in a given resistor in the same time** as the AC does. AC ammeters and voltmeters read **rms values**.



$$I_{rms} = I_0 / \sqrt{2} \approx 0.707 I_0$$

mean over a full cycle = 0

Sinusoidal AC: the + and - areas cancel, so the mean over a full cycle is zero; $I_{rms} = 0.707 I_0$

Peak, mean and rms

$$I_{rms} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0 \quad \text{rms current (Derivation 1)}$$

$$V_{rms} = \frac{V_0}{\sqrt{2}} \quad \text{rms voltage}$$

$$\bar{i}_T = 0, \quad \bar{i}_{T/2} = \frac{2I_0}{\pi} \approx 0.637 I_0 \quad \text{Mean over full cycle; over half cycle}$$

Solved Example

Mains supply is 220 V, 50 Hz. Find the peak voltage and ω .

Solution: $V_0 = \sqrt{2} \times 220 \approx \mathbf{311 \text{ V}}$; $\omega = 2\pi \times 50 \approx \mathbf{314 \text{ rad/s.}}$

Common Trap

"220 V AC" is the **rms** value; the peak is 311 V. Mean of AC over a cycle is zero, but mean **power** is not.

7.3 Phasors; AC through R, L and C

A **phasor** is a vector that rotates anticlockwise about the origin with angular speed ω . Its length equals the peak value and its **projection on the vertical axis gives the instantaneous value**. The angle between the voltage and current phasors is the **phase difference ϕ** .

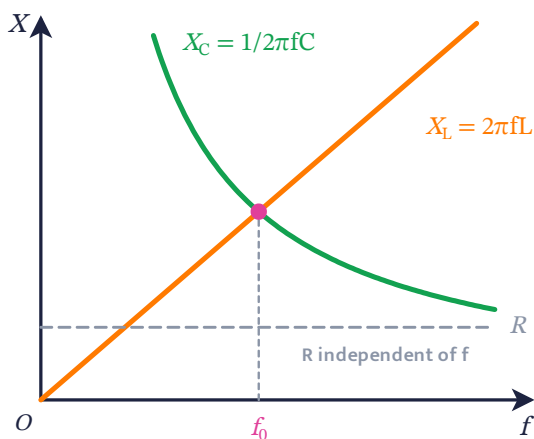
Pure resistor: with $v = V_0 \sin \omega t$, Ohm's law gives $i = \frac{V_0}{R} \sin \omega t$. The current is **in phase** with the voltage, and the average power is $\bar{P} = I_{rms}^2 R = V_{rms} I_{rms}$ (since the mean of $\sin^2 \omega t$ over a cycle is $1/2$).

Reactances

$$X_L = \omega L = 2\pi \nu L, \quad i = \frac{V_0}{X_L} \sin\left(\omega t - \frac{\pi}{2}\right) \quad \text{Inductive reactance } (\Omega); \text{ current lags voltage by } \pi/2; \bar{P} = 0 \quad (\text{Derivation 2})$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi \nu C}, \quad i = \frac{V_0}{X_C} \sin\left(\omega t + \frac{\pi}{2}\right) \quad \text{Capacitive reactance } (\Omega); \text{ current leads voltage by } \pi/2; \bar{P} = 0 \quad (\text{Derivation 3})$$

Property	Resistor R	Inductor L	Capacitor C
Opposition	R	$X_L = \omega L$	$X_C = 1 / \omega C$
Variation with ν	independent	$X_L \propto \nu$	$X_C \propto 1 / \nu$
For DC ($\nu = 0$)	R	0 (passes DC easily)	∞ (blocks DC)
Phase of i w.r.t. v	in phase	lags by $\pi/2$	leads by $\pi/2$
Power factor; average power	1; $I_{rms}^2 R$	0; zero	0; zero



X_L rises linearly with f ; X_C falls as $1/f$ (rectangular hyperbola)

Solved Example

Find the rms current when (i) a 44 mH inductor, (ii) a 100 μ F capacitor is connected to a 220 V, 50 Hz supply.

Solution: (i) $X_L = 2\pi(50)(0.044) = 13.8 \Omega \Rightarrow$

$$I = 220 / 13.8 \approx \mathbf{15.9 \text{ A.}} \quad (\text{ii}) \quad X_C = \frac{1}{2\pi(50)(10^{-4})} = 31.8 \Omega \Rightarrow$$

$$I = 220 / 31.8 \approx \mathbf{6.9 \text{ A.}}$$

Memory Trick

CIVIL – in C, I leads V; V leads I in L.

7.4 Series LCR Circuit, Impedance and Resonance

In a series LCR circuit the **same current** flows through R, L and C, but the voltages V_R , V_L , V_C are out of phase, so they are added **as phasors**, not algebraically. The total opposition is the **impedance** Z (unit ohm). At the **resonant frequency** $X_L = X_C$, so $Z = R$ is minimum and the current is maximum (Derivation 4).

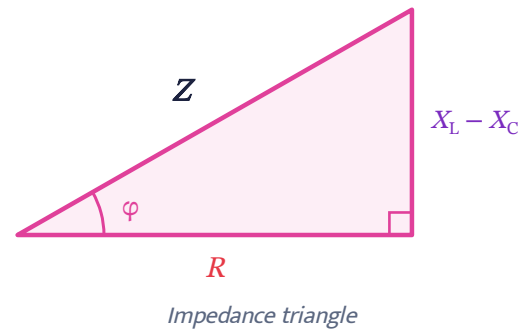
Series LCR

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad \text{Impedance; } I = V/Z$$

$$\tan \phi = \frac{X_L - X_C}{R}, \quad \cos \phi = \frac{R}{Z} \quad \text{Phase of } V \text{ ahead of } I$$

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad \nu_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{Resonance: } Z = R, I \text{ maximum}$$

$$\cos \phi = R/Z, \quad \tan \phi = (X_L - X_C)/R$$



Condition	Nature of circuit	Phase
$X_L > X_C$ ($\omega > \omega_0$)	inductive	current lags voltage by ϕ
$X_L < X_C$ ($\omega < \omega_0$)	capacitive	current leads voltage by ϕ
$X_L = X_C$ ($\omega = \omega_0$)	purely resistive (resonance)	in phase, $\phi = 0$

Solved Example

A series LCR circuit has $R = 30 \, \Omega$, $X_L = 80 \, \Omega$, $X_C = 40 \, \Omega$ and is connected to 220 V (rms). Find Z , I , the power factor and the voltage across each element.

Solution: $Z = \sqrt{30^2 + (80 - 40)^2} = 50 \, \Omega$; $I = 220 / 50 = 4.4 \, \text{A}$; $\cos \phi = 30 / 50 = 0.6$ (inductive, current lags). $V_R = 132 \, \text{V}$, $V_L = 352 \, \text{V}$, $V_C = 176 \, \text{V}$. Check: $\sqrt{132^2 + (352 - 176)^2} = 220 \, \text{V} \checkmark$ (not $132 + 352 + 176$).

Solved Example

A series LCR circuit with $L = 5.0 \, \text{H}$, $C = 80 \, \mu\text{F}$, $R = 40 \, \Omega$ is connected to a 230 V variable-frequency supply. Find the resonant angular frequency and the rms current at resonance.

Solution: $\omega_0 = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = \frac{1}{0.02} = 50 \, \text{rad/s}$ ($\nu_0 \approx 7.96 \, \text{Hz}$). At resonance $Z = R \Rightarrow I = 230 / 40 = 5.75 \, \text{A}$ (peak 8.13 A).

Common Trap

- Never add rms voltages: $V \neq V_R + V_L + V_C$. Use $V = \sqrt{V_R^2 + (V_L - V_C)^2}$.
- Resonance needs **both L and C**; an RL or RC circuit cannot resonate.

NCERT Line to Remember

Resonance is used in the **tuning circuit of a radio or TV**: C is varied until ν_0 equals the frequency of the wanted station, which then gives the maximum current.

7.5 Power in AC Circuits; Power Factor

The average power over a cycle is $\bar{P} = V_{rms} I_{rms} \cos \phi$ (Derivation 5). $V_{rms} I_{rms}$ is the **apparent power** and $\cos \phi$ is the **power factor** $= R/Z$.

Circuit	ϕ	Power Factor $\cos \phi$	Average power
Pure R, or LCR at resonance	0	1	maximum = VI
Pure L or pure C	$\pi / 2$	0	zero (wattless)
Series LCR / RL / RC	$\tan^{-1} \frac{X_L - X_C}{R}$	R / Z	$VI \cos \phi = I^2 R$



Wattless current

- Resolve I into $I \cos \phi$ (along V) and $I \sin \phi$ ($\perp$ to V).
- $I \sin \phi$ consumes **no power** – it is the **wattless current**. In a pure L or C the whole current is wattless ($\phi = 90^\circ$).
- A **choke coil** (large L, tiny R) controls AC with almost no power loss.



Solved Example

A circuit draws 5 A from a 220 V AC supply and consumes 550 W. Find the power factor and the phase difference.

Solution: $\cos \phi = \frac{P}{VI} = \frac{550}{220 \times 5} = 0.5 \Rightarrow \phi = 60^\circ$.



VS Sir's Exam Tip

Use $\frac{V_0 I_0}{2} \cos \phi$ with **peak** values or $VI \cos \phi$ with **rms** values – never mix them.



NCERT Line to Remember

A low power factor means a large current for the same power and hence a large $I^2 R$ loss in transmission lines. Power factor of inductive loads is improved by connecting a suitable **capacitor in parallel**.

7.6 Transformer

A **transformer** changes an alternating voltage (step-up or step-down) using **mutual induction**. A **primary** coil (N_p turns) and a **secondary** coil (N_s turns) are wound on a **laminated soft-iron core** so that the same flux links both (Derivation 7).



Transformer equations

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} \quad \text{Ideal transformer } (V_p I_p = V_s I_s)$$

$$\eta = \frac{\text{output power}}{\text{input power}} \times 100\% \quad \text{Efficiency}$$



Step-up vs step-down

- Step-up:** $N_s > N_p \rightarrow V$ increases, I decreases.
- Step-down:** $N_s < N_p \rightarrow V$ decreases, I increases.
- Frequency and power are **not** increased.

Energy loss	Cause	Remedy
Flux leakage	all the flux of the primary does not link the secondary	wind the two coils one over the other
Copper (Joule) loss	heating $I^2 R$ in the resistance of the windings	thick copper wire for high-current coil
Eddy-current loss	currents induced in the iron core heat it	use a laminated core
Hysteresis loss	repeated magnetisation and demagnetisation of the core	core of soft iron (low hysteresis)

Solved Example

A step-down transformer changes 2200 V to 220 V. Its primary has 3000 turns and the load draws 11 A. Find N_s and I_p (ideal).

Solution: $N_s = N_p \frac{V_s}{V_p} = 3000 \times \frac{220}{2200} = 300$.

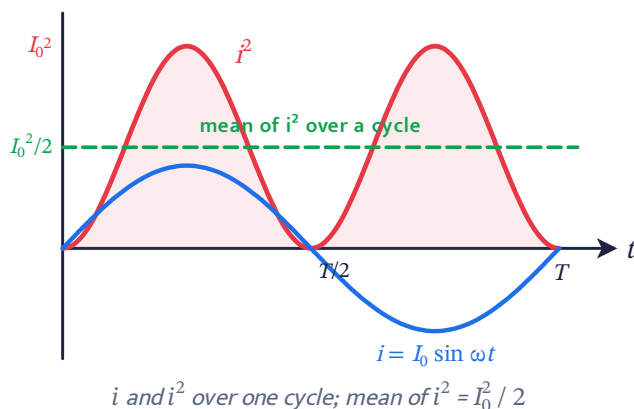
$I_p = \frac{V_s I_s}{V_p} = \frac{220 \times 11}{2200} = 1.1 \text{ A}.$

Common Trap

- A transformer does **not** work on steady DC – the flux does not change, so no emf is induced.
- Power is transmitted at high voltage: for fixed $P = VI$, high V means low I , so line loss $I^2 R$ is small.

Important derivations ★**VS Sir's Exam Tip**

Board examiners give marks step by step: **diagram** → **starting equation** → **key step** → **final boxed result** → **one line of conclusion**. The marks shown in brackets are how a typical RBSE answer is marked.

**Why "root mean square"?**

- **Square** the current: i^2 is always positive.
- Take its **mean** over a cycle: $\overline{i^2} = I_0^2 / 2$.
- Take the square **root**: $I_{rms} = I_0 / \sqrt{2}$.

Derivation ★ [3 marks]

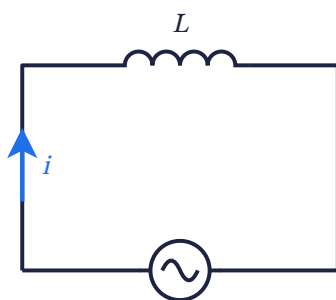
Derivation 1. Define the rms value of alternating current and show that $I_{rms} = I_0 / \sqrt{2}$. [RBSE 2025]

Solution: 1. Definition: the rms value of AC is the steady current which produces the same heat in a resistor in the same time as the AC. (½)

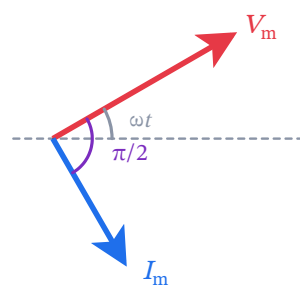
2. Let $i = I_0 \sin \omega t$ flow through R . Heat in time dt : $dH = i^2 R dt$; heat in one cycle $H = \int_0^T I_0^2 R \sin^2 \omega t dt$. (½)

3. $\int_0^T \sin^2 \omega t dt = \int_0^T \frac{1 - \cos 2\omega t}{2} dt = \frac{T}{2} - 0 = \frac{T}{2}$ ($\cos 2\omega t$ averages to zero over a cycle). So $H = \frac{I_0^2 R T}{2}$. (1)

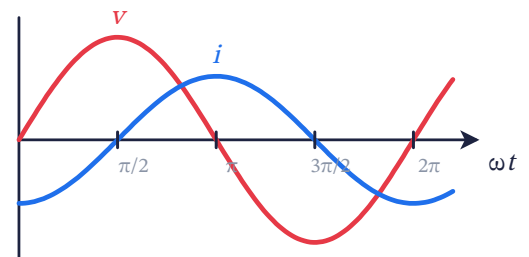
4. If I_{rms} (DC) gives the same heat: $I_{rms}^2 R T = \frac{I_0^2 R T}{2} \Rightarrow I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707 I_0$. Similarly $V_{rms} = V_0 / \sqrt{2}$. (1)



(a) circuit



(b) phasor diagram

(c) i lags v by $\pi/2$

AC through a pure inductor: current lags voltage by $\pi/2$

🔧 Derivation ★ [4 marks]

Derivation 2. An AC voltage $v = V_0 \sin \omega t$ is applied to a pure inductor L . Find the current, show that it lags the voltage by $\pi/2$, define inductive reactance and show that the average power is zero. [RBSE 2024]

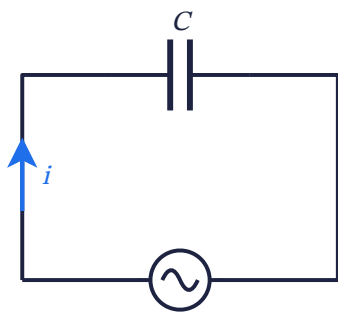
Solution: 1. The self-induced emf is $-L \frac{di}{dt}$. By Kirchhoff's loop rule: $V_0 \sin \omega t - L \frac{di}{dt} = 0 \Rightarrow \frac{di}{dt} = \frac{V_0}{L} \sin \omega t$. (½)

2. Integrating: $i = -\frac{V_0}{\omega L} \cos \omega t$ (constant = 0, as the current has no steady part). (½)

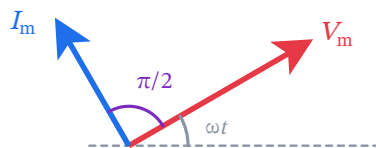
3. Since $-\cos \omega t = \sin(\omega t - \frac{\pi}{2})$: $i = I_0 \sin(\omega t - \frac{\pi}{2})$, $I_0 = \frac{V_0}{\omega L}$. So the current **lags** the voltage by $\pi/2$ (phasor diagram, Fig. b). (1)

4. $X_L = \omega L = 2\pi \nu L$ is the **inductive reactance** (unit ohm); $X_L \propto \nu$, and $X_L = 0$ for DC. (½)

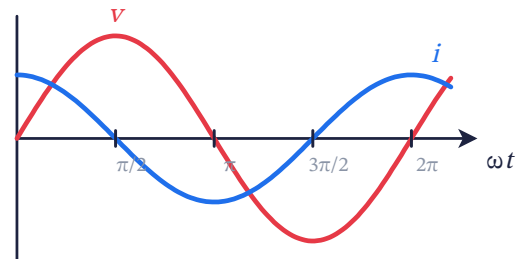
5. Power: $p = vi = V_0 I_0 \sin \omega t \sin(\omega t - \frac{\pi}{2}) = -\frac{V_0 I_0}{2} \sin 2\omega t$. The average of $\sin 2\omega t$ over a cycle is zero $\Rightarrow \bar{P} = 0$. Energy stored in the magnetic field in one quarter cycle is returned to the source in the next. (1) + diagram (½)



$v = v_m \sin \omega t$
(a) circuit



(b) phasor diagram



(c) i leads v by $\pi/2$

AC through a pure capacitor: current leads voltage by $\pi/2$

🔧 Derivation ★ [4 marks]

Derivation 3. An AC voltage $v = V_0 \sin \omega t$ is applied across a capacitor C . Find the current, its phase relative to the voltage, the capacitive reactance, and draw the phasor diagram. [RBSE 2025]

Solution: 1. At any instant the charge on the capacitor is $q = Cv = CV_0 \sin \omega t$. (½)

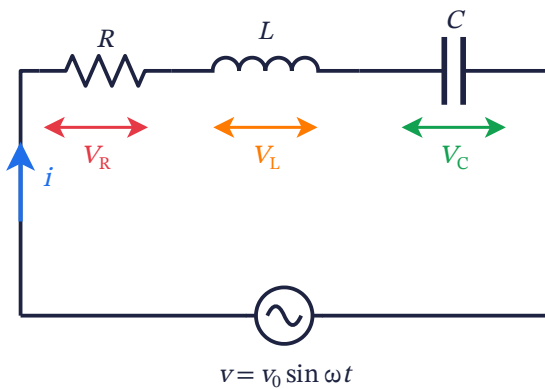
2. Current $i = \frac{dq}{dt} = \omega C V_0 \cos \omega t$. (½)

3. Since $\cos \omega t = \sin(\omega t + \frac{\pi}{2})$: $i = I_0 \sin(\omega t + \frac{\pi}{2})$, $I_0 = \frac{V_0}{1/\omega C}$. The current **leads** the voltage by $\pi/2$. (1)

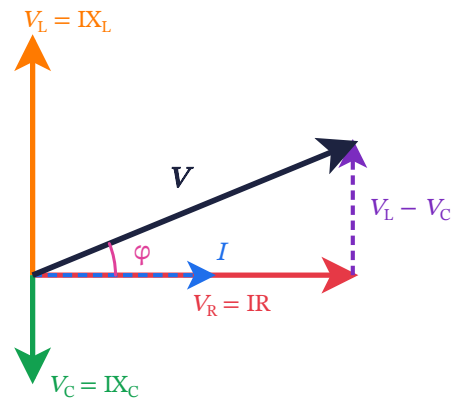
4. $X_C = \frac{1}{\omega C} = \frac{1}{2\pi \nu C}$ is the **capacitive reactance** (unit ohm). $X_C \propto 1/\nu$; for DC ($\nu = 0$) $X_C = \infty$, so a capacitor **blocks** DC. (½)

5. Power: $p = vi = \frac{V_0 I_0}{2} \sin 2\omega t$, whose average over a cycle is **zero**. (½)

6. Phasor diagram (Fig. b): I_0 phasor is 90° **ahead** (anticlockwise) of the V_0 phasor. (1)



Series LCR circuit

Phasor diagram ($X_L > X_C$): V leads I by ϕ

🔧 Derivation ★ [3 marks]

Derivation 4 (Part 1). Using a phasor diagram, find the impedance of a series LCR circuit and the phase difference between voltage and current. [RBSE 2026*]

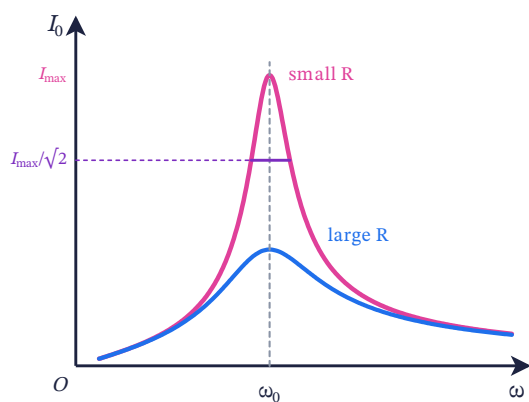
Solution: 1. The same current $i = I_0 \sin \omega t$ flows through R, L and C ; take the current phasor as reference. (½)

2. $V_R = IR$ is **in phase** with I ; $V_L = IX_L$ **leads** I by 90° ; $V_C = IX_C$ **lags** I by 90° . (1)

3. V_L and V_C are opposite, so their resultant ($V_L - V_C$) is perpendicular to V_R : $V^2 = V_R^2 + (V_L - V_C)^2$. (½)

4. $V = I \sqrt{R^2 + (X_L - X_C)^2} \Rightarrow Z = \frac{V}{I} = \sqrt{R^2 + (X_L - X_C)^2}$. (½)

5. From the phasor triangle: $\tan \phi = \frac{X_L - X_C}{R}$ (V leads I if $X_L > X_C$, lags if $X_L < X_C$). (½)



Current vs ω : maximum at ω_0 ; smaller R gives a sharper peak

🔧 Derivation ★ [4 marks with Part 1]

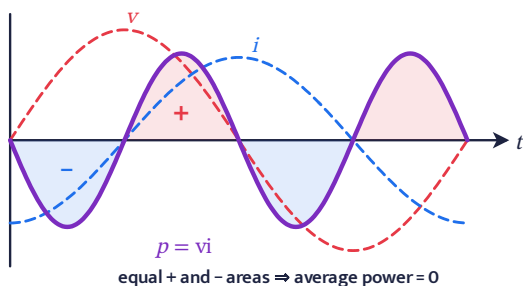
Derivation 4 (Part 2). What is resonance in a series LCR circuit? Find the resonant frequency and draw the I - ω graph. [RBSE 2025]

Solution: 1. $I = V / Z$ is **maximum** when Z is minimum, i.e. when $X_L = X_C$; this state is **resonance**. (½)

2. $\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}, \nu_0 = \frac{1}{2\pi\sqrt{LC}}$. (1)

3. At resonance $Z = R, I_{\max} = V / R, \phi = 0$ and power factor = 1. (½)

4. Graph (left): I rises to a peak at ω_0 and then falls. (1) For 4 marks, also write Part 1 (Z).



Pure inductor: $p = vi$ is + and - alternately, so $\bar{P} = 0$

🔧 Derivation ★ [3 marks]

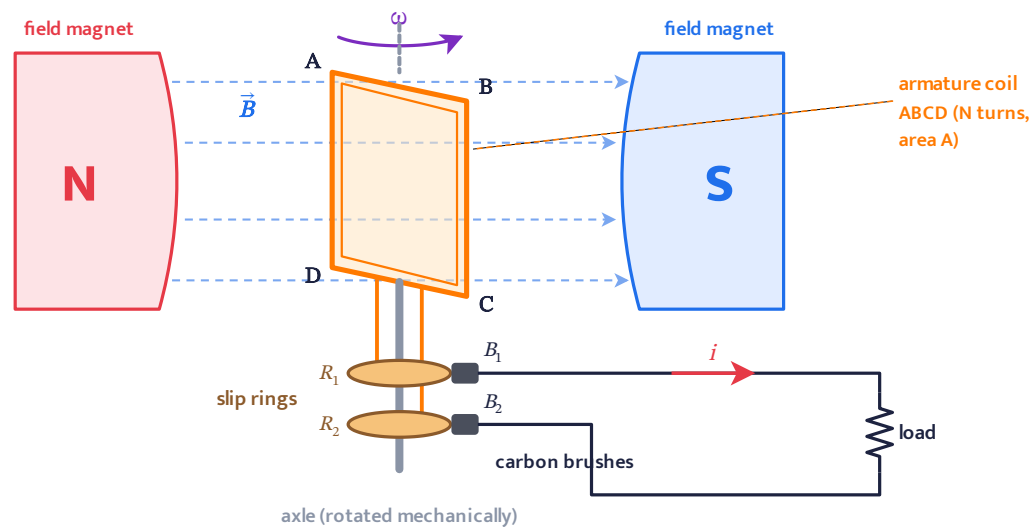
Derivation 5. Show that the average power in an AC circuit is $V_{rms} I_{rms} \cos \phi$. What is the power factor of a pure capacitor and of a series LCR at resonance? [RBSE 2023]

Solution: 1. $v = V_0 \sin \omega t, i = I_0 \sin(\omega t + \phi) \Rightarrow p = V_0 I_0 \sin \omega t \sin(\omega t + \phi) = \frac{V_0 I_0}{2} [\cos \phi - \cos(2\omega t + \phi)]$. (1)

2. Mean of $\cos(2\omega t + \phi)$ over a cycle = 0 $\Rightarrow$

$\bar{P} = \frac{V_0 I_0}{2} \cos \phi = V_{rms} I_{rms} \cos \phi$. (1)

3. Pure C: $\phi = \pi/2 \Rightarrow \cos \phi = 0$; LCR at resonance: $\phi = 0 \Rightarrow \cos \phi = 1$. (1)



AC generator: armature coil ABCD rotating between the poles, slip rings R_1, R_2 and brushes B_1, B_2

Derivation ★ [3 marks]

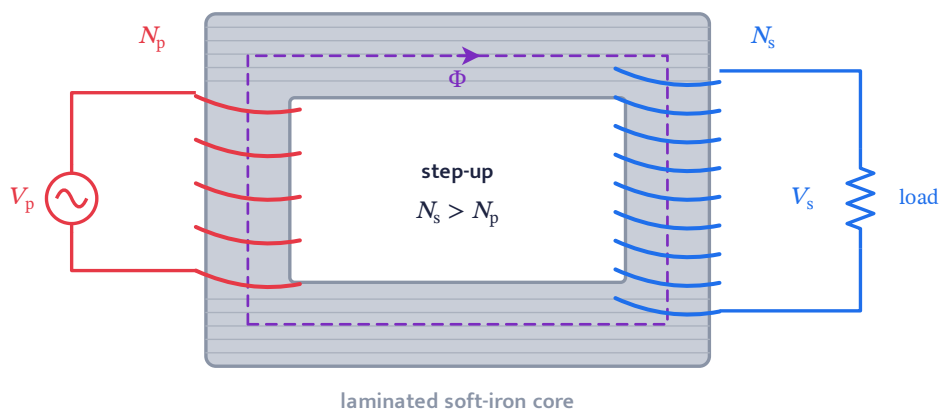
Derivation 6. Draw a labelled diagram of an AC generator. State its principle and derive the expression for the induced emf. [RBSE 2024]

Solution: 1. Principle: electromagnetic induction – when a coil rotates in a uniform magnetic field, the flux linked with it changes and an emf is induced. (½)

2. Construction: armature coil (N turns, area A) on an axle; strong field magnets; two **slip rings** joined to the ends of the coil; two carbon **brushes** pressing on the rings connect the load. (Labelled diagram) (1)

3. Theory: at angle $\theta = \omega t$ between the normal to the coil and $\vec{B}$: $\Phi = NBA \cos \omega t$. Induced emf $\varepsilon = -\frac{d\Phi}{dt} \Rightarrow \varepsilon = NBA\omega \sin \omega t = \varepsilon_0 \sin \omega t$ with $\varepsilon_0 = NBA\omega$. (1)

4. The emf reverses every half rotation, so the output is AC whose frequency equals the frequency of rotation. Mechanical energy $\rightarrow$ electrical energy. (½)



Step-up transformer: the changing flux Φ in the laminated core links both coils

 Derivation ★ [3 marks]

Derivation 7. State the principle of a transformer and obtain the relation between the voltages, currents and numbers of turns of an ideal transformer. [RBSE 2026*]

Solution: 1. Principle: mutual induction – an alternating current in the primary produces a changing flux in the core, which induces an emf in the secondary. (½)

2. Two coils (N_p, N_s turns) on a laminated soft-iron core (diagram). (½)

3. If Φ is the flux in each turn (no leakage): $\varepsilon_s = -N_s \frac{d\Phi}{dt}, \varepsilon_p = -N_p \frac{d\Phi}{dt}$; with negligible resistance $V_p = \varepsilon_p, V_s = \varepsilon_s$. (½)

4. Dividing: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$. (½)

5. Ideal (no loss): $V_p I_p = V_s I_s \Rightarrow \frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$. (½)

6. $N_s > N_p$: step-up ($V \uparrow, I \downarrow$); $N_s < N_p$: step-down ($V \downarrow, I \uparrow$). (½)

Board question bank **Section A – Multiple choice (½ mark each)**

[RBSE year] = asked in that BSER paper (reworded). Answers at the end of the chapter.

- 1 [RBSE 2025] An AC generator converts
 - (a) mechanical energy into electrical energy
 - (b) electrical energy into mechanical energy
 - (c) chemical energy into electrical energy
 - (d) electrical energy into heat
- 2 [RBSE 2023] The frequency of the AC mains supply in India is
 - (a) 50 Hz
 - (b) 100 Hz
 - (c) 60 Hz
 - (d) 220 Hz
- 3 The peak value of a 220 V (rms) AC supply is about
 - (a) 220 V
 - (b) 440 V
 - (c) 311 V
 - (d) 156 V
- 4 In a pure capacitive AC circuit, the current
 - (a) is in phase with the voltage
 - (b) leads the voltage by π
 - (c) leads the voltage by $\pi/2$
 - (d) lags the voltage by $\pi/2$
- 5 The reactance of an inductor of $\frac{1}{\pi}$ H at 50 Hz is
 - (a) 50 Ω
 - (b) 314 Ω
 - (c) 1 Ω
 - (d) 100 Ω
- 6 In a series LCR circuit, the rms voltages across R, L and C are 80 V, 100 V and 40 V. The source voltage is
 - (a) 140 V
 - (b) 100 V
 - (c) 220 V
 - (d) 20 V
- 7 The resonant angular frequency of a series LCR circuit with $L = 0.5$ H and $C = 8 \mu\text{F}$ is
 - (a) 250 rad/s
 - (b) 1000 rad/s
 - (c) 500 rad/s
 - (d) 50 rad/s
- 8 [RBSE 2026*] The current in an AC circuit is wattless when the phase difference between current and voltage is
 - (a) 0°
 - (b) 60°
 - (c) 90°
 - (d) 45°

- 9 The power factor of a series LCR circuit at resonance is
 (a) 0 (b) 1 (c) 0.5 (d) $1/\sqrt{2}$
- 10 The core of a transformer is laminated to reduce
 (a) eddy-current loss (b) flux leakage
 (c) copper loss (d) hysteresis loss

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2025] In a pure inductor, the current lags the voltage by ____.	$\pi/2$ (90°)
[RBSE 2024] The mean value of alternating current over a complete cycle is ____.	zero
The rms value of AC is ____ times its peak value.	$1/\sqrt{2} \approx 0.707$
The reactance of a capacitor for DC is ____.	infinite
The resonant frequency of a series LCR circuit is ____.	$\frac{1}{2\pi\sqrt{LC}}$
A transformer works on the principle of ____.	mutual induction

Very short answer (1 mark)

Question	Answer
Define the rms value of AC.	The steady current that produces the same heat in a resistor in the same time as the AC; $I_{rms} = I_0 / \sqrt{2}$.
Define impedance. Give its SI unit.	Total opposition of an LCR circuit to AC, $Z = V_{rms} / I_{rms} = \sqrt{R^2 + (X_L - X_C)^2}$; unit ohm (Ω) .
What is wattless current?	The current component $I \sin \phi$ ($\perp$ to V) which consumes no power; e.g. current in a pure L or C.
Why does a capacitor block DC?	For DC, $\nu = 0 \Rightarrow X_C = 1 / 2\pi\nu C = \infty$.
Why can a transformer not step up DC?	Steady DC gives constant flux, so no emf is induced in the secondary.
Define power factor. What is its value for a pure resistor?	$\cos \phi = (\text{average power})/(\text{apparent power}) = R/Z$; for pure R it is 1 .

Short answer – Section B (1½ marks)

Short answer (1½ marks)

[RBSE 2023] Write the power factor of (i) a purely capacitive circuit, (ii) a series LCR circuit at resonance.

Solution: (i) $\phi = \pi/2 \Rightarrow \cos \phi = 0$ (no power consumed). (¾) (ii) $X_L = X_C \Rightarrow Z = R \Rightarrow \cos \phi = R/Z = 1$ (maximum power). (¾)

Short answer (1½ marks)

[RBSE 2023] Write any three energy losses in a transformer and how they are reduced.

Solution: (1) **Eddy-current loss** → laminated core. (½) (2) **Copper loss** ($I^2 R$) → thick copper windings. (½) (3) **Hysteresis loss** → soft-iron core (or **flux leakage** → coils wound over each other). (½)

 Short answer (1½ marks)

[RBSE 2024] For $I = 4\sin\omega t$ A and $V = 200\sin(\omega t + \pi/3)$ V, show that $I_0 = \sqrt{2} I_{rms}$ and find the average power.

Solution: $I_0 = 4$ A, $I_{rms} = 4 / \sqrt{2} = 2.83$ A $\Rightarrow \sqrt{2} \times 2.83 = 4$ A = I_0 ✓ (from Derivation 1). (½) $\phi = \pi/3$ (V leads I). (½)
 $\bar{P} = \frac{V_0 I_0}{2} \cos\phi = \frac{200 \times 4}{2} \times \frac{1}{2} = 200$ W. (½)

 Short answer (1½ marks)

[RBSE 2024] A series LCR circuit has $R = 24 \Omega$ and $X_L = X_C = 110 \Omega$. Find its impedance. Also, what is the average power in a pure inductor?

Solution: $Z = \sqrt{24^2 + (110 - 110)^2} = 24 \Omega$ (resonance, $Z = R$). (1) Pure inductor: $\phi = \pi/2 \Rightarrow \bar{P} = VI\cos 90^\circ = 0$. (½)

 Short answer (1½ marks)

A $100 \mu\text{F}$ capacitor and a 40Ω resistor are in series across a 220 V, 50 Hz supply. Find the impedance and the rms current.

Solution: $X_C = \frac{1}{2\pi(50)(10^{-4})} = 31.8 \Omega$. (½) $Z = \sqrt{40^2 + 31.8^2} \approx 51.1 \Omega$. (½) $I = 220 / 51.1 \approx 4.3$ A. (½)

 Answer Key & Quick Solutions

1(a) 2(a) 3(c) 4(c) 5(d) 6(b) 7(c) 8(c) 9(b) 10(a)

- (a) The coil is rotated mechanically; the induced emf supplies electrical energy.
- (a) India uses 220 V, 50 Hz (USA uses 60 Hz).
- (c) $V_0 = \sqrt{2} \times 220 \approx 311$ V.
- (c) For C, $i = I_0 \sin(\omega t + \pi/2)$.
- (d) $X_L = 2\pi\nu L = 2\pi(50)(1/\pi) = 100 \Omega$.
- (b) $V = \sqrt{80^2 + (100 - 40)^2} = 100$ V.
- (c) $\omega_0 = 1 / \sqrt{0.5 \times 8 \times 10^{-6}} = 1 / (2 \times 10^{-3}) = 500$ rad/s.
- (c) $\bar{P} = VI\cos 90^\circ = 0$ (pure L or C).
- (b) $Z = R \Rightarrow \cos \phi = R/Z = 1$.
- (a) Insulated laminations break the paths of eddy currents.

 Last-minute revision

- AC generator: $\varepsilon = NBA\omega \sin\omega t$; mechanical $\rightarrow$ electrical; India 220 V, 50 Hz.
- Mean over a cycle = 0; $I_{rms} = I_0 / \sqrt{2} = 0.707 I_0$; meters read rms.
- L: $X_L = \omega L$, i lags by $\pi/2$. C: $X_C = 1 / \omega C$, i leads by $\pi/2$ (CIVIL). Both: $\bar{P} = 0$.
- $Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\tan\phi = (X_L - X_C) / R$, $\cos\phi = R / Z$.
- Resonance: $\omega_0 = 1 / \sqrt{LC}$, $Z = R$ (min), I max, $\cos\phi = 1$; used in radio tuning.
- $\bar{P} = V_{rms} I_{rms} \cos\phi$; wattless current $I \sin\phi$ ($\phi = 90^\circ$).
- Transformer: mutual induction; $V_s / V_p = N_s / N_p = I_p / I_s$; no DC.
- Losses: flux leakage, copper, eddy (laminated), hysteresis (soft iron).

CHAPTER

8

Electromagnetic Waves

17

Day 12

Board marks: 2

PYQ frequency ★★☆☆

RBSE Class 12

Displacement current

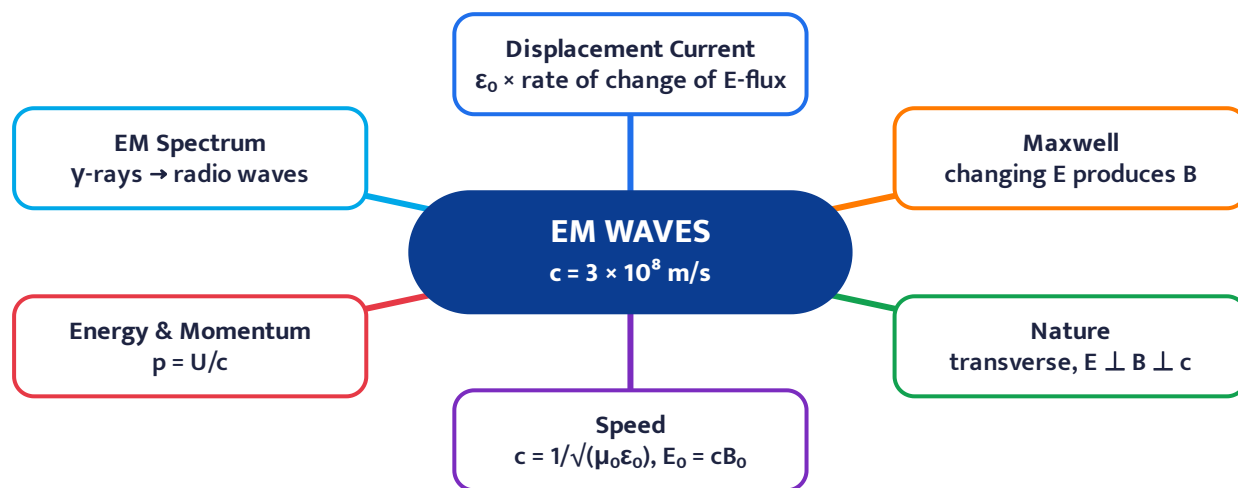
Maxwell's idea

Nature of EM waves

 $c = 1/\sqrt{(\mu_0\epsilon_0)}$

Energy & momentum

EM spectrum & uses



Chapter at a glance

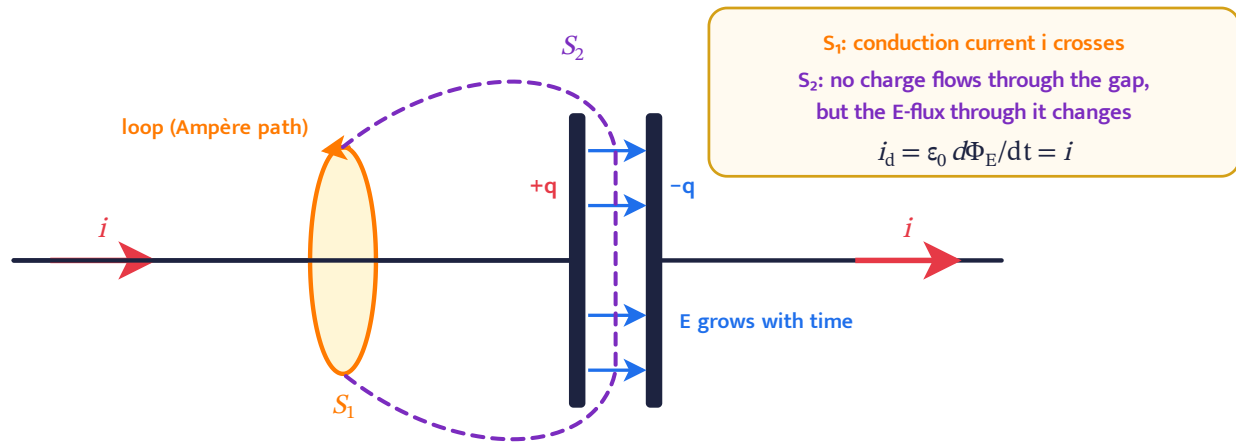
Board snapshot

- **Weightage: 2 marks** – a short chapter: usually one MCQ or fill-in (½ mark) plus one 1½-mark short answer. It is **asked every year** and is easy to score – do not skip it.
- **2025:** MCQ – device that produces microwaves (klystron / magnetron / Gunn diode); 1½ marks – two characteristics of electromagnetic waves.
- **2024:** MCQ – displacement current $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$; 1½ marks – name three electromagnetic waves.
- **2023:** MCQ – FM broadcast band (88–108 MHz); a very-short question on EM waves. **2026 (CWSN):** fill-in on displacement current.
- **Most likely in 2027:** (1) characteristics / properties of EM waves; (2) names, sources and uses of the EM-spectrum bands (microwave, UV, IR, X-ray); (3) displacement current – definition and expression.

8.1 Displacement Current

Faraday showed that a **changing magnetic field produces an electric field**. Maxwell argued the converse: a **changing electric field produces a magnetic field**. He noticed that Ampère's circuital law fails for a charging capacitor: for the same loop, a flat surface S_1 is crossed by the current i , but a pot-shaped surface S_2 passing through the gap between the plates is crossed by **no** current – so $\oint \vec{B} \cdot d\vec{l}$ would be both $\mu_0 i$ and 0.

- **Displacement current (i_d):** the current that arises from a **time-varying electric flux**, $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$. It is not a flow of charge, yet it produces a magnetic field exactly like a conduction current.
- **Total current = conduction current + displacement current**, and it is continuous everywhere in the circuit. In the wires $i_d = 0$; in the gap $i_c = 0$ and $i_d = i$.



Charging capacitor: the conduction current i in the wires continues as the displacement current i_d in the gap

Displacement current

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt} \quad \text{Displacement current (A)}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_c + \epsilon_0 \frac{d\Phi_E}{dt} \right) \quad \text{Ampère–Maxwell law}$$

$$i_d = C \frac{dV}{dt} = \epsilon_0 A \frac{dE}{dt} \quad \text{Parallel-plate capacitor}$$

Solved Example

A capacitor with plates of area 0.02 m^2 is charged by a current of 0.15 A . Find the displacement current and the rate of change of E between the plates.

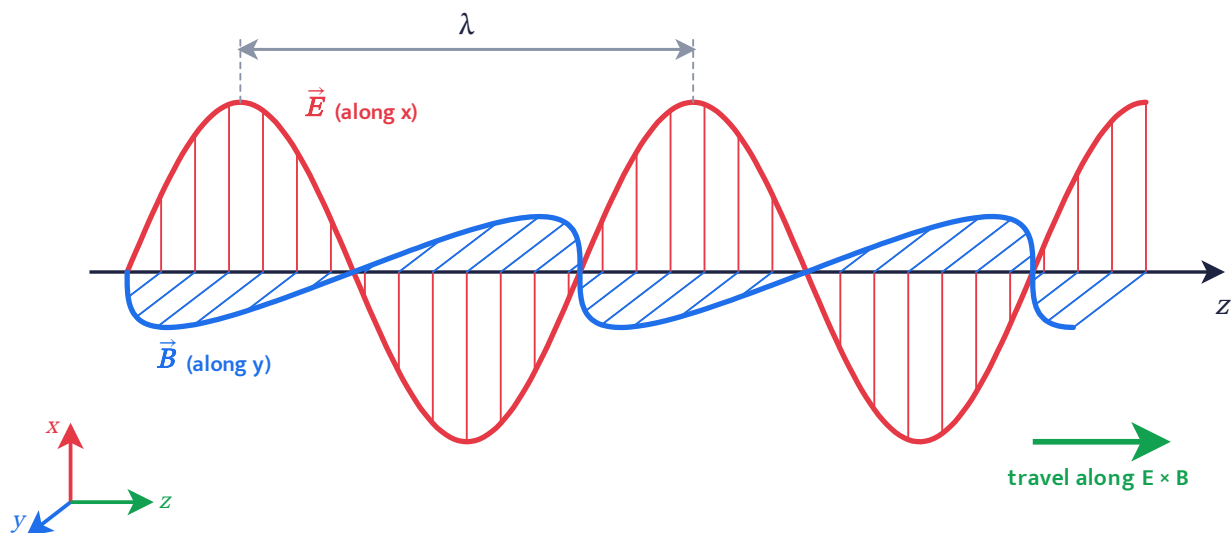
Solution: $i_d = i = 0.15 \text{ A}$.

$$\frac{dE}{dt} = \frac{i_d}{\epsilon_0 A} = \frac{0.15}{8.85 \times 10^{-12} \times 0.02} \approx 8.5 \times 10^{11} \text{ V m}^{-1} \text{ s}^{-1}.$$

Maxwell's equations (qualitative)	Form	Meaning
Gauss's law (electricity)	$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$	charges are the sources of E
Gauss's law (magnetism)	$\oint \vec{B} \cdot d\vec{A} = 0$	no magnetic monopoles
Faraday's law	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$	changing B produces E
Ampère–Maxwell law	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$	current and changing E produce B

8.2 Electromagnetic Waves – Sources and Nature

A charge at rest produces only an electric field; a charge moving with **constant velocity** produces steady E and B but **no wave**. An **accelerated (oscillating) charge** radiates: the changing E produces a changing B , which in turn produces a changing E ... and the disturbance travels through space as an **electromagnetic wave** with no need of a medium. **The frequency of the wave equals the frequency of oscillation of the charge.**



A plane EM wave travelling along z : E_x and B_y oscillate in phase, perpendicular to each other and to the direction of travel



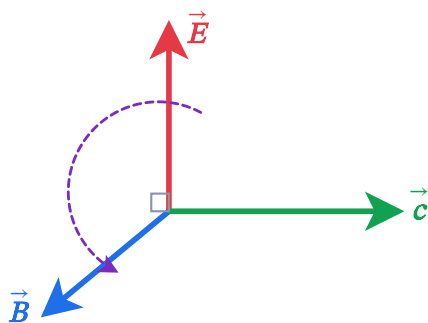
EM wave equations

$$E_x = E_0 \sin(kz - \omega t), \quad B_y = B_0 \sin(kz - \omega t) \quad \text{Plane wave along } +z$$

$$k = \frac{2\pi}{\lambda}, \quad \omega = 2\pi\nu, \quad c = \nu\lambda = \frac{\omega}{k} \quad \text{Wave relations}$$

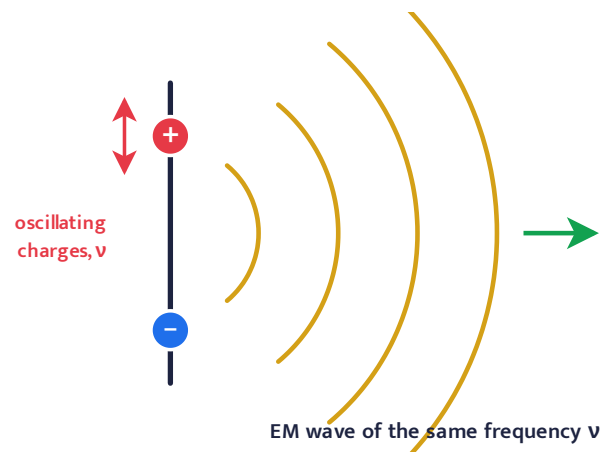
$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m s}^{-1} \quad \text{Speed in vacuum – light is an EM wave}$$

$$\frac{E_0}{B_0} = c, \quad v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} \quad \text{Ratio of fields; speed in a medium}$$



direction of travel = direction of $\vec{E} \times \vec{B}$

$\vec{E}$, $\vec{B}$ and the direction of travel form a right-handed set



An oscillating (accelerated) charge is the source of EM waves



Characteristics of EM waves

- **Transverse** waves: $\vec{E} \perp \vec{B} \perp$ direction of propagation; travel is along $\vec{E} \times \vec{B}$.
- E and B oscillate **in phase** with the same frequency, and $E_0 = cB_0$.
- They need **no material medium**; in vacuum all travel with the same speed $c = 3 \times 10^8 \text{ m/s}$ whatever the wavelength.
- They are **not deflected** by electric or magnetic fields (they carry no charge) and carry **energy and momentum**.
- In a medium the frequency is unchanged while speed and wavelength decrease; $n = \sqrt{\mu_r \epsilon_r}$.

 Solved Example

A plane EM wave of frequency 25 MHz travels in vacuum along x. At a point $\vec{E} = 6.3 \hat{j}$ V/m. Find λ and $\vec{B}$ there.

Solution: $\lambda = c / \nu = 3 \times 10^8 / 25 \times 10^6 = 12 \text{ m}$.

$B = E / c = 6.3 / 3 \times 10^8 = 2.1 \times 10^{-8} \text{ T}$. As $\hat{j} \times \hat{k} = \hat{i}$, $\vec{B}$ is along $+\hat{k}$.

 History (NCERT)

- **Maxwell** predicted EM waves and showed light is one (1865).
- **Hertz (1887)** produced and detected them in the laboratory.
- **J. C. Bose** produced waves of 25 mm to 5 mm wavelength; **Marconi** sent them over many kilometres.

8.3 Energy, Intensity and Momentum

An EM wave carries energy in both its fields, **shared equally** on average between E and B. It also carries momentum, so it exerts a **radiation pressure** on a surface (a comet's tail points away from the Sun for this reason).

 Energy and momentum of EM waves

$$u_E = \frac{1}{2} \epsilon_0 E^2, \quad u_B = \frac{B^2}{2\mu_0}, \quad \bar{u}_E = \bar{u}_B \quad \text{Energy densities (J m}^{-3}\text{)}$$

$$I = \bar{u} c = \frac{1}{2} c \epsilon_0 E_0^2 \quad \text{Intensity (W m}^{-2}\text{); point source } I = \frac{P}{4\pi r^2}$$

$$p = \frac{U}{c} \text{ (absorbed), } p = \frac{2U}{c} \text{ (reflected)} \quad \text{Momentum delivered to a surface}$$

 Solved Example

A 100 W point source radiates uniformly. Find the intensity and the peak electric and magnetic fields at 2 m.

Solution: $I = \frac{P}{4\pi r^2} = \frac{100}{4\pi \times 4} \approx 2.0 \text{ W m}^{-2}$. $E_0 = \sqrt{\frac{2I}{\epsilon_0 c}} \approx 39 \text{ V/m}$; $B_0 = E_0 / c \approx 1.3 \times 10^{-7} \text{ T}$.

 Common Trap

- $E_0 = cB_0$ does **not** mean that E carries more energy – the two energy densities are **equal**.
- Reflection gives double the momentum transfer ($2U / c$), absorption gives U / c .

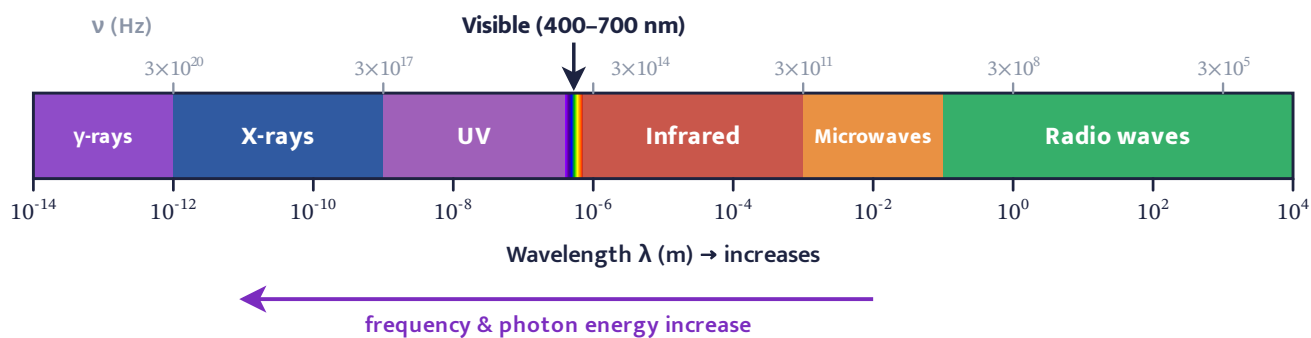
 Solved Example

A wave is $E_x = 30 \sin(2 \times 10^7 z - 6 \times 10^{15} t)$ V/m (SI units). Find λ , ν , B_0 and the region of the spectrum.

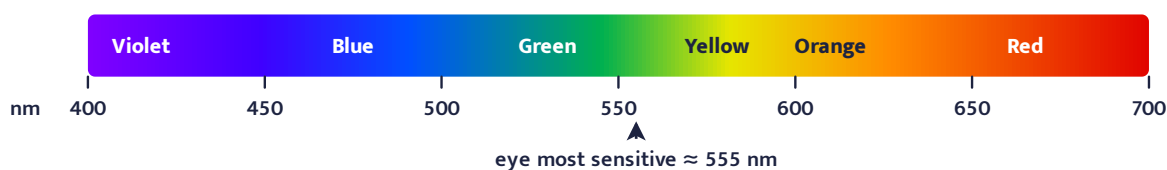
Solution: $\lambda = 2\pi / k \approx 314 \text{ nm}$; $\nu = \omega / 2\pi \approx 9.5 \times 10^{14} \text{ Hz}$; check $\omega / k = 3 \times 10^8 \text{ m/s} \checkmark$. $B_0 = E_0 / c = 10^{-7} \text{ T}$, along y. $314 \text{ nm} < 400 \text{ nm} \Rightarrow \text{ultraviolet}$.

8.4 The Electromagnetic Spectrum

EM waves are classified by wavelength (or frequency). The bands overlap and have no sharp boundaries; they differ only in λ and ν – all have the same nature and the same speed in vacuum.



The EM spectrum on a logarithmic scale: wavelength increases to the right, frequency and energy to the left



Visible light: violet ($\approx$ 400 nm) to red ($\approx$ 700 nm)

Band	Wavelength	Production	Detection
Radio	> 0.1 m	accelerated electrons in aerials / oscillating circuits	receiver's aerial
Microwave	0.1 m – 1 mm	klystron, magnetron, Gunn diode	point-contact diodes
Infrared	1 mm – 700 nm	vibrating atoms and molecules; hot bodies	thermopile, bolometer, IR film
Visible	700 – 400 nm	electrons jumping between atomic levels	eye, photocell, photographic film
Ultraviolet	400 – 1 nm	inner-shell electrons; Sun; UV lamps	photocell, photographic film
X-rays	1 nm – 10^{-3} nm	X-ray tube: fast electrons striking a metal target	photographic film, Geiger tube
γ -rays	$< 10^{-3}$ nm	radioactive decay of nuclei	as for X-rays

Band	Uses and Facts
Radio	Radio and TV broadcasting, cellular phones. AM band 530–1710 kHz; TV 54–890 MHz; FM band 88–108 MHz (asked in 2023).
Microwave	Radar , aircraft navigation, speed guns; microwave oven – the frequency matches the natural frequency of water molecules, so food gets heated.
Infrared	" Heat waves "; keep the Earth warm (greenhouse effect); physiotherapy; TV remote controls; night-vision and satellite photography.
Ultraviolet	Mostly absorbed by the ozone layer ; water purifiers (kill germs), LASIK eye surgery; harmful in large doses.
X-rays / γ-rays	X-rays: medical diagnosis (bone fractures); γ -rays: destroying cancer cells. Both damage living tissue.

 Memory Trick

Increasing wavelength: **γ-rays, X-rays, UV, Visible, IR, Microwave, Radio**. Read backwards for increasing frequency and photon energy. Inside the visible band: **VIBGYOR** – violet has the shortest, red the longest wavelength.

 Common Trap

- All EM waves have the **same speed in vacuum**; they differ in λ and ν .
- Sound and ultrasonic waves are **not** EM waves – they need a medium.

8.5 Important Derivations and Results ★ **Derivation ★ [1½ marks]**

Derivation 1. Show that the displacement current between the plates of a charging parallel-plate capacitor equals the conduction current in the connecting wires.

Solution: (i) Let the plates (area A) carry charge q at time t . Field between them: $E = \frac{q}{\epsilon_0 A}$. (½)

(ii) Electric flux through the gap: $\Phi_E = EA = \frac{q}{\epsilon_0}$. (½)

(iii) Displacement current: $i_d = \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 \cdot \frac{1}{\epsilon_0} \frac{dq}{dt} = \frac{dq}{dt} = i$, the conduction current in the wires. (½)

(iv) So the total current is continuous and Ampère's law becomes $\oint \vec{B} \cdot d\vec{l} = \mu_0(i_c + i_d)$ – the same value for surfaces S_1 and S_2 . (½)

 Derivation ★ [1½ marks]

Result 2. Obtain the speed of EM waves in vacuum in terms of μ_0 and ϵ_0 and calculate its value. How does it change in a medium?

Solution: (i) Maxwell's equations give a wave whose speed is $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$. (½)

(ii) $c = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}} = 3 \times 10^8 \text{ m/s}$ – the speed of light, so light is an EM wave. (½)

(iii) In a medium $v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} < c$, and in the wave $\frac{E_0}{B_0} = c$. (½)

 Derivation ★ [1½ marks]

Derivation 3. Show that in an EM wave the average energy densities of the electric and magnetic fields are equal.

Solution: (i) Electric energy density $u_E = \frac{1}{2} \epsilon_0 E^2$; magnetic energy density $u_B = \frac{B^2}{2\mu_0}$. (½)

(ii) In the wave $B = \frac{E}{c}$ and $c^2 = \frac{1}{\mu_0 \epsilon_0}$, so $u_B = \frac{E^2}{2\mu_0 c^2} = \frac{1}{2} \epsilon_0 E^2 = u_E$. (1)

(iii) Averages: $\bar{u}_E = \bar{u}_B = \frac{1}{4} \epsilon_0 E_0^2$, so $\bar{u} = \frac{1}{2} \epsilon_0 E_0^2$ (half in each field) and intensity $I = \bar{u} c$. (½)

 VS Sir's Exam Tip

- For "two characteristics" or "properties" questions write **two or three crisp points** (transverse; $E \perp B \perp$ direction; speed c in vacuum; no medium needed) – 1½ marks are for ½ mark per point.
- For "name EM waves" give the **name + source or use** – e.g. microwave (radar), UV (water purifier), X-ray (bone fracture).

Board question bank



Section A – Multiple choice (½ mark each)

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- 1 [RBSE 2025] Which of the following devices is used to produce microwaves?
 - (a) thermopile
 - (b) klystron valve
 - (c) Geiger tube
 - (d) photocell
- 2 [RBSE 2024] The displacement current in terms of the electric flux Φ_E is
 - (a) $\mu_0 \frac{d\Phi_E}{dt}$
 - (b) $\epsilon_0 \frac{d\Phi_B}{dt}$
 - (c) $\epsilon_0 \frac{d\Phi_E}{dt}$
 - (d) $\frac{1}{\epsilon_0} \frac{d\Phi_E}{dt}$
- 3 [RBSE 2023] The frequency range of the FM broadcast band is
 - (a) 530 – 1710 kHz
 - (b) 200 – 300 MHz
 - (c) 88 – 108 MHz
 - (d) 1 – 3 MHz
- 4 The peak electric field of an EM wave in vacuum is 6 V/m. The peak magnetic field is
 - (a) 5×10^7 T
 - (b) 2×10^{-6} T
 - (c) 2×10^{-8} T
 - (d) 1.8×10^9 T
- 5 A 5 μ F capacitor is charged so that the voltage across it rises at 2×10^5 V/s. The displacement current between its plates is
 - (a) 1 A
 - (b) 0.1 A
 - (c) 2.5×10^{-11} A
 - (d) 10 A
- 6 An EM wave travels along +x with its electric field along +y. Its magnetic field is along
 - (a) -z
 - (b) +z
 - (c) +y
 - (d) -x
- 7 The speed of an EM wave in a medium of relative permittivity 4 and relative permeability 1 is
 - (a) 3×10^8 m/s
 - (b) 7.5×10^7 m/s
 - (c) 1.5×10^8 m/s
 - (d) 6×10^8 m/s
- 8 Which of the following has the highest frequency?
 - (a) microwaves
 - (b) infrared
 - (c) X-rays
 - (d) radio waves
- 9 The harmful ultraviolet radiation from the Sun is mostly absorbed by
 - (a) the ozone layer
 - (b) carbon dioxide
 - (c) water vapour
 - (d) the ionosphere
- 10 A beam carrying energy 6 J is completely absorbed by a surface. The momentum delivered to the surface is
 - (a) 1.8×10^9 kg m/s
 - (b) 2×10^{-8} kg m/s
 - (c) 2×10^8 kg m/s
 - (d) 4×10^{-8} kg m/s

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2026] Displacement current is produced by a changing _____ flux.	electric
EM waves are produced by _____ charges.	accelerated
EM waves are _____ waves ($E \perp B \perp$ direction of travel).	transverse
The speed of EM waves in vacuum is $c =$ _____ in terms of μ_0 and ϵ_0 .	$\frac{1}{\sqrt{\mu_0 \epsilon_0}}$
Infrared waves are also called _____ waves.	heat
Radio waves have the _____ wavelength in the EM spectrum.	largest

Very short answer (1 mark)

Question	Model answer
Who first produced EM waves in the laboratory?	Heinrich Hertz (1887).
Which EM waves are used to sterilise (purify) water?	Ultraviolet rays.
Write the relation between E_0 and B_0 of an EM wave in vacuum.	$E_0 / B_0 = c$.
Give two uses of microwaves.	Radar; microwave oven (also mobile / satellite communication).
Why are infrared waves called heat waves?	They are readily absorbed by water, CO_2 etc., raising the thermal motion of molecules – producing heating.
Do EM waves need a medium for propagation?	No – they travel through vacuum.

Short answer (Section B, 1½ marks)

 Short answer (1½ marks)

[RBSE 2025] Write two characteristics of electromagnetic waves.

Solution: (i) EM waves are **transverse**: the electric field $\vec{E}$ and magnetic field $\vec{B}$ oscillate in phase, perpendicular to each other and to the direction of propagation. (¾) (ii) They need **no material medium** and travel in vacuum with the same speed $c = 3 \times 10^8$ m/s (and $E_0 = cB_0$). (¾)

 Short answer (1½ marks)

[RBSE 2024] Name three electromagnetic waves and give one use of each.

Solution: **Microwaves** – radar / microwave oven. (½) **Ultraviolet rays** – water purifiers (killing germs). (½) **X-rays** – medical diagnosis (bone fracture). (½) (Also: infrared – remote control; γ -rays – cancer treatment; radio waves – broadcasting.)

 Short answer (1½ marks)

What is displacement current? Write its expression and state why it was introduced by Maxwell.

Solution: The current that arises due to a **time-varying electric flux** is the displacement current, $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$. (1)

Maxwell introduced it to make Ampère's circuital law consistent for a charging capacitor, where there is no conduction current in the gap but a magnetic field is still present. (½)

 Short answer (1½ marks)

The magnetic field of a plane EM wave in vacuum has peak value 2.4×10^{-7} T. Find the peak electric field and the frequency if the wavelength is 3 m.

Solution: $E_0 = cB_0 = 3 \times 10^8 \times 2.4 \times 10^{-7} = 72$ V/m. (1) $\nu = c / \lambda = 3 \times 10^8 / 3 = 10^8$ Hz = 100 MHz (FM band). (½)

 Last-minute revision

- $i_d = \epsilon_0 \frac{d\Phi_E}{dt}$; total current = $i_c + i_d$ is continuous; in a capacitor gap $i_d = i$.
- Changing $E \rightarrow B$ (Maxwell) and changing $B \rightarrow E$ (Faraday); accelerated charges radiate EM waves.
- EM waves: transverse, $E \perp B \perp$ travel, in phase, no medium, same speed $c = 1 / \sqrt{\mu_0 \epsilon_0}$, $E_0 = cB_0$.
- Energy shared equally by E and B; $I = \frac{1}{2} \epsilon_0 c E_0^2$; momentum U / c (absorbed), $2U / c$ (reflected).
- Spectrum (decreasing λ): radio, microwave, IR, visible, UV, X-ray, γ .
- Microwave – radar, oven (klystron, magnetron, Gunn diode); IR – heat, remote; UV – water purifier, ozone absorbs; X-ray – bones; γ – cancer.
- FM band 88–108 MHz; AM band 530–1710 kHz.

 Answer Key & Quick Solutions

1(b) 2(c) 3(c) 4(c) 5(a) 6(b) 7(c) 8(c) 9(a) 10(b)

1. (b) Klystron, magnetron and Gunn diode produce microwaves; the others are detectors.
2. (c) $i_d = \epsilon_0 d\Phi_E / dt$ – rate of change of electric flux.
3. (c) FM band: 88–108 MHz (AM: 530–1710 kHz).
4. (c) $B_0 = E_0 / c = 6 / (3 \times 10^8) = 2 \times 10^{-8}$ T.
5. (a) $i_d = C dV / dt = 5 \times 10^{-6} \times 2 \times 10^5 = 1$ A.
6. (b) Direction of travel is along $\vec{E} \times \vec{B}$: $\hat{j} \times \hat{k} = \hat{i} \Rightarrow B$ along +z.
7. (c) $v = c / \sqrt{\mu_r \epsilon_r} = 3 \times 10^8 / 2 = 1.5 \times 10^8$ m/s.
8. (c) Order of increasing frequency: radio < microwave < infrared < visible < UV < X-rays < γ -rays.
9. (a) The ozone layer (≈ 40 –50 km) absorbs most of the UV.
10. (b) $p = U / c = 6 / (3 \times 10^8) = 2 \times 10^{-8}$ kg m/s (for reflection it would be double).

CHAPTER

9

Ray Optics and Optical Instruments

17

Day 13–15

Board marks: 7

PYQ frequency ★★★

RBSE Class 12

Mirror formula

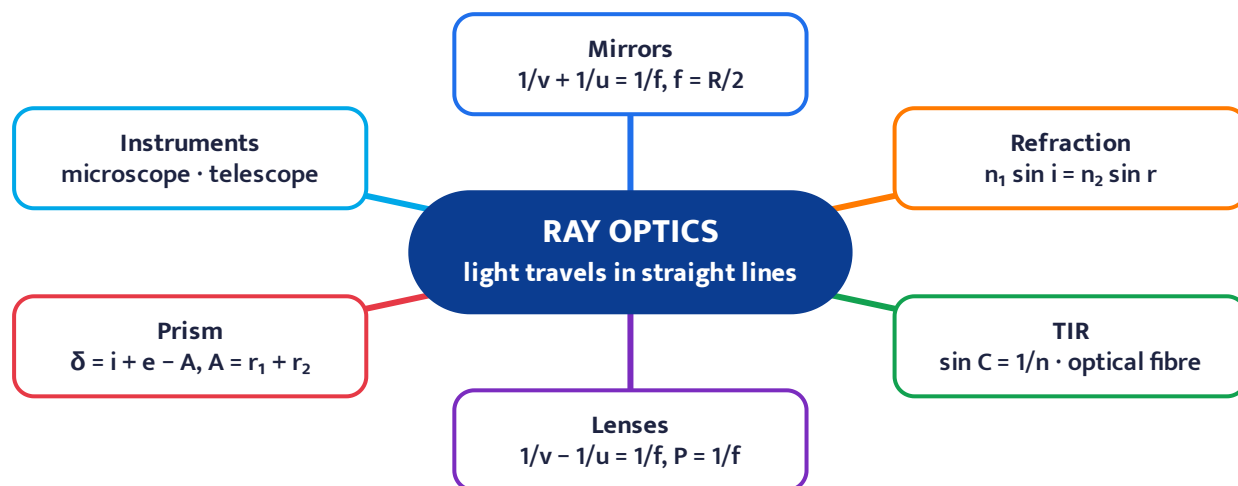
Refraction & TIR

Lens maker's formula

Power & combination

Prism: minimum deviation

Microscope & telescope



Chapter at a glance

Board snapshot

- **Marks: 7 of 56 – the highest-weight chapter.** Every recent paper has 1–3 objective items, 1–2 Section B numericals and a 4-mark Section D question (with internal choice).
- **2025:** MCQ focal length of two lenses in contact; B: object 15 cm from concave mirror ($R = 20$ cm); B: $5000 \text{ \AA}$ light from air to water; D: prism minimum-deviation formula OR power of lens + compound microscope.
- **2024:** MCQ negative magnification; MCQ compound microscope $m = m_o \times m_e$; fill-in f for $R = 24$ cm; B: telescope $M = 9$, length 100 cm; D: TIR + mirror formula OR lateral shift + lens maker's formula.
- **2023:** MCQ convex lens; fill-in primary rainbow; VSA focal length of concave mirror; D: mirror formula OR prism.
- **2026 (CWSN):** TIR; D: lens power & $f = R/2$ OR simple microscope. **2020:** lens in water; presbyopia; reflecting telescope.
- **Most likely in 2027:** mirror formula or lens maker's formula derivation (D); prism $n = \frac{\sin[(A + \delta_m)/2]}{\sin(A/2)}$; telescope/microscope magnifying power; power of lenses in contact.

9.1 Reflection by Spherical Mirrors

Laws of reflection: (i) the angle of incidence equals the angle of reflection; (ii) the incident ray, the reflected ray and the **normal** at the point of incidence lie in the same plane. For a spherical mirror the normal at any point is along the **radius** through that point. The centre of the mirror surface is the **pole (P)**, the line through P and the centre of curvature C is the **principal axis**, and the point where rays parallel to the axis meet (or appear to diverge from) after reflection is the **principal focus (F)**.

💡 Cartesian sign convention

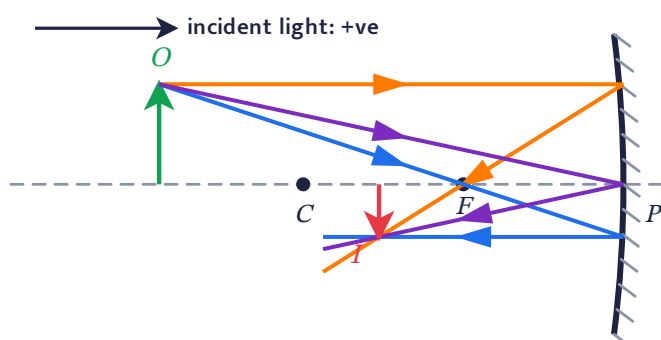
- All distances are measured **from the pole** (optical centre for a lens).
- Distances in the direction of **incident light** are positive; opposite to it negative.
- Heights above the principal axis +ve, below -ve.
- Real object: $u < 0$. Concave mirror $f < 0$, convex mirror $f > 0$.

📐 Mirror formulas

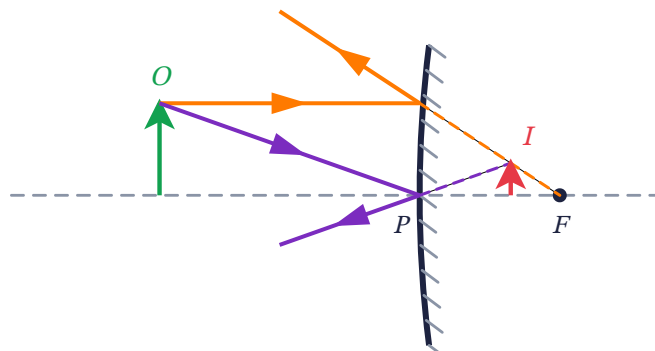
$$f = \frac{R}{2} \quad \text{Focal length (paraxial rays)}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{Mirror formula}$$

$$m = \frac{h'}{h} = -\frac{v}{u} \quad \text{Magnification: } m < 0 \rightarrow \text{real \& inverted; } m > 0 \rightarrow \text{virtual \& erect}$$



Concave mirror, object beyond C: image between F and C – real, inverted, diminished



Convex mirror: image always virtual, erect, diminished (rear-view mirror)

Object (concave mirror)	Image position	Nature and size
At infinity	At F	real, inverted, point-sized
Beyond C	Between F and C	real, inverted, diminished
At C	At C	real, inverted, same size ($m = -1$)
Between C and F	Beyond C	real, inverted, magnified
At F	At infinity	real, inverted, highly magnified
Between F and P	Behind the mirror	virtual, erect, magnified

🔧 Board numerical (1½ marks)

[RBSE 2025] An object is placed 15 cm in front of a concave mirror of radius of curvature 20 cm. Find the position and nature of the image.

Solution: $f = -R/2 = -10$ cm, $u = -15$ cm. $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{10} + \frac{1}{15} = -\frac{1}{30} \Rightarrow v = -30$ cm (in front of the mirror). (1)

$m = -\frac{v}{u} = -\frac{-30}{-15} = -2 \Rightarrow$ image is **real, inverted and twice the size** of the object. (½)

! Common Trap

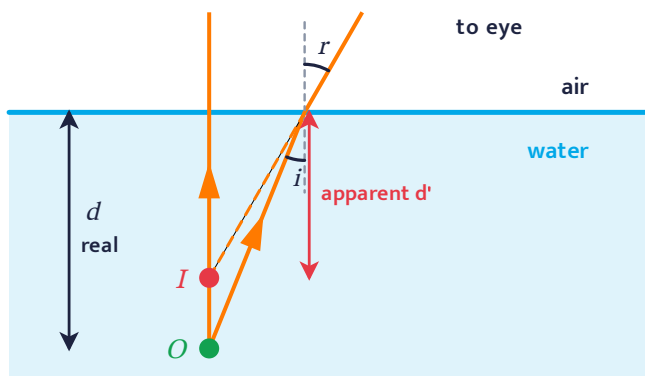
- Forgetting signs: for a concave mirror put f **negative** ($-R/2$) before substituting.
- A mirror's focal length does **not** change in water – reflection does not depend on the medium.

🎯 VS Sir's Exam Tip

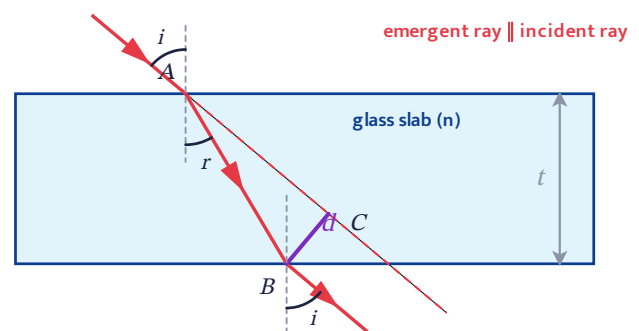
- Always write the **sign convention line** first, then the formula, then substitution with units – each earns marks.
- State the nature of the image from the sign of m (real/virtual, erect/inverted, size).

9.2 Refraction of Light

Refraction is the change in the direction of light when it passes obliquely from one medium to another. **Snell's law:** $\frac{\sin i}{\sin r} = n_{21} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$, where n_{21} is the **refractive index** of medium 2 with respect to medium 1. Absolute refractive index $n = c/v$ (no unit). On refraction the **frequency stays the same**, while speed and wavelength change ($\lambda_{\text{medium}} = \lambda_0 / n$). Light bends **towards the normal** on entering an optically denser medium.



Real and apparent depth: $d' = d / n$ (viewed near-normally)



Glass slab: emergent ray is parallel to the incident ray but shifted sideways by d (lateral shift)

🔧 Refraction formulas

$$n = \frac{c}{v}, \quad n_{21} = \frac{n_2}{n_1}, \quad n_{12} = \frac{1}{n_{21}} \quad \text{Refractive index relations}$$

$$\text{apparent depth } d' = \frac{d}{n}, \quad \text{shift} = d \left(1 - \frac{1}{n} \right)$$

Normal shift
(object appears raised)

$$d = \frac{t \sin(i - r)}{\cos r}$$

Lateral shift by a slab of thickness t

$$\nu = \frac{c}{\lambda_0} = \text{same in all media} \quad \text{Frequency is decided by the source}$$

🔧 Board numerical (1½ marks)

[RBSE 2025] Light of wavelength $5000 \text{ \AA}$ in air enters water ($n = 4/3$). Find its frequency, speed and wavelength in water.

Solution: Frequency $\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{5 \times 10^{-7}} = 6 \times 10^{14} \text{ Hz}$ – same in water. (½)

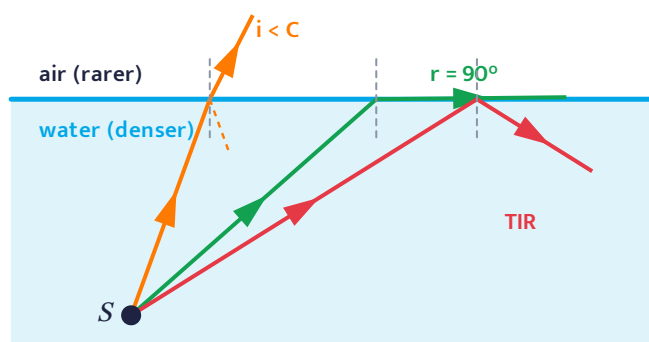
Speed $v = \frac{c}{n} = \frac{3 \times 10^8}{4/3} = 2.25 \times 10^8 \text{ m/s}$. (½)

Wavelength $\lambda_w = \frac{\lambda}{n} = \frac{5000}{4/3} = 3750 \text{ \AA}$. (½)

9.3 Total Internal Reflection (TIR)

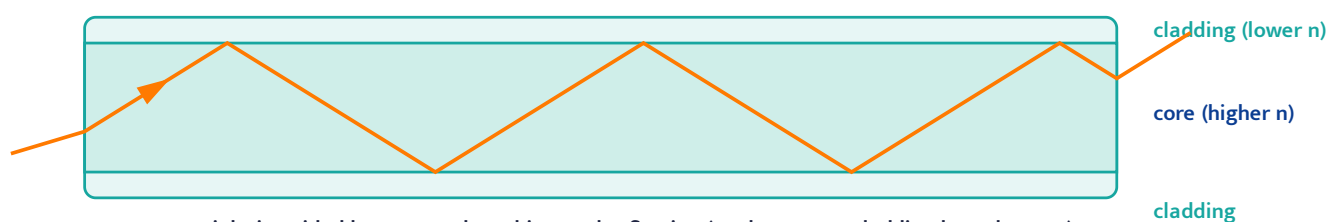
When light goes from a **denser to a rarer** medium, it bends away from the normal. The angle of incidence in the denser medium for which the angle of refraction is 90° is the **critical angle (C)**. If $i > C$, no refraction takes place and the whole light is reflected back into the denser medium – this is **total internal reflection**. **Conditions:** (i) light must travel from a denser to a rarer medium; (ii) angle of incidence must exceed the critical angle.

$$\sin C = \frac{1}{n} = \frac{n_{\text{rarer}}}{n_{\text{denser}}}$$



As i grows beyond C , refraction stops and light is totally reflected

Medium	n	C
Water	1.33	48.6°
Crown glass	1.5	41.8°
Diamond	2.42	24.4°



Light is guided by repeated total internal reflection (angle at core–cladding boundary $> C$)

Optical fibre: core of higher refractive index inside cladding of lower refractive index – light travels by repeated TIR

Applications of TIR

- **Optical fibres** – telecommunication, endoscopy, light pipes.
- **Totally reflecting prisms** (45° – 90° – 45°) bend light by 90° or 180° – periscopes, binoculars.
- **Mirage** in deserts and the **sparkling of diamond** ($C \approx 24.4^\circ$).

Common Trap

- TIR is impossible from rarer to denser, whatever the angle.
- Glass–water boundary: $\sin C = n_w / n_g = 8/9$ (not $2/3$).
- Larger $n \Rightarrow$ smaller critical angle.

9.4 Refraction at Spherical Surfaces and by Lenses

A **thin lens** is bounded by two spherical surfaces (or one spherical and one plane). Applying refraction at a single spherical surface twice gives the **lens maker's formula**, and from it the **thin lens formula**. R is positive if the centre of curvature lies on the side of the outgoing light. Convex lens: $f > 0$ (converging); concave lens: $f < 0$ (diverging).

Lens formulas

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad \text{Refraction at a spherical surface}$$

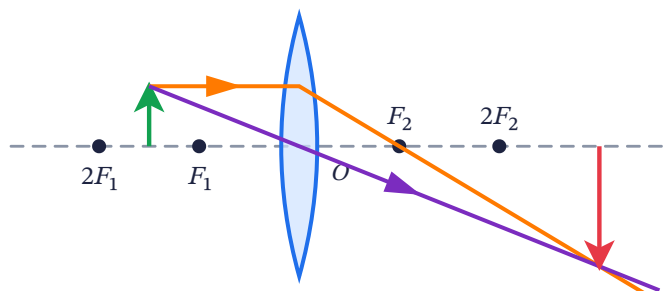
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \quad m = \frac{v}{u} \quad \text{Thin lens formula and magnification}$$

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}, \quad P = P_1 + P_2 \quad \text{Thin lenses in contact}$$

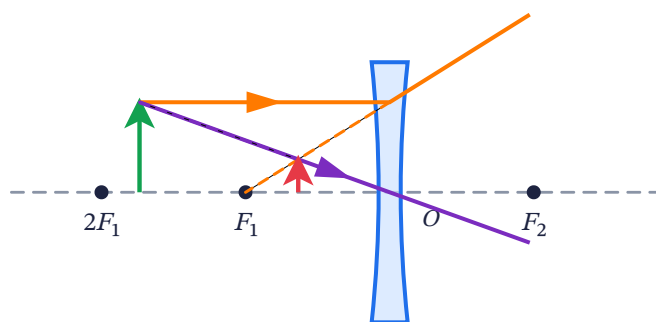
$$\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \text{Lens maker's formula}$$

$$P = \frac{1}{f(\text{m})} \text{ dioptre (D)} \quad \text{Power: convex +, concave -; } 1 \text{ D} = 1 \text{ m}^{-1}$$

$$m = m_1 \times m_2 \quad \text{Magnification of a combination}$$



Convex lens, object between F_1 and $2F_2$: real, inverted, magnified image beyond $2F_2$



Concave lens: image always virtual, erect, diminished

Board numerical (1½ marks)

[RBSE 2020] A convex glass lens ($n_g = 3/2$) has focal length 25 cm in air. Find its focal length in water ($n_w = 4/3$).

Solution: By lens maker's formula, $\frac{f_w}{f_a} = \frac{\frac{n_g}{n_w} - 1}{n_g - 1} = \frac{\frac{0.5}{4/3} - 1}{3/2 - 1} = \frac{0.5}{0.125} = 4$ (1)

$\Rightarrow f_w = 4 \times 25 = 100 \text{ cm}$ (still converging, but weaker). (½)

Solved Example

A convex lens of focal length 25 cm is placed in contact with a concave lens of focal length 20 cm. Find the power, focal length and nature of the combination.

Solution: $P = P_1 + P_2 = \frac{1}{0.25} + \frac{1}{-0.20} = 4 - 5 = -1 \text{ D} \Rightarrow F = \frac{1}{P} = -1 \text{ m} = -100 \text{ cm}$: a **diverging** (concave-like) combination.

Common Trap

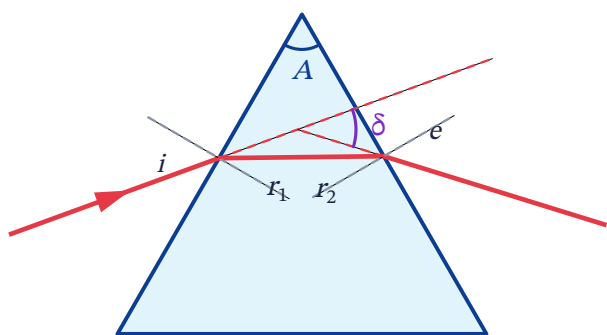
- Lens in a liquid of **equal** n : $f \rightarrow \infty$ (lens disappears); in a **denser** liquid a convex lens becomes **diverging**.
- Power must use f in **metres**: $f = 25 \text{ cm} \Rightarrow P = 1/0.25 = +4 \text{ D}$.

Memory Trick

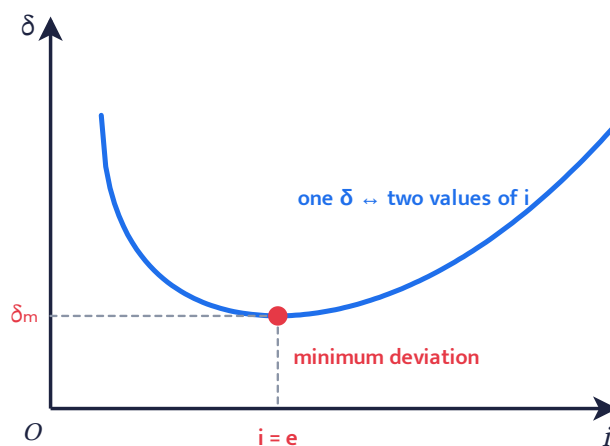
- Mirror: $\frac{1}{v} + \frac{1}{u}, m = -\frac{v}{u}$. Lens: $\frac{1}{v} - \frac{1}{u}, m = +\frac{v}{u}$.
- Covering half a lens: full image, only less bright.

9.5 Refraction through a Prism

A ray passing through a prism refracts twice and bends towards the **base**. The angle between the two refracting faces is the **angle of prism (A)**; the angle between the incident ray and the emergent ray is the **angle of deviation (δ)**. As i increases, δ first decreases, becomes **minimum (δ_m)** and then increases. At minimum deviation $i = e, r_1 = r_2 = A/2$ and the ray inside an equilateral prism is parallel to the base.



Path of a ray through a prism

 δ - i graph: δ is minimum when $i = e$ **Prism Formulas**

$$r_1 + r_2 = A, \quad \delta = i + e - A \quad \text{General relations}$$

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Refractive index from minimum deviation

$$\delta = (n - 1)A \quad \text{Thin prism (small } A)$$

$$i = \frac{A + \delta_m}{2}$$

Angle of incidence at minimum deviation

**Solved Example**

The angle of minimum deviation of an equilateral glass prism is 30° . Find the refractive index and the angle of incidence at minimum deviation.

Solution: $A = 60^\circ$; $n = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{1/\sqrt{2}}{1/2} = \sqrt{2} \approx 1.414$; $i = \frac{A + \delta_m}{2} = 45^\circ$.

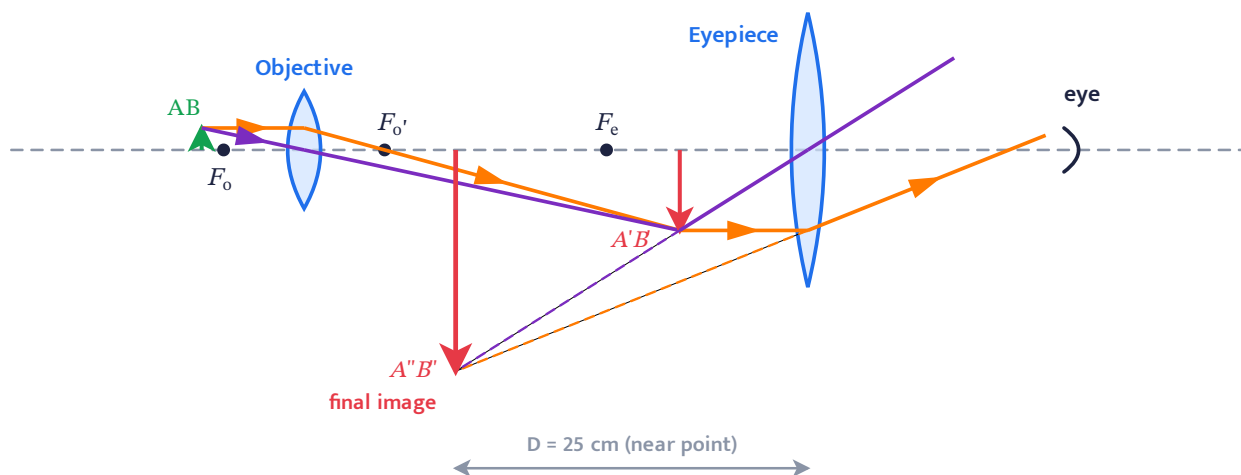
**Asked earlier by RBSE**

- These topics are **outside the current (rationalised) NCERT book** but RBSE has asked them: **Dispersion** – splitting of white light into colours; $n_{\text{violet}} > n_{\text{red}}$, so violet deviates most.
- Rainbow (2023):** formed by refraction, dispersion and internal reflection in raindrops. **Primary rainbow** – one internal reflection, red on the outer (top) side, violet inner; **secondary** – two internal reflections, colours reversed, fainter.
- Presbyopia (2020):** with age the eye loses its power of accommodation, the near point recedes; corrected by a **convex lens** (bifocals if myopia is also present).

9.6 Optical Instruments

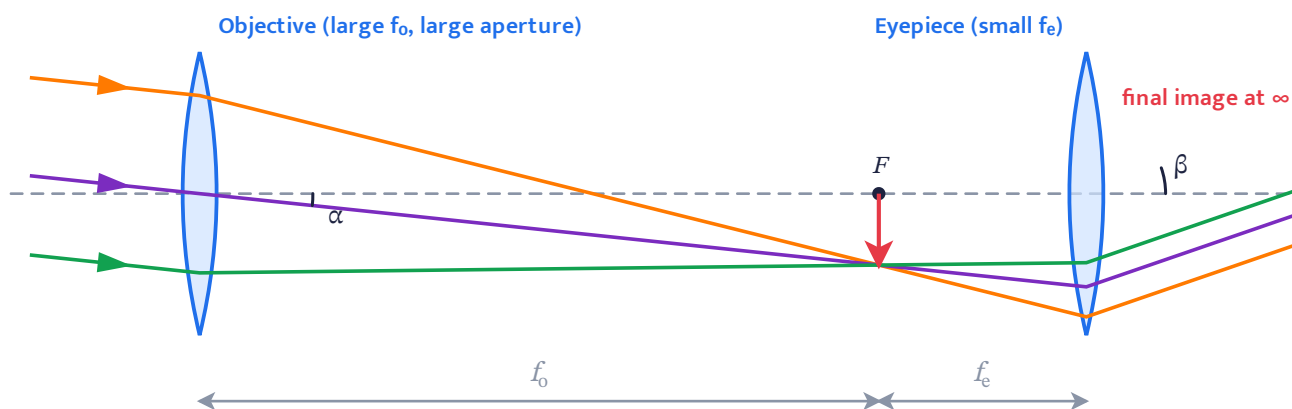
Magnifying power (angular magnification) of an instrument = angle subtended at the eye by the final image $\div$ angle subtended by the object at the unaided eye (object at near point $D = 25$ cm for a microscope; at its actual distance for a telescope). The final image may be at the **near point** (maximum strain-free clarity, maximum m) or at **infinity** (**normal adjustment**, relaxed eye).

Instrument	Final image at near point D	Normal adjustment (image at ∞)
Simple microscope	$m = 1 + \frac{D}{f}$	$m = \frac{D}{f}$
Compound microscope	$m = m_o m_e = \frac{v_o}{ u_o } \left(1 + \frac{D}{f_e} \right)$	$m \approx \frac{L}{f_o} \cdot \frac{D}{f_e}$
Astronomical telescope	$m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$	$m = \frac{f_o}{f_e}, L = f_o + f_e$

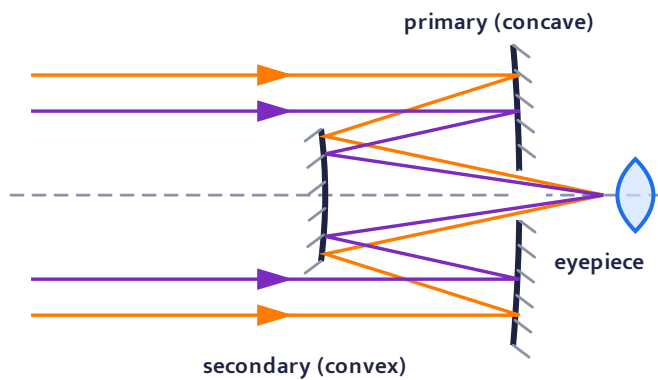


Compound microscope: objective forms a real, inverted, magnified image $A'B'$ inside F_e ; the eyepiece acts as a simple microscope and forms the final virtual image $A''B''$

Tube length L = distance between the second focal point of the objective and the first focal point of the eyepiece. In a microscope both lenses have **short** focal lengths ($f_o < f_e$); in a telescope the objective has a **large focal length and large aperture**, the eyepiece a short focal length.



Astronomical (refracting) telescope in normal adjustment: $m = \beta / \alpha = f_o / f_e$



[RBSE 2020] Cassegrain reflecting telescope

Advantages of a reflecting telescope

- No **chromatic aberration** (a mirror does not disperse light).
- A **parabolic** mirror removes spherical aberration.
- Mirror is lighter and supported over its whole back – very large aperture possible $\Rightarrow$ more light and better resolution.

Board numerical (1½ marks)

[RBSE 2024] An astronomical telescope in normal adjustment has magnifying power 9 and length 100 cm. Find the focal lengths of the objective and the eyepiece.

Solution: $\frac{f_o}{f_e} = 9$ and $f_o + f_e = 100$ cm (½) $\Rightarrow 9f_e + f_e = 100 \Rightarrow f_e = 10$ cm (½), $f_o = 90$ cm (½).

Solved Example

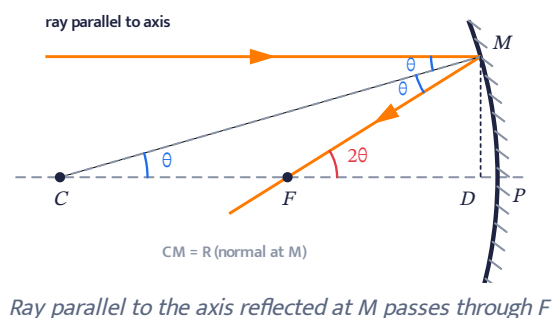
A compound microscope has $f_o = 1$ cm, $f_e = 5$ cm and tube length $L = 20$ cm. Find its magnifying power in normal adjustment ($D = 25$ cm).

Solution: $m = \frac{L}{f_o} \times \frac{D}{f_e} = \frac{20}{1} \times \frac{25}{5} = 100$.

Common Trap

- Telescope: f_o **large**, f_e small. Microscope: **both** small, $f_o < f_e$. Interchanging the lenses of a telescope gives $m < 1$.
- A large objective aperture in a telescope increases light gathering and **resolving power**, not the magnifying power.

Important Derivations ★



Derivation ★ [2 marks]

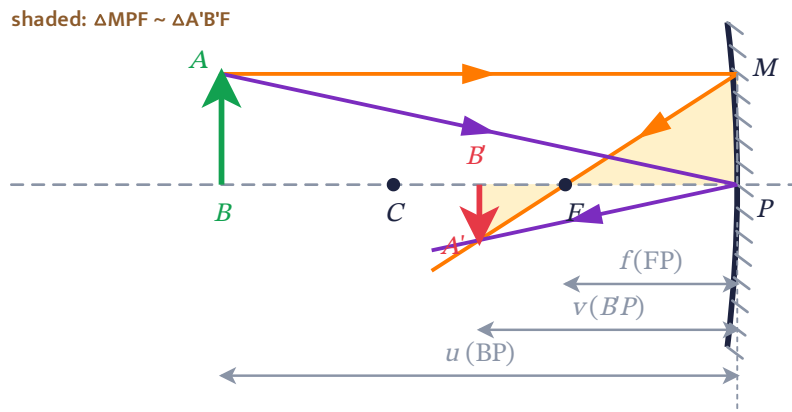
[RBSE 2026] Prove that for a spherical mirror $f = R / 2$.

Solution: 1. A ray parallel to the axis strikes the mirror at M; CM is the normal. $\angle MCP = \theta =$ angle of incidence = angle of reflection, so $\angle MFP = 2\theta$. (½)

2. $MD \perp$ axis: $\tan \theta = \frac{MD}{CD}$, $\tan 2\theta = \frac{MD}{FD}$. (½)

3. For small θ (paraxial): $\tan \theta \approx \theta \Rightarrow \frac{MD}{FD} = 2 \frac{MD}{CD} \Rightarrow FD = \frac{CD}{2}$. (½)

4. D is very close to P: $FD \approx FP = f$, $CD \approx CP = R \Rightarrow f = R / 2$. (½)



Mirror formula: object AB beyond C, real image A'B'; rays AM (parallel $\rightarrow$ through F) and AP (at the pole)

🔧 Derivation ★ [3 marks]

[RBSE 2023, 2024] Derive the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for a concave mirror.

Solution: 1. Draw the ray diagram (above): $AM \parallel$ axis reflects through F; AP reflects with $\angle APB = \angle A'PB'$. For paraxial rays, MP is taken as a straight line $\perp$ axis, so $MP = AB$. (½)

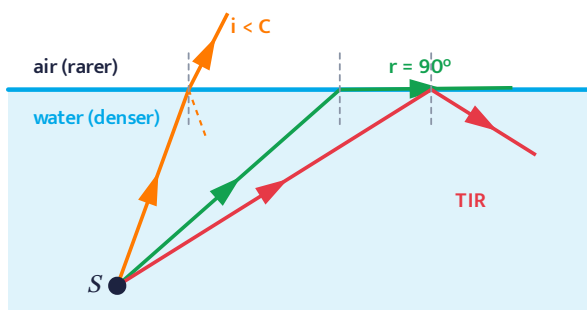
2. $\triangle A'B'F \sim \triangle MPF$ (right-angled, vertically opposite angles at F): $\frac{B'A'}{PM} = \frac{BF}{FP} \Rightarrow \frac{B'A'}{BA} = \frac{BF}{FP} \dots (1)$ (½)

3. $\triangle A'B'P \sim \triangle ABP$ (equal angles at P): $\frac{B'A'}{BA} = \frac{BP}{BP} \dots (2)$ (½)

4. From (1) and (2): $\frac{BF}{FP} = \frac{BP}{BP}$, i.e. $\frac{BP - FP}{FP} = \frac{BP}{BP}$. (½)

5. Sign convention: $B'P = -v$, $FP = -f$, $BP = -u \Rightarrow \frac{-v + f}{-f} = \frac{-v}{-u} \Rightarrow \frac{v - f}{f} = \frac{v}{u}$. (½)

6. $\frac{v}{f} - 1 = \frac{v}{u}$; dividing by v: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$. (½)



At $i = C$ the refracted ray grazes the surface ($r = 90^\circ$)

🔧 Derivation ★ [2 marks]

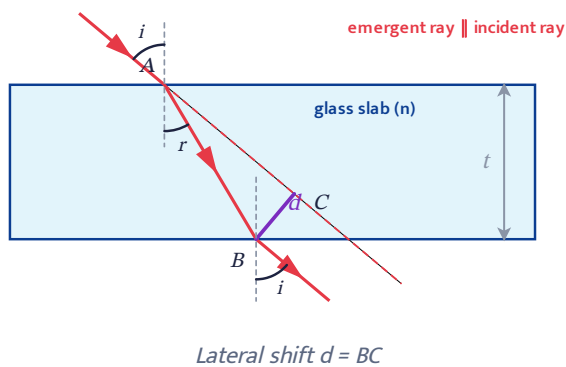
[RBSE 2024, 2026] What is total internal reflection? Obtain the relation between critical angle and refractive index.

Solution: 1. Definition of TIR and its two conditions (see 9.3). (½)

2. Light goes from denser (n_1) to rarer (n_2). By Snell's law $n_1 \sin i = n_2 \sin r$. (½)

3. At $i = C$, $r = 90^\circ$: $n_1 \sin C = n_2 \Rightarrow \sin C = \frac{n_2}{n_1}$. (½)

4. If the rarer medium is air ($n_2 = 1$, $n_1 = n$): $\sin C = \frac{1}{n}$, i.e. $n = \frac{1}{\sin C}$. (½)



Derivation ★ [2 marks]

[RBSE 2024] Derive an expression for the lateral shift of a ray passing through a glass slab of thickness t .

Solution: 1. Ray enters at A (angle i), refracts at r , leaves at B; the emergent ray is parallel to the incident ray (same media on both sides). (½)

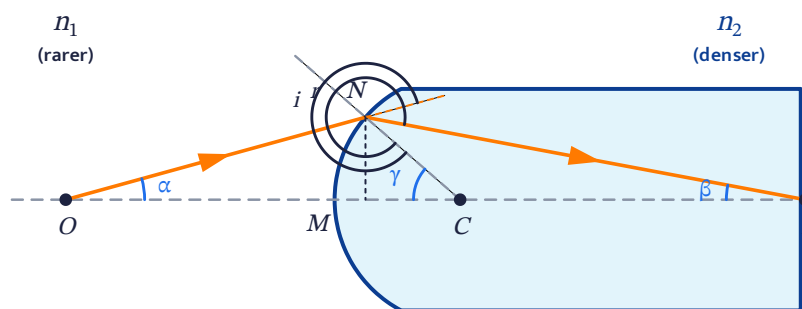
2. In $\triangle ANB$ (N = foot of normal at A on the lower face):

$$AB = \frac{t}{\cos r}. \quad (½)$$

3. Draw $BC \perp$ extended incident ray; $\angle BAC = i - r$. In $\triangle ABC$:

$$d = BC = AB \sin(i - r). \quad (½)$$

$$4. \quad d = \frac{t \sin(i - r)}{\cos r}; \quad d \text{ increases with } t, i \text{ and } n. \quad (½)$$



Refraction at a convex spherical surface: object O in medium n_1 , real image I in medium n_2 (angles exaggerated)

Derivation ★ [3 marks]

Derive the relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ for refraction at a convex spherical surface.

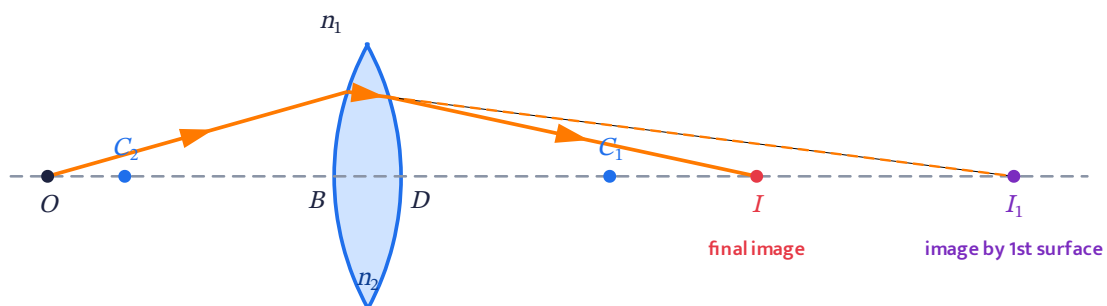
Solution: 1. Ray ON from O refracts at N towards I ; NC is the normal. Let $\angle NOM = \alpha$, $\angle NCM = \gamma$, $\angle NIM = \beta$. For paraxial rays the foot of the perpendicular from N is M and $\alpha \approx \frac{MN}{OM}$, $\gamma \approx \frac{MN}{MC}$, $\beta \approx \frac{MN}{MI}$. (½)

2. Exterior angles: in $\triangle NOC$, $i = \alpha + \gamma$; in $\triangle NCI$, $\gamma = r + \beta \Rightarrow r = \gamma - \beta$. (½)

3. Snell's law for small angles: $n_1 i = n_2 r \Rightarrow n_1(\alpha + \gamma) = n_2(\gamma - \beta)$. (½)

$$4. \quad n_1 \left(\frac{MN}{OM} + \frac{MN}{MC} \right) = n_2 \left(\frac{MN}{MC} - \frac{MN}{MI} \right) \Rightarrow \frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}. \quad (½)$$

$$5. \text{ Signs: } OM = -u, MI = +v, MC = +R \Rightarrow \frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}. \quad (1)$$

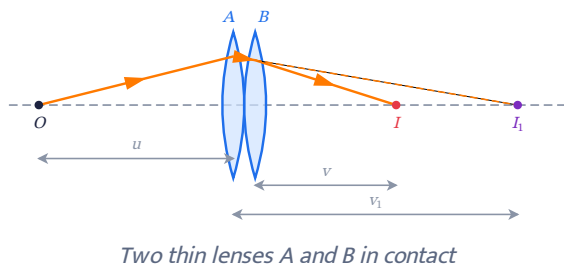


Lens maker's formula: surface 1 alone would form image I_1 ; surface 2 refracts the rays to form the final image I

Derivation ★ [4 marks]

[RBSE 2024] Derive the lens maker's formula for a thin double-convex lens and hence the thin lens formula. (Lens of index n_2 in a medium of index n_1 .)

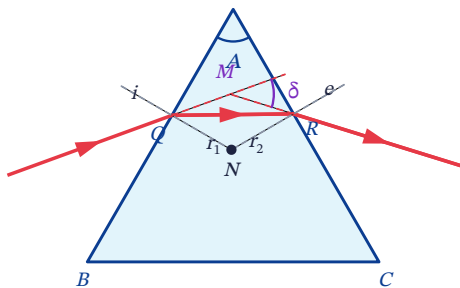
- Solution:** 1. First surface ABC (radius R_1) alone forms image I_1 of O: $\frac{n_1}{OB} + \frac{n_2}{BI_1} = \frac{n_2 - n_1}{BC_1} \dots (1) \text{ (}\frac{1}{2}\text{)}$
2. For the second surface ADC, I_1 acts as a virtual object and final image is I: $-\frac{n_2}{DI_1} + \frac{n_1}{DI} = \frac{n_1 - n_2}{DC_2} \dots (2) \text{ (}\frac{1}{2}\text{)}$
3. Thin lens: $BI_1 \approx DI_1$. Adding (1) and (2): $\frac{n_1}{OB} + \frac{n_1}{DI} = (n_2 - n_1) \left(\frac{1}{BC_1} + \frac{1}{DC_2} \right) \dots (3) \text{ (}\frac{1}{2}\text{)}$
4. Signs: $OB = -u, DI = +v, BC_1 = +R_1, DC_2 = -R_2 \Rightarrow \frac{1}{v} - \frac{1}{u} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), n_{21} = \frac{n_2}{n_1} \dots (4) \text{ (1)}$
5. For $u = \infty, v = f$: $\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ (lens maker's formula); putting in (4): $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ (thin lens formula). (1)



Derivation ★ [2 marks]

[RBSE 2025, 2026] Define power of a lens. Obtain the focal length (power) of two thin lenses in contact.

- Solution:** 1. Power $P = \frac{1}{f(\text{m})}$; SI unit dioptre (D). (½)
2. Lens A forms image I_1 of O: $\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \dots (1) \text{ (}\frac{1}{2}\text{)}$
3. I_1 is the object for B (thin, in contact): $\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \dots (2) \text{ (}\frac{1}{2}\text{)}$
4. Adding: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{F} \Rightarrow \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}, P = P_1 + P_2$; also $m = m_1 m_2$. (½)



Prism ABC: incident ray at Q, emergent at R; normals meet at N

Derivation ★ [3 marks]

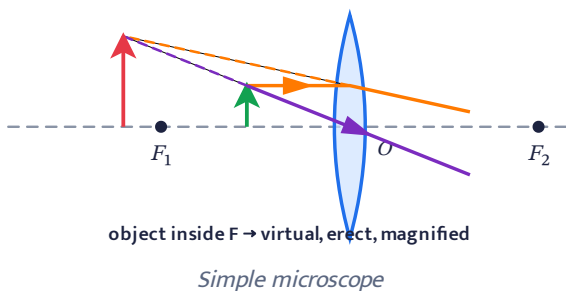
[RBSE 2023, 2025] Derive $n = \frac{\sin[(A + \delta_m)/2]}{\sin(A/2)}$ for a prism.

- Solution:** 1. In quadrilateral AQNR, $\angle Q = \angle R = 90^\circ \Rightarrow \angle A + \angle QNR = 180^\circ$. In $\triangle QNR$: $r_1 + r_2 + \angle QNR = 180^\circ \Rightarrow r_1 + r_2 = A \dots (1) \text{ (1)}$
2. Deviation = sum of deviations at the two faces: $\delta = (i - r_1) + (e - r_2) \Rightarrow \delta = i + e - A \dots (2) \text{ (}\frac{1}{2}\text{)}$
3. At minimum deviation $i = e \Rightarrow r_1 = r_2 = r$: from (1) $r = \frac{A}{2}$; from (2) $i = \frac{A + \delta_m}{2}$. (1)
4. Snell's law $n = \frac{\sin i}{\sin r} \Rightarrow n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$. (½)

Derivation ★ [2 marks]

[RBSE 2026] Obtain the magnifying power of a simple microscope when the image is at (a) the near point, (b) infinity.

- Solution:** (a) Image at near point: $v = -D$. Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow m = \frac{v}{u} = 1 - \frac{v}{f} \Rightarrow m = 1 + \frac{D}{f}$. (1)
- (b) Object at near point subtends $\theta_0 = \frac{h}{D}$. With the object at F, image at ∞ subtends $\theta_i = \frac{h}{f} \Rightarrow m = \frac{\theta_i}{\theta_0} = \frac{D}{f}$. (1)



 Derivation ★ [3 marks]

[RBSE 2024, 2025] Draw the ray diagram of a compound microscope (figure in 9.6) and derive its magnifying power.

Solution: 1. Labelled ray diagram: objective forms real, inverted, magnified image A'B' within F_e ; eyepiece forms final virtual image A''B''. (1)

2. Total magnifying power $m = m_o \times m_e$. (½)

3. Objective: $m_o = \frac{h'}{h} = \frac{v_o}{|u_o|}$; since the object is just beyond F_o and A'B' is near F_e , similar triangles give $m_o \approx \frac{L}{f_o}$ (L = tube length). (½)

4. Eyepiece acts as a simple microscope: $m_e = 1 + \frac{D}{f_e}$ (image at D) or $\frac{D}{f_e}$ (image at ∞). (½)

5. $m = \frac{L}{f_o} \left(1 + \frac{D}{f_e}\right)$ (near point) and $m = \frac{L}{f_o} \cdot \frac{D}{f_e}$ (normal adjustment). (½)

 Derivation ★ [3 marks]

Draw the ray diagram of an astronomical telescope in normal adjustment (figure in 9.6) and derive its magnifying power.

Solution: 1. Labelled diagram: parallel rays from a distant object at angle α form a real, inverted image of height h at the common focus F; the eyepiece sends the rays out parallel at angle β (final image at ∞). (1)

2. Since the object is far away, angle at the unaided eye = α (same as at the objective): $\tan \alpha \approx \alpha = \frac{h}{f_o}$. (½)

3. At the eyepiece: $\tan \beta \approx \beta = \frac{h}{f_e}$. (½)

4. $m = \frac{\beta}{\alpha} = \frac{f_o}{f_e}$, length $L = f_o + f_e$. (If the final image is at D: $m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D}\right)$). (1)

 How to write a derivation for full marks

- Draw a **neat labelled ray diagram with arrows** first – it alone carries ½–1 mark.
- Number the equations (1), (2)... and write the **sign convention step** explicitly.
- Box the final result and state any approximation used (paraxial rays, thin lens, small angles).

Board question bank **Section A – Multiple choice (½ mark each)**

Questions tagged [RBSE year] are from official BSER papers (reworded). Answers at the end of the chapter.

1 [RBSE 2025] Two thin lenses of focal lengths f_1 and f_2 are placed in contact. The focal length of the combination is

(a) $\frac{f_1 f_2}{f_1 + f_2}$

(b) $f_1 + f_2$

(c) $\frac{f_1 + f_2}{f_1 f_2}$

(d) $\sqrt{f_1 f_2}$

2 [RBSE 2024] If the magnification produced by a spherical mirror is negative, the image is

(a) always diminished

(b) virtual and inverted

(c) virtual and erect

(d) real and inverted

3 [RBSE 2024] The magnifying power of a compound microscope is (m_o, m_e = magnifications of objective and eyepiece)

(a) $m_e - m_o$

(b) m_o / m_e

(c) $m_o \times m_e$

(d) $m_o + m_e$

- 4 The critical angle for glass ($n = 1.5$) – air boundary is about
 (a) 60° (b) 24.4° (c) 48.6° (d) 41.8°
- 5 The power of a concave lens of focal length 50 cm is
 (a) -50 D (b) -2 D (c) -0.02 D (d) $+2$ D
- 6 When light passes from air into glass, the quantity that does not change is
 (a) direction (for oblique incidence)
 (b) speed
 (c) frequency
 (d) wavelength
- 7 Optical fibres work on the principle of
 (a) total internal reflection (b) dispersion
 (c) diffraction (d) refraction only
- 8 The angle of minimum deviation of an equilateral prism is 30° . Its refractive index is
 (a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\sqrt{3}$ (d) 1.5
- 9 An astronomical telescope has $f_o = 100$ cm and $f_e = 5$ cm. Its magnifying power in normal adjustment is
 (a) 0.05 (b) 20 (c) 105 (d) 500
- 10 The focal length of a concave mirror in air is f . When it is dipped in water ($n = 4/3$), its focal length becomes
 (a) $4f$ (b) $4f/3$ (c) $3f/4$ (d) f

Fill in the blanks (½ mark)

Question	Answer
[RBSE 2024] The focal length of a spherical mirror of radius of curvature 24 cm is _____.	12 cm ($f = R/2$)
Total internal reflection occurs when light goes from a _____ to a _____ medium.	denser, rarer
The SI unit of power of a lens is _____.	diopetre (D) = m^{-1}
At minimum deviation, the angle of incidence is _____ the angle of emergence.	equal to ($i = e$)
In normal adjustment the length of an astronomical telescope is _____.	$f_o + f_e$
In a primary rainbow the _____ colour is on the outer side.	red

Very short answer (1 mark)

Question	Answer
[RBSE 2023] Write the relation between the focal length and the radius of curvature of a concave mirror.	$f = \frac{R}{2}$
[RBSE 2026] Define critical angle.	Angle of incidence in the denser medium for which the angle of refraction is 90° .
Why does a diamond sparkle?	Small critical angle (24.4°) $\Rightarrow$ repeated TIR inside it.
When does a convex lens behave as a diverging lens?	When placed in a medium of higher refractive index than its own.
Why is the objective of a telescope of large aperture?	To gather more light and increase resolving power.
[RBSE 2020] Which lens corrects presbyopia?	Convex (converging) lens.

Short answer – Section B (1½ marks)

 Short answer (1½ marks)

A biconvex lens of glass ($n = 1.5$) has both radii of curvature 20 cm. Find its focal length and power in air.

Solution: $R_1 = +20, R_2 = -20; \frac{1}{f} = (1.5 - 1)\left(\frac{1}{20} + \frac{1}{20}\right) = \frac{1}{20} \Rightarrow f = 20 \text{ cm (1); } P = \frac{1}{0.20} = +5 \text{ D. (½)}$

 Short answer (1½ marks)

A ray is incident at 60° on a glass slab ($n = \sqrt{3}$) of thickness 3 cm. Find the lateral shift.

Solution: $\sin r = \frac{\sin 60^\circ}{\sqrt{3}} = \frac{1}{2} \Rightarrow r = 30^\circ \text{ (½); } d = \frac{t \sin(i - r)}{\cos r} = \frac{3 \times \sin 30^\circ}{\cos 30^\circ} = \frac{1.5}{0.866} = \sqrt{3} \approx 1.73 \text{ cm. (1)}$

 Short answer (1½ marks)

A simple microscope uses a lens of focal length 5 cm. Find its magnifying power when the image is formed (a) at the near point (25 cm), (b) at infinity.

Solution: (a) $m = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6 \text{ (1); (b) } m = \frac{D}{f} = 5 \text{ (½)}$

 Short answer (1½ marks)

Give three advantages of a reflecting telescope over a refracting telescope.

Solution: (i) No chromatic aberration (mirror does not disperse light). (½) (ii) Spherical aberration removed by a parabolic mirror. (½) (iii) Lighter mirror, supported over the whole back $\Rightarrow$ large aperture, more light and higher resolving power. (½)

 Answer Key & Quick Solutions

1(a) 2(d) 3(c) 4(d) 5(b) 6(c) 7(a) 8(a) 9(b) 10(d)

1. (a) $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow F = \frac{f_1 f_2}{f_1 + f_2}$.

2. (d) $m < 0 \Rightarrow$ image inverted and (for a mirror) real.

3. (c) The eyepiece magnifies the image formed by the objective: $m = m_o m_e$.

4. (d) $\sin C = 1 / 1.5 = 0.667 \Rightarrow C \approx 41.8^\circ$.

5. (b) $P = 1/f(m) = 1/(-0.5) = -2 \text{ D.}$

6. (c) Frequency is fixed by the source; speed and wavelength both decrease by n .
7. (a) Light is guided by repeated TIR at the core–cladding boundary.
8. (a) $n = \sin 45^\circ / \sin 30^\circ = \sqrt{2}$.
9. (b) $m = f_o/f_e = 100/5 = 20$ (tube length 105 cm).
10. (d) $f = R/2$ depends only on the mirror's shape, not on the medium.



Last-minute revision

- Mirror: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $m = -\frac{v}{u}$, $f = \frac{R}{2}$; concave $f < 0$.
- Snell: $n_1 \sin i = n_2 \sin r$; frequency never changes; $\lambda_m = \lambda_0 / n$.
- TIR: denser $\rightarrow$ rarer and $i > C$; $\sin C = 1/n$ (glass 41.8° , water 48.6° , diamond 24.4°).
- Lateral shift $d = t \sin(i - r) / \cos r$; apparent depth d/n .
- Lens maker: $\frac{1}{f} = (n_{21} - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$; lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $m = \frac{v}{u}$.
- $P = 1/f$ (m) in dioptre; in contact $P = P_1 + P_2$.
- Prism: $A = r_1 + r_2$, $\delta = i + e - A$, $n = \frac{\sin[(A + \delta_m)/2]}{\sin(A/2)}$.
- Simple microscope $1 + D/f$ or D/f ; compound $\frac{L}{f_o} \cdot \frac{D}{f_e}$; telescope f_o/f_e , $L = f_o + f_e$.

CHAPTER

10

Wave Optics

17

Day 16–17

Board marks: 5

PYQ frequency ★★☆☆

RBSE Class 12

Wavefront & Huygens

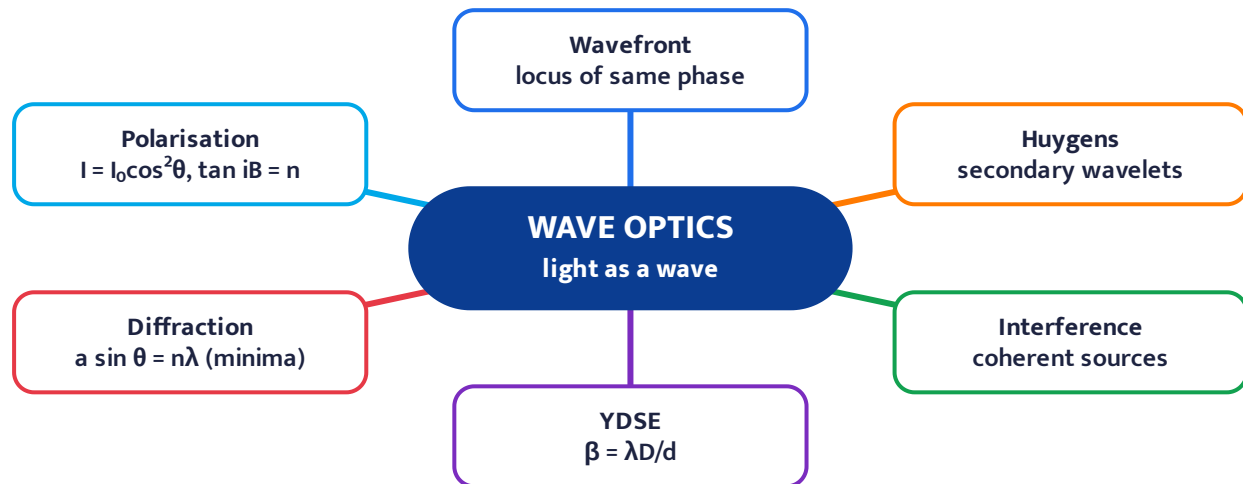
Coherent sources

Young's double slit

Single-slit diffraction

Polarisation

Malus & Brewster laws



Chapter at a glance

Board snapshot

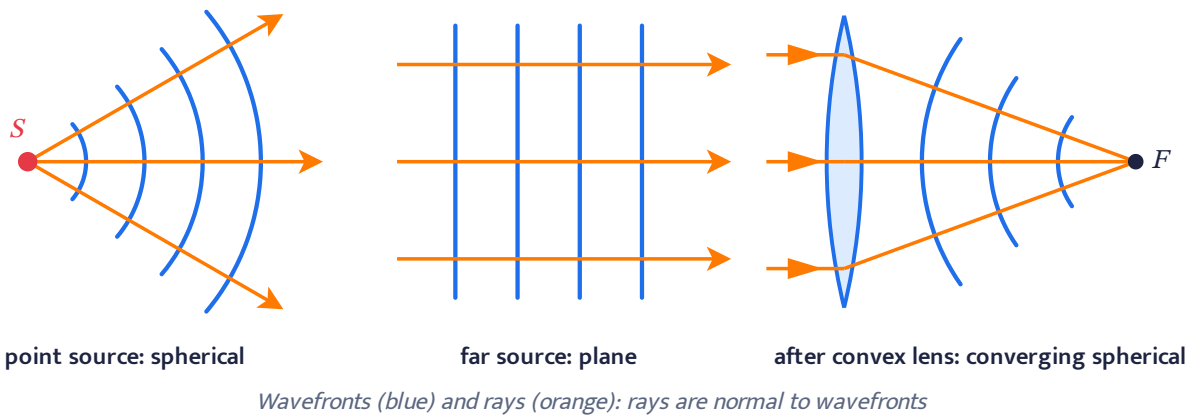
- **Marks: 5 of 56** (1–2 objective items, 1–2 Section B, sometimes a 3-mark Section C). **Huygens' construction for Snell's law: asked in 2024, 2025 and 2026.**
- **2025:** MCQ transverse nature of light shown by polarisation; fill-in intensity through a polaroid for unpolarised light; B: laws of refraction using Huygens' principle.
- **2024:** MCQ sunlight is unpolarised; VSA coherent sources; B: Snell's law by Huygens; B: define interference and polarisation.
- **2023:** MCQ phase difference ↔ path difference; VSA Fresnel distance; C (3 marks): YDSE and fringe width $\beta = \lambda D / d$. **2026 (CWSN):** Snell's law by Huygens; polaroids. **2020:** interference vs diffraction; wavefront and Huygens' principle.
- **Most likely in 2027:** Snell's law (and reflection) by Huygens; YDSE fringe width derivation; definitions of interference/diffraction/polarisation; polaroid–Malus numerical; $\Delta x = \lambda \phi / 2\pi$.

10.1 Wavefront and Huygens' Principle

A **wavefront** is the locus of all points that oscillate **in the same phase** at a given instant. The direction of propagation (the **ray**) is always **perpendicular to the wavefront**. A point source gives **spherical** wavefronts; a very distant source (the Sun) gives a **plane** wavefront; a line source gives a cylindrical one.

Huygens' principle

- Every point of a wavefront acts as a source of **secondary wavelets** that spread in the forward direction with the speed of the wave in that medium.
- The **forward envelope** (common tangent) of these wavelets after time τ is the **new wavefront**.



! Common Trap

- In a denser medium light is **slower** ($v_2 < v_1$) and bends towards the normal – the wave theory is confirmed (the particle theory wrongly predicted a faster speed).
- On refraction **frequency does not change**; ν and λ change in the ratio $n: \lambda_2 = \lambda_1 \nu_2 / \nu_1$.

10.2 Coherent Sources and Interference

Interference is the redistribution of light energy in space (alternate bright and dark regions) when two or more **coherent** waves overlap. **Coherent sources** emit waves of the **same frequency** with a **constant phase difference**. Two separate bulbs are incoherent – their phase difference changes randomly in $\sim 10^{-10}$ s, so intensities just add. Coherent sources are obtained from **one** source, e.g. the two slits of YDSE. Energy is only **redistributed**, never lost: the average intensity stays $I_1 + I_2$.

Superposition of two waves

$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$	Resultant intensity, $\phi =$ phase difference	$\phi = \frac{2\pi}{\lambda} \Delta x \Rightarrow \Delta x = \frac{\lambda}{2\pi} \phi$	Phase difference $\leftrightarrow$ path difference
Bright: $\Delta x = n\lambda, \phi = 2n\pi$	Constructive: $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$	Dark: $\Delta x = (2n - 1)\frac{\lambda}{2}$	Destructive: $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$
$I = 4I_0 \cos^2 \frac{\phi}{2}$	Equal sources $I_1 = I_2 = I_0$: $I_{\max} = 4I_0, I_{\min} = 0$	$\frac{I_{\max}}{I_{\min}} = \left(\frac{r+1}{r-1}\right)^2$	$r = a_1 / a_2 = \sqrt{I_1 / I_2}$ (amplitude ratio)

🔧 Board numerical (1½ marks)

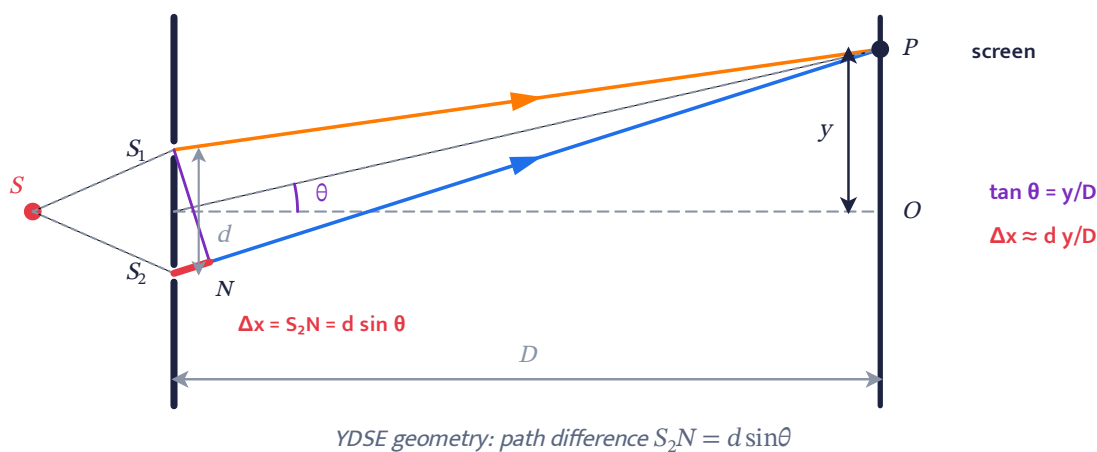
Two coherent sources have intensities in the ratio 9 : 1. Find $I_{\max} / I_{\min}$. Also find the phase difference for a path difference of $\lambda / 4$.

Solution: $r = \sqrt{9} = 3 \Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{3+1}{3-1}\right)^2 = 4:1. (1)$

$\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2} \text{ rad } (90^\circ). (1/2)$

10.3 Young's Double-Slit Experiment (YDSE)

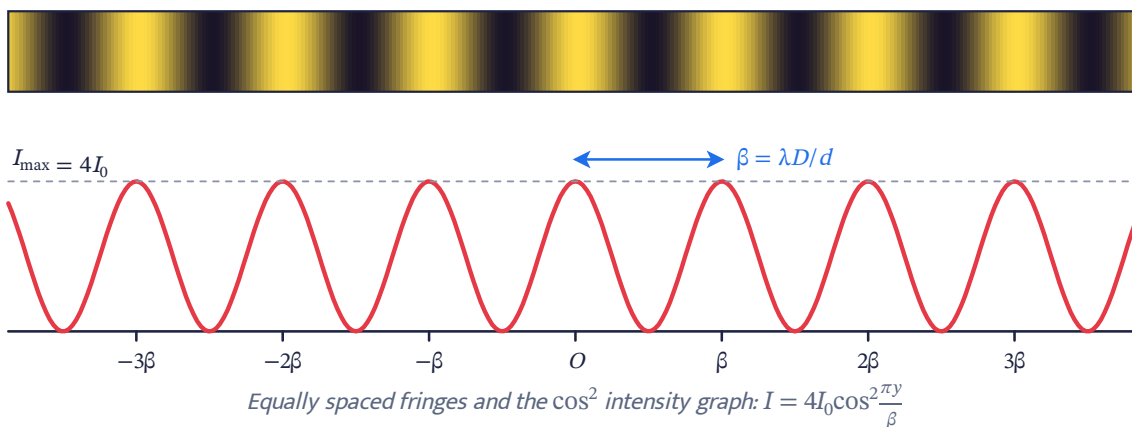
Light from a narrow slit S falls on two close slits S_1, S_2 (separation d) which act as coherent sources. A screen at distance D ($D \gg d$) shows equally spaced bright and dark **fringes**. The central point O (equidistant from both slits) is always **bright**.



YDSE formulas

$\Delta x = \frac{yd}{D}$	Path difference at height y	$y_n = \frac{n\lambda D}{d}$	n -th bright fringe ($n = 0, 1, 2, \dots$)
$y'_n = \frac{(2n-1)\lambda D}{2d}$	n -th dark fringe ($n = 1, 2, \dots$)	$\beta = \frac{\lambda D}{d}$	Fringe width (bright = dark)
$\theta_0 = \frac{\beta}{D} = \frac{\lambda}{d}$	Angular fringe width	$\beta_{\text{medium}} = \frac{\beta}{n}$	Whole set-up inside a medium of index n

central bright fringe (O)



Board numerical (1½ marks)

In a YDSE, $d = 0.5 \text{ mm}$, $D = 1 \text{ m}$ and $\lambda = 600 \text{ nm}$. Find (i) the fringe width, (ii) the distance of the 3rd dark fringe from the centre, (iii) the fringe width if the whole apparatus is immersed in water ($n = 4/3$).

Solution: (i) $\beta = \frac{\lambda D}{d} = \frac{600 \times 10^{-9} \times 1}{0.5 \times 10^{-3}} = 1.2 \text{ mm} \cdot (\frac{1}{2})$

(ii) $y'_3 = \frac{(2 \times 3 - 1)}{2} \beta = 2.5 \times 1.2 = 3.0 \text{ mm} \cdot (\frac{1}{2})$

(iii) $\beta' = \frac{\beta}{n} = 1.2 \times \frac{3}{4} = 0.9 \text{ mm} \cdot (\frac{1}{2})$

! Common Trap

- β depends on λ, D and d : $\beta \propto \lambda D / d$. Bringing the slits **closer** (smaller d) **widens** the fringes.
- With **white light** the central fringe is white; the coloured fringes on either side have violet nearer and red farther from the centre ($\beta_{red} > \beta_{violet}$).

🎯 VS Sir's Exam Tip

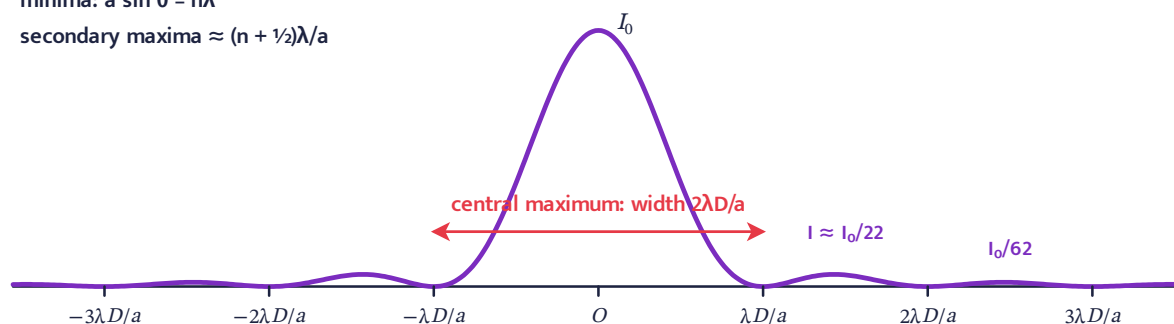
- Convert everything to metres first: $\text{nm} = 10^{-9}$ m, $\text{mm} = 10^{-3}$ m, $\text{\AA} = 10^{-10}$ m.
- Number of fringes in a given width $\propto 1 / \lambda$.

10.4 Diffraction of Light

Diffraction is the bending of light around the edges of an obstacle or slit whose size is comparable to the wavelength, so that light enters the geometrical shadow. For a **single slit** of width a lit by a plane wave, the pattern on a distant screen has a wide, bright **central maximum** with weak **secondary maxima** on both sides.

minima: $a \sin \theta = n\lambda$

secondary maxima $\approx (n + \frac{1}{2})\lambda/a$



Single-slit intensity pattern: central maximum twice as wide as the others

📐 Diffraction Formulas

$a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2, \dots$) **Minima** (dark)

$a \sin \theta \approx (n + \frac{1}{2})\lambda$ Secondary maxima (weak)

$$2\theta = \frac{2\lambda}{a}$$

Angular width of central maximum

$$W = \frac{2\lambda D}{a}$$

Linear width of central maximum on screen at D

$$Z_F = \frac{a^2}{\lambda}$$

Fresnel distance: ray optics is valid for distances $\ll Z_F$

🔧 Board numerical (1½ marks)

Light of wavelength 500 nm falls normally on a slit of width 0.1 mm. Find the width of the central maximum on a screen 1 m away.

Solution: $W = \frac{2\lambda D}{a} = \frac{2 \times 500 \times 10^{-9} \times 1}{10^{-4}} (1) = 10^{-2} \text{ m} = 10 \text{ mm}. (1/2)$

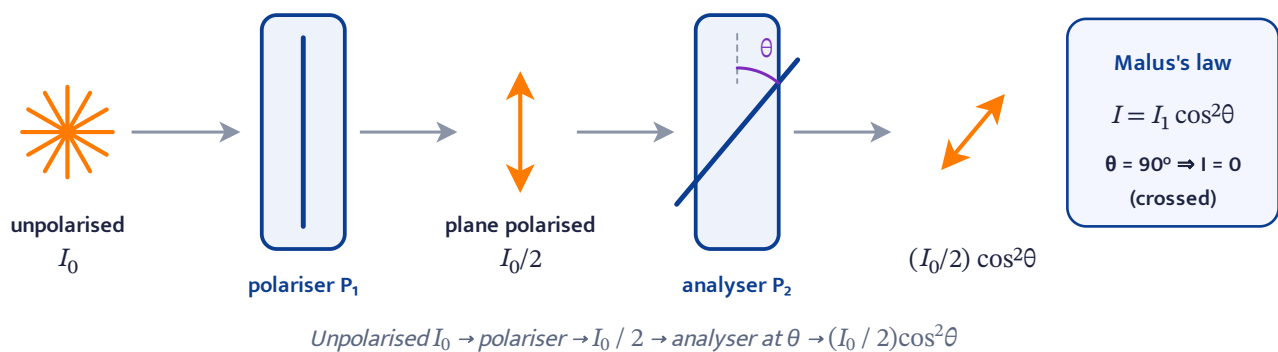
Feature	Interference (YDSE)	Diffraction (single slit)
Cause	two coherent sources	different parts of one wavefront
Fringe width	all equal	central maximum twice the others
Intensity	all bright fringes equal; dark = zero	falls rapidly with order

! Common Trap

- Narrower slit $\Rightarrow$ **wider** central maximum ($W \propto 1/a$); red light gives a wider pattern than blue.
- $a \sin \theta = n\lambda$ gives **minima** in single-slit diffraction but $d \sin \theta = n\lambda$ gives **maxima** in YDSE – do not mix them up.

10.5 Polarisation

Light is a **transverse** EM wave: the electric vector $\vec{E}$ oscillates perpendicular to the direction of travel. In **unpolarised** light (Sun, bulb) $\vec{E}$ vibrates in all directions of the transverse plane. If the vibrations are confined to **one direction** the light is **plane polarised**. **Polarisation** is the process of restricting the vibrations to one plane; it proves that light waves are **transverse** (longitudinal sound waves cannot be polarised). A **polaroid** transmits only the component of $\vec{E}$ along its **pass axis**.



! Polarisation laws

$$I = \frac{I_0}{2}$$

Unpolarised light through one polaroid (any orientation)

$$I = I_1 \cos^2 \theta$$

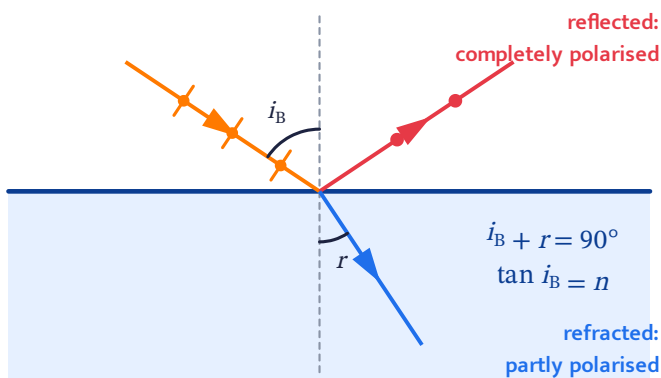
Malus's law: polarised light I_1 , analyser at angle θ

$$\tan i_B = n$$

Brewster's law: reflected light completely polarised at i_B

$$i_B + r = 90^\circ$$

At i_B reflected ray $\perp$ refracted ray



Polarisation by reflection at Brewster's angle

! Uses of polaroids

- Sunglasses cut glare from roads and water.
- Camera filters; windows of trains and aircraft (control of light).
- 3-D movies; LCD displays.

🔧 Board numerical (1½ marks)

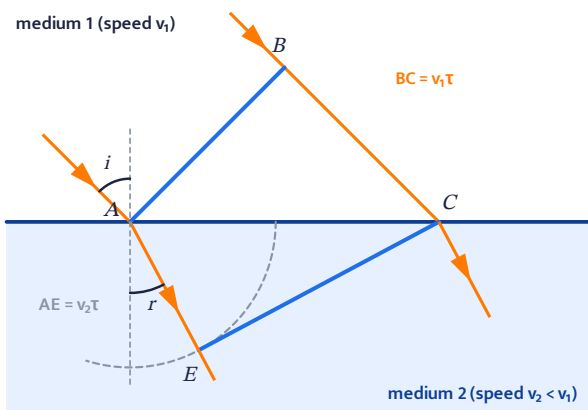
Unpolarised light of intensity I_0 passes through two polaroids whose pass axes make 30° . Find the final intensity. What is it when the axes are crossed?

Solution: After the first: $I_1 = \frac{I_0}{2}$. (½) After the second: $I = I_1 \cos^2 30^\circ = \frac{I_0}{2} \times \frac{3}{4} = \frac{3I_0}{8}$. (1) Crossed ($\theta = 90^\circ$): $I = 0$.

🎯 VS Sir's Exam Tip

- Rotating an analyser through 360° in plane polarised light gives **two maxima and two zeros**; in unpolarised light the intensity does **not** change.
- For full marks always write the reason: "polaroid passes only the component of $\vec{E}$ parallel to its pass axis".

Important derivations ★



Huygens' construction for refraction

🔧 Derivation ★ [3 marks]

[RBSE 2024, 2025, 2026] Using Huygens' principle, derive the laws of refraction (Snell's law).

Solution: 1. A plane wavefront AB is incident on the plane surface separating medium 1 (speed v_1) and medium 2 (speed $v_2 < v_1$), making angle i with it. (½)

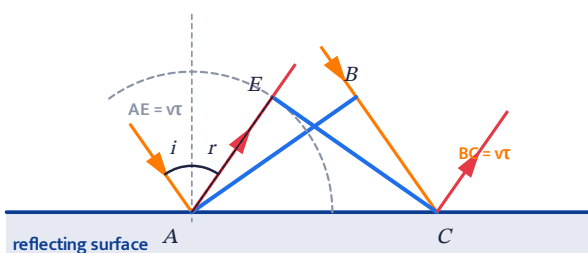
2. While B travels $BC = v_1\tau$ to reach C, the secondary wavelet from A spreads in medium 2 up to $AE = v_2\tau$. The tangent CE from C to this wavelet is the **refracted wavefront**. (1)

3. In right-angled $\triangle ABC$: $\sin i = \frac{BC}{AC} = \frac{v_1\tau}{AC}$. In $\triangle AEC$:

$$\sin r = \frac{AE}{AC} = \frac{v_2\tau}{AC} \quad (½)$$

4. Dividing: $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n_{21} = \text{constant}$ (**Snell's law**).

Incident ray, refracted ray and normal lie in the same plane. (1)



Huygens' construction for reflection

🔧 Derivation ★ [2 marks]

Using Huygens' principle, prove the law of reflection ($i = r$).

Solution: 1. Plane wavefront AB meets the reflecting surface at A first. While B travels $BC = v\tau$ to the surface, the wavelet from A spreads in the **same** medium to $AE = v\tau$; hence $AE = BC$. CE (tangent from C) is the reflected wavefront. (1)

2. Right-angled triangles ABC and AEC have a common hypotenuse AC and $BC = AE \Rightarrow$ they are congruent $\Rightarrow \angle BAC = \angle ECA$, i.e. $i = r$. (½)

3. Incident ray, reflected ray and normal lie in one plane. (½)

Derivation ★ [3 marks]

[RBSE 2023] Describe Young's double-slit experiment and derive the expression for fringe width $\beta = \lambda D / d$ (figure in 10.3).

Solution: 1. Slits S_1, S_2 (separation d), lit by the same source, are coherent. Screen at distance $D \gg d$. O is equidistant from both, P is at height y from O, $\angle POO' = \theta$. (½)

2. Draw $S_1N \perp S_2P$. Since $D \gg d$, $S_1P \approx NP$, so path difference $\Delta x = S_2P - S_1P \approx S_2N = d \sin \theta$. For small θ , $\sin \theta \approx \tan \theta = \frac{y}{D} \Rightarrow \Delta x = \frac{yd}{D}$. (1)

3. Bright fringe: $\Delta x = n\lambda \Rightarrow y_n = \frac{n\lambda D}{d}$. Dark fringe: $\Delta x = (2n-1)\frac{\lambda}{2} \Rightarrow y'_n = \frac{(2n-1)\lambda D}{2d}$. (1)

4. Fringe width = distance between two consecutive bright (or dark) fringes: $\beta = y_{n+1} - y_n \Rightarrow \beta = \frac{\lambda D}{d}$. All fringes have equal width. (½)

Derivation ★ [3 marks]

Show that the resultant intensity of two coherent waves of equal intensity I_0 is $I = 4I_0 \cos^2 \frac{\phi}{2}$ (graph in 10.3).

Solution: 1. Let the waves at P be $y_1 = a \cos \omega t$ and $y_2 = a \cos(\omega t + \phi)$, ϕ = phase difference. (½)

2. Superposition: $y = y_1 + y_2 = 2a \cos \frac{\phi}{2} \cos \left(\omega t + \frac{\phi}{2} \right)$, so the resultant amplitude is $A = 2a \cos \frac{\phi}{2}$. (1)

3. Intensity $\propto (\text{amplitude})^2$ and $I_0 \propto a^2 \Rightarrow I = 4I_0 \cos^2 \frac{\phi}{2}$. (½)

4. $\phi = 2n\pi$ ($\Delta x = n\lambda$): $I_{\max} = 4I_0$; $\phi = (2n-1)\pi$: $I_{\min} = 0$. (1)

Derivation ★ [2 marks]

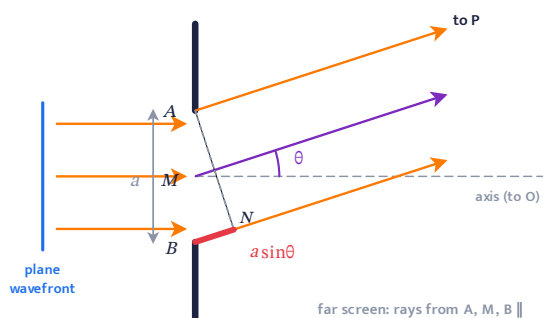
Derive Malus's law (figure in 10.5).

Solution: 1. Polariser P_1 gives plane polarised light of amplitude E_0 along its pass axis; intensity $I_1 \propto E_0^2$. The analyser P_2 has its axis at angle θ to that of P_1 . (½)

2. E_0 is resolved: component $E_0 \cos \theta$ along the axis of P_2 (transmitted), $E_0 \sin \theta$ perpendicular to it (blocked). (½)

3. Intensity $\propto (\text{amplitude})^2 \Rightarrow I \propto E_0^2 \cos^2 \theta \Rightarrow I = I_1 \cos^2 \theta$. (½)

4. For unpolarised light the average of $\cos^2 \theta$ over all orientations is $\frac{1}{2}$, so one polaroid gives $I = \frac{I_0}{2}$. (½)



Single slit: path difference between the edge wavelets is $a \sin \theta$

Derivation ★ [3 marks]

Explain diffraction at a single slit and obtain the condition for minima; find the width of the central maximum.

Solution: 1. Every point of the slit AB (width a) acts as a source of wavelets. At the centre ($\theta = 0$) all wavelets arrive in phase $\Rightarrow$ **central maximum**. (½)

2. At angle θ the path difference between wavelets from the edges A and B is $BN = a \sin \theta$. (½)

3. Divide the slit into two halves AM and MB. Corresponding points are $\frac{a}{2}$ apart, with path difference $\frac{a}{2} \sin \theta$. If $a \sin \theta = \lambda$ this is $\frac{\lambda}{2}$: every wavelet of AM cancels one of MB $\Rightarrow$ **first minimum**. Dividing into $2n$ parts gives $a \sin \theta = n\lambda$. (1)

4. Central maximum lies between the first minima $\theta = \pm \frac{\lambda}{a}$: angular width $\frac{2\lambda}{a}$, linear width $W = \frac{2\lambda D}{a}$. (1)

 Derivation ★ [2 marks]

State and prove Brewster's law (figure in 10.5).

Solution: Law: when unpolarised light is incident on a transparent surface at the polarising (Brewster) angle i_B , the reflected light is **completely plane polarised**, and $\tan i_B = n$. (½)

Proof: At i_B the reflected and refracted rays are mutually perpendicular: $i_B + r = 90^\circ \Rightarrow r = 90^\circ - i_B$. (½)

Snell's law: $n = \frac{\sin i_B}{\sin r} = \frac{\sin i_B}{\sin(90^\circ - i_B)} = \frac{\sin i_B}{\cos i_B} \Rightarrow n = \tan i_B$. (1)

 How to write for full marks

- Huygens' construction: draw AB, BC, AE, CE and angles i, r **with labels** – the figure carries about 1 mark. Write " $BC = v_1 \tau, AE = v_2 \tau$ " explicitly.
- Definitions (interference, polarisation, coherent sources) are 1-mark answers: use the bold keywords from the notes.

Board question bank **Section A – Multiple choice (½ mark each)**

Questions tagged [RBSE year] are from official BSER papers (reworded). Answers at the end of the chapter.

- [RBSE 2025] The phenomenon that shows the transverse nature of light waves is
(a) interference (b) refraction (c) diffraction (d) polarisation
- [RBSE 2024] Light coming directly from the Sun is
(a) circularly polarised (b) plane polarised
(c) unpolarised (d) completely polarised
- [RBSE 2023] If the phase difference between two waves is ϕ , the path difference Δx is
(a) $\frac{\lambda}{2\pi} \phi$ (b) $\frac{\pi}{\lambda} \phi$ (c) $\frac{2\pi}{\lambda} \phi$ (d) $\lambda \phi$
- In YDSE, $d = 1$ mm, $D = 2$ m and $\lambda = 500$ nm. The fringe width is
(a) 1 mm (b) 2 mm (c) 0.25 mm (d) 0.5 mm
- The wavefront of light from a very distant source such as the Sun is
(a) elliptical (b) cylindrical (c) spherical (d) plane
- The polarising (Brewster) angle for a glass of refractive index $\sqrt{3}$ is
(a) 30° (b) 45° (c) 90° (d) 60°
- Plane polarised light of intensity I_0 falls on a polaroid whose axis makes 60° with the plane of vibration. The transmitted intensity is
(a) $\frac{I_0}{4}$ (b) $\frac{3I_0}{4}$ (c) $\frac{I_0}{2}$ (d) $\frac{I_0}{8}$
- Two coherent sources have intensities in the ratio 4 : 1. The ratio $I_{\max} : I_{\min}$ is
(a) 4 : 1 (b) 5 : 3 (c) 9 : 1 (d) 3 : 1

- 9 Light of wavelength 600 nm falls on a slit of width 0.2 mm. The width of the central maximum on a screen 2 m away is
 (a) 6 mm (b) 12 mm (c) 24 mm (d) 1.2 mm
- 10 In YDSE the distance between the slits is halved and the screen distance is doubled. The fringe width becomes
 (a) unchanged (b) 4 times (c) 2 times (d) 1/4 times

Fill in the blanks (½ mark)

Question	Answer
[RBSE 2025] When unpolarised light passes through a polaroid, the intensity becomes _____ of the incident intensity.	half ($I_0 / 2$)
The locus of points vibrating in the same phase is called a _____.	wavefront
A ray of light is _____ to the wavefront.	perpendicular
In YDSE, a bright fringe is formed where the path difference is _____.	$n\lambda$ (integral multiple of λ)
At Brewster's angle the reflected and refracted rays are _____ to each other.	perpendicular (90°)
Sound waves cannot be polarised because they are _____.	longitudinal

Very short answer (1 mark)

Question	Answer
[RBSE 2024] What are coherent sources?	Sources emitting waves of the same frequency with a constant phase difference.
[RBSE 2023] Write the expression for the Fresnel distance.	$z_F = \frac{a^2}{\lambda}$
[RBSE 2020] What is a wavefront?	Locus of all points vibrating in the same phase.
State Brewster's law.	$\tan i_B = n$
How does the fringe width change if red light is replaced by blue light in YDSE?	Decreases ($\beta \propto \lambda, \lambda_{blue} < \lambda_{red}$).

Short answer – Section B (1½ marks)

Short answer (1½ marks)

[RBSE 2024] Define (i) interference and (ii) polarisation of light.

Solution: (i) **Interference:** the **redistribution of light energy** in space, giving alternate bright and dark regions, when two **coherent** waves superpose. (¾)

(ii) **Polarisation:** the process of **restricting the vibrations** of the electric vector of light to **a single plane** perpendicular to the direction of propagation; it shows that light is a **transverse** wave. (¾)

 Short answer (1½ marks)

[RBSE 2020] Write any three differences between interference and diffraction.

Solution: (i) Interference: superposition of waves from **two coherent sources**; diffraction: superposition of wavelets from **different parts of the same wavefront**. (½)

(ii) Interference: **all fringes of equal width**; diffraction: central maximum is **twice** as wide as the others. (½)

(iii) Interference: bright fringes **equally intense**, dark fringes perfectly dark; diffraction: intensity **falls rapidly** with order, minima not perfectly dark. (½)

 Short answer (1½ marks)

In YDSE the slits are 0.2 mm apart and the screen is 1 m away. Light of wavelength 600 nm is used. Find the fringe width and the distance of the 3rd bright fringe from the centre.

Solution: $\beta = \frac{\lambda D}{d} = \frac{6 \times 10^{-7} \times 1}{2 \times 10^{-4}} = 3 \text{ mm}$ (1); $y_3 = 3\beta = 9 \text{ mm}$. (½)

 Answer Key & Quick Solutions

1(d) 2(c) 3(a) 4(a) 5(d) 6(d) 7(a) 8(c) 9(b) 10(b)

1. (d) Only transverse waves can be polarised.

2. (c) Its electric vector vibrates in all directions of the transverse plane.

3. (a) $\phi = \frac{2\pi}{\lambda} \Delta x \Rightarrow \Delta x = \frac{\lambda \phi}{2\pi}$.

4. (a) $\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 2}{10^{-3}} = 10^{-3} \text{ m} = 1 \text{ mm}$.

5. (d) A small part of a very large sphere is practically a plane.

6. (d) $\tan i_B = \sqrt{3} \Rightarrow i_B = 60^\circ$.

7. (a) $I = I_0 \cos^2 60^\circ = \frac{I_0}{4}$.

8. (c) Amplitude ratio 2 : 1 $\Rightarrow \frac{(2+1)^2}{(2-1)^2} = 9$.

9. (b) $W = \frac{2\lambda D}{a} = \frac{2 \times 6 \times 10^{-7} \times 2}{2 \times 10^{-4}} = 1.2 \times 10^{-2} \text{ m} = 12 \text{ mm}$.

10. (b) $\beta \propto D/d \Rightarrow \beta' = \frac{2}{1/2} \beta = 4\beta$.

**Last-minute revision**

- Wavefront: same phase; ray $\perp$ wavefront. Huygens: secondary wavelets, forward envelope = new wavefront.
- Refraction: $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = n_{21}$; reflection $i = r$; frequency never changes.
- Coherent: same frequency, constant phase difference. $\Delta x = \frac{\lambda}{2\pi} \phi$.
- Bright: $\Delta x = n\lambda$; dark: $(2n-1)\frac{\lambda}{2}$; $I = 4I_0 \cos^2 \frac{\phi}{2}$; $\frac{I_{\max}}{I_{\min}} = \left(\frac{r+1}{r-1}\right)^2$.
- YDSE: $\beta = \frac{\lambda D}{d}$ (all equal); in a medium β/n ; $y_n = \frac{n\lambda D}{d}$.
- Single slit: minima $a \sin \theta = n\lambda$; central width $\frac{2\lambda D}{a}$; Fresnel distance $\frac{a^2}{\lambda}$.
- Polarisation $\Rightarrow$ transverse waves. Unpolarised $\rightarrow$ polaroid: $I_0/2$; Malus $I = I_1 \cos^2 \theta$; Brewster $\tan i_B = n$, $i_B + r = 90^\circ$.

CHAPTER

11

Dual Nature of Radiation and Matter

17

Day 18

Board marks: 4

PYQ frequency ★★☆☆

RBSE Class 12

Work function

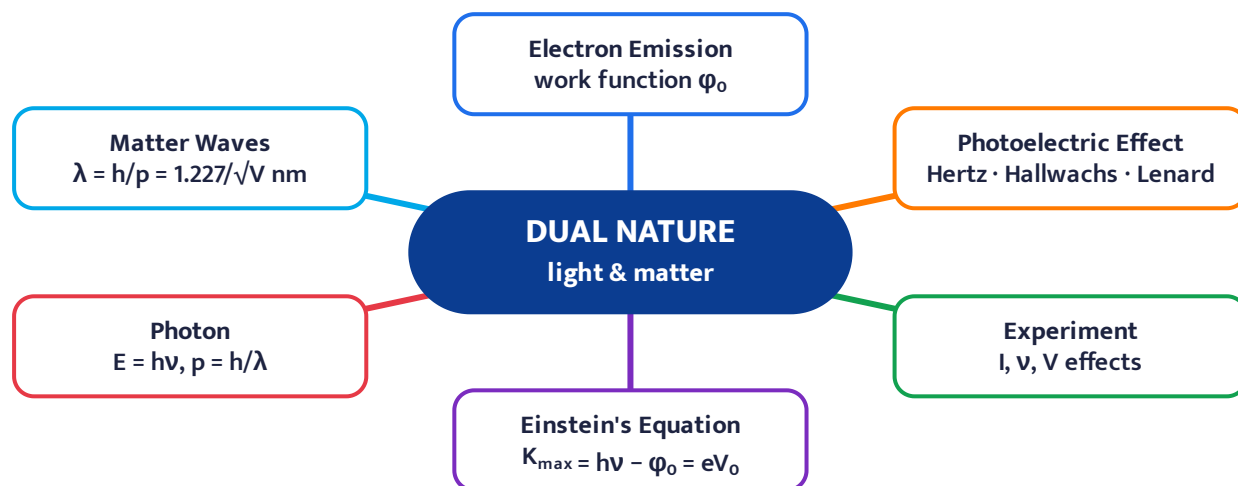
Photoelectric effect

Stopping potential

Einstein's equation

Photon

de Broglie waves



Chapter at a glance

Board snapshot

- **Marks: 4 of 56** (1–2 objective items, 1–2 Section B numericals, sometimes a 3-mark Section C). **Photoelectric effect, Einstein's equation and the V_0 - ν graph are the favourites.**
- **2025:** MCQ photon 18 eV, work function 8 eV → stopping potential 10 V; fill-in Lenard used ultraviolet light; C (3 marks): photoelectric effect, set-up and current–potential graph OR Einstein's equation + V_0 - ν graph for two metals.
- **2024:** MCQ $K_{max} = 1.8$ eV → $V_0 = 1.8$ V; MCQ photon momentum $p = h / \lambda$; fill-in $\lambda = 1.227 / \sqrt{V}$ nm; VSA threshold frequency; B: energy per photon of a 20 W source ($E = P / n$).
- **2023:** MCQ de Broglie wavelength; B: work function, stopping potential; B: threshold frequency of caesium ($\phi_0 = 2.14$ eV → 5.16×10^{14} Hz). **2026 (CWSN):** limitations of wave theory; de Broglie λ . **2020:** aim of Davisson–Germer; why wave theory fails.
- **Most likely in 2027:** set-up + I–V graph (3 marks); Einstein's equation + V_0 - ν graph; limitations of wave theory; photon-number or K_{max} numerical; $\lambda = h / p$.

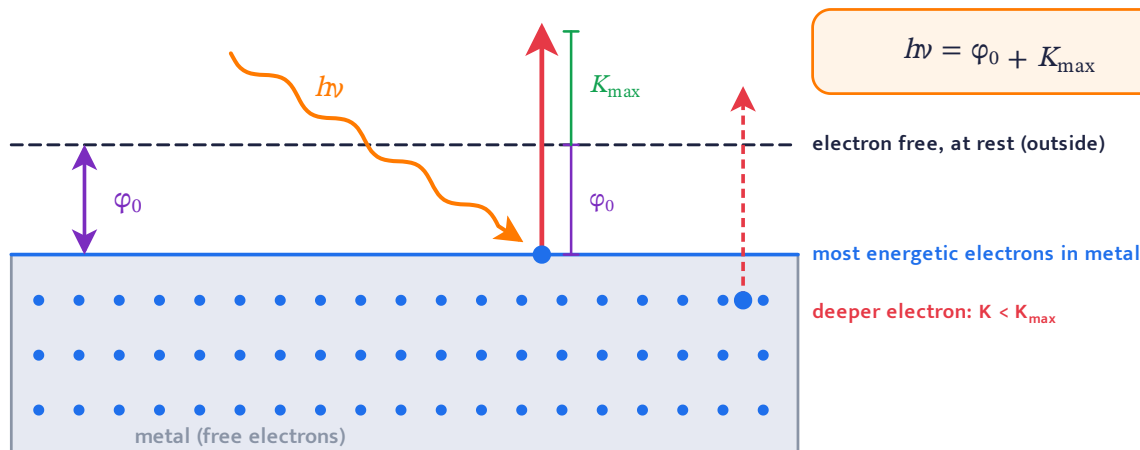
11.1 Electron Emission and Work Function

Free electrons cannot normally leave a metal – the positive ions pull them back. The **minimum energy** needed to remove an electron from the metal surface is the **work function** ϕ_0 . Unit: **electron volt** ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$). It depends on the **metal and the nature of its surface**.

Ways of supplying ϕ_0

- **Thermionic emission** – by heating the metal.
- **Field emission** – a very strong electric field ($\sim 10^8$ V/m).
- **Photoelectric emission** – light of suitable frequency.

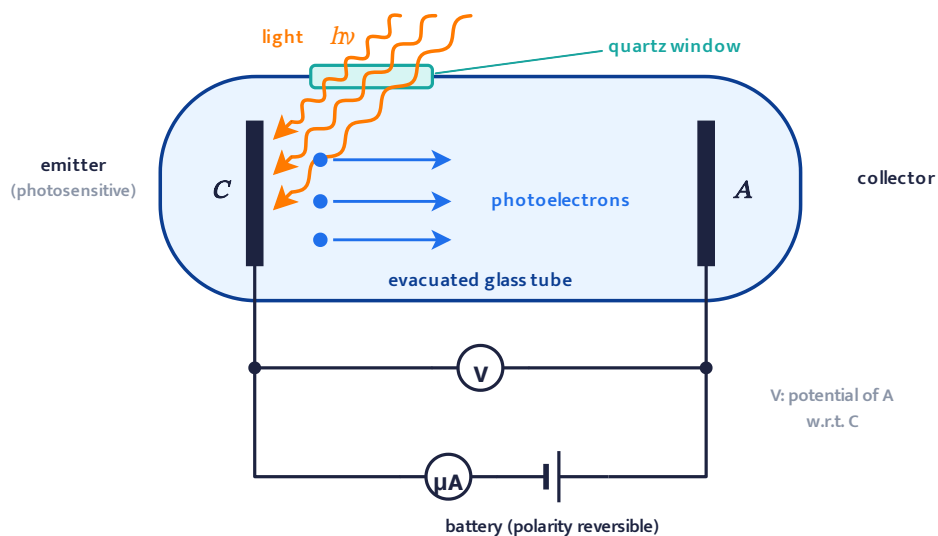
Metal	ϕ_0 (eV)	Metal	ϕ_0 (eV)
Cs	2.14	Al	4.28
K	2.30	Cu	4.65
Na	2.75	Ag	4.70
Ca	3.20	Ni	5.15
Mo	4.17	Pt	5.65



Photon energy first pays the work function; the rest becomes kinetic energy

11.2 Photoelectric Effect

Hertz (1887) observed that ultraviolet light on the emitter plate helps a spark to jump more easily. **Hallwachs and Lenard (1886–1902)** studied the effect in an evacuated tube using **ultraviolet light** and showed that negatively charged particles – **electrons** – are emitted. The emission of electrons from a metal surface when light of suitable frequency falls on it is the **photoelectric effect**; the electrons are **photoelectrons** and the current they produce is the **photocurrent**.



Set-up: light through the quartz window ejects electrons from C; the potential of A is varied; photocurrent is read on the microammeter

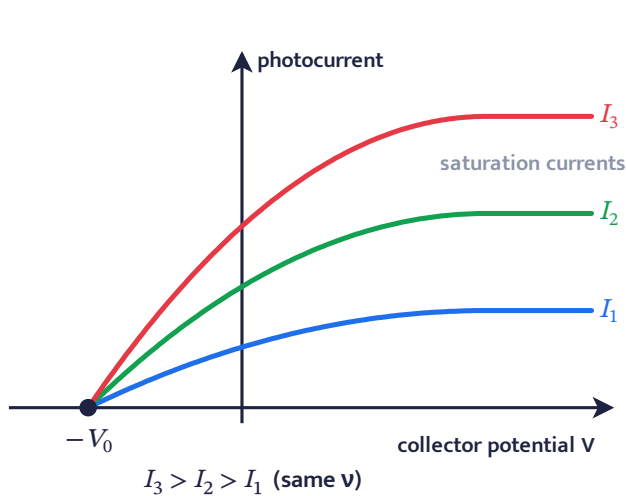
Note

The window is **quartz** because ordinary glass absorbs ultraviolet light. Each metal has a **threshold frequency** ν_0 : below it there is **no emission**, however intense the light is.

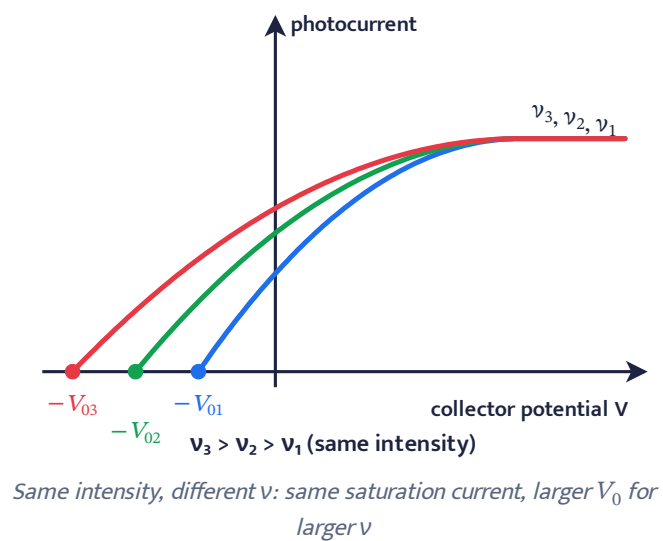
11.3 Experimental Study – Intensity, Potential, Frequency

(a) Intensity: at fixed ν and V , the photocurrent is **proportional to intensity**. **(b) Potential:** as the collector potential rises, the current increases and becomes constant – the **saturation current**. Making A negative repels electrons; at $-V_0$ the current just becomes zero. V_0 is the **stopping potential**: the minimum negative potential of the collector that stops the fastest photoelectron. **(c) Frequency:** V_0 (and K_{max}) **increases linearly with ν** and is **independent of intensity**.

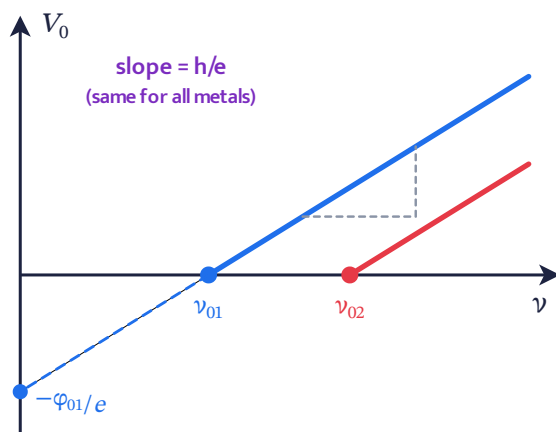
$$eV_0 = K_{max} = \frac{1}{2}mv_{max}^2$$



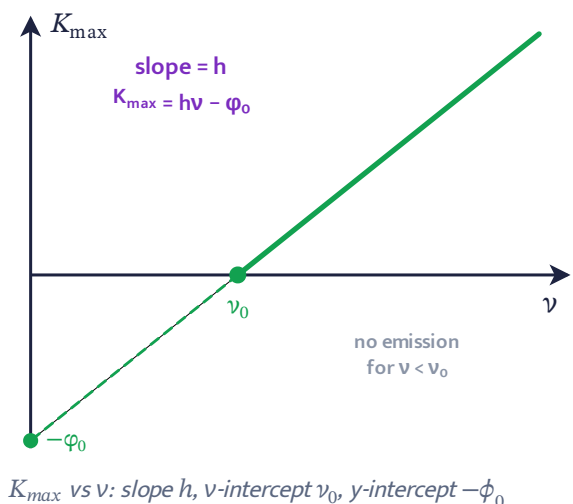
Same ν , different intensities: same V_0 , saturation current $\propto$ intensity



Same intensity, different ν : same saturation current, larger V_0 for larger ν



V_0 vs ν for two metals: parallel lines, slope h/e



K_{max} vs ν : slope h , ν -intercept ν_0 , y -intercept $-\phi_0$

Laws of photoelectric emission

- For $\nu > \nu_0$, photocurrent (number of photoelectrons per second) $\propto$ **intensity**.
- K_{max} and V_0 **do not depend on intensity**; they increase linearly with frequency.
- Below the **threshold frequency** there is no emission, whatever the intensity.
- Emission is **instantaneous** (delay $\leq 10^{-9}$ s) even for very dim light.

! Why the wave theory fails (limitations of wave theory)

- Wave theory: K_{max} should increase with **intensity** (larger amplitude, more energy). Experiment: K_{max} is **independent of intensity**.
- Wave theory: emission at **any frequency** if light is intense enough. Experiment: a sharp **threshold frequency**.
- Wave theory: a **time lag** (minutes to hours for dim light) to collect energy. Experiment: emission is **instantaneous**.

11.4 Einstein's Photoelectric Equation

Einstein (1905): light travels as discrete packets of energy – **quanta (photons)**, each of energy $h\nu$. An electron absorbs **one whole photon** in one interaction. A part ϕ_0 is spent in leaving the metal and the rest becomes kinetic energy, maximum (K_{max}) for electrons at the surface.

Photoelectric formulas

$$K_{max} = h\nu - \phi_0 = eV_0 \quad \text{Einstein's photoelectric equation}$$

$$\phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

Work function, threshold frequency and wavelength

$$V_0 = \frac{h}{e} \nu - \frac{\phi_0}{e} \quad \text{Straight line, slope } h/e \text{ (same for all metals)}$$

$$E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$$

Shortcut ($hc = 1240 \text{ eV nm}$)

🔑 Board numerical (2½ marks)

Light of frequency $6.0 \times 10^{14} \text{ Hz}$ falls on a caesium surface ($\phi_0 = 2.14 \text{ eV}$). Find (i) K_{max} , (ii) the stopping potential, (iii) the threshold frequency. ($h = 6.63 \times 10^{-34} \text{ J s}$)

Solution: $h\nu = \frac{6.63 \times 10^{-34} \times 6.0 \times 10^{14}}{1.6 \times 10^{-19}} = 2.49 \text{ eV}$. (i) $K_{max} = 2.49 - 2.14 = 0.35 \text{ eV}$. (1)

(ii) $V_0 = K_{max} / e = 0.35 \text{ V}$. (½)

(iii) $\nu_0 = \frac{\phi_0}{h} = \frac{2.14 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} = 5.16 \times 10^{14} \text{ Hz}$. (½)

🔑 Solved Example

For a metal the stopping potential is 1.20 V at $8.0 \times 10^{14} \text{ Hz}$ and 2.86 V at $1.2 \times 10^{15} \text{ Hz}$. Find Planck's constant and the work function.

Solution: Slope $\frac{h}{e} = \frac{2.86 - 1.20}{(12 - 8) \times 10^{14}} = 4.15 \times 10^{-15} \text{ V s} \Rightarrow h = 4.15 \times 10^{-15} \times 1.6 \times 10^{-19} = 6.64 \times 10^{-34} \text{ J s}$. (1)

$\phi_0 = h\nu_1 - eV_1 = (4.15 \times 10^{-15} \times 8 \times 10^{14}) - 1.20 = 3.32 - 1.20 \approx 2.12 \text{ eV}$. (1)

! Common Trap

- Doubling **intensity** doubles the current but leaves V_0 and K_{max} **unchanged**.
- Doubling **frequency** does **not** double K_{max} :
 $K_2 = 2h\nu - \phi_0 > 2K_1$.

🧠 Memory Trick

"Intensity → how many; frequency → how fast." Intensity fixes the number of photoelectrons (current); frequency fixes their maximum speed (V_0).

11.5 Particle Nature of Light – the Photon

- In its interaction with matter, radiation behaves as a stream of **photons**, each of energy $E = h\nu = \frac{hc}{\lambda}$ and momentum $p = \frac{h\nu}{c} = \frac{h}{\lambda}$ (from $E = pc$ for a massless particle).
- Photons are **electrically neutral**, have **zero rest mass** and travel at speed c ; they are not deflected by electric or magnetic fields.
- Photon energy depends only on **frequency**. Increasing **intensity** increases the **number** of photons per second, not the energy of each.
- In collisions, **energy and momentum are conserved**, but the number of photons need not be.



Photon formulas

$$n = \frac{P}{h\nu} = \frac{P\lambda}{hc}$$

Photons emitted per second by a source of power P

$$E = \frac{P}{n}$$

Energy of one photon from n photons/s



Board numerical (1½ marks)

A 20 W source emits 5×10^{19} photons per second. Find the energy of one photon (in J and eV) and the wavelength of the light.

Solution: $E = \frac{P}{n} = \frac{20}{5 \times 10^{19}} = 4 \times 10^{-19} \text{ J} = \frac{4 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.5 \text{ eV}. (1)$

$$\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4 \times 10^{-19}} \approx 4.97 \times 10^{-7} \text{ m } (\approx 500 \text{ nm}). (1/2)$$

11.6 Wave Nature of Matter – de Broglie Relation

If radiation shows particle nature, then **moving material particles** should show wave nature. **de Broglie (1924)** proposed that a particle of momentum p has an associated wave – the **matter wave** – of wavelength $\lambda = \frac{h}{p}$. Matter waves are **not** electromagnetic waves. They are observable only when λ is comparable to atomic dimensions – for an electron, yes; for a cricket ball, $\lambda \sim 10^{-34} \text{ m}$ is far too small.



de Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

de Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mK}}$$

In terms of kinetic energy K

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Charge q accelerated from rest through V

$$\lambda_e = \frac{1.227}{\sqrt{V}} \text{ nm}$$

Electron accelerated through V volts



Board numerical (1½ marks)

Find the de Broglie wavelength of (i) an electron accelerated through 100 V, (ii) a 0.12 kg ball moving at 20 m s^{-1} .

Solution: (i) $\lambda = \frac{1.227}{\sqrt{100}} = 0.123 \text{ nm}$ (comparable to atomic spacing). (1)

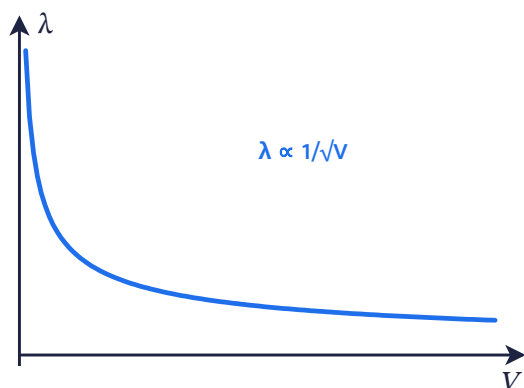
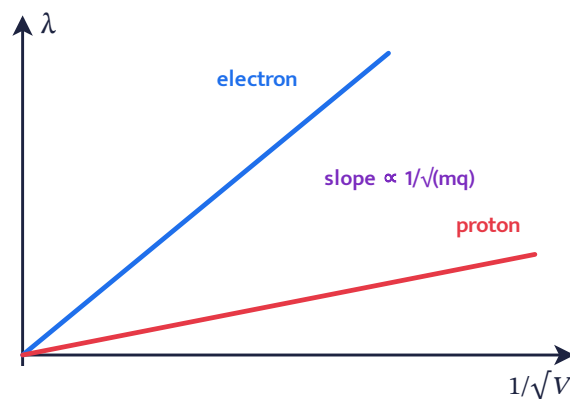
(ii) $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.12 \times 20} = 2.8 \times 10^{-34} \text{ m}$ – undetectable. (1/2)

VS Sir's Exam Tip

- Same V : $\lambda \propto \frac{1}{\sqrt{mq}} \Rightarrow \lambda_p : \lambda_\alpha = 2\sqrt{2} : 1$.
- Same K : $\lambda \propto \frac{1}{\sqrt{m}}$ – the electron has the longest λ .
Same $\lambda \Rightarrow$ same **momentum**.

Common Trap

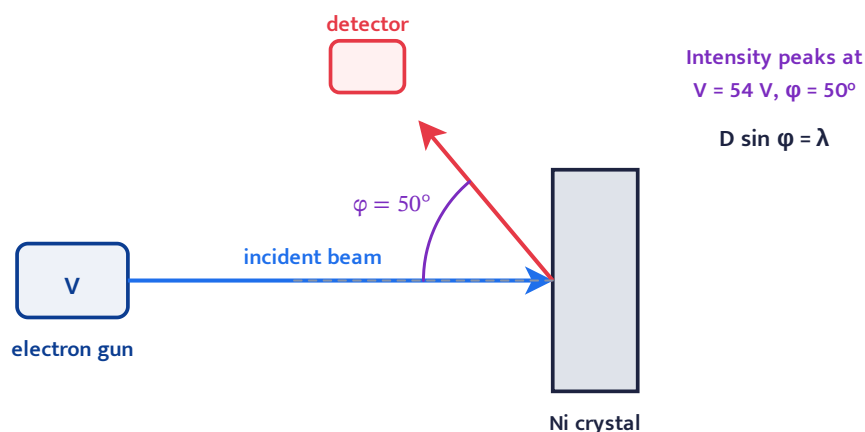
- $\lambda \propto 1 / \sqrt{K}$: four times $K \Rightarrow \lambda$ becomes **half**, not one-fourth.
- For a photon $\lambda = hc / E$; for a particle $\lambda = h / \sqrt{2mK}$ – do not mix.

(a) λ versus V (b) λ versus $1/\sqrt{V}$ – straight lines

$\lambda \propto 1/\sqrt{V}$ for charged particles accelerated through V

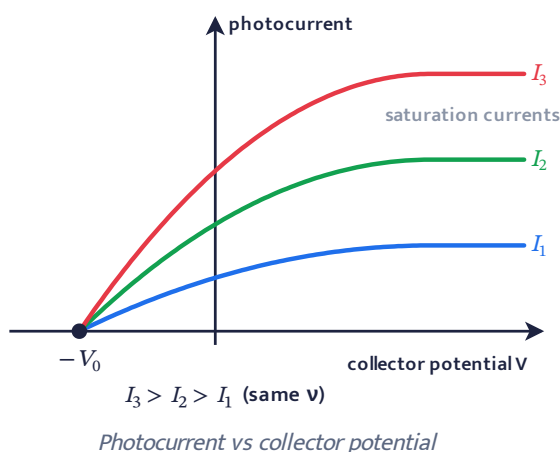
Asked earlier by RBSE

- **Davisson–Germer experiment (2020: "aim")**. It is no longer a main-text topic of the rationalised NCERT book, but RBSE has asked its **aim: to verify the wave nature of electrons** by electron diffraction from a nickel crystal.
- A beam of electrons accelerated through 54 V is scattered from a Ni crystal; the scattered intensity is maximum at $\phi = 50^\circ$. With the surface atomic-row spacing $D = 0.215 \text{ nm}$: $\lambda = D \sin \phi = 0.215 \sin 50^\circ \approx \mathbf{0.165 \text{ nm}}$, in agreement with de Broglie's $\frac{1.227}{\sqrt{54}} = \mathbf{0.167 \text{ nm}}$.



Davisson–Germer arrangement (schematic): electron gun, Ni crystal and movable detector

Important derivations ★



Derivation ★ [3 marks]

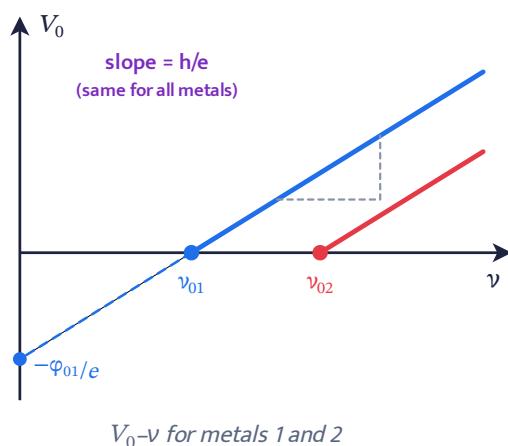
[RBSE 2025] What is the photoelectric effect? Describe the experimental set-up (figure in 11.2) and explain the current-potential graph.

Solution: 1. Photoelectric effect: emission of electrons from a metal surface when light of suitable frequency falls on it. (½)

2. Set-up: an evacuated tube with a quartz window, emitter plate C and collector A, a battery with reversing key, a microammeter and a voltmeter. Light on C ejects photoelectrons; the potential of A is varied. (½)

3. Graph: as A is made more positive the current rises and then becomes constant – the **saturation current** (all photoelectrons reach A); it is proportional to the intensity. (1)

4. For negative potentials the current falls and becomes zero at $-V_0$, the **stopping potential**: $eV_0 = K_{max}$. V_0 depends on frequency, not on intensity. (1)



Derivation ★ [3 marks]

[RBSE 2025] State Einstein's photoelectric equation. Draw the V_0 - ν graph for two metals and state what it shows.

Solution: 1. Postulate: light consists of photons of energy $h\nu$; an electron absorbs **one whole photon**. (½)

2. Energy conservation: $h\nu = \phi_0 + K_{max}$, where $\phi_0 = h\nu_0$. (½)

3. With $K_{max} = eV_0$: $V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$ – a straight line. (1)

4. Graph: the lines are **parallel**, slope = h/e (same for all metals) $\Rightarrow h = e \times \text{slope}$; they cut the ν -axis at the threshold frequencies $\nu_{01} < \nu_{02}$ (metal 2 has the larger work function); no emission for $\nu < \nu_0$. (1)

Derivation ★ [2 marks]

Show that the de Broglie wavelength of an electron accelerated from rest through a potential difference V is

$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm (graph in 11.6).}$$

Solution: 1. de Broglie: $\lambda = \frac{h}{p}$. (½)

2. Work done by the field = kinetic energy gained: $eV = \frac{p^2}{2m} \Rightarrow p = \sqrt{2meV}$. (½)

3. $\lambda = \frac{h}{\sqrt{2meV}}$. (½)

4. Putting $h = 6.63 \times 10^{-34} \text{ J s}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$: $\lambda \approx \frac{1.23 \times 10^{-9}}{\sqrt{V}} \text{ m} \Rightarrow \lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$ (with precise constants). (½)

How to write for full marks

- Draw the set-up / graph with **labelled axes** (V or v on x, current or V_0 on y) – it carries $\frac{1}{2}$ –1 mark.
- In numericals write: formula $\rightarrow$ substitution with units $\rightarrow$ answer. Use $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ and $hc = 1240 \text{ eV nm}$.

Board question bank

Section A – Multiple choice ($\frac{1}{2}$ mark each)

Questions tagged [RBSE year] are from official BSER papers (reworded). Answers at the end of the chapter.

- [RBSE 2025] Photons of energy 18 eV fall on a metal of work function 8 eV. The stopping potential is
(a) 8 V (b) 10 V (c) 18 V (d) 26 V
- [RBSE 2024] The maximum kinetic energy of photoelectrons is 1.8 eV. The stopping potential is
(a) 0.9 V (b) 3.6 V (c) 1.8 V (d) 1.8 kV
- [RBSE 2024] The momentum of a photon of wavelength λ is
(a) $\frac{hc}{\lambda}$ (b) $\frac{\lambda}{h}$ (c) $\frac{h}{\lambda}$ (d) $h\lambda$
- [RBSE 2023, 2026] The de Broglie wavelength of a particle of mass m moving with speed v is
(a) $\frac{h}{mv}$ (b) hmv (c) $\frac{h}{mv^2}$ (d) $\frac{mv}{h}$
- The threshold wavelength of a metal of work function 2.48 eV is
(a) 400 nm (b) 250 nm (c) 620 nm (d) 500 nm
- The intensity of light falling on a photocell is doubled, the frequency remaining the same. Then
(a) both remain unchanged
(b) saturation current unchanged, stopping potential doubles
(c) saturation current doubles, stopping potential unchanged
(d) both double
- The de Broglie wavelength of an electron accelerated through 100 V is about
(a) 12.3 nm (b) 0.0123 nm (c) 1.23 nm (d) 0.123 nm
- The slope of the stopping potential versus frequency graph is
(a) h (b) h/e (c) e/h (d) ϕ_0/e
- A proton and an α -particle are accelerated from rest through the same potential difference. The ratio $\lambda_p:\lambda_\alpha$ is
(a) $2\sqrt{2}:1$ (b) 2:1 (c) $\sqrt{2}:1$ (d) $1:2\sqrt{2}$
- A 100 W source emits light of wavelength 663 nm. The number of photons emitted per second is about
(a) 3.3×10^{20} (b) 6.6×10^{20} (c) 3.3×10^{18} (d) 1.5×10^{20}

Fill in the blanks (½ mark)

Question	Answer
[RBSE 2025] Lenard used _____ light in his photoelectric experiments.	ultraviolet
[RBSE 2024] The de Broglie wavelength of an electron accelerated through V volts is _____/ $\sqrt{V}$ nm.	1.227
Photoelectric emission occurs only if the frequency of light is greater than the _____ frequency.	threshold
The stopping potential depends on the _____ of light but not on its _____.	frequency, intensity
The rest mass of a photon is _____.	zero
1 eV = _____ J.	1.6×10^{-19}

Very short answer (1 mark)

Question	Answer
[RBSE 2024] Define threshold frequency.	Minimum frequency of light below which no photoelectric emission occurs from a metal.
Write Einstein's photoelectric equation.	$K_{max} = h\nu - \phi_0$
Write the de Broglie relation.	$\lambda = \frac{h}{p} = \frac{h}{mv}$
On what does the photocurrent (saturation) depend – intensity or frequency?	Intensity (directly proportional).
An electron and a proton have the same kinetic energy. Which has the longer de Broglie wavelength?	Electron ($\lambda \propto 1/\sqrt{m}$).
[RBSE 2020] Write the aim of the Davisson–Germer experiment.	To verify the wave nature of electrons (electron diffraction by a crystal).

Short answer – Section B (1½ marks)

 Short answer (1½ marks)

[RBSE 2023] Define (i) work function and (ii) stopping potential.

Solution: (i) **Work function (ϕ_0):** the **minimum energy** required to liberate an electron from the surface of a metal; unit eV. (¾)

(ii) **Stopping potential (V_0):** the **minimum negative potential** of the collector relative to the emitter at which the photocurrent just becomes zero; $eV_0 = K_{max}$. (¾)

 Short answer (1½ marks)

[RBSE 2023] The work function of caesium is 2.14 eV. Find its threshold frequency.

Solution: $\nu_0 = \frac{\phi_0}{h} = \frac{2.14 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$ (1) = 5.16×10^{14} Hz. (½)

 Short answer (1½ marks)

[RBSE 2026, 2020] Write three limitations of the wave theory in explaining the photoelectric effect.

Solution: (i) Wave theory predicts K_{max} should increase with **intensity**; experiment shows it is **independent of intensity**. (½)

(ii) It predicts emission at **any frequency** for strong enough light; experiment shows a **threshold frequency**. (½)

(iii) It predicts a **time lag** to accumulate energy; emission is **instantaneous**. (½)

 Short answer (1½ marks)

A 20 W source emits 5×10^{19} photons per second. Find the energy of each photon.

Solution: $E = \frac{P}{n} = \frac{20}{5 \times 10^{19}}$ (1) = 4×10^{-19} J (2.5 eV). (½)

 Short answer (1½ marks)

Find the de Broglie wavelength of an electron accelerated through 150 V.

Solution: $\lambda = \frac{1.227}{\sqrt{V}}$ nm = $\frac{1.227}{\sqrt{150}}$ (1) = 0.100 nm = 1 Å. (½)

 Answer Key & Quick Solutions

1(b) 2(c) 3(c) 4(a) 5(d) 6(c) 7(d) 8(b) 9(a) 10(a)

1. (b) $K_{max} = 18 - 8 = 10$ eV $\Rightarrow V_0 = 10$ V.

2. (c) $eV_0 = K_{max} \Rightarrow V_0 = 1.8$ V.

3. (c) $p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$ ($\frac{hc}{\lambda}$ is the energy).

4. (a) $\lambda = \frac{h}{p} = \frac{h}{mv}$.

5. (d) $\lambda_0 = \frac{1240}{2.48} = 500$ nm.

6. (c) Intensity fixes the number of photons (current); V_0 depends only on ν .

7. (d) $\lambda = \frac{1.227}{\sqrt{100}} = 0.123$ nm.

8. (b) $V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e} \Rightarrow \text{slope} = h/e$.

9. (a) $\lambda \propto \frac{1}{\sqrt{mq}}$; $m_\alpha q_\alpha = 4m \times 2e = 8m_p e \Rightarrow \text{ratio} = \sqrt{8} = 2\sqrt{2}$.

10. (a) $E = \frac{hc}{\lambda} = 3.0 \times 10^{-19}$ J; $n = \frac{100}{3.0 \times 10^{-19}} \approx 3.3 \times 10^{20} \text{ s}^{-1}$.

 Last-minute revision

- Work function ϕ_0 : minimum energy to free an electron (eV); $\phi_0 = h\nu_0 = hc/\lambda_0$.
- Photocurrent (saturation) $\propto$ intensity; V_0 and K_{max} depend on **ν only**; no emission below ν_0 ; emission instantaneous.
- Einstein: $K_{max} = h\nu - \phi_0 = eV_0$; V_0 - ν line: slope h/e , intercept ν_0 .
- Wave theory fails: intensity effect, threshold, time lag.
- Photon: $E = h\nu$, $p = h/\lambda$, rest mass 0, neutral; $n = P/h\nu$; $hc = 1240$ eV nm.
- de Broglie: $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$; electron $\frac{1.227}{\sqrt{V}}$ nm; Davisson–Germer confirmed electron waves.

CHAPTER

12

Atoms

17

Day 19

Board marks: 4

PYQ frequency ★★☆☆

RBSE Class 12

 α -scattering & nuclear model

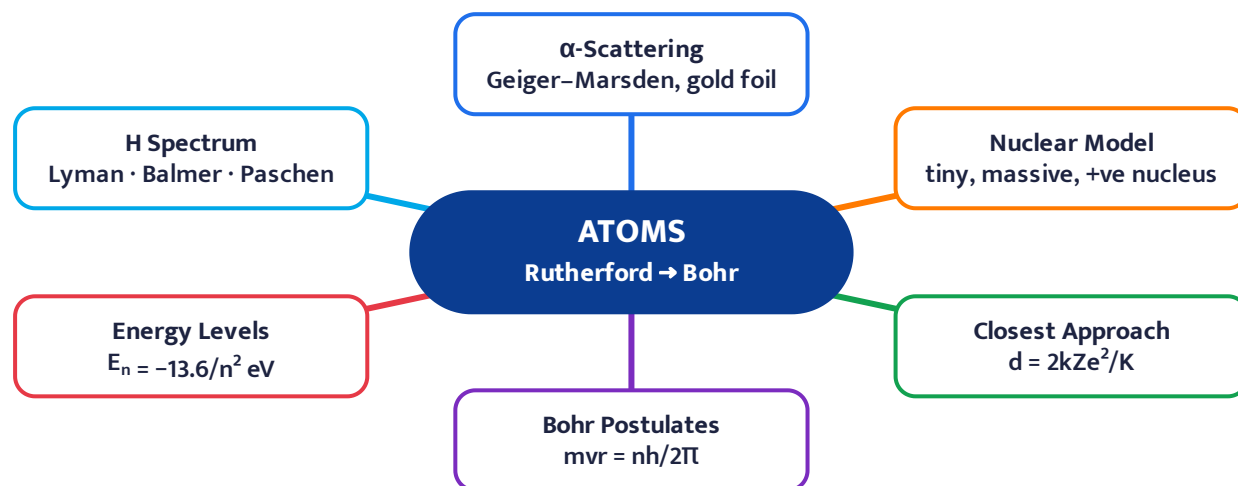
Closest approach

Hydrogen spectrum

Bohr's postulates

Radius & energy of orbits

de Broglie explanation



Chapter at a glance

Board snapshot

- **Weightage: 4 marks** – typically 1–2 MCQ / fill-in questions (½–1 mark) plus one 1½-mark short answer; the Bohr-orbit derivations can be asked for 3 marks.
- **2025:** MCQ – ionisation energy from the ground state (13.6 eV); MCQ – angular momentum is quantised; 1½ marks – radius of the stable orbit and total energy of the electron (Bohr).
- **2024:** MCQ – α -scattering angle $\approx 0^\circ$ at large impact parameter; MCQ – first excitation energy 10.2 eV; fill-in – $r_2 = 4r_1 = 2 \times 10^{-10}$ m; VSA – ionisation energy; 1½ marks – Bohr's second postulate from de Broglie's hypothesis.
- **2023:** for total energy –13.6 eV find KE (+13.6 eV) and PE (–27.2 eV); two drawbacks of Rutherford's model + which Balmer series lies in the visible region. **2026 CWSN:** Bohr quantisation. **2020:** de Broglie explanation, standing wave for $n=3$.
- **Most likely in 2027:** (1) radius and energy of the n th Bohr orbit (asked 2025, and the 2023 KE/PE numerical); (2) Bohr's quantisation from de Broglie waves (2024, 2020, 2026); (3) drawbacks of Rutherford's model / ionisation and excitation energy (10.2 eV, 13.6 eV).

12.1 α -Particle Scattering and Rutherford's Nuclear Model

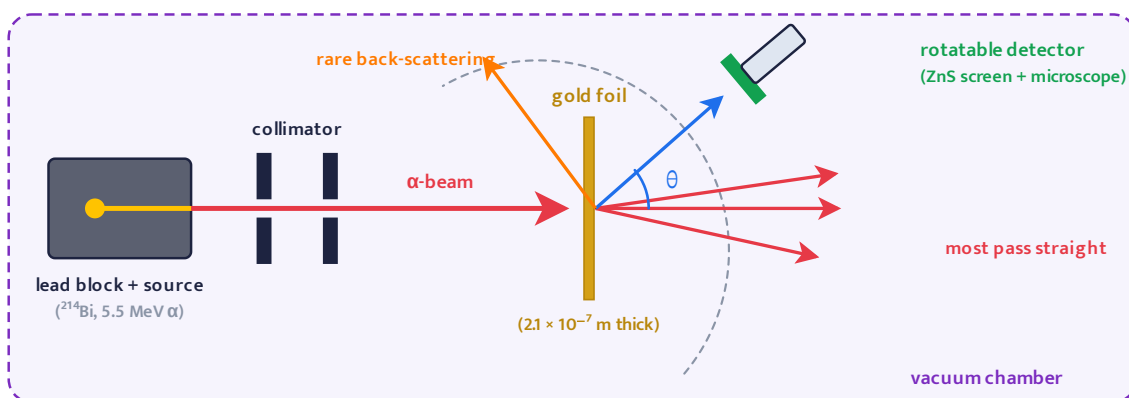
In **Thomson's model** the atom was a sphere of positive charge with electrons embedded in it. To test it, **Geiger and Marsden** (on Rutherford's advice) directed a narrow beam of **5.5 MeV α -particles** at a thin **gold foil** and counted the particles scattered in different directions with a rotating ZnS screen and microscope, all in vacuum.

Observations and conclusions

- **Most** α -particles pass straight through $\Rightarrow$ the atom is mostly **empty space**.
- Only **about 1 in 8000** is deflected by more than 90° , a few rebound ($\theta \approx 180^\circ$) $\Rightarrow$ a strong repulsion from a tiny, heavy, **positively charged** core.

Rutherford's nuclear model (1911)

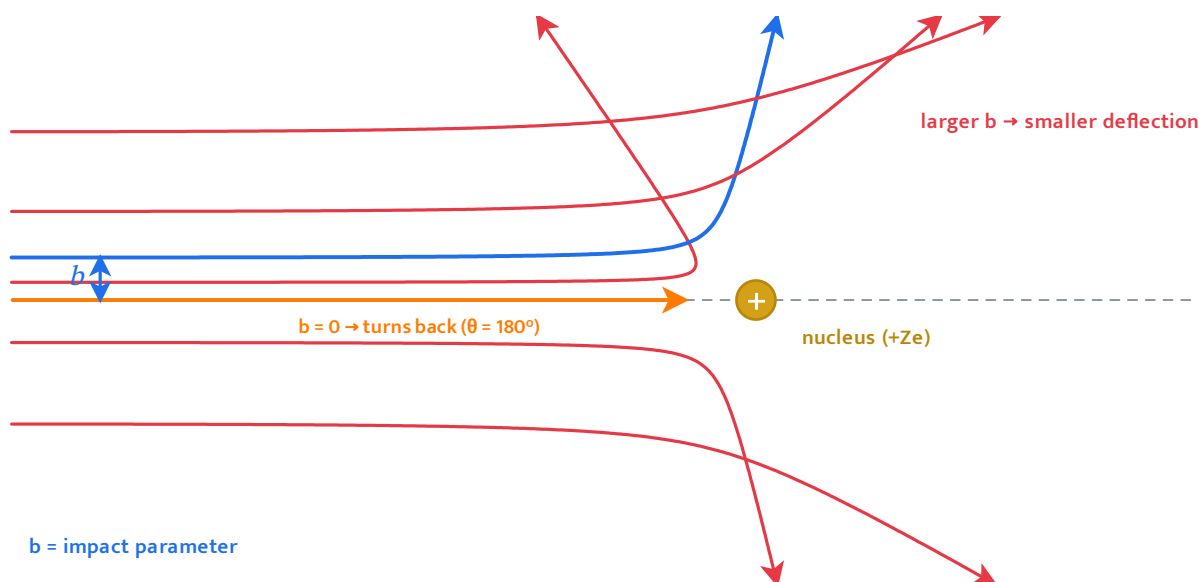
- All the positive charge and nearly all the mass lie in a tiny **nucleus** (size $\sim 10^{-15}$ m); the atom is $\sim 10^{-10}$ m.
- Electrons revolve around the nucleus; the electrostatic force supplies the centripetal force.



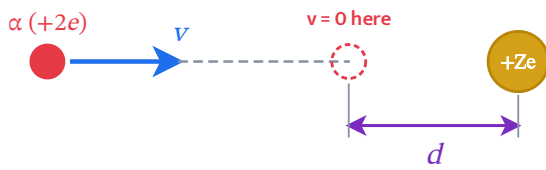
Geiger-Marsden experiment (top view)

12.2 Impact Parameter and Distance of Closest Approach

The **impact parameter** b is the perpendicular distance of the initial velocity line of the α -particle from the centre of the nucleus. Small $b \Rightarrow$ large deflection; $b = 0$ (head-on) $\Rightarrow$ the particle is turned back ($\theta = 180^\circ$); **large** $b \Rightarrow \theta \approx 0^\circ$. Rutherford's calculation gives $b = \frac{d}{2} \cot \frac{\theta}{2}$, with d defined below. The number scattered at angle θ is $N \propto \frac{1}{\sin^4(\theta/2)}$.



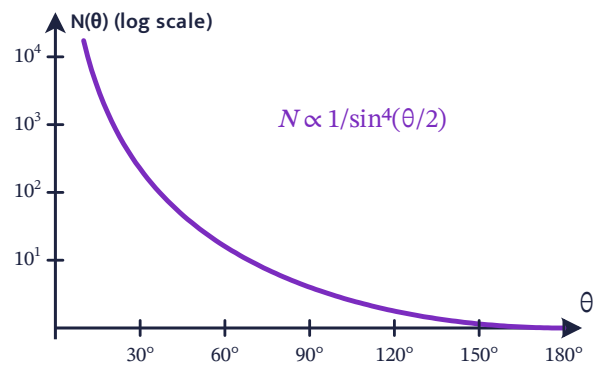
Paths of α -particles for different impact parameters b



$$\frac{1}{2}mv^2 = k(2e)(Ze)/d$$

all KE → electrostatic PE

Head-on approach: all KE becomes PE at distance d



Scattered count falls steeply with θ

Scattering formulas

$$d = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$$

Distance of closest approach;
K = KE of α -particle (derived in 12.8)

$$b = \frac{d}{2} \cot \frac{\theta}{2}$$

Impact parameter and scattering angle

Solved Example

Find the distance of closest approach of a 7.7 MeV α -particle to a gold nucleus ($Z = 79$). Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.

Solution: $d = \frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{7.7 \times 10^6 \times 1.6 \times 10^{-19}} = 2.95 \times 10^{-14} \text{ m}$
($\approx 30 \text{ fm}$).

Common Trap

- $d \propto \frac{1}{K}$ and $d \propto Z$: doubling the KE halves the distance.
- K in MeV: write $K = 7.7 \times 10^6 \times 1.6 \times 10^{-19} \text{ J}$ – do not forget to convert eV to joule.

12.3 Drawbacks of Rutherford's Model

Two drawbacks [asked 2023]

- Instability:** an orbiting electron is accelerated, so by classical electromagnetic theory it should radiate energy continuously and spiral into the nucleus almost instantly. Atoms, however, are stable.
- Line spectrum:** the model predicts a continuous spectrum (the spiralling electron changes frequency continuously), but atoms emit **discrete line spectra**.

VS Sir's Exam Tip

- Write both drawbacks as one line each, then add "Bohr removed them using quantised stable orbits." That earns the full 1½ marks.

12.4 Spectrum of the Hydrogen Atom

Hydrogen gas in a discharge tube emits light of only certain wavelengths – a **line (emission) spectrum**. Balmer found the visible lines fitted a formula; Rydberg generalised it for all series:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad R = 1.097 \times 10^7 \text{ m}^{-1}, \quad n_i > n_f$$

Series	Ends at n_f	Starts From n_i	Region
Lyman	1	2, 3, 4 ...	ultraviolet
Balmer	2	3, 4, 5 ...	visible
Paschen	3	4, 5, 6 ...	infrared
Brackett	4	5, 6, 7 ...	infrared
Pfund	5	6, 7, 8 ...	far infrared



Balmer series (visible): H_α 656.3 nm, H_β 486.1 nm, H_γ 434.1 nm, H_δ 410.2 nm; series limit 364.6 nm

Solved Example

Find the longest wavelength of the Balmer series.

Solution: Longest $\lambda \Rightarrow n_i = 3, n_f = 2$:

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.097 \times 10^7 \times \frac{5}{36} \Rightarrow \lambda = 6.563 \times 10^{-7} \text{ m} = 656.3 \text{ nm (red, } H_\alpha).$$

Solved Example

Find the shortest wavelength of the Lyman series (series limit).

Solution: $n_i \rightarrow \infty, n_f = 1$: $\frac{1}{\lambda} = R \Rightarrow \lambda = \frac{1}{1.097 \times 10^7} = 91.2 \text{ nm (UV)}.$

12.5 Bohr's Model of the Hydrogen Atom

Bohr's three postulates (1913)

- **1. Stable orbits:** an electron revolves only in certain circular **stationary orbits** without radiating energy; the Coulomb force gives the centripetal force.
- **2. Quantisation of angular momentum:** allowed orbits are those in which the angular momentum is an integral multiple of $\frac{h}{2\pi}$: $L = mvr = \frac{nh}{2\pi}, n = 1, 2, 3, \dots$
- **3. Emission / absorption:** a photon is emitted (or absorbed) when the electron jumps between two orbits: $h\nu = E_i - E_f$.

Hydrogen-like results

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} = n^2 a_0, \quad a_0 = 0.529 \text{ \AA}$$

Radius of nth orbit; $r \propto n^2$ (derived in 12.8)

$$v_n = \frac{e^2}{2\epsilon_0 n h} = \frac{2.19 \times 10^6}{n} \text{ m s}^{-1}$$

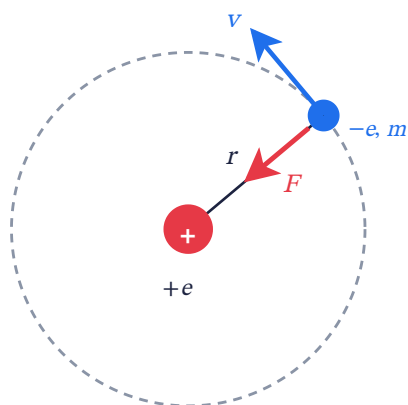
Orbital speed; $v \propto 1/n$

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2} \text{ eV}$$

Total energy; $E \propto 1/n^2$

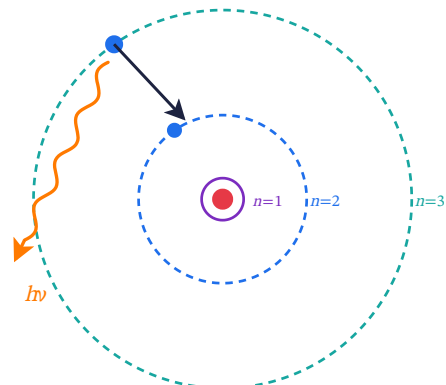
$$\text{KE} = -E, \quad \text{PE} = 2E$$

Relations between KE, PE and total energy



$$F = mv^2/r = e^2 / (4\pi\epsilon_0 r^2)$$

Forces on the electron in the n th orbit



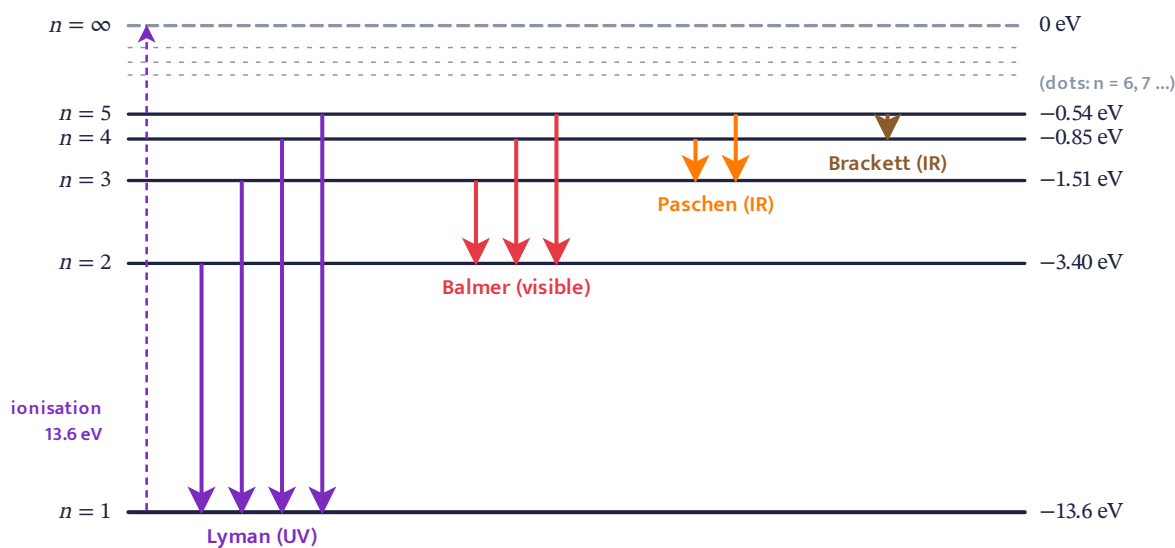
$$r_n \propto n^2; r_1 : r_2 : r_3 = 1 : 4 : 9$$

Allowed orbits $r_n \propto n^2$ and a jump $3 \rightarrow 2$

! Common Trap

- Total energy is **negative** (bound electron). The orbit with larger n has **higher (less negative)** energy:
 $E_3 = -1.51 \text{ eV} > E_1 = -13.6 \text{ eV}$.
- For a hydrogen-like ion of atomic number Z : $r_n = \frac{n^2 a_0}{Z}$ and $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$ (e.g. He^+ ground state -54.4 eV).

12.6 Energy Levels, Ionisation and Excitation



Energy-level diagram of hydrogen showing the spectral series (not to scale)

- Ground state** ($n = 1, -13.6 \text{ eV}$) is the lowest energy state; states with $n > 1$ are **excited states**.
- Ionisation energy**: the minimum energy needed to remove the electron from the ground state to infinity = **13.6 eV** for hydrogen.
- First excitation energy**: energy to take the electron from $n = 1$ to $n = 2$: $13.6 - 3.4 = 10.2 \text{ eV}$.
- With n excited states populated, the number of possible spectral lines is $\frac{n(n-1)}{2}$.

Solved Example

[RBSE 2023] The total energy of an electron in an orbit of hydrogen is -13.6 eV. Find its kinetic and potential energy.

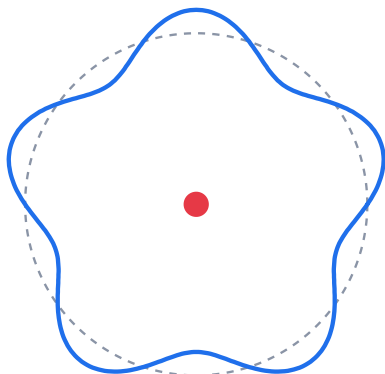
Solution: $KE = -E = +13.6$ eV; $PE = 2E = -27.2$ eV (check: $13.6 - 27.2 = -13.6$ eV).

Solved Example

Find the energy and wavelength of the photon emitted in the transition $n = 3 \rightarrow 2$.

Solution: $E = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.89$ eV; $\lambda = \frac{1240}{1.89} \approx 656$ nm.

12.7 de Broglie's Explanation and Limitations of Bohr's Model



$$2\pi r_n = n\lambda \quad (n = 5 \text{ here})$$

$$\lambda = h/mv \Rightarrow mvr = nh/2\pi$$

Standing wave of an electron on the orbit: $2\pi r = n\lambda$

Treating the electron as a **matter wave** of wavelength $\lambda = \frac{h}{mv}$, an orbit is allowed only when the circumference holds a whole number of wavelengths – a **standing wave** forms. This gives $2\pi r = n\lambda$, i.e. $mvr = \frac{nh}{2\pi}$ – Bohr's second postulate (full derivation 12.8).



Limitations of Bohr's model

- Works only for **H-like one-electron** systems (H, He^+, Li^{2+}); fails for multi-electron atoms such as helium.
- Cannot explain the **relative intensities** of spectral lines or the fine structure.
- Mixes classical and quantum ideas (orbits with definite r and v).

12.8 Important Derivations ★

Derivation ★ [3 marks]

Derivation 1 (Part 1) – [RBSE 2025]. Using Bohr's postulates, obtain the expressions for the radius of the n th stable orbit and the speed of the electron in the hydrogen atom.

Solution: (i) **Diagram:** electron ($-e$, mass m) moves in a circle of radius r with speed v around the nucleus ($+e$) (figure above). (½)

(ii) The Coulomb attraction supplies the centripetal force: $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \Rightarrow v^2 = \frac{e^2}{4\pi\epsilon_0 mr} \dots (1)$ (½)

(iii) Bohr's quantum condition: $mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr} \dots (2)$ (½)

(iv) Squaring (2) and equating with (1): $\frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{e^2}{4\pi\epsilon_0 mr} \dots (1/2)$

(v) Hence $r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} = n^2 a_0$, where $a_0 = \frac{h^2 \epsilon_0}{\pi m e^2} = 0.529 \text{ \AA}$ (Bohr radius). (½)

(vi) Putting r_n in (2): $v_n = \frac{e^2}{2\epsilon_0 nh} \approx \frac{2.19 \times 10^6}{n} \text{ m s}^{-1}$. (½)

 Derivation ★ [3 marks]

Derivation 1 (Part 2) – [RBSE 2025]. Obtain the expression for the total energy of the electron in the n th orbit and show that it is -13.6 eV for $n = 1$.

Solution: (i) Kinetic energy: $K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\epsilon_0 r}$ (using (1) of Part 1). (½)

(ii) Potential energy of the electron–nucleus pair: $U = -\frac{e^2}{4\pi\epsilon_0 r}$. (½)

(iii) Total energy $E = K + U = -\frac{e^2}{8\pi\epsilon_0 r}$ (so $K = -E, U = 2E$). (1)

(iv) Substituting $r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2}$: $E_n = -\frac{m e^4}{8 \epsilon_0^2 h^2} \cdot \frac{1}{n^2}$. (½)

(v) Numerically $\frac{m e^4}{8 \epsilon_0^2 h^2} = 2.18 \times 10^{-18} \text{ J} = 13.6 \text{ eV}$, so $E_n = -\frac{13.6}{n^2} \text{ eV}$; for $n = 1, E_1 = -13.6 \text{ eV}$. (½)

 Derivation ★ [3 marks]

Derivation 2 – [RBSE 2024, 2020, 2026 CWSN]. Using de Broglie's hypothesis, explain Bohr's second postulate of quantisation of angular momentum.

Solution: (i) de Broglie: a moving electron behaves as a wave of wavelength $\lambda = \frac{h}{mv}$. (½)

(ii) In a circular orbit the wave travels round and returns to the starting point. For a **stable orbit** the wave must reinforce itself, forming a **standing wave** (figure 12.7), so the circumference must hold a whole number of wavelengths. (1)

(iii) Condition: $2\pi r = n\lambda$, $n = 1, 2, 3 \dots$ (for $n = 3$, the circumference contains three wavelengths). (½)

(iv) Put $\lambda = \frac{h}{mv}$: $2\pi r = \frac{nh}{mv} \Rightarrow mvr = \frac{nh}{2\pi}$. (½)

(v) This is Bohr's second postulate: angular momentum is quantised in units of $\frac{h}{2\pi}$. If the circumference is not a whole number of wavelengths, the wave destructively interferes with itself and the orbit is not allowed. (½)

 Derivation ★ [3 marks]

Derivation 3. Obtain an expression for the distance of closest approach of an α -particle of kinetic energy K moving head-on towards a nucleus of atomic number Z .

Solution: (i) (figure in 12.2) α -particle charge = $+2e$, nucleus charge = $+Ze$. At infinite distance, energy = K ($PE = 0$). (½)

(ii) The repulsion slows it down; at the distance of closest approach d it stops momentarily, so $KE = 0$ and all energy is potential: $U = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{d}$. (1)

(iii) By conservation of energy: $K = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{d}$. (½)

(iv) $d = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{K}$. Since $d \propto 1/K$, a more energetic α -particle gets closer. (1)

 Derivation ★ [3 marks]

Derivation 4. Using Bohr's third postulate, obtain the Rydberg formula for the hydrogen spectrum and an expression for the Rydberg constant.

Solution: (i) Energy of the n th level: $E_n = -\frac{m e^4}{8 \epsilon_0^2 h^2 n^2}$. (½)

(ii) Electron jumps from n_i to n_f ($n_i > n_f$): photon energy $h\nu = E_{n_i} - E_{n_f} = \frac{m e^4}{8 \epsilon_0^2 h^2} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$. (1)

(iii) With $\nu = \frac{c}{\lambda}$: $\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$. (½)

(iv) $R = \frac{m e^4}{8 \epsilon_0^2 h^3 c} = 1.097 \times 10^7 \text{ m}^{-1}$, in agreement with the experimental value. (1)

VS Sir's Exam Tip

- In Part 1 always write the two starting equations (1) force balance and (2) $mvr = nh / 2\pi$ – the examiner awards a mark for each before you reach r_n .
- Remember the shortcut: $K = -E, U = 2E$. It solves the 2023 type numerical in one line.

Board question bank

Section A – Multiple choice (½ mark each)

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- [RBSE 2025]** The energy required to ionise a hydrogen atom from its ground state is
 (a) 13.6 eV (b) 27.2 eV (c) 3.4 eV (d) 10.2 eV
- [RBSE 2025]** According to Bohr, the angular momentum of the electron in the n th orbit is
 (a) $\frac{n^2 h}{2\pi}$ (b) $\frac{h}{2\pi n}$ (c) $\frac{nh}{2\pi}$ (d) nh
- [RBSE 2024]** In Rutherford's α -scattering experiment, for a very large impact parameter the angle of scattering is nearly
 (a) 45° (b) 90° (c) 180° (d) 0°
- [RBSE 2024]** The first excitation energy of the hydrogen atom is
 (a) 3.4 eV (b) 10.2 eV (c) 13.6 eV (d) 12.09 eV
- The ratio of the radii of the third and the first Bohr orbits of hydrogen is
 (a) 9 : 1 (b) 3 : 1 (c) 1 : 9 (d) 27 : 1
- The ratio of the longest wavelengths of the Lyman and the Balmer series is
 (a) 27 : 5 (b) 3 : 4 (c) 1 : 4 (d) 5 : 27
- The energy of the photon emitted when the electron of a hydrogen atom falls from $n = 3$ to $n = 2$ is
 (a) 0.66 eV (b) 1.89 eV (c) 10.2 eV (d) 12.09 eV
- The total energy of an electron in a Bohr orbit is -3.4 eV. Its kinetic energy is
 (a) -6.8 eV (b) $+3.4$ eV (c) $+6.8$ eV (d) -3.4 eV
- If the speed of the electron in the first Bohr orbit is v , its speed in the second orbit is
 (a) v (b) $v / 4$ (c) $v / 2$ (d) $2v$
- The ground-state energy of the He^+ ion ($Z = 2$) is
 (a) -6.8 eV (b) -54.4 eV (c) -27.2 eV (d) -13.6 eV

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2024] The radius of the second Bohr orbit of hydrogen is $r_2 = 4r_1 = \underline{\hspace{2cm}}$ m.	2×10^{-10}
The Balmer series of hydrogen lies in the $\underline{\hspace{2cm}}$ region.	visible
The energy of the electron in the n th Bohr orbit is proportional to $\underline{\hspace{2cm}}$.	$1 / n^2$ (negative)
According to classical theory an orbiting electron should $\underline{\hspace{2cm}}$ energy and spiral into the nucleus.	radiate
Large-angle scattering of α -particles occurs when the impact parameter is $\underline{\hspace{2cm}}$.	small
The total energy of the electron in an orbit is $\underline{\hspace{2cm}}$ of its kinetic energy.	negative

Very short answer (1 mark)

Question	Model answer
[RBSE 2024] Define ionisation energy.	The minimum energy required to remove the electron from the ground state of an atom to infinity (13.6 eV for hydrogen).
What is meant by impact parameter?	The perpendicular distance of the initial velocity line of the α -particle from the centre of the nucleus.
Name the quantity that is quantised in Bohr's model.	Angular momentum of the electron, $L = \frac{nh}{2\pi}$.
Why do most α -particles pass undeflected through a thin gold foil?	The atom is mostly empty space; the nucleus occupies a tiny volume.
Which series of hydrogen lies in the ultraviolet region?	The Lyman series ($n_f = 1$).
State the value of the Rydberg constant.	$R = 1.097 \times 10^7 \text{ m}^{-1}$.

Short answer (Section B, 1½ marks)

 Short answer (1½ marks)

[RBSE 2025] Write the expressions for the radius of the stable orbit and the total energy of the electron in the hydrogen atom according to Bohr's model.

Solution: Radius: $r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} = n^2 a_0$ with $a_0 = 0.529 \text{ Å}$. (¾) Total energy: $E_n = -\frac{me^4}{8\epsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2} \text{ eV}$ (negative: bound electron). (¾)

 Short answer (1½ marks)

[RBSE 2023] Write two drawbacks of Rutherford's atomic model. Which series of hydrogen lies in the visible region?

Solution: (i) An accelerated orbiting electron should radiate energy and fall into the nucleus – atoms would be unstable. (½) (ii) It predicts a continuous spectrum, whereas atoms give discrete line spectra. (½) The **Balmer series** lies in the visible region. (½)

 Short answer (1½ marks)

[RBSE 2020] In a hydrogen atom the electron wave forms a standing wave on the third allowed orbit. How many de Broglie wavelengths fit in its circumference, and what is the angular momentum?

Solution: Standing-wave condition $2\pi r = n\lambda$ with $n = 3$: **3 wavelengths** fit. (¾) Angular momentum $L = \frac{3h}{2\pi} = 3 \times 1.055 \times 10^{-34} = 3.16 \times 10^{-34} \text{ J s}$. (¾)

 Short answer (1½ marks)

Calculate the wavelength of the first line (longest) of the Lyman series. ($R = 1.097 \times 10^7 \text{ m}^{-1}$)

Solution: $n_f = 1, n_i = 2$: $\frac{1}{\lambda} = R \left(1 - \frac{1}{4} \right) = \frac{3R}{4}$ (½) $\Rightarrow \lambda = \frac{4}{3 \times 1.097 \times 10^7}$ (½) $= 1.215 \times 10^{-7} \text{ m} = 121.5 \text{ nm}$. (½)

 Short answer (1½ marks)

How many different spectral lines can a hydrogen atom emit when its electron is in the $n = 4$ level? Which transition gives the shortest wavelength?

Solution: Number of lines $= \frac{4 \times 3}{2} = 6$ ($4 \rightarrow 3, 4 \rightarrow 2, 4 \rightarrow 1, 3 \rightarrow 2, 3 \rightarrow 1, 2 \rightarrow 1$). (1) The largest energy gap, **$4 \rightarrow 1$** , gives the shortest wavelength: $\lambda = 97.2 \text{ nm}$. (½)

 Answer Key & Quick Solutions

1(a) 2(c) 3(d) 4(b) 5(a) 6(d) 7(b) 8(b) 9(c) 10(b)

- (a)** The ground-state energy is -13.6 eV , so 13.6 eV takes the electron to infinity ($E = 0$).
- (c)** Second postulate: $mvr = \frac{nh}{2\pi}$ – angular momentum is quantised.
- (d)** Far from the nucleus the repulsion is weak, so the particle is hardly deflected: $b = \frac{d}{2} \cot \frac{\theta}{2}$ is large when $\theta \rightarrow 0^\circ$.
- (b)** $E_2 - E_1 = -3.4 - (-13.6) = 10.2 \text{ eV}$.
- (a)** $r_n \propto n^2 \Rightarrow r_3:r_1 = 9:1$.
- (d)** $\lambda_L = \frac{4}{3R}, \lambda_B = \frac{36}{5R} \Rightarrow \text{ratio} = \frac{4}{3} \times \frac{5}{36} = \frac{5}{27}$.
- (b)** $13.6 \left(\frac{1}{4} - \frac{1}{9} \right) = 13.6 \times 5/36 = 1.89 \text{ eV}$.
- (b)** $K = -E = +3.4 \text{ eV}$ (and $U = 2E = -6.8 \text{ eV}$).
- (c)** $v_n \propto \frac{1}{n} \Rightarrow v_2 = v / 2$.
- (b)** $E_1 = -13.6 Z^2 = -13.6 \times 4 = -54.4 \text{ eV}$.

 Last-minute revision

- α -scattering: most pass straight (empty atom); ~ 1 in 8000 turns back (tiny, heavy, +ve nucleus).
- Closest approach $d = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$; large $b \Rightarrow \theta \approx 0^\circ, b = 0 \Rightarrow \theta = 180^\circ$.
- Rutherford drawbacks: electron should spiral in (instability); predicts continuous, not line, spectrum.
- Bohr: stable orbits; $mvr = \frac{nh}{2\pi}; h\nu = E_i - E_f$.
- $r_n = 0.529 n^2 \text{ Å}; v_n = 2.19 \times 10^6 / n \text{ m/s}; E_n = -13.6 / n^2 \text{ eV}; K = -E, U = 2E$.
- Ionisation energy 13.6 eV ; first excitation energy 10.2 eV ; He^+ : $E_n = -13.6 Z^2 / n^2$.
- Series: Lyman (UV, $n_f = 1$), Balmer (visible, 2), Paschen, Brackett, Pfund (IR); $1/\lambda = R(1/n_f^2 - 1/n_i^2)$.
- de Broglie: $2\pi r = n\lambda \Rightarrow mvr = nh / 2\pi$ (standing wave).

CHAPTER

13

Nuclei

17

Day 20

Board marks: 3

PYQ frequency ★★☆☆

RBSE Class 12

Size & density

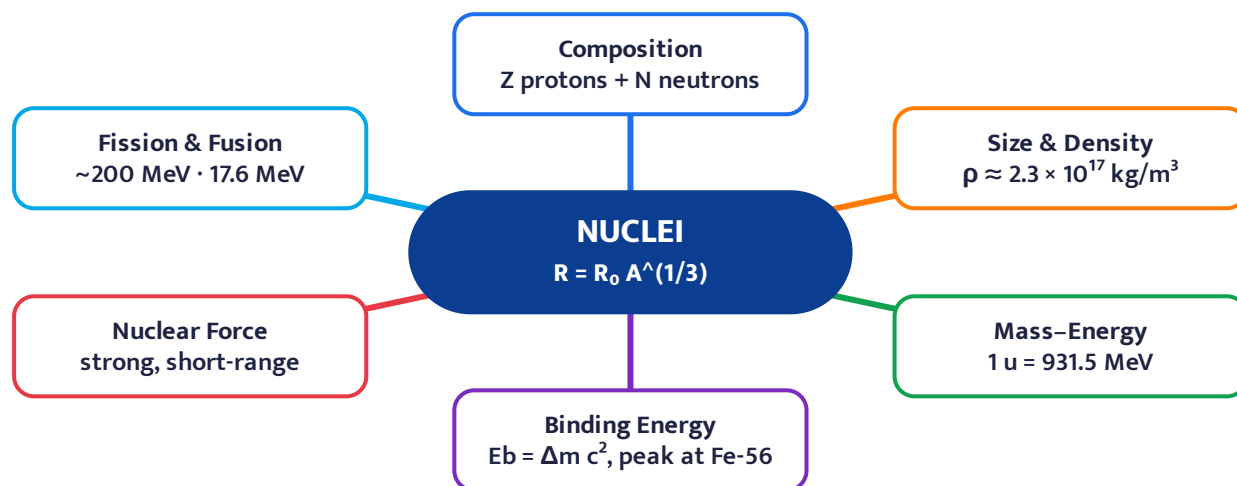
Isotopes, isobars

Mass defect

B.E. per nucleon

Nuclear force

Fission & fusion



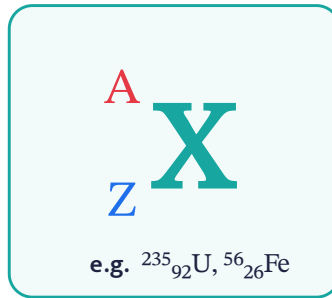
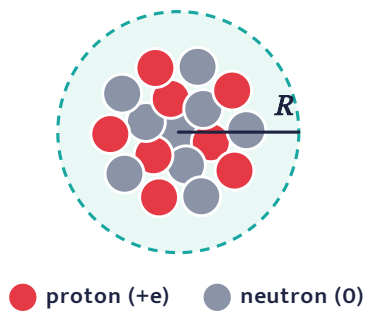
Chapter at a glance

Board snapshot

- **Weightage: 3 marks** – usually one MCQ / fill-in and one 1½-mark short answer (nuclear force, fission, fusion, mass defect). Descriptive – no long derivation, but diagrams (B.E./A graph) matter.
- **2025:** MCQ – binding energy equals mass defect $\times c^2$; fill-in – neutron was discovered by Chadwick; 1½ marks – nuclear force (properties) + B.E. per nucleon graph.
- **2024:** MCQ – isotopes; VSA – significance of $E = mc^2$; 1½ marks – nuclear fission and nuclear fusion.
- **2023:** 1½ marks – radioactive decay law and half-life for $\lambda = 0.693 \text{ min}^{-1}$ ($T_{1/2} = 1 \text{ min}$) – outside the current NCERT, see 13.7; 1½ marks – fusion, fission and mass defect. **2020:** α, β, γ decay chain of a nucleus ${}_{92}^{235}\text{D}$.
- **Most likely in 2027:** (1) nuclear force properties + B.E./nucleon curve (2025; the curve is the base for fission/fusion answers); (2) nuclear fission and fusion with equations (2023, 2024); (3) mass defect and binding energy numerical.

13.1 Composition of the Nucleus

The nucleus contains **protons** (charge +e) and **neutrons** (no charge), together called **nucleons**. The neutron was discovered by **Chadwick (1932)**. Notation: ${}_Z^AX$ with **Z** = atomic number (protons), **A** = mass number (nucleons), and neutron number $N = A - Z$. Nuclear masses are expressed in the **atomic mass unit**: $1 \text{ u} = \frac{1}{12}$ of the mass of one ${}^{12}\text{C}$ atom = $1.66 \times 10^{-27} \text{ kg}$.



$A = \text{mass number} = Z + N$

$Z = \text{atomic number (protons)}$

$N = A - Z$ (neutrons)

nucleus $\approx 10^{-15}$ m (fm)

atom $\approx 10^{-10}$ m

→ atom is 10^5 times bigger

A nucleus ${}^A_Z X$: Z protons + $(A - Z)$ neutrons

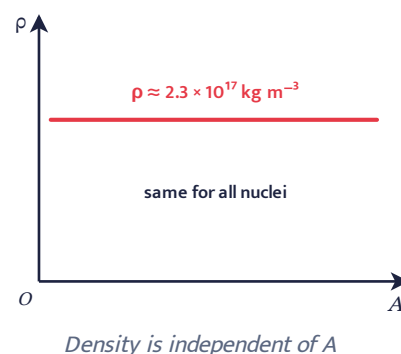
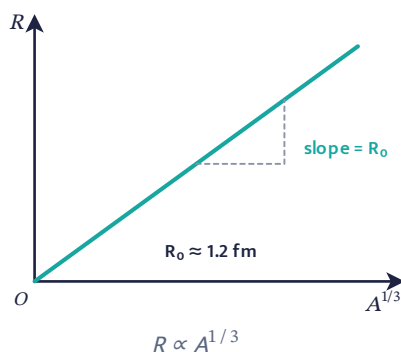
Term	Meaning	Example
Isotopes	same Z , different A (same chemical properties)	${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$
Isobars	same A , different Z	${}^3_1\text{H}$, ${}^3_2\text{He}$
Isotones	same neutron number N	${}^{198}_{80}\text{Hg}$, ${}^{197}_{79}\text{Au}$

! Common Trap

- Isotopes differ in the number of **neutrons**, not protons – so their chemical properties are the same but nuclear (mass) properties differ.
- Mass number A is the number of nucleons – it is **not** the mass in u exactly (e.g. ${}^{35}_{17}\text{Cl}$ has $A = 35$, mass = 34.97 u).

13.2 Size and Density of the Nucleus

Experiments show that the nuclear radius depends on the mass number as $R = R_0 A^{1/3}$ with $R_0 = 1.2$ fm (1 fm = 10^{-15} m, called a fermi). Since $R^3 \propto A$, the **volume is proportional to the number of nucleons**, i.e. every nucleon occupies the same volume – so the **density is the same for all nuclei**, about 2.3×10^{17} kg m $^{-3}$ (about 10^{14} times that of water).



Size and density

$$R = R_0 A^{1/3} \quad R_0 = 1.2 \times 10^{-15} \text{ m}$$

$$\rho = \frac{3m_u}{4\pi R_0^3} \approx 2.3 \times 10^{17} \text{ kg m}^{-3} \quad \text{Nuclear density (derived in 13.8)}$$

Solved Example

Find the radius of the nucleus of ${}^{64}_{30}\text{Zn}$. How does it compare with ${}^8_4\text{Be}$?

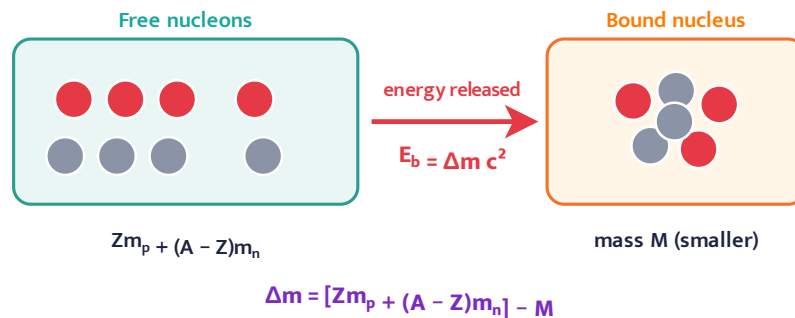
Solution: $R = 1.2 \times 64^{1/3} = 1.2 \times 4 = 4.8$ fm.

$$\frac{R_{\text{Zn}}}{R_{\text{Be}}} = \left(\frac{64}{8}\right)^{1/3} = 2.$$

13.3 Mass–Energy Equivalence, Mass Defect and Binding Energy

Einstein's relation $E = mc^2$ means that **mass and energy are equivalent**: a mass m corresponds to energy mc^2 . Energy of 1 u: $1.66 \times 10^{-27} \times (3 \times 10^8)^2 = 1.49 \times 10^{-10} \text{ J} = 931.5 \text{ MeV}$.

- **Mass defect (Δm)**: the difference between the total mass of the separate nucleons and the actual mass of the nucleus: $\Delta m = [Zm_p + (A - Z)m_n] - M$. The nucleus is always **lighter** than its free nucleons.
- **Binding energy (E_b)**: the energy equivalent of the mass defect – the minimum energy required to split the nucleus into free nucleons: $E_b = \Delta m c^2$. **B.E. per nucleon** = E_b / A measures stability.



Mass defect: the bound nucleus is lighter; the difference appears as binding energy

Solved Example

Find the binding energy and B.E. per nucleon of ${}^4_2\text{He}$. ($m_H = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$, $M_{\text{He}} = 4.002603 \text{ u}$)

Solution:

$$\Delta m = 2(1.007825) + 2(1.008665) - 4.002603 = 0.030377 \text{ u}. E_b = 0.030377 \times 931.5 = \mathbf{28.3 \text{ MeV}}; \text{ per nucleon} = 28.3/4 = \mathbf{7.07 \text{ MeV}}.$$

Solved Example

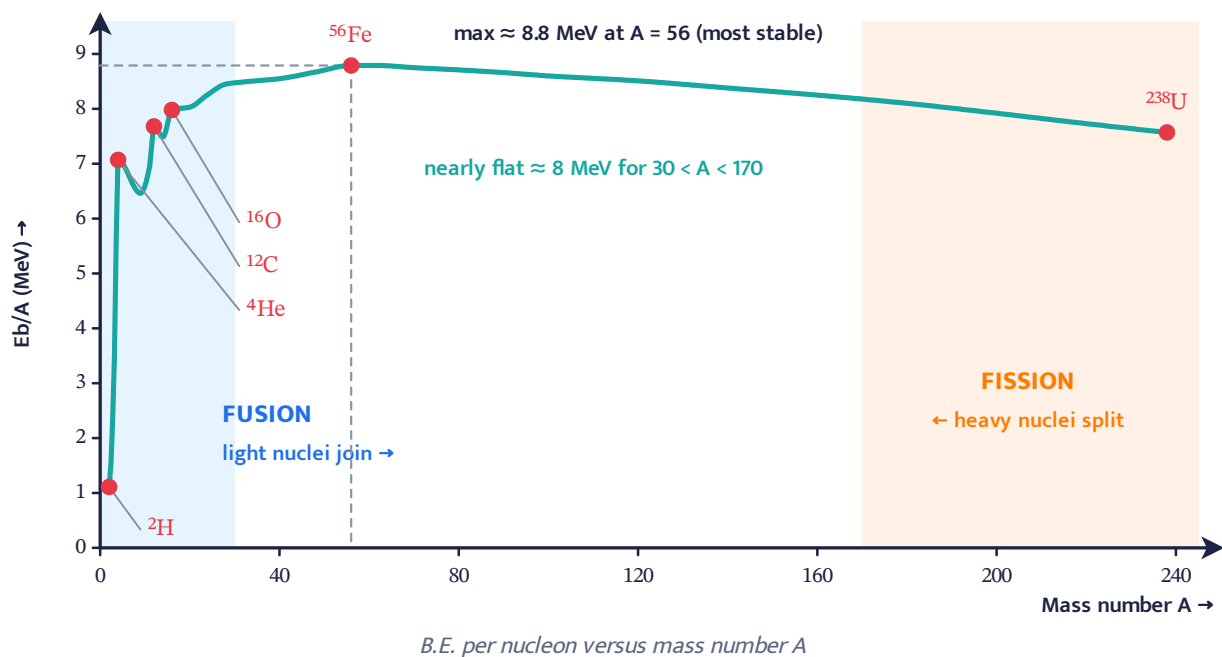
The deuteron (${}^2_1\text{H}$, 2.014102 u) is formed from a proton and a neutron. Find its binding energy.

Solution: $\Delta m = 1.007825 + 1.008665 - 2.014102 = 0.002388 \text{ u} \Rightarrow E_b = 0.002388 \times 931.5 = \mathbf{2.22 \text{ MeV}}$.

Common Trap

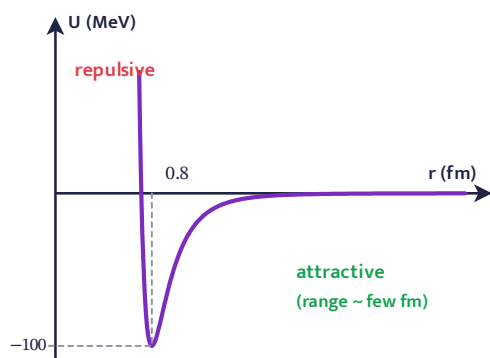
- Use **atomic** masses consistently (as given in tables), not bare nuclear masses; the electron masses cancel when you use m_H for the Z protons.
- $1 \text{ u} \triangleq 931.5 \text{ MeV}$ – multiply Δm in u by 931.5 to get MeV directly.

13.4 Binding Energy per Nucleon Curve



- Average B.E. per nucleon is about **8 MeV** for $30 < A < 170$ – nearly constant (saturation of nuclear force).
- It is **maximum** (≈ 8.75 MeV) **near A = 56** (iron) – the most stable nuclei.
- For **light nuclei** ($A < 20$) it is small and rises steeply (${}^2\text{H}$: 1.1 MeV; ${}^4\text{He}$ is a peak: 7.07 MeV).
- For **heavy nuclei** it falls slowly to about 7.6 MeV for ${}^{238}\text{U}$ (growing Coulomb repulsion).
- **Energy is released** when a heavy nucleus **splits** (fission) or two light nuclei **join** (fusion) – in both cases the products have higher B.E. per nucleon.

13.5 Nuclear Force



Potential energy of a pair of nucleons versus separation

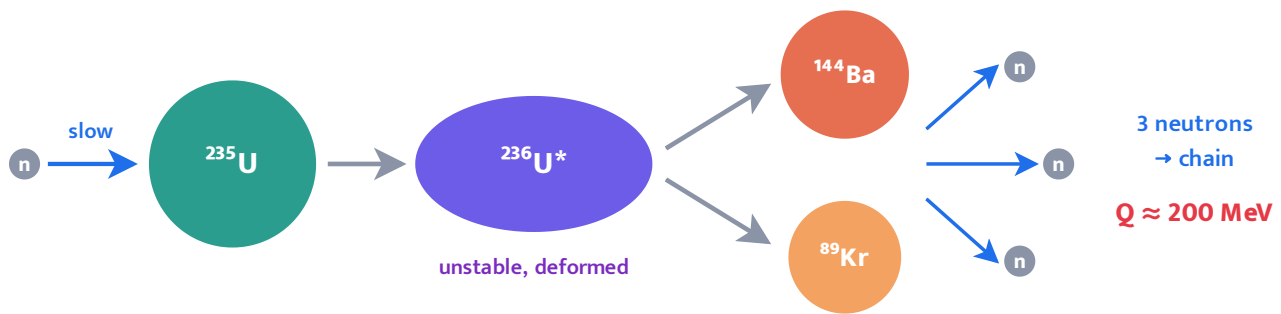


Properties of the nuclear force

- **Strongest** of the known forces – much stronger than the electromagnetic force.
- **Short range** – a few fermi ($\sim 2-3$ fm); negligible beyond.
- **Charge-independent**: same for p-p, n-n and p-n pairs.
- Attractive beyond ~ 0.8 fm, **repulsive** below ~ 0.8 fm; it is **not a central force** (depends on spin).

13.6 Nuclear Energy: Fission and Fusion

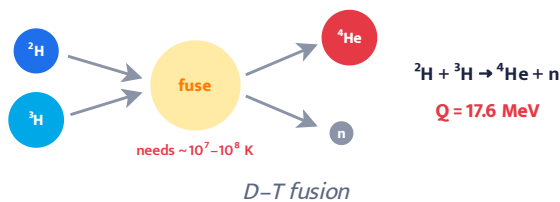
Nuclear fission: the splitting of a heavy nucleus into two (or more) medium-mass nuclei, with release of energy, after absorbing a neutron. Example: ${}_{92}^{235}\text{U} + {}_0^1\text{n} \rightarrow {}_{92}^{236}\text{U}^* \rightarrow {}_{56}^{144}\text{Ba} + {}_{36}^{89}\text{Kr} + 3{}_0^1\text{n} + Q$, with **$Q \approx 200$ MeV per fission**.



Fission of ^{235}U : the 3 emitted neutrons can cause further fissions (chain reaction)

- **Chain reaction:** neutrons released cause fission of more nuclei. **Uncontrolled** – atom bomb; **controlled** (nuclear reactor) – a **moderator** (heavy water, graphite) slows the neutrons and **control rods** (cadmium, boron) absorb the extra neutrons.
- **Nuclear fusion:** two light nuclei combine into a heavier nucleus with release of energy, e.g.
 $^2_1\text{H} + ^3_1\text{H} \rightarrow ^4_2\text{He} + ^1_0\text{n} + 17.6 \text{ MeV}$. It needs a temperature of about 10^7 – 10^8 K to overcome the Coulomb repulsion between the nuclei (thermonuclear reaction).
- The **energy of the Sun and stars** comes from fusion: in the Sun, hydrogen is converted to helium ($4\ ^1_1\text{H} \rightarrow ^4_2\text{He} + 2e^+ + 2\nu + 26.7 \text{ MeV}$).

	Fission	Fusion
Process	heavy nucleus splits	light nuclei join
Condition	slow neutron; normal temperature	$\sim 10^7$ – 10^8 K and high pressure
Energy per event	$\approx 200 \text{ MeV}$	$\approx 17.6 \text{ MeV}$ (D–T)
Energy per unit mass	smaller	larger
Use	reactor, atom bomb	stars, hydrogen bomb



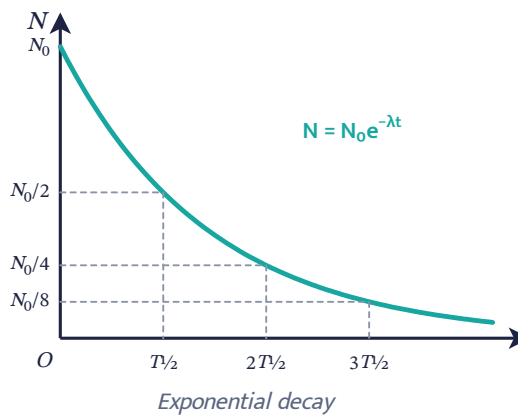
Solved Example

Calculate the energy released in $^2_1\text{H} + ^3_1\text{H} \rightarrow ^4_2\text{He} + \text{n}$.
(Masses: 2.014102, 3.016049, 4.002603, 1.008665 u)

Solution:

$$\Delta m = (2.014102 + 3.016049) - (4.002603 + 1.008665) = 0.018883 \text{ u} \Rightarrow Q = 0.018883 \times 931.5 = \mathbf{17.6 \text{ MeV}}$$

13.7 Radioactivity – Asked Earlier by RBSE



Asked earlier by RBSE

- Radioactive decay is **outside the current rationalised NCERT book**, but it appeared in RBSE 2023 and 2020 – learn this compact version:
- Decay law:** $N = N_0 e^{-\lambda t}$ (λ = decay constant); activity $R = \lambda N$ (unit becquerel, 1 Bq = 1 decay/s).
- Half-life:** $T_{1/2} = \frac{0.693}{\lambda}$; after n half-lives $N = N_0 / 2^n$.
- α -decay:** $Z - 2, A - 4$; **β^- -decay:** $Z + 1, A$ same; **γ -decay:** Z, A unchanged (photon emitted).

Solved Example

[RBSE 2023] For a radioactive sample $\lambda = 0.693 \text{ min}^{-1}$. Find its half-life.

Solution: $T_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{0.693 \text{ min}^{-1}} = 1 \text{ min}.$

Solved Example

A nucleus ${}_{92}^{235}\text{X}$ emits one α -particle and then two β^- -particles. Find the Z and A of the final nucleus (pattern of the 2020 question).

Solution: After α : $Z = 90, A = 231$. After two β^- : $Z = 90 + 2 = 92, A = 231$. The final nucleus ${}_{92}^{231}\text{Y}$ is an isotope of the parent X .

13.8 Important Derivations ★

Derivation ★ [3 marks]

Derivation 1. Show that the density of nuclear matter is independent of the mass number A , and estimate its value.

Solution: (i) Nuclear radius: $R = R_0 A^{1/3}$ with $R_0 = 1.2 \text{ fm}$. (½)

(ii) Volume of the nucleus (a sphere): $V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^3 A$. (½)

(iii) Mass of the nucleus $\approx A m_u$, where $m_u = 1.66 \times 10^{-27} \text{ kg}$ is the mass of one nucleon. (½)

(iv) Density $\rho = \frac{\text{mass}}{\text{volume}} = \frac{A m_u}{\frac{4}{3}\pi R_0^3 A} = \frac{3 m_u}{4\pi R_0^3}$ – **A cancels**, so ρ is the same for all nuclei. (1)

(v) $\rho = \frac{3 \times 1.66 \times 10^{-27}}{4\pi (1.2 \times 10^{-15})^3} \approx 2.3 \times 10^{17} \text{ kg m}^{-3}$. (½)

 Derivation ★ [3 marks]

Derivation 2 – [RBSE 2023, 2025]. Define mass defect and binding energy. Obtain the binding energy of a nucleus ${}_Z^AX$ and explain why the nucleus is stable.

Solution: (i) **Mass defect:** $\Delta m = [Zm_p + (A - Z)m_n] - M$, where M is the mass of the nucleus (figure in 13.3). (1)

(ii) By Einstein's relation the missing mass appears as energy when the nucleons bind together: $E_b = \Delta m c^2 = \Delta m \times 931.5 \text{ MeV}$ (Δm in u). (1)

(iii) **Binding energy** is the energy needed to separate the nucleus into free nucleons; it is the energy released when they bind. (½)

(iv) Because $M < Zm_p + (A - Z)m_n$ the nucleus has lower energy than its free nucleons, so energy must be supplied to break it – the nucleus is stable. **Larger $E_b / A \Rightarrow$ more stable.** (½)

 Derivation ★ [3 marks]

Derivation 3 – [RBSE 2023]. Derive the radioactive decay law and the relation between half-life and decay constant. (outside the current NCERT)

Solution: (i) The number of nuclei decaying per unit time is proportional to the number N present: $-\frac{dN}{dt} = \lambda N$ (λ = decay constant). (1)

(ii) Rearranging and integrating from N_0 at $t = 0$ to N at time t : $\int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt \Rightarrow \ln \frac{N}{N_0} = -\lambda t$. (½)

(iii) $N = N_0 e^{-\lambda t}$ – the decay law. (½)

(iv) Half-life: at $t = T_{1/2}$, $N = N_0 / 2 \Rightarrow \frac{1}{2} = e^{-\lambda T_{1/2}} \Rightarrow \lambda T_{1/2} = \ln 2 \Rightarrow T_{1/2} = \frac{0.693}{\lambda}$. (1)

 VS Sir's Exam Tip

- For "nuclear force + B.E. curve" (2025) draw the curve first, mark Fe-56 at the peak ($\approx 8.75 \text{ MeV}$), then write 4 properties of the force – the graph and the properties carry the marks.
- In fission / fusion answers always write a balanced equation with the Q -value (200 MeV; 17.6 MeV) and say why energy is released (higher B.E. per nucleon of the products).

Board question bank **Section A – Multiple choice (½ mark each)**

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- [RBSE 2025] The binding energy of a nucleus is equal to

(a) $\frac{1}{2} \times \text{mass defect} \times c^2$	(b) mass defect / c^2
(c) mass defect $\times c^2$	(d) mass defect $\times c$
- [RBSE 2024] Atoms with the same atomic number but different mass numbers are called

(a) isomers	(b) isobars	(c) isotones	(d) isotopes
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- The number of neutrons in ${}_{92}^{235}\text{U}$ is

(a) 143	(b) 327	(c) 235	(d) 92
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- The ratio of the radii of the nuclei of mass numbers 64 and 8 is

(a) 2 : 1	(b) 1 : 2	(c) 8 : 1	(d) 4 : 1
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- 5 The density of nuclear matter
 (a) decreases with mass number (b) depends on atomic number only
 (c) is the same for all nuclei (d) increases with mass number
- 6 The energy equivalent of 1 atomic mass unit is about
 (a) 1.6 MeV (b) 9.315 MeV (c) 931.5 MeV (d) 93.15 MeV
- 7 The mass defect of a nucleus is 0.03 u. Its binding energy is about
 (a) 2.79 MeV (b) 31 MeV (c) 279 MeV (d) 27.9 MeV
- 8 The binding energy per nucleon is maximum for nuclei of mass number about
 (a) 56 (b) 120 (c) 238 (d) 4
- 9 The range of the nuclear force is of the order of
 (a) 10^{-10} m (b) 1 m (c) 10^{-8} m (d) 10^{-15} m
- 10 The source of energy of the Sun is
 (a) nuclear fission (b) chemical burning
 (c) radioactive decay (d) nuclear fusion

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2025] The neutron was discovered by ____.	Chadwick (1932)
The nuclear force is a ____ range force.	short
Nuclei having the same mass number but different atomic number are called ____.	isobars
1 u is equivalent to ____ MeV.	931.5
Nuclear fusion takes place only at very high ____.	temperature
In a nuclear reactor the neutrons are slowed down by a ____.	moderator

Very short answer (1 mark)

Question	Model answer
[RBSE 2024] What does the equation $E = mc^2$ signify?	Mass–energy equivalence: a mass m is equivalent to energy mc^2 ; mass can be converted into energy and vice versa.
Define mass defect.	The difference between the sum of the masses of the free nucleons and the mass of the nucleus.
What is a fermi? Give its value in metre.	A unit of nuclear length: $1 \text{ fm} = 10^{-15} \text{ m}$.
Name the particle that carries no charge and is present in the nucleus.	Neutron.
What is the function of control rods in a nuclear reactor?	They absorb the excess neutrons and control the rate of the chain reaction (cadmium, boron).
Which nucleus has the highest binding energy per nucleon?	Iron, ${}^{56}_{26}\text{Fe}$ ($\approx 8.75 \text{ MeV}$).

Short answer (Section B, 1½ marks)

 Short answer (1½ marks)

[RBSE 2025] Write any three properties of the nuclear force. Draw the graph of binding energy per nucleon against mass number.

Solution: Properties: (i) strongest known force, much greater than the electromagnetic force; (ii) short range (a few fm); (iii) charge-independent (p–p, n–n, p–n same). (¼ each, ¾) **Graph:** as in 13.4 – rises steeply for $A < 20$, flat peak $\approx 8.75 \text{ MeV}$ at $A \approx 56$, slowly falls to $\approx 7.6 \text{ MeV}$ at $A = 238$; axes labelled. (¾)

 Short answer (1½ marks)

[RBSE 2024, 2023] What is nuclear fission? What is nuclear fusion? Give one example of each.

Solution: Fission: a heavy nucleus splits into two medium nuclei with release of energy: ${}^{235}_{92}\text{U} + n \rightarrow {}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 3n + 200 \text{ MeV}$. (¾) **Fusion:** two light nuclei combine into a heavier one with release of energy: ${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + n + 17.6 \text{ MeV}$ (needs $\sim 10^7 \text{ K}$). (¾)

 Short answer (1½ marks)

[RBSE 2023] Explain mass defect. Why is the energy released in fusion per nucleon larger than in fission?

Solution: Mass defect $\Delta m = (\text{mass of free nucleons}) - (\text{mass of the nucleus})$; $E_b = \Delta m c^2$. (¾) Per nucleon: fusion (D–T) gives $17.6/5 \approx 3.5 \text{ MeV}$, fission of ${}^{235}\text{U}$ gives $200/236 \approx 0.85 \text{ MeV}$ – because the B.E./nucleon curve is much steeper for light nuclei than for heavy ones. (¾)

 Short answer (1½ marks)

Find the binding energy per nucleon of ${}^{56}_{26}\text{Fe}$. ($m_H = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$, $M_{\text{Fe}} = 55.934939 \text{ u}$)

Solution: $\Delta m = 26(1.007825) + 30(1.008665) - 55.934939 = 0.528461 \text{ u}$ (½); $E_b = 0.528461 \times 931.5 = 492.3 \text{ MeV}$ (½); $E_b / A = 492.3 / 56 = 8.79 \text{ MeV}$ (½).

**Short answer (1½ marks)**

A radioactive sample has an initial activity 800 Bq and half-life 10 min. What is its activity after 30 min?

Solution: Number of half-lives $n = 30/10 = 3$ (½) $\Rightarrow R = R_0 / 2^3 = 800 / 8$ (½) = **100 Bq**. (½)

**Answer Key & Quick Solutions**

1(c) 2(d) 3(a) 4(a) 5(c) 6(c) 7(d) 8(a) 9(d) 10(d)

1. (c) $E_b = \Delta m c^2$ (Einstein's mass-energy relation).
2. (d) Same Z, different A $\Rightarrow$ isotopes (e.g. ${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$).
3. (a) $N = A - Z = 235 - 92 = 143$.
4. (a) $R \propto A^{1/3} \Rightarrow (64/8)^{1/3} = 2$.
5. (c) $\rho = 3m_u / 4\pi R_0^3$ – independent of A.
6. (c) $1.66 \times 10^{-27} \times (3 \times 10^8)^2 = 1.49 \times 10^{-10} \text{ J} = 931.5 \text{ MeV}$.
7. (d) $E_b = 0.03 \times 931.5 = 27.9 \text{ MeV}$.
8. (a) Peak $\approx 8.75 \text{ MeV}$ at $A \approx 56$ (iron).
9. (d) Short-range force: a few fermi, $1 \text{ fm} = 10^{-15} \text{ m}$.
10. (d) Hydrogen nuclei fuse into helium at $\sim 10^7 \text{ K}$, releasing 26.7 MeV per helium nucleus.

**Last-minute revision**

- ${}^A_Z\text{X}$: Z protons, $N = A - Z$ neutrons; neutron discovered by Chadwick (1932).
- Isotopes same Z; isobars same A; isotones same N.
- $R = 1.2A^{1/3} \text{ fm}$; density $\approx 2.3 \times 10^{17} \text{ kg m}^{-3}$, same for all nuclei.
- $1 \text{ u} = 931.5 \text{ MeV}$; $\Delta m = [Zm_p + (A - Z)m_n] - M$; $E_b = \Delta m c^2$.
- B.E./A: $\approx 8 \text{ MeV}$ flat, peak 8.75 MeV at Fe-56; light nuclei fuse, heavy nuclei split.
- Nuclear force: strongest, short range (fm), charge-independent, repulsive below 0.8 fm.
- Fission $\approx 200 \text{ MeV}$ (chain reaction: moderator + control rods); fusion D-T 17.6 MeV ($\sim 10^7 \text{ K}$).
- Decay (old syllabus): $N = N_0 e^{-\lambda t}$, $T_{1/2} = 0.693 / \lambda$; α : Z-2, A-4; β^- : Z+1.

CHAPTER

14

Semiconductor Electronics

17

Day 21–22

Board marks: 4

PYQ frequency ★★★

RBSE Class 12

Energy bands

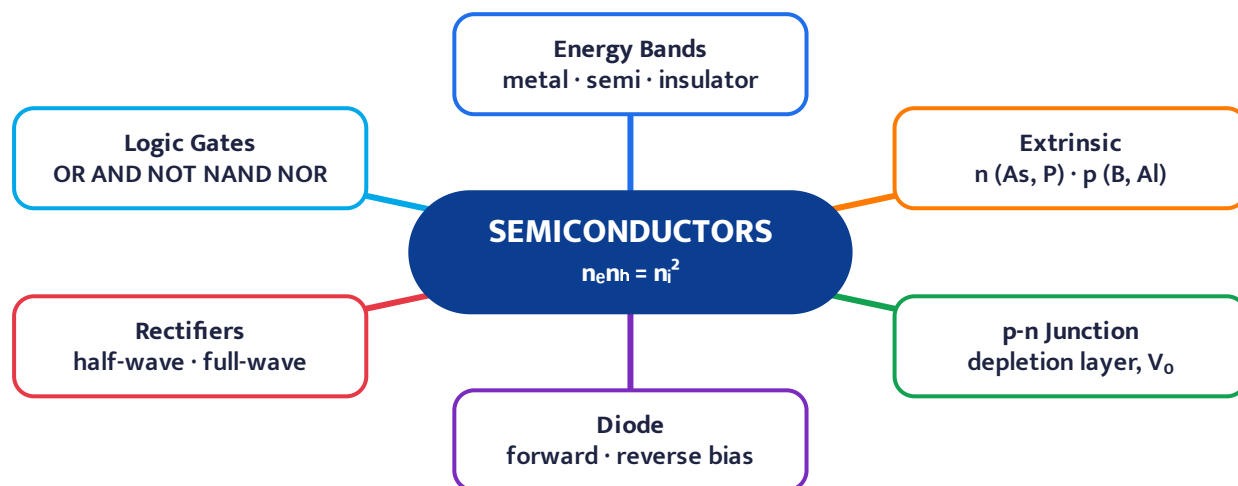
n- and p-type

p-n junction

Diode & rectifiers

Zener, LED, solar cell

Logic gates (earlier)



Chapter at a glance

Board snapshot

- **Weightage: 4 marks** – the most question-rich chapter of the last unit: expect one MCQ, one fill-in and a 3- or 4-mark descriptive question (bands, p-n junction, rectifier).
- **2025:** MCQ – majority carriers in n-type (electrons); MCQ – conductivity of an intrinsic semiconductor at 0 K (zero); fill-in – resistivity range 10^{-5} – 10^6 Ω m; 1½ marks – insulator vs semiconductor using energy bands.
- **2024:** MCQ – an inorganic semiconductor (CdS, Ge); fill-in – the two types of extrinsic semiconductor; 3 marks – bands of conductor / insulator / semiconductor + n-type band diagram **OR** identify a rectifier and explain.
- **2023:** MCQ – logic-gate output (outside current NCERT); fill-in – resistivity of semiconductors; 4 marks – full-wave rectifier with waveforms **OR** intrinsic semiconductor + p-n junction formation + Zener and p-n diode symbols. **2026 CWSN:** 3 marks – bands **OR** p-n junction. **2020:** Zener regulator; intrinsic vs extrinsic.
- **Most likely in 2027:** (1) energy bands + n-/p-type band diagrams (asked 2024, 2025, 2026); (2) p-n junction formation, depletion layer and barrier potential; (3) full-wave rectifier circuit with input/output waveforms.

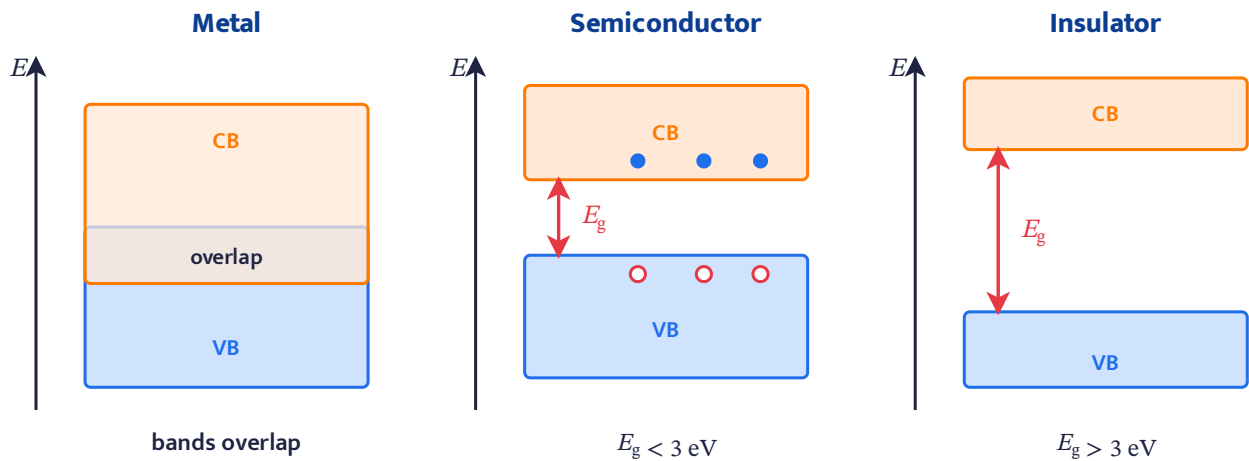
14.1 Classification of Solids: Metals, Semiconductors, Insulators

	Metals (conductors)	Semiconductors	Insulators
Resistivity ρ (Ω m)	10^{-2} – 10^{-8}	10^{-5} – 10^6	10^{11} – 10^{19}
Conductivity σ (S m $^{-1}$)	10^2 – 10^8	10^{-6} – 10^5	10^{-11} – 10^{-19}
Examples	Cu, Ag, Al	Si, Ge, GaAs, CdS	glass, diamond, rubber

- **Elemental** semiconductors: Si, Ge. **Compound inorganic:** CdS, GaAs, CdSe, InP. **Organic:** anthracene, doped phthalocyanines, polypyrrole.
- For a semiconductor resistivity **falls** with temperature (α negative), unlike a metal.

14.2 Energy Bands

In a crystal the closely spaced atomic levels merge into bands. The band formed from the outermost (valence) electrons is the **valence band (VB)**; above it lies the **conduction band (CB)**; the gap between them is the **forbidden energy gap** E_g .



Energy bands of a metal, a semiconductor and an insulator

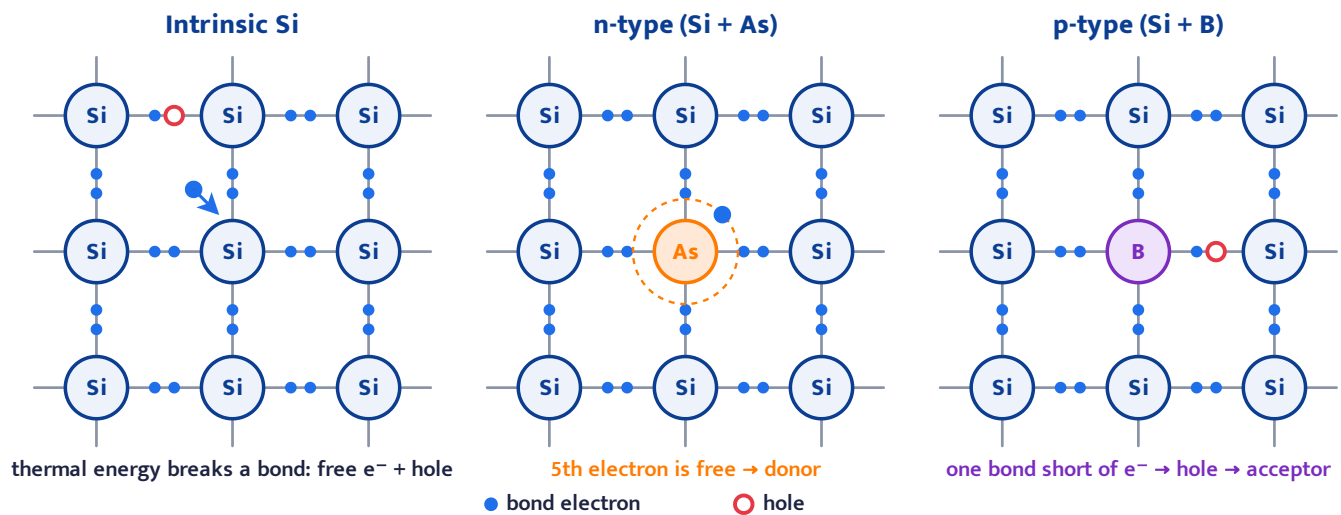
	Metal	Semiconductor	Insulator
Energy gap	none (VB and CB overlap)	$E_g < 3 \text{ eV}$ (Si 1.1, Ge 0.67)	$E_g > 3 \text{ eV}$
Conduction band	partly filled	empty at 0 K; few electrons at room temp.	empty always

VS Sir's Exam Tip

Insulator vs semiconductor (2025): the only difference is the **size of E_g** – large ($> 3 \text{ eV}$) so electrons cannot reach CB at room temperature; small ($\sim 1 \text{ eV}$) so thermal energy lifts a few electrons across.

14.3 Intrinsic Semiconductors

A pure semiconductor is an **intrinsic semiconductor**. When a covalent bond breaks, a **free electron** and a vacancy – a **hole** – are created, so $n_e = n_h = n_i$. At **0 K** all bonds are intact: no free carriers, conductivity **zero** (it behaves as an insulator); conductivity rises with temperature. Total current $I = I_e + I_h$, and $\sigma = e(n_e\mu_e + n_h\mu_h)$.

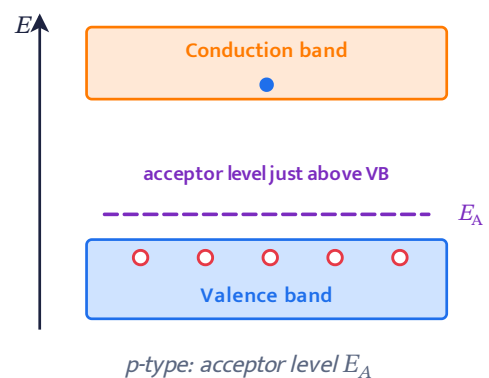
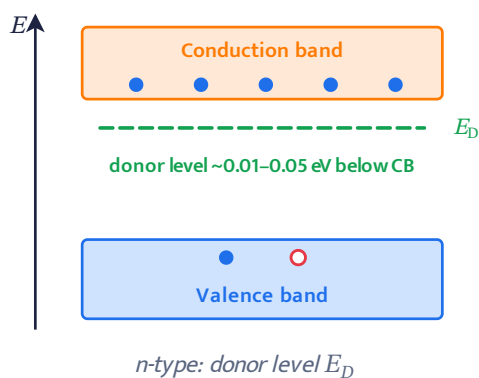


Si lattice: intrinsic (left), n-type with a donor As atom (middle), p-type with an acceptor B atom (right)

14.4 Extrinsic Semiconductors: n-type and p-type

Adding a small amount of a suitable impurity (about 1 part in 10^7) is called **doping**; it gives an **extrinsic semiconductor**.

	n-type	p-type
Dopant	pentavalent – As, P, Sb (donor)	trivalent – B, Al, In, Ga (acceptor)
Majority / minority	electrons / holes	holes / electrons
Impurity level	donor level E_D just below CB	acceptor level E_A just above VB
Relation	$n_e \gg n_h$	$n_h \gg n_e$
Charge	electrically neutral	electrically neutral



Carrier relations

$$n_e n_h = n_i^2 \quad \text{Mass-action law, valid for doped samples as well}$$

$$\sigma = e(n_e \mu_e + n_h \mu_h) \quad \text{Conductivity}$$

Solved Example

Si has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$. It is doped with As so that $n_e \approx 10^{21} \text{ m}^{-3}$. Find n_h .

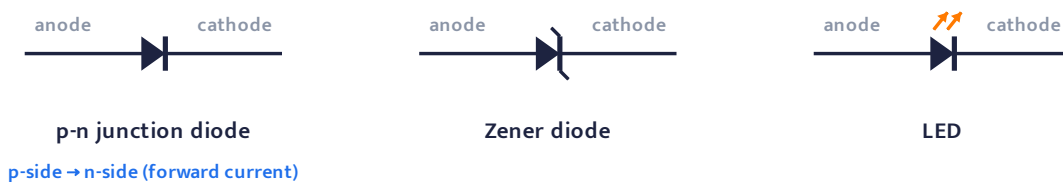
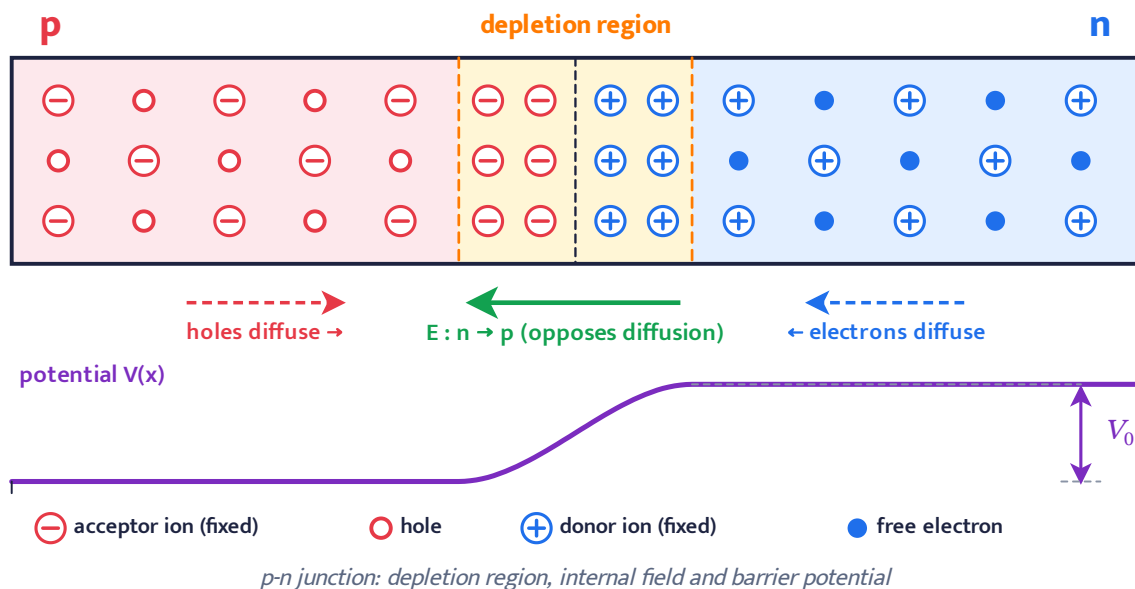
Solution: $n_h = \frac{n_i^2}{n_e} = \frac{(1.5 \times 10^{16})^2}{10^{21}} = 2.25 \times 10^{11} \text{ m}^{-3}$ – holes become the minority carriers.

! Common Trap

- n-type and p-type crystals are **not charged** – the donor / acceptor atoms are neutral, so the total charge stays zero. "n" means negative carriers, not a negative crystal.
- Even in n-type material some holes exist (thermally generated), and $n_e n_h = n_i^2$ still holds.

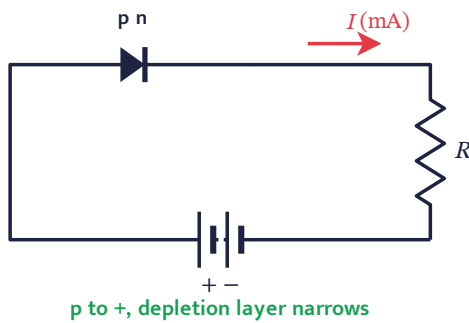
14.5 The p-n Junction

A p-n junction is formed when a p-type and an n-type region are made in the same crystal. Because of the concentration difference, **electrons diffuse** from n to p and **holes diffuse** from p to n (diffusion current). Near the junction the n-side is left with **positive donor ions** and the p-side with **negative acceptor ions** – a thin region **depleted of mobile carriers**, the **depletion layer** ($\sim 10^{-6}$ m), with an internal field from n to p. This field causes a **drift current** opposing diffusion; at equilibrium they cancel. The potential difference across the layer is the **barrier potential** V_0 (≈ 0.7 V for Si, 0.3 V for Ge).

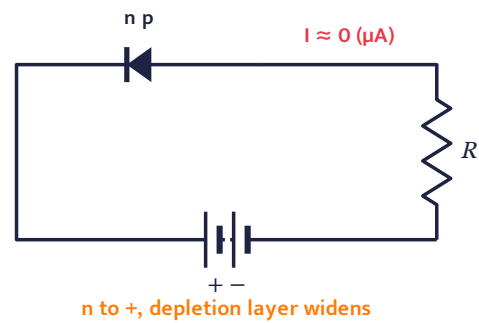


Circuit symbols: the arrow points from p (anode) to n (cathode)

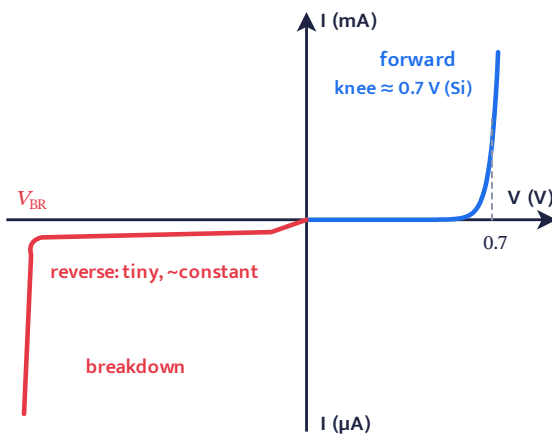
14.6 Diode: Forward and Reverse Bias



Forward bias (p to +): barrier lowers to $V_0 - V$



Reverse bias (p to -): barrier rises to $V_0 + V$



Diode I–V characteristic (different scales for mA and μA)



Reading the characteristic

- **Forward:** depletion layer **narrows**; current is negligible until the **knee (threshold) voltage** ($\sim 0.7 \text{ V Si}$, $\sim 0.3 \text{ V Ge}$), then rises rapidly (mA).
- **Reverse:** layer **widens**; tiny (μA) current due to minority carriers, almost independent of voltage, until **breakdown**.
- So a diode **conducts in one direction only** – the basis of rectification.

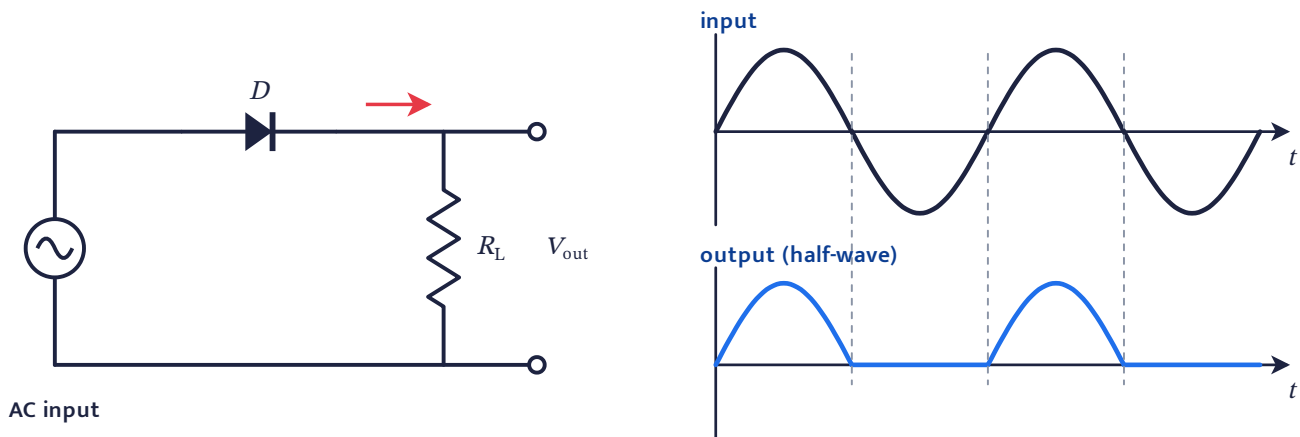
Solved Example

A Si diode (drop 0.7 V) is connected in forward bias with a 5 V cell and a 430Ω resistor in series. Find the current.

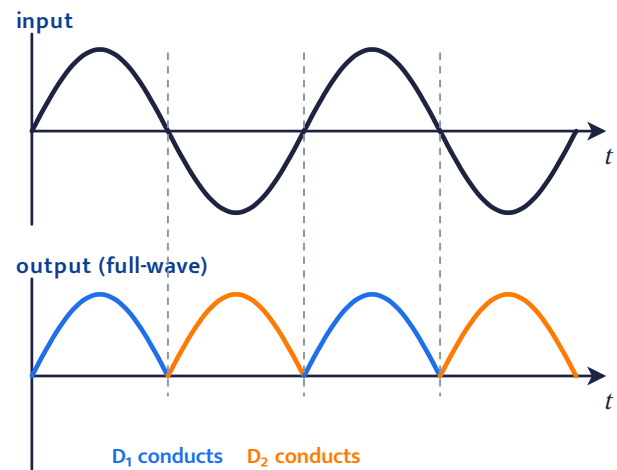
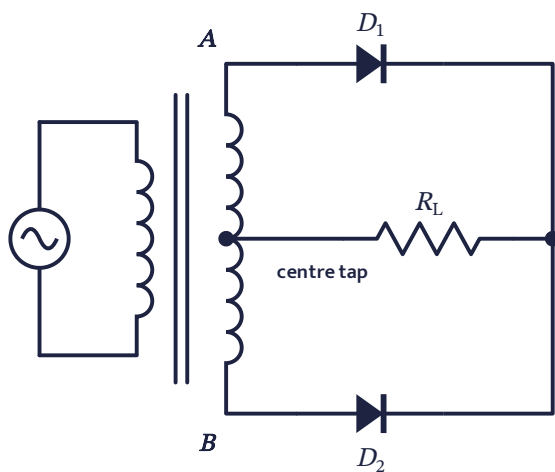
Solution: $I = \frac{5 - 0.7}{430} = \frac{4.3}{430} = 10 \text{ mA}.$

14.7 Rectifiers

A **rectifier** converts alternating voltage into direct (unidirectional) voltage using the one-way conduction of a diode.



Half-wave rectifier: output only in the positive half-cycles



Full-wave (centre-tap) rectifier: D_1 and D_2 conduct alternately; current through R_L is always in the same direction

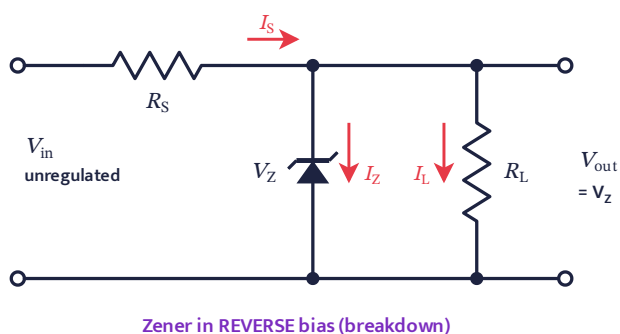
	Half-wave	Full-wave
Diodes	1	2 (centre-tap) or 4 (bridge)
Output pulses	every positive half-cycle only	every half-cycle
Output frequency (input 50 Hz)	50 Hz	100 Hz

Note

A capacitor connected across R_L (a **filter**) charges at the peaks and discharges slowly, smoothing the pulsating output into nearly steady dc.

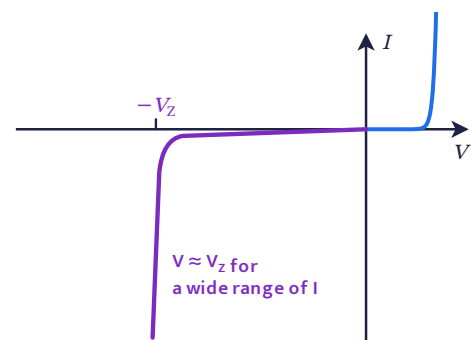
14.8 Zener Diode and Optoelectronic Devices

A **Zener diode** is a heavily doped p-n junction designed to work in **reverse breakdown**. Beyond the Zener voltage V_Z the voltage across it stays almost constant while the current changes over a wide range – hence its use as a **voltage regulator**.



Zener in REVERSE bias (breakdown)

Zener regulator: $V_{out} = V_Z$, $I_S = I_Z + I_L$



Zener characteristic (reverse bias)

 Solved Example

In a Zener regulator $V_{in} = 15\text{ V}$, $R_S = 250\ \Omega$, $V_Z = 10\text{ V}$, $R_L = 1\text{ k}\Omega$. Find I_S , I_L , I_Z .

Solution: $I_S = \frac{15 - 10}{250} = 20\text{ mA}$; $I_L = \frac{10}{1000} = 10\text{ mA}$; $I_Z = I_S - I_L = 10\text{ mA}$.

Device	Bias	Principle
Photodiode	reverse	light creates e–h pairs $\Rightarrow$ current $\propto$ intensity
LED	forward	e–h recombination emits photons, $h\nu \approx E_g$
Solar cell	none	light generates an emf (photovoltaic)

14.9 Logic Gates – Asked Earlier by RBSE **Asked earlier by RBSE**

- Digital electronics / logic gates are **outside the current rationalised NCERT book**, but a gate-output MCQ came in RBSE 2023. Learn the truth table below – nothing more is needed.
- OR:** $Y = A + B$; **AND:** $Y = A \cdot B$; **NOT:** $Y = \bar{A}$; **NAND** = NOT(AND); **NOR** = NOT(OR).

A	B	OR	AND	NAND	NOR
0	0	0	0	1	1
0	1	1	0	1	0
1	0	1	0	1	0
1	1	1	1	0	0

14.10 Important Derivations and Descriptive Answers ★ **Derivation ★ [3 marks]**

Answer 1 – [RBSE 2024, 2025, 2026 CWSN]. On the basis of energy bands distinguish between a conductor, an insulator and a semiconductor, and draw the energy-band diagram of an n-type semiconductor.

Solution: (i) **Conductor:** VB and CB overlap (or CB is partly filled); no forbidden gap, so free electrons are always available; large conductivity. (½)

(ii) **Insulator:** large forbidden gap $E_g > 3\text{ eV}$; CB empty and VB full; electrons cannot cross the gap at room temperature; conductivity ≈ 0 . (½)

(iii) **Semiconductor:** small gap $E_g < 3\text{ eV}$ (Si 1.1 eV, Ge 0.67 eV); at 0 K it is an insulator, but at room temperature thermal energy takes some electrons to CB leaving holes in VB. (1)

(iv) **n-type:** doping with a pentavalent donor creates a donor level E_D just below the CB; donor electrons are easily excited into the CB, so electrons are majority carriers (diagram in 14.4). (1)

 Derivation ★ [4 marks]

Answer 2 – [RBSE 2023, 2026 CWSN]. Explain the formation of a p-n junction (depletion layer and barrier potential). Draw the symbols of a p-n junction diode and a Zener diode.

Solution: (i) Joining p- and n-regions: electrons ($n \rightarrow p$) and holes ($p \rightarrow n$) **diffuse** across the junction because of the concentration gradient; this is the diffusion current. (1)

(ii) The electrons arriving on the p-side recombine with holes, as do the holes on the n-side; so near the junction fixed **positive donor ions** (n-side) and **negative acceptor ions** (p-side) are left uncovered. (1)

(iii) This **depletion layer** (no mobile carriers, width $\sim 10^{-6}\text{ m}$) sets up an internal electric field from n to p and a potential difference V_0 – the **barrier potential** (Si $\approx 0.7\text{ V}$). (1)

(iv) The field drives a **drift current** (minority carriers) opposite to diffusion; equilibrium is reached when the two are equal and the net current is zero. (½)

(v) **Symbols** (figure in 14.5): diode – triangle pointing $p \rightarrow n$ with a bar; Zener – the bar bent at both ends. (½)

 Derivation ★ [4 marks]

Answer 3 – [RBSE 2023, 2024]. Draw the circuit of a full-wave (centre-tap) rectifier, explain its working and sketch the input and output waveforms.

- Solution:** (i) **Circuit** (figure in 14.7): centre-tapped transformer secondary, two diodes D_1, D_2 with anodes at the two ends, cathodes joined and connected to load R_L whose other end goes to the centre tap. **(1)**
- (ii) **Positive half-cycle:** end A is positive, so D_1 is forward biased and conducts; D_2 is reverse biased. Current flows through R_L from the joined cathodes to the centre tap. **(1)**
- (iii) **Negative half-cycle:** end B is positive, so D_2 conducts and D_1 is off; current through R_L again flows in the **same direction**. **(1)**
- (iv) **Waveforms:** input sine wave; output = a series of positive pulses in **both** half-cycles, i.e. pulsating dc of frequency **2f** (100 Hz for 50 Hz input). A filter capacitor smooths it. **(1)**

 Derivation ★ [3 marks]

Answer 4 – [RBSE 2020]. With a circuit diagram explain how a Zener diode works as a voltage regulator.

- Solution:** (i) **Circuit** (figure in 14.8): the Zener diode is connected in **reverse bias** across the load R_L , in series with a resistor R_S and the unregulated input V_{in} ($V_{in} > V_Z$). **(1)**
- (ii) In breakdown the voltage across the Zener stays $\approx V_Z$, so the output $V_{out} = V_Z$. **(½)**
- (iii) If V_{in} rises, the extra current flows through the Zener (I_Z increases) and the extra voltage drops across R_S ; V_{out} stays constant. **(½)**
- (iv) If the load current I_L increases, I_Z decreases so that $I_S = I_Z + I_L$ remains the same and V_{out} is again constant. **(1)**

 VS Sir's Exam Tip

- For rectifier questions the **labelled circuit + both waveforms** carry about half the marks – draw the output pulses on the same time axis as the input.
- For "intrinsic vs extrinsic" (2020) use a 3-row table: purity / doping, n_e vs n_h , conductivity at room temperature.

Board question bank **Section A – Multiple choice (½ mark each)**

[RBSE year] = asked in that official BSER paper. Answers and solutions at the end of the chapter.

- [RBSE 2025]** In an n-type semiconductor the majority charge carriers are
 (a) protons (b) holes (c) neutrons (d) electrons
- [RBSE 2025]** At 0 K the conductivity of an intrinsic semiconductor is
 (a) maximum (b) infinite
 (c) equal to that of a metal (d) zero
- [RBSE 2024]** Which of the following is an inorganic semiconductor?
 (a) doped phthalocyanine (b) CdS
 (c) anthracene (d) polypyrrole

- 4 [RBSE 2023] The output of an AND gate is 1 when its inputs are
 (a) both 1 (b) both 0 (c) $A = 1, B = 0$ (d) $A = 0, B = 1$
- 5 The energy gap of silicon is about
 (a) 0.67 eV (b) 0.01 eV (c) 1.1 eV (d) 5.4 eV
- 6 Boron is added to pure silicon. The resulting semiconductor is
 (a) an insulator
 (b) negatively charged
 (c) n-type, with electrons as majority carriers
 (d) p-type, with holes as majority carriers
- 7 On forward biasing a p-n junction diode, the width of the depletion layer
 (a) decreases (b) does not change
 (c) becomes infinite (d) increases
- 8 A full-wave rectifier is fed with a 50 Hz ac. The frequency of the output ripple is
 (a) 200 Hz (b) 100 Hz (c) 50 Hz (d) 25 Hz
- 9 A Zener diode is used as a voltage regulator when it is operated in
 (a) reverse breakdown region (b) forward bias above the knee
 (c) forward bias below the knee (d) zero bias
- 10 For an intrinsic semiconductor $n_i = 2 \times 10^{16} \text{ m}^{-3}$. After doping $n_e = 10^{22} \text{ m}^{-3}$. The hole concentration is
 (a) $2 \times 10^{16} \text{ m}^{-3}$ (b) $4 \times 10^{10} \text{ m}^{-3}$ (c) $4 \times 10^{14} \text{ m}^{-3}$ (d) 10^{22} m^{-3}

Fill in the blanks (½ mark)

Statement	Answer
[RBSE 2025, 2023] The resistivity of semiconductors lies in the range _____ $\Omega \text{ m}$.	10^{-5} to 10^6
[RBSE 2024] The two types of extrinsic semiconductors are _____ and _____.	n-type, p-type
The resistivity of a semiconductor _____ with increase in temperature.	decreases
A pentavalent impurity in a semiconductor is called a _____ impurity.	donor
A diode is used as a rectifier in _____ bias.	forward
The barrier potential of a silicon p-n junction is about _____ V.	0.7

Very short answer (1 mark)

Question	Model answer
What is an intrinsic semiconductor?	A pure semiconductor with $n_e = n_h$ (e.g. pure Si or Ge).
What is doping?	Adding a small amount of a suitable impurity to a pure semiconductor to increase its conductivity.
What is the depletion layer?	The thin region near a p-n junction which has no mobile charge carriers, only fixed ions.
Name the device that converts ac into dc.	Rectifier (a p-n junction diode).
Why is a Zener diode heavily doped?	To make the depletion layer very thin so that breakdown occurs at a low, sharp voltage V_Z .
Why is an LED made of a material with $E_g \approx 2$ eV for visible light?	Emitted photon energy $h\nu \approx E_g$; 2 eV corresponds to $\lambda \approx 1240/2 = 620$ nm (visible).

Short answer (Section B, 1½ marks)

 Short answer (1½ marks)

[RBSE 2025] Distinguish between an insulator and a semiconductor on the basis of energy bands.

Solution: (i) Insulator: $E_g > 3$ eV, CB empty, VB full; no conduction. (½) (ii) Semiconductor: $E_g < 3$ eV (≈ 1 eV); thermal energy moves some electrons to CB, giving conduction that increases with temperature. (½) (iii) At 0 K both are insulating; the resistivity of an insulator $\sim 10^{11} - 10^{19} \Omega \text{ m}$, of a semiconductor $\sim 10^{-5} - 10^6 \Omega \text{ m}$. (½)

 Short answer (1½ marks)

[RBSE 2020] Distinguish between intrinsic and extrinsic semiconductors (three points).

Solution: (i) Intrinsic: pure; extrinsic: doped with a trivalent / pentavalent impurity. (ii) Intrinsic: $n_e = n_h$; extrinsic: $n_e \gg n_h$ (n-type) or $n_h \gg n_e$ (p-type). (iii) Intrinsic: low conductivity depending strongly on temperature; extrinsic: much higher conductivity, controlled by doping. (½ each)

 Short answer (1½ marks)

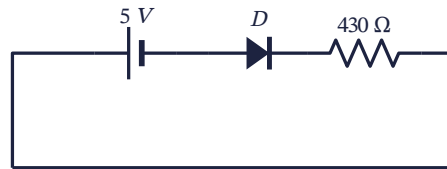
Draw the I–V characteristic of a p-n junction diode in forward and reverse bias and define the knee voltage.

Solution: Graph as in 14.6: forward current negligible then exponential rise; reverse current tiny and constant (μA) until breakdown; axes labelled. (1) **Knee (threshold) voltage:** the forward voltage beyond which the current rises rapidly (≈ 0.7 V Si, 0.3 V Ge). (½)

 Short answer (1½ marks)

In a half-wave rectifier the input is 50 Hz. What is the output frequency and why is the output not pure dc?

Solution: The diode conducts only in positive half-cycles, giving one pulse per cycle $\Rightarrow$ output frequency = **50 Hz**. (1) The output is a train of pulses (pulsating dc, with ac ripple); a filter is needed to smooth it. (½)



Si diode in forward bias with a 430 Ω resistor

🔧 Short answer (1½ marks)

Find the current in the circuit shown (Si diode, 0.7 V drop).

Solution: The diode is forward biased (anode towards the + terminal). $(\frac{1}{2}) I = \frac{5 - 0.7}{430} (\frac{1}{2}) = 10 \text{ mA. } (\frac{1}{2})$

✓ Answer Key & Quick Solutions

1(d) 2(d) 3(b) 4(a) 5(c) 6(d) 7(a) 8(b) 9(a) 10(b)

1. (d) Pentavalent donor atoms supply free electrons, so electrons are the majority carriers.
2. (d) At 0 K all bonds are intact: no free electrons or holes, so it behaves as an insulator.
3. (b) CdS (like Ge, Si, GaAs) is inorganic; the others are organic semiconductors.
4. (a) AND: $Y = A \cdot B = 1$ only if $A = B = 1$.
5. (c) Si: 1.1 eV; Ge: 0.67 eV; diamond (insulator): ~ 5.4 eV.
6. (d) Boron is trivalent (acceptor): each atom creates a hole $\Rightarrow$ p-type; the crystal remains neutral.
7. (a) The applied field opposes the internal field, narrowing the layer and lowering the barrier.
8. (b) Both half-cycles give pulses $\Rightarrow$ output frequency $= 2 \times 50 = 100$ Hz.
9. (a) In breakdown the voltage across it is constant while the current varies.
10. (b) $n_h = n_i^2 / n_e = (4 \times 10^{32}) / 10^{22} = 4 \times 10^{10} \text{ m}^{-3}$.

🧠 Last-minute revision

- Resistivity: metals 10^{-8} – 10^{-2} ; semiconductors 10^{-5} – 10^6 ; insulators 10^{11} – $10^{19} \Omega \text{ m}$.
- E_g : metals 0; semiconductors < 3 eV (Si 1.1, Ge 0.67); insulators > 3 eV.
- Intrinsic: $n_e = n_h = n_i$; at 0 K conductivity = 0. $n_e n_h = n_i^2$.
- n-type: pentavalent donor, electrons majority, E_D below CB. p-type: trivalent acceptor, holes majority, E_A above VB.
- p-n junction: diffusion $\rightarrow$ depletion layer $\rightarrow$ barrier V_0 (Si 0.7 V, Ge 0.3 V); forward narrows, reverse widens.
- Half-wave: 1 diode, output f; full-wave: 2 diodes, output 2f; filter = capacitor.
- Zener: reverse breakdown, heavily doped, $V_{out} = V_Z, I_S = I_Z + I_L$.
- LED forward, $h\nu \approx E_g$; photodiode reverse; solar cell no bias. Gates (old): OR, AND, NOT, NAND, NOR.

DAY
25

RBSE CLASS 12 PHYSICS • FULL PAPER

Model Paper 1

SS-40 • Physics

3 h 15 min

56 marks

18 questions

2025 pattern

Original paper in the 2025 BSER pattern. Write it in one sitting with a clock, then mark yourself with the marking scheme that follows.

Instructions to candidates

 Paper details

- Subject: **Physics** (theory) • Class 12 • Code SS-40
- Time: **3 hours 15 minutes** • Maximum marks: **56**
- **18** questions, all compulsory; internal choice only in Q.14–18.
- Q.1: 18 MCQ (½ each) • Q.2: 10 blanks (½ each) • Q.3: 10 very short (1 each)
- Q.4–13: short answer (1½ each) • Q.14–16: 3 marks • Q.17–18: 4 marks

 Constants you may use

- $e = 1.6 \times 10^{-19} \text{ C}$, $c = 3 \times 10^8 \text{ m/s}$,
 $h = 6.63 \times 10^{-34} \text{ J s}$
- $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$
- $m_e = 9.1 \times 10^{-31} \text{ kg}$, $1 \text{ u} = 931.5 \text{ MeV}/c^2$,
 $hc = 12400 \text{ eV Å}$
- Draw neat labelled diagrams wherever needed. Write the units in every numerical answer.
- If the Hindi and English versions differ, the Hindi version is valid.

Blueprint of this paper (chapter-wise marks = RBSE weightage)

Ch	Chapter	Marks	Question numbers
1	Electric Charges and Fields	4	1(1), 1(2), 2(1), 3(1), 4
2	Electrostatic Potential and Capacitance	3	1(3), 3(2), 5
3	Current Electricity	4	1(4), 2(2), 14
4	Moving Charges and Magnetism	4	1(5), 2(3), 15
5	Magnetism and Matter	3	1(6), 3(3), 6
6	Electromagnetic Induction	4	1(7), 1(8), 2(4), 3(4), 7
7	Alternating Current	5	1(9), 2(5), 17
8	Electromagnetic Waves	2	1(10), 2(6), 3(5)
9	Ray Optics and Optical Instruments	7	1(11), 3(6), 8, 18
10	Wave Optics	5	1(12), 2(7), 3(7), 9, 10
11	Dual Nature of Radiation and Matter	4	1(13), 2(8), 16
12	Atoms	4	1(14), 1(15), 2(9), 3(8), 11
13	Nuclei	3	1(16), 3(9), 12
14	Semiconductor Electronics	4	1(17), 1(18), 2(10), 3(10), 13
Total		56	

The question paper

Section A (Q.1–3 • 24 marks)



Q.1 Choose the correct option ($18 \times \frac{1}{2} = 9$)

Write the letter and the answer in your answer-book. Items are numbered 1–18.

- 1 A charge of $8.85 \mu\text{C}$ is placed at the centre of a cube. The total electric flux through the cube is ($\epsilon_0 = 8.85 \times 10^{-12} \text{ SI}$)
 (a) $8.85 \times 10^6 \text{ N m}^2/\text{C}$ (b) $1 \times 10^{12} \text{ N m}^2/\text{C}$ (c) $1 \times 10^{-6} \text{ N m}^2/\text{C}$ (d) $1 \times 10^6 \text{ N m}^2/\text{C}$
- 2 The torque on an electric dipole placed in a uniform electric field is maximum when the angle between $\vec{p}$ and $\vec{E}$ is
 (a) 180° (b) 45° (c) 0° (d) 90°
- 3 The energy stored in a $10 \mu\text{F}$ capacitor charged to 100 V is
 (a) 5 J (b) 0.5 J (c) 0.05 J (d) 0.1 J
- 4 A uniform wire of resistance R is uniformly stretched to double its length (its volume stays constant). Its new resistance is
 (a) $R/2$ (b) $4R$ (c) $R/4$ (d) $2R$
- 5 A straight wire 0.5 m long carries 2 A and lies perpendicular to a uniform magnetic field of 0.4 T . The force on it is
 (a) 0.4 N (b) 0.2 N (c) 0.8 N (d) 1.6 N
- 6 The magnetic susceptibility of a diamagnetic substance is
 (a) small and negative (b) large and positive
 (c) exactly zero (d) small and positive
- 7 A conducting rod 0.5 m long moves at 4 m/s perpendicular to a magnetic field of 0.2 T (and perpendicular to its own length). The emf induced across it is
 (a) 2 V (b) 0.8 V (c) 0.4 V (d) 0.1 V
- 8 The unit henry of self-inductance is equivalent to
 (a) J A (b) V A^{-1} (c) Wb A^{-1} (d) Wb A
- 9 An AC supply has an rms voltage of 220 V . Its peak voltage is nearly
 (a) 440 V (b) 155 V (c) 311 V (d) 220 V
- 10 In an electromagnetic wave travelling in vacuum, the ratio E_0 / B_0 is equal to
 (a) c^2 (b) $1/c$
 (c) speed of light c (d) μ_0
- 11 The power of a convex lens of focal length 20 cm is
 (a) -5 D (b) $+5 \text{ D}$ (c) $+0.2 \text{ D}$ (d) $+20 \text{ D}$

- 12 In Young's double-slit experiment $\lambda = 6000 \text{ \AA}$, $D = 1 \text{ m}$ and $d = 1 \text{ mm}$. The fringe width is
 (a) 1.2 mm (b) 0.06 mm (c) 6 mm (d) 0.6 mm
- 13 The energy of a photon of wavelength $4000 \text{ \AA}$ is
 (a) 0.31 eV (b) 1.55 eV (c) 6.2 eV (d) 3.1 eV
- 14 The energy of the electron in the second orbit ($n = 2$) of a hydrogen atom is
 (a) -6.8 eV (b) -13.6 eV (c) -1.51 eV (d) -3.4 eV
- 15 The number of different spectral lines possible when an electron falls from $n = 4$ to the ground state is
 (a) 10 (b) 3 (c) 4 (d) 6
- 16 With $R_0 = 1.2 \text{ fm}$, the radius of a nucleus of mass number 125 is
 (a) 3.0 fm (b) 6.0 fm (c) 12 fm (d) 30 fm
- 17 When a p-n junction is forward biased, the width of its depletion region
 (a) decreases (b) remains unchanged
 (c) becomes infinite (d) increases
- 18 The forbidden energy gap of silicon at room temperature is about
 (a) 0.67 eV (b) 0 eV (c) 5 eV (d) 1.1 eV

Q.2 Fill in the blanks ($10 \times \frac{1}{2} = 5$)

- The electric field inside a charged conducting spherical shell is _____.
- The SI unit of electrical conductivity is _____.
- In a cyclotron the time period of revolution of a charged particle does not depend on its _____.
- The direction of the induced emf is given by _____ law.
- At resonance in a series LCR circuit the impedance is _____ and equal to R.
- The ozone layer absorbs _____ radiation coming from the Sun.
- If white light is used in Young's experiment, the central fringe is _____.
- The saturation photocurrent is directly proportional to the _____ of the incident light.
- The Balmer series of the hydrogen spectrum lies in the _____ region.
- For forward bias, the p-side of a junction diode is connected to the _____ terminal of the battery.

Q.3 Very short answer questions ($10 \times 1 = 10$)

- Why can two electric field lines never cross each other?
- Define one farad.
- State Gauss's law for magnetism.
- Write two factors on which the self-inductance of a long solenoid depends.
- Name the electromagnetic radiation used in water purifiers to kill germs.
- Define the critical angle.
- State Malus's law.
- Write the expression for the energy of the electron in the n -th orbit of a hydrogen atom.
- Define the atomic mass unit (u).
- What is meant by doping of a semiconductor?

Section B (Q.4–13 • short answer • $10 \times 1\frac{1}{2} = 15$ marks)

- Q.4** Find the electric field at a distance of 20 cm from an infinitely long straight wire carrying a uniform linear charge density of 10 nC/m. Give its direction. [1½]
- Q.5** Three capacitors of $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are joined in series across a 12 V battery. Find the equivalent capacitance and the charge on each capacitor. [1½]
- Q.6** Define (i) magnetic declination, (ii) angle of dip and (iii) horizontal component of the earth's magnetic field. [1½]
- Q.7** A coil of 100 turns and area $0.02\ \text{m}^2$ is held perpendicular to a magnetic field of 0.5 T. The field is reduced to zero in 0.1 s. Find the average induced emf. [1½]
- Q.8** An object is placed 30 cm in front of a convex mirror of focal length 20 cm. Find the position of the image and the magnification. [1½]
- Q.9** What is a wavefront? State Huygens' principle. What is the shape of the wavefront due to a point source in a homogeneous medium? [1½]
- Q.10** A single slit of width 0.2 mm is illuminated by light of wavelength $5000\ \text{\AA}$. The screen is 1 m away. Find the width of the central maximum. [1½]
- Q.11** Calculate the wavelength of the first line (H_{α} , $n = 3 \rightarrow 2$) of the Balmer series of hydrogen. ($R = 1.097 \times 10^7\ \text{m}^{-1}$) [1½]
- Q.12** Calculate the binding energy of the deuteron. Masses: $m({}_1^1\text{H}) = 1.007825\ \text{u}$, $m_n = 1.008665\ \text{u}$, $m({}_1^2\text{H}) = 2.014102\ \text{u}$; $1\ \text{u} = 931.5\ \text{MeV}/c^2$. [1½]
- Q.13** Write three differences between an n-type and a p-type semiconductor. [1½]

Section C (Q.14–16 • long answer • $3 \times 3 = 9$ marks)

- Q.14** (a) Draw a Wheatstone bridge and, using Kirchhoff's laws, obtain its balance condition. [3]
- OR —
- (b) Two cells of emf 3 V (internal resistance $1\ \Omega$) and 2 V (internal resistance $0.5\ \Omega$) are connected in series (aiding) with an external resistance of $3.5\ \Omega$. Derive the equivalent emf and internal resistance of the series combination and find the current and the terminal voltage of each cell.
- Q.15** (a) State Ampere's circuital law. Use it to obtain the magnetic field at a distance r from a long straight wire carrying current I . [3]
- OR —
- (b) Obtain an expression for the force per unit length between two long parallel current-carrying wires. Hence define the ampere.
- Q.16** (a) State three experimental features of the photoelectric effect that the wave theory cannot explain, and show how Einstein's photoelectric equation explains them. [3]

— OR —

- (b) Derive the de Broglie wavelength of an electron accelerated from rest through a potential difference V , in the form $\lambda = 1.227 / \sqrt{V}\ \text{nm}$. Find the wavelength for $V = 150\ \text{V}$.

Section D (Q.17–18 • essay type • $2 \times 4 = 8$ marks)

Q.17 (a) A series LCR circuit is connected to a source $v = v_0 \sin \omega t$. With the help of a phasor diagram obtain the expressions for the impedance of the circuit and the phase angle between voltage and current. **[4]**

— OR —

(b) Derive the expression for the average power of an AC circuit, $\bar{P} = V_{rms} I_{rms} \cos \phi$. A series circuit has $R = 30 \, \Omega$, $X_L = 100 \, \Omega$, $X_C = 60 \, \Omega$ and is connected to a 200 V rms supply. Find the impedance, rms current, power factor and average power.

Q.18 (a) Derive the lens maker's formula $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ for a thin convex lens of refractive index n placed in air. **[4]**

— OR —

(b) Draw a labelled ray diagram of a compound microscope forming the final image at the near point and derive its magnifying power. Find it for $f_o = 1.0 \, \text{cm}$, $f_e = 5.0 \, \text{cm}$, tube length $L = 20 \, \text{cm}$ and $D = 25 \, \text{cm}$.

— End of paper —

Solutions and marking scheme

How to use the marking scheme

- Step marks are shown as (½), (1) at the end of each step, the way an examiner awards them. A missing unit or a missing diagram costs ½ mark.
- Any other correct method and answer earns full marks. Total your score: Section A /24, B /15, C /9, D /8.

✓ Q.1 – Answer key and quick solutions

1(d) 2(d) 3(c) 4(b) 5(a) 6(a) 7(c) 8(c) 9(c) 10(c) 11(b) 12(d) 13(d) 14(d) 15(d)
16(b) 17(a) 18(d)

- (d) Gauss's law: $\Phi = Q / \epsilon_0 = 8.85 \times 10^{-6} / 8.85 \times 10^{-12} = 10^6 \text{ N m}^2/\text{C}$.
- (d) $\tau = pE \sin \theta$ is largest for $\sin \theta = 1$, i.e. $\theta = 90^\circ$.
- (c) $U = \frac{1}{2} CV^2 = \frac{1}{2} (10 \times 10^{-6}) (100)^2 = 0.05 \text{ J}$.
- (b) $R = \rho l / A$ and lA is constant: $l \rightarrow 2l, A \rightarrow A / 2$, so $R' = \rho(2l) / (A / 2) = 4R$.
- (a) $F = BIl \sin 90^\circ = 0.4 \times 2 \times 0.5 = 0.4 \text{ N}$.
- (a) Diamagnetic materials get weakly magnetised opposite to the applied field, so χ is small and negative (about -10^{-5}).
- (c) $\varepsilon = Blv = 0.2 \times 0.5 \times 4 = 0.4 \text{ V}$.
- (c) $L = \Phi / I$, so $1 \text{ H} = 1 \text{ Wb/A} (= 1 \text{ V s/A})$.
- (c) $V_0 = \sqrt{2} V_{rms} = 1.414 \times 220 \approx 311 \text{ V}$.
- (c) For an EM wave $E_0 = cB_0$, so the ratio is the speed of light.
- (b) $P = 1 / f(m) = 1 / 0.20 = +5 \text{ D}$ (convex lens: positive).
- (d) $\beta = \lambda D / d = (6 \times 10^{-7})(1) / (10^{-3}) = 6 \times 10^{-4} \text{ m} = 0.6 \text{ mm}$.
- (d) $E = 12400 / 4000 = 3.1 \text{ eV}$ (using $hc = 12400 \text{ eV \AA}$).
- (d) $E_n = -13.6 / n^2$; $E_2 = -13.6 / 4 = -3.4 \text{ eV}$.
- (d) Number of lines $= \frac{n(n-1)}{2} = \frac{4 \times 3}{2} = 6$.
- (b) $R = R_0 A^{1/3} = 1.2 \times 125^{1/3} = 1.2 \times 5 = 6.0 \text{ fm}$.
- (a) The applied field opposes the barrier field, pushes majority carriers towards the junction and narrows the depletion layer.
- (d) Si has $E_g \approx 1.1 \text{ eV}$ (Ge: 0.67 eV ; insulators: more than about 3 eV).

Q.2 – Fill in the blanks: answers (½ each)

No.	Ch	Answer
1	1	zero
2	3	$\Omega^{-1} \text{ m}^{-1}$ (S m^{-1})
3	4	speed (and the radius of its path)
4	6	Lenz's
5	7	minimum
6	8	ultraviolet
7	10	white (bright)
8	11	intensity
9	12	visible
10	14	positive

Q.3 – Very short answers (1 mark each)

No.	Ch	Answer
1	1	At the point of intersection the field would have two directions, but the electric field at a point has only one definite direction.
2	2	The capacitance of a capacitor which needs 1 coulomb of charge to raise its potential difference by 1 volt.
3	5	The net magnetic flux through any closed surface is zero, $\oint \vec{B} \cdot d\vec{S} = 0$ (isolated magnetic monopoles do not exist).
4	6	Number of turns per unit length (and total turns), area of cross-section, length and permeability of the core ($L = \mu_0 n^2 Al$). Any two.
5	8	Ultraviolet rays.
6	9	The angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90° .
7	10	When plane-polarised light of intensity I_0 passes through an analyser, the transmitted intensity is $I = I_0 \cos^2 \theta$, θ being the angle between the pass-axes.
8	12	$E_n = -\frac{13.6}{n^2} \text{ eV}$
9	13	1 u is $\frac{1}{12}$ of the mass of one carbon-12 atom: $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$.
10	14	Adding a small, controlled amount of a suitable impurity to a pure semiconductor to increase its conductivity.

Q.4–13 – Short answers (1½ marks each)

 Q.4 • Ch 1 • 1½ marks

Find the electric field at a distance of 20 cm from an infinitely long straight wire carrying a uniform linear charge density of 10 nC/m. Give its direction.

Solution: $E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$ (½)

$E = \frac{2 \times 9 \times 10^9 \times 10 \times 10^{-9}}{0.20}$ (½)

$E = 900 \text{ N/C}$ (½); directed radially **outward** from the wire (for positive λ).

 Q.5 • Ch 2 • 1½ marks

Three capacitors of $2 \mu\text{F}$, $3 \mu\text{F}$ and $6 \mu\text{F}$ are joined in series across a 12 V battery. Find the equivalent capacitance and the charge on each capacitor.

Solution: Series: $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = 1$ (½)

$C = 1 \mu\text{F}$ (½)

Same charge on each: $Q = CV = 1 \mu\text{F} \times 12 \text{ V} = 12 \mu\text{C}$ (½)

(Check: $V_1 + V_2 + V_3 = 6 + 4 + 2 = 12 \text{ V}$.)

 Q.6 • Ch 5 • 1½ marks

Define (i) magnetic declination, (ii) angle of dip and (iii) horizontal component of the earth's magnetic field.

Solution: (i) **Declination:** the angle between the magnetic meridian and the geographic meridian at a place. (½)

(ii) **Dip (inclination):** the angle which the earth's total magnetic field makes with the horizontal at a place. (½)

(iii) **Horizontal component:** the component of the earth's field along the horizontal in the magnetic meridian,

$B_H = B_e \cos \delta$. (½)

 Q.7 • Ch 6 • 1½ marks

A coil of 100 turns and area 0.02 m^2 is held perpendicular to a magnetic field of 0.5 T . The field is reduced to zero in 0.1 s . Find the average induced emf.

Solution: $\varepsilon = -N \frac{\Delta\Phi}{\Delta t} = N \frac{A \Delta B}{\Delta t}$ (½)

$$\varepsilon = \frac{100 \times 0.02 \times (0.5 - 0)}{0.1} \text{ (½)}$$

$$\varepsilon = 10 \text{ V (½)}$$

 Q.8 • Ch 9 • 1½ marks

An object is placed 30 cm in front of a convex mirror of focal length 20 cm . Find the position of the image and the magnification.

Solution: $f = +20 \text{ cm}$, $u = -30 \text{ cm}$. $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} + \frac{1}{30} = \frac{5}{60}$ (½)

$$v = +12 \text{ cm (behind the mirror, virtual) (½)}$$

$$m = -\frac{v}{u} = -\frac{12}{-30} = +0.4 \text{ (erect, diminished) (½)}$$

 Q.9 • Ch 10 • 1½ marks

What is a wavefront? State Huygens' principle. What is the shape of the wavefront due to a point source in a homogeneous medium?

Solution: Wavefront: the locus of all points that vibrate in the same phase at a given instant. (½)

Huygens' principle: every point on a wavefront acts as a source of secondary wavelets; the new wavefront at a later time is the forward envelope (common tangent surface) of these wavelets. (½)

Point source in a homogeneous medium: **spherical** wavefronts. (½)

 Q.10 • Ch 10 • 1½ marks

A single slit of width 0.2 mm is illuminated by light of wavelength 5000 Å . The screen is 1 m away. Find the width of the central maximum.

Solution: First minimum: $a \sin \theta = \lambda$, so $\theta \approx \lambda / a = \frac{5 \times 10^{-7}}{2 \times 10^{-4}} = 2.5 \times 10^{-3} \text{ rad}$ (½)

$$\text{Width of central maximum} = 2\theta D = \frac{2\lambda D}{a} \text{ (½)}$$

$$= \frac{2 \times 5 \times 10^{-7} \times 1}{2 \times 10^{-4}} = 5 \times 10^{-3} \text{ m} = 5 \text{ mm (½)}$$

 Q.11 • Ch 12 • 1½ marks

Calculate the wavelength of the first line (H_{α} , $n = 3 \rightarrow 2$) of the Balmer series of hydrogen. ($R = 1.097 \times 10^7 \text{ m}^{-1}$)

Solution: $\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R \cdot \frac{5}{36}$ (½)

$$\lambda = \frac{36}{5 \times 1.097 \times 10^7} \text{ (½)}$$

$$\lambda = 6.56 \times 10^{-7} \text{ m} = 6563 \text{ Å (red, visible) (½)}$$

Q.12 • Ch 13 • 1½ marks

Calculate the binding energy of the deuteron. Masses: $m({}_1^1\text{H}) = 1.007825 \text{ u}$, $m_n = 1.008665 \text{ u}$, $m({}_1^2\text{H}) = 2.014102 \text{ u}$; $1 \text{ u} = 931.5 \text{ MeV}/c^2$.

Solution: $\Delta m = m({}_1^1\text{H}) + m_n - m({}_1^2\text{H}) = 1.007825 + 1.008665 - 2.014102 = 0.002388 \text{ u}$ (½)

$E_b = \Delta m \times 931.5 = 0.002388 \times 931.5$ (½)

$E_b \approx 2.22 \text{ MeV}$ (about 1.11 MeV per nucleon) (½)

Q.13 • Ch 14 • 1½ marks

Write three differences between an n-type and a p-type semiconductor.

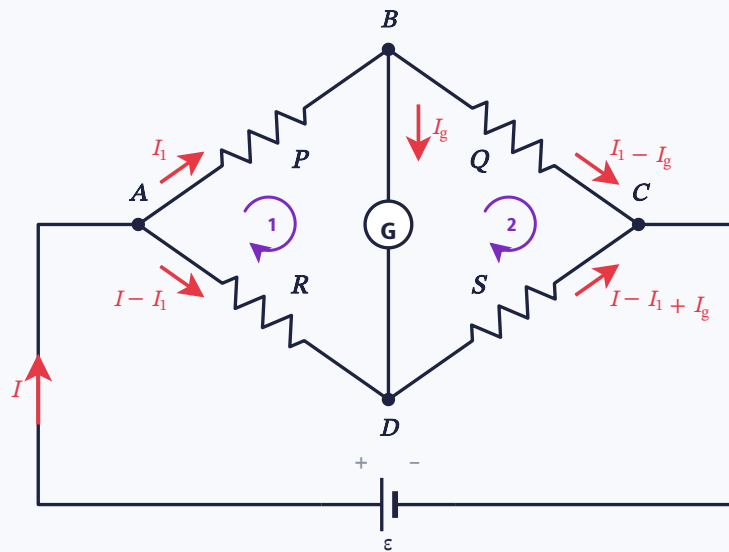
Solution: (1) **Dopant:** n-type – pentavalent (P, As, Sb); p-type – trivalent (B, Al, In). (½)

(2) **Majority carriers:** n-type – electrons (minority: holes); p-type – holes (minority: electrons). (½)

(3) **Impurity level:** n-type – donor level just below the conduction band; p-type – acceptor level just above the valence band. (½)

Q.14–16 – Long answers (3 marks each)**Q.14 (a) • Ch 3 • 3 marks**

Draw a Wheatstone bridge and, using Kirchhoff's laws, obtain its balance condition.



Solution: (i) Diagram: P, Q, R, S in arms AB, BC, AD, DC; galvanometer G between B and D; cell between A and C. (½)

(ii) Junction rule: current I splits at A into I_1 (in P) and $I - I_1$ (in R); at B, I_g goes through G, so Q carries $I_1 - I_g$ and S carries $I - I_1 + I_g$. (½)

(iii) Loop rule for loop ABDA: $-I_1P - I_gG + (I - I_1)R = 0$ (½)

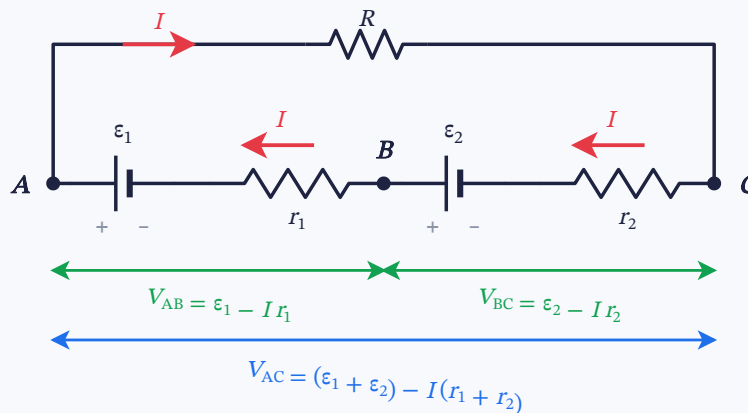
(iv) Loop rule for loop BCDB: $-(I_1 - I_g)Q + (I - I_1 + I_g)S + I_gG = 0$ (½)

(v) At balance $I_g = 0$: the two loop equations give $I_1P = (I - I_1)R$ and $I_1Q = (I - I_1)S$. (½)

(vi) Dividing: $\frac{P}{Q} = \frac{R}{S}$ (½)

 Q.14 (b) • Ch 3 • 3 marks

Two cells of emf 3 V (internal resistance 1 Ω) and 2 V (internal resistance 0.5 Ω) are connected in series (aiding) with an external resistance of 3.5 Ω . Derive the equivalent emf and internal resistance of the series combination and find the current and the terminal voltage of each cell.



Solution: Derivation: with current I in the circuit, $V_{AB} = \epsilon_1 - Ir_1$ and $V_{BC} = \epsilon_2 - Ir_2$. (½)

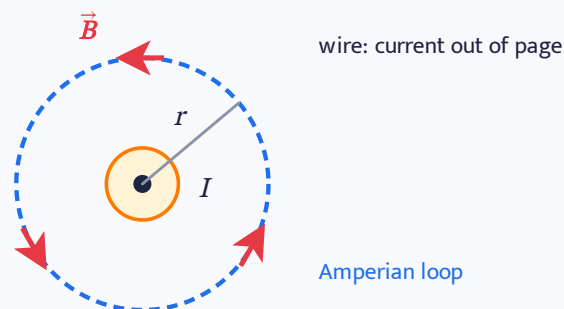
$V_{AC} = V_{AB} + V_{BC} = (\epsilon_1 + \epsilon_2) - I(r_1 + r_2)$, comparing with $V = \epsilon_{eq} - Ir_{eq}$: $\epsilon_{eq} = \epsilon_1 + \epsilon_2$, $r_{eq} = r_1 + r_2$ (1)

Numerical: $\epsilon_{eq} = 3 + 2 = 5$ V, $r_{eq} = 1 + 0.5 = 1.5$ Ω . $I = \frac{\epsilon_{eq}}{R + r_{eq}} = \frac{5}{3.5 + 1.5} = 1$ A (½)

$V_1 = \epsilon_1 - Ir_1 = 3 - 1(1) = 2$ V and $V_2 = \epsilon_2 - Ir_2 = 2 - 1(0.5) = 1.5$ V (1)

 Q.15 (a) • Ch 4 • 3 marks

State Ampere's circuital law. Use it to obtain the magnetic field at a distance r from a long straight wire carrying current I .



Solution: Law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$ - the line integral of $\vec{B}$ around any closed loop equals μ_0 times the current enclosed. (½)

Loop: choose a circular Amperian loop of radius r , coaxial with the wire (figure). (½)

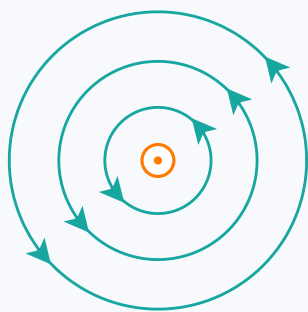
Symmetry: by symmetry $\vec{B}$ is tangential and has the same magnitude B at every point of the loop, so $\vec{B} \parallel d\vec{l}$. (½)

Integral: $\oint B dl = B(2\pi r)$ and $I_{enc} = I$, so $B(2\pi r) = \mu_0 I$. (1)

$B = \frac{\mu_0 I}{2\pi r}$, direction by the right-hand grip rule (circles around the wire). (½)

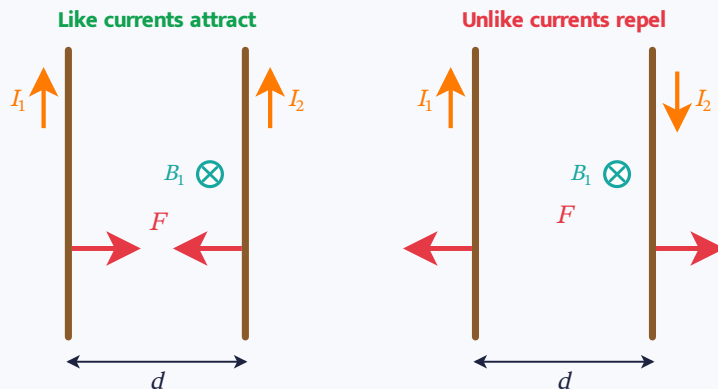
 Q.15 (b) • Ch 4 • 3 marks

Obtain an expression for the force per unit length between two long parallel current-carrying wires. Hence define the ampere.



$I \odot$ out of page: B anticlockwise

$$B = \mu_0 I / 2\pi r$$



$$F/l = \mu_0 I_1 I_2 / 2\pi d$$

Solution: Wires 1 and 2 carry currents I_1, I_2 at separation d . (**figure**)

Field of wire 1 at wire 2: $B_1 = \frac{\mu_0 I_1}{2\pi d}$ (perpendicular to wire 2). ($\frac{1}{2}$)

Force on a length l of wire 2: $F = I_2 l B_1$ (as $\vec{B}_1 \perp$ wire). ($\frac{1}{2}$)

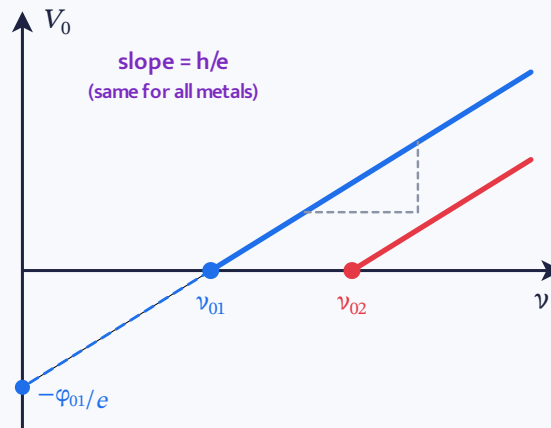
$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d} \quad (\frac{1}{2})$$

Same-direction currents attract; opposite currents repel. ($\frac{1}{2}$)

Ampere: if $I_1 = I_2 = 1$ A and $d = 1$ m give $F / l = 2 \times 10^{-7}$ N/m, then each current is 1 ampere (the current which, in two infinitely long parallel wires 1 m apart in vacuum, produces 2×10^{-7} N per metre). (**1**)

 Q.16 (a) • Ch 11 • 3 marks

State three experimental features of the photoelectric effect that the wave theory cannot explain, and show how Einstein's photoelectric equation explains them.



Solution: Features (½ each):

- (i) Below a **threshold frequency** ν_0 no emission occurs, however intense the light (wave theory: any frequency should work if intensity is large). (½)
- (ii) K_{max} (and stopping potential) depends on **frequency**, not on intensity (wave theory: more intensity $\Rightarrow$ larger energy). (½)
- (iii) Emission is **instantaneous** (about 10^{-9} s) even for dim light (wave theory: a time lag to store energy). (½)

Einstein's explanation: light is a stream of photons of energy $h\nu$; one photon gives all its energy to one electron:

$$h\nu = \phi_0 + K_{max} \quad (½)$$

$K_{max} = h(\nu - \nu_0)$ with $\nu_0 = \phi_0 / h$ explains (i) and (ii); intensity only fixes the number of photons, hence the photocurrent; a photon–electron collision is one-step, explaining (iii). (1)

 Q.16 (b) • Ch 11 • 3 marks

Derive the de Broglie wavelength of an electron accelerated from rest through a potential difference V , in the form $\lambda = 1.227 / \sqrt{V}$ nm. Find the wavelength for $V = 150$ V.

Solution: de Broglie: $\lambda = \frac{h}{p}$. (½)

$$\text{Kinetic energy gained } K = eV = \frac{p^2}{2m} \Rightarrow p = \sqrt{2meV} \quad (½)$$

$$\lambda = \frac{h}{\sqrt{2meV}} \quad (½)$$

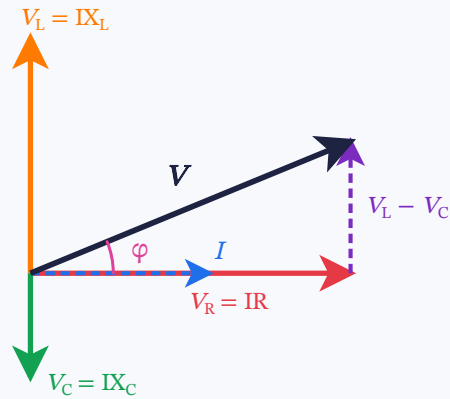
$$\text{Putting } h = 6.63 \times 10^{-34}, m = 9.1 \times 10^{-31}, e = 1.6 \times 10^{-19}: \frac{h}{\sqrt{2me}} = 1.227 \times 10^{-9} \Rightarrow \lambda = \frac{1.227}{\sqrt{V}} \text{ nm} \quad (½)$$

$$V = 150 \text{ V}: \lambda = \frac{1.227}{\sqrt{150}} = \frac{1.227}{12.25} \approx 0.100 \text{ nm} = 1.0 \text{ Å} \quad (1)$$

Q.17–18 – Essay type (4 marks each)

🔧 Q.17 (a) • Ch 7 • 4 marks

A series LCR circuit is connected to a source $v = v_0 \sin \omega t$. With the help of a phasor diagram obtain the expressions for the impedance of the circuit and the phase angle between voltage and current.



Solution: 1. Circuit and phasor diagram (figure: $X_L > X_C$). (1)

2. The same current $i = i_0 \sin \omega t$ flows through R, L, C. Voltage amplitudes: $V_R = i_0 R$ (in phase with i), $V_L = i_0 X_L$ (leads i by 90°), $V_C = i_0 X_C$ (lags i by 90°). (1)

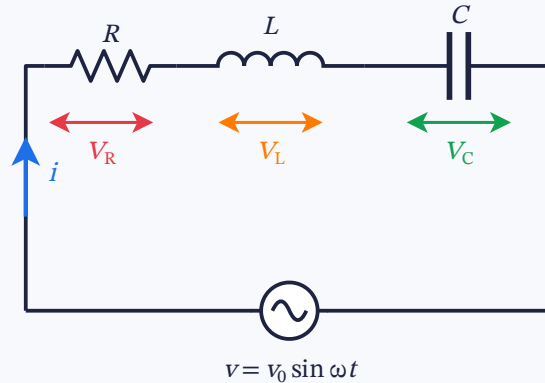
3. $\vec{V}_L$ and $\vec{V}_C$ are opposite, so their resultant is $V_L - V_C$, perpendicular to V_R . Resultant source amplitude:

$$v_0^2 = V_R^2 + (V_L - V_C)^2 = i_0^2 [R^2 + (X_L - X_C)^2]. \quad (1)$$

4. $Z = \frac{v_0}{i_0} = \sqrt{R^2 + (X_L - X_C)^2}$, where $X_L = \omega L$, $X_C = 1 / \omega C$. Phase angle: $\tan \phi = \frac{X_L - X_C}{R}$; $\phi > 0$ (voltage leads) if $X_L > X_C$, $\phi < 0$ if $X_C > X_L$, $\phi = 0$ at resonance. (1)

 Q.17 (b) • Ch 7 • 4 marks

Derive the expression for the average power of an AC circuit, $\bar{P} = V_{rms} I_{rms} \cos \phi$. A series circuit has $R = 30 \Omega$, $X_L = 100 \Omega$, $X_C = 60 \Omega$ and is connected to a 200 V rms supply. Find the impedance, rms current, power factor and average power.



Solution: Derivation: $v = v_0 \sin \omega t$, $i = i_0 \sin(\omega t + \phi)$. Instantaneous power $p = vi = v_0 i_0 \sin \omega t \sin(\omega t + \phi)$ (½)
 $= \frac{v_0 i_0}{2} [\cos \phi - \cos(2\omega t + \phi)]$ (½)

The average of $\cos(2\omega t + \phi)$ over a cycle is zero, so $\bar{P} = \frac{v_0 i_0}{2} \cos \phi = V_{rms} I_{rms} \cos \phi$ (1)

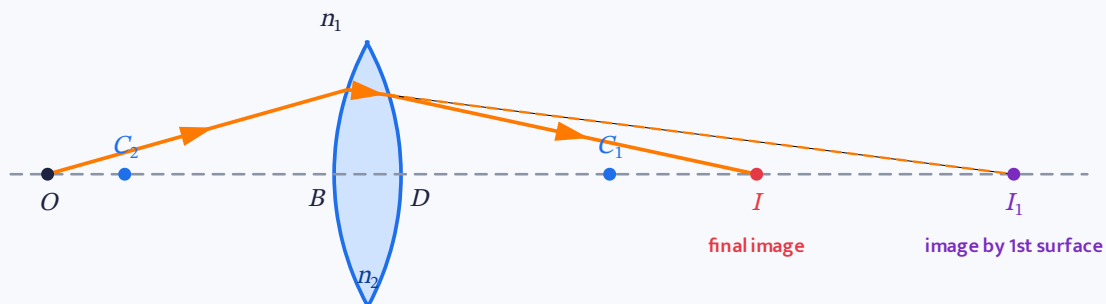
Numerical: $Z = \sqrt{30^2 + (100 - 60)^2} = \sqrt{900 + 1600} = 50 \Omega$ (½)

$I_{rms} = V / Z = 200 / 50 = 4 \text{ A}$, $\cos \phi = R / Z = 30 / 50 = 0.6$ (½)

$\bar{P} = 200 \times 4 \times 0.6 = 480 \text{ W}$ (check: $I^2 R = 16 \times 30 = 480 \text{ W}$) (1)

 Q.18 (a) • Ch 9 • 4 marks

Derive the lens maker's formula $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ for a thin convex lens of refractive index n placed in air.



Solution: 1. Ray diagram (figure): object O, first surface (radius R_1) forms image I_1 , second surface (radius R_2) forms final image I; paraxial rays, thin lens so distances are measured from the optical centre. (1)

2. Refraction at surface 1 (air $\rightarrow$ glass), using $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$: $\frac{n}{v_1} - \frac{1}{u} = \frac{n - 1}{R_1}$... (1) (1)

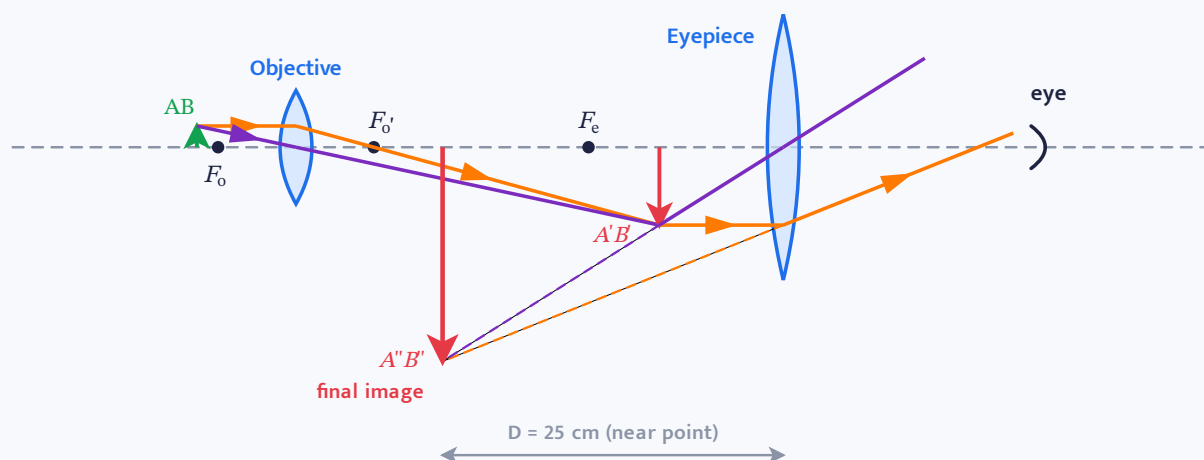
3. Refraction at surface 2 (glass $\rightarrow$ air), with I_1 as virtual object: $\frac{1}{v} - \frac{n}{v_1} = \frac{1 - n}{R_2}$... (2) (1)

4. Adding (1) and (2): $\frac{1}{v} - \frac{1}{u} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$. (½)

5. For $u = \infty$, $v = f$: $\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ (½)

 Q.18 (b) • Ch 9 • 4 marks

Draw a labelled ray diagram of a compound microscope forming the final image at the near point and derive its magnifying power. Find it for $f_o = 1.0$ cm, $f_e = 5.0$ cm, tube length $L = 20$ cm and $D = 25$ cm.



Solution: 1. Labelled diagram (figure): objective f_o forms a real, inverted, magnified image $A'B'$ just inside F_e of the eyepiece; the eyepiece acts as a simple microscope giving the final virtual image $A''B''$ at D. (1)

2. Objective: object just beyond f_o , so $u_o \approx f_o$ and $v_o \approx L$; $m_o = \frac{v_o}{u_o} \approx \frac{L}{f_o}$ (magnitude). (1)

3. Eyepiece (image at near point): $m_e = 1 + \frac{D}{f_e}$. (½)

4. $M = m_o m_e = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$ (image inverted; magnitude taken) (½)

5. $M = \frac{20}{1.0} \left(1 + \frac{25}{5.0} \right) = 20 \times 6 = 120$ (1)

Score sheet

Section	Maximum	Your marks
A (Q.1–3)	24	
B (Q.4–13)	15	
C (Q.14–16)	9	
D (Q.17–18)	8	
Total	56	

Score bands

48+ excellent – polish diagrams and units • **40–47** good – recheck every numerical • **below 40** – revise the derivations and formula boxes of the chapters where you lost marks, then retry this paper after 3 days.

DAY
26

RBSE CLASS 12 PHYSICS • FULL PAPER

Model Paper 2

SS-40 • Physics

3 h 15 min

56 marks

18 questions

2025 pattern

Original paper in the 2025 BSER pattern. Write it in one sitting with a clock, then mark yourself with the marking scheme that follows.

Instructions to candidates

 Paper details

- Subject: **Physics** (theory) • Class 12 • Code SS-40
- Time: **3 hours 15 minutes** • Maximum marks: **56**
- **18** questions, all compulsory; internal choice only in Q.14–18.
- Q.1: 18 MCQ (½ each) • Q.2: 10 blanks (½ each) • Q.3: 10 very short (1 each)
- Q.4–13: short answer (1½ each) • Q.14–16: 3 marks • Q.17–18: 4 marks

 Constants you may use

- $e = 1.6 \times 10^{-19} \text{ C}$, $c = 3 \times 10^8 \text{ m/s}$,
 $h = 6.63 \times 10^{-34} \text{ J s}$
- $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$
- $m_e = 9.1 \times 10^{-31} \text{ kg}$, $1 \text{ u} = 931.5 \text{ MeV}/c^2$,
 $hc = 12400 \text{ eV Å}$
- Draw neat labelled diagrams wherever needed. Write the units in every numerical answer.
- If the Hindi and English versions differ, the Hindi version is valid.

Blueprint of this paper (chapter-wise marks = RBSE weightage)

Ch	Chapter	Marks	Question numbers
1	Electric Charges and Fields	4	1(1), 2(1), 14
2	Electrostatic Potential and Capacitance	3	1(2), 3(1), 4
3	Current Electricity	4	1(3), 1(4), 2(2), 3(2), 5
4	Moving Charges and Magnetism	4	1(5), 2(3), 15
5	Magnetism and Matter	3	1(6), 3(3), 6
6	Electromagnetic Induction	4	1(7), 1(8), 2(4), 3(4), 7
7	Alternating Current	5	1(9), 2(5), 17
8	Electromagnetic Waves	2	1(10), 2(6), 3(5)
9	Ray Optics and Optical Instruments	7	1(11), 3(6), 8, 18
10	Wave Optics	5	1(12), 2(7), 3(7), 9, 10
11	Dual Nature of Radiation and Matter	4	1(13), 1(14), 2(8), 3(8), 11
12	Atoms	4	1(15), 1(16), 2(9), 3(9), 12
13	Nuclei	3	1(17), 3(10), 13
14	Semiconductor Electronics	4	1(18), 2(10), 16
Total		56	

The question paper

Section A (Q.1–3 • 24 marks)



Q.1 Choose the correct option ($18 \times \frac{1}{2} = 9$)

Write the letter and the answer in your answer-book. Items are numbered 1–18.

- 1 Two point charges q and q are at distance r and exert a force F on each other. If both charges and the distance are doubled ($2q, 2q, 2r$), the new force is
(a) $2F$ (b) $F/2$ (c) F (d) $4F$
- 2 A parallel-plate air capacitor has capacitance C_0 . If the space between the plates is completely filled with a dielectric of constant $K = 4$, the capacitance becomes
(a) $4C_0$ (b) $2C_0$ (c) $C_0 / 4$ (d) C_0
- 3 Kirchhoff's junction rule is a consequence of the conservation of
(a) momentum (b) angular momentum
(c) electric charge (d) energy
- 4 Resistors of $6\ \Omega$ and $3\ \Omega$ are connected in parallel across a 6 V battery of negligible internal resistance. The current drawn from the battery is
(a) 3 A (b) 9 A (c) 1 A (d) 2 A
- 5 A proton moves with a speed of $2 \times 10^6\text{ m/s}$ perpendicular to a magnetic field of 0.5 T . The magnetic force on it is
(a) $1.6 \times 10^{-19}\text{ N}$ (b) $1.6 \times 10^{-13}\text{ N}$ (c) $3.2 \times 10^{-13}\text{ N}$ (d) $8 \times 10^{-14}\text{ N}$
- 6 According to Curie's law, the magnetic susceptibility of a paramagnetic substance is
(a) independent of temperature
(b) directly proportional to absolute temperature
(c) proportional to T^2
(d) inversely proportional to absolute temperature
- 7 Which of the following devices works on the principle of eddy currents?
(a) Induction furnace (b) Electric bulb
(c) Electric bell (d) Dry cell
- 8 The current in a coil of self-inductance 2 H changes uniformly from 3 A to 1 A in 0.5 s . The magnitude of the induced emf is
(a) 16 V (b) 4 V (c) 8 V (d) 2 V
- 9 In a purely capacitive AC circuit, the current
(a) leads the voltage by 90° (b) lags the voltage by 45°
(c) is in phase with the voltage (d) lags the voltage by 90°

- 10 The speed of an electromagnetic wave in a medium of relative permittivity 4 and relative permeability 1 is
 (a) 1.5×10^8 m/s (b) 3×10^8 m/s (c) 0.75×10^8 m/s (d) 6×10^8 m/s
- 11 The speed of light in glass of refractive index 1.5 is
 (a) 1.5×10^8 m/s (b) 4.5×10^8 m/s (c) 2×10^8 m/s (d) 3×10^8 m/s
- 12 Plane-polarised light of intensity I_0 is incident on a polaroid whose pass-axis makes 60° with the plane of polarisation. The transmitted intensity is
 (a) $I_0 / 4$ (b) I_0 (c) $I_0 / 2$ (d) $3I_0 / 4$
- 13 If the momentum of a moving particle is doubled, its de Broglie wavelength
 (a) remains the same (b) becomes double
 (c) becomes half (d) becomes four times
- 14 The threshold wavelength of a metal of work function 2.48 eV is (use $hc = 12400$ eV Å)
 (a) 10000 Å (b) 3000 Å (c) 2500 Å (d) 5000 Å
- 15 In Rutherford's α -particle scattering experiment most α -particles pass through the gold foil undeviated because
 (a) electrons attract α -particles (b) the nucleus is very large
 (c) α -particles are very heavy (d) the atom is mostly empty space
- 16 The radius of the second Bohr orbit of hydrogen is ($a_0 = 0.53$ Å)
 (a) 1.06 Å (b) 0.265 Å (c) 2.12 Å (d) 4.77 Å
- 17 Among the following nuclei, the binding energy per nucleon is greatest for
 (a) ^{56}Fe (b) ^2H (c) ^4He (d) ^{235}U
- 18 The depletion region of a p-n junction contains
 (a) both electrons and holes in large numbers
 (b) immobile ions
 (c) only holes
 (d) only free electrons

Q.2 Fill in the blanks ($10 \times \frac{1}{2} = 5$)

- The net charge of an electric dipole is _____.
- The relation between current and drift velocity in a conductor is $I =$ _____.
- The magnetic force on a charged particle moving parallel to the magnetic field is _____.
- The SI unit of magnetic flux is _____.
- The power factor of a pure inductor is _____.
- Infrared waves are also called _____ waves.
- In single-slit diffraction the width of the central maximum is _____ the width of a secondary maximum.
- Einstein received the Nobel Prize for his explanation of the _____.
- The Lyman series of hydrogen lies in the _____ region.
- With rise in temperature the resistivity of a semiconductor _____.

Q.3 Very short answer questions ($10 \times 1 = 10$)

1	Write the expression for the capacitance of an isolated spherical conductor of radius R.
2	Define current density. Write its SI unit.
3	Define magnetic susceptibility.
4	Write the expression for the energy stored in an inductor of inductance L carrying current I.
5	Name the part of the electromagnetic spectrum used in radar.
6	What is the nature of the image formed by a concave lens?
7	State Brewster's law.
8	Define work function of a metal.
9	State Bohr's frequency condition.
10	What are isobars? Give one example.

Section B (Q.4–13 • short answer • $10 \times 1\frac{1}{2} = 15$ marks)

Q.4 Two charges $+4 \mu\text{C}$ and $-4 \mu\text{C}$ are placed 20 cm apart in air. Find the electric potential energy of the system and the electric potential at the midpoint of the line joining them. [1½]

Q.5 A copper wire of cross-section 1 mm^2 carries a current of 3.2 A. If the free-electron density of copper is $8.5 \times 10^{28} \text{ m}^{-3}$, find the drift velocity of electrons. [1½]

Q.6 A bar magnet of magnetic moment 2 A m^2 is placed in a uniform magnetic field of 0.5 T at 30° to the field. Find the torque on it and its potential energy. [1½]

Q.7 Derive the expression for the motional emf $\varepsilon = Blv$ of a rod of length l moving with speed v perpendicular to a uniform field B . Find the emf for $l = 0.4 \text{ m}$, $v = 5 \text{ m/s}$, $B = 0.25 \text{ T}$. [1½]

Q.8 A convex lens of focal length 10 cm forms an image of an object placed 15 cm from it. Find the image distance, magnification and nature of the image. [1½]

Q.9 Write three differences between interference and diffraction of light. [1½]

Q.10 In a Young's double-slit experiment the slits are 0.5 mm apart, the screen is 1 m away and the light has wavelength $5000 \text{ \AA}$. Find the fringe width and the distance of the third bright fringe from the centre. [1½]

Q.11 Light of wavelength $2000 \text{ \AA}$ falls on a metal of work function 4.2 eV. Find the maximum kinetic energy of the photoelectrons and the stopping potential. ($hc = 12400 \text{ eV \AA}$) [1½]

Q.12 Calculate the wavelength of the first line of the Lyman series of hydrogen ($n = 2 \rightarrow 1$). ($R = 1.097 \times 10^7 \text{ m}^{-1}$) [1½]

Q.13 Write any three properties of the nuclear force. [1½]

Section C (Q.14–16 • long answer • $3 \times 3 = 9$ marks)

Q.14 (a) Derive an expression for the electric field at a point on the axial line of an electric dipole. [3]

— OR —

(b) Using Gauss's law, find the electric field due to a uniformly charged thin spherical shell (i) outside, (ii) inside it.

Q.15 (a) Derive an expression for the torque on a rectangular current loop placed in a uniform magnetic field. Hence define the magnetic moment of the loop. [3]

— OR —

(b) Describe the principle, construction and working of a moving-coil galvanometer. Define its current sensitivity.

Q.16 (a) Explain the formation of the depletion region and the barrier potential in a p-n junction. How do the width of the depletion region and the current change in forward and reverse bias? [3]

— OR —

(b) What is a Zener diode? With a circuit diagram explain how it acts as a voltage regulator.

Section D (Q.17–18 • essay type • $2 \times 4 = 8$ marks)

Q.17 (a) Derive an expression for the current in a pure inductor connected to an AC source. Show that the current lags the voltage by $\pi / 2$, define inductive reactance and show that the average power is zero. [4]

— OR —

(b) A series LCR circuit has $R = 20 \Omega$, $L = 0.1 \text{ H}$ and $C = 10 \mu\text{F}$ and is connected to a 100 V (rms) variable-frequency source. Find the resonant angular frequency and frequency, the current at resonance, the voltage across the inductor at resonance and the quality factor. State the condition for resonance.

Q.18 (a) Derive the relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ for refraction at a convex spherical surface separating two media. [4]

— OR —

(b) Derive the formula $n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$ for a prism. A prism of angle 60° produces a minimum deviation of 37° ; find the refractive index of its material.

— End of paper —

Solutions and marking scheme

How to use the marking scheme

- Step marks are shown as (½), (1) at the end of each step, the way an examiner awards them. A missing unit or a missing diagram costs ½ mark.
- Any other correct method and answer earns full marks. Total your score: Section A /24, B /15, C /9, D /8.

✓ Q.1 – Answer key and quick solutions

1(c) 2(a) 3(c) 4(a) 5(b) 6(d) 7(a) 8(c) 9(a) 10(a) 11(c) 12(a) 13(c) 14(d) 15(d) 16(c)

17(a) 18(b)

$$1. (c) F' = k \frac{(2q)(2q)}{(2r)^2} = k \frac{4q^2}{4r^2} = F.$$

$$2. (a) C = KC_0 = 4C_0.$$

3. (c) The total current entering a junction equals the total leaving it because charge cannot pile up at a point: conservation of charge.

$$4. (a) R_{eq} = \frac{6 \times 3}{6 + 3} = 2 \Omega, I = V / R_{eq} = 6 / 2 = 3 \text{ A}.$$

$$5. (b) F = qvB = 1.6 \times 10^{-19} \times 2 \times 10^6 \times 0.5 = 1.6 \times 10^{-13} \text{ N}.$$

6. (d) $\chi = C / T$; $\chi \propto 1 / T$ for a paramagnetic substance.

7. (a) In an induction furnace the changing flux induces strong eddy currents in the metal, which heats and melts it.

$$8. (c) \varepsilon = L \frac{|\Delta I|}{\Delta t} = 2 \times \frac{2}{0.5} = 8 \text{ V}.$$

9. (a) $i = i_0 \sin(\omega t + \pi / 2)$; the current leads the voltage by $\pi / 2$.

$$10. (a) v = \frac{c}{\sqrt{\varepsilon_r \mu_r}} = \frac{3 \times 10^8}{\sqrt{4}} = 1.5 \times 10^8 \text{ m/s}.$$

$$11. (c) v = c / n = 3 \times 10^8 / 1.5 = 2 \times 10^8 \text{ m/s}.$$

$$12. (a) I = I_0 \cos^2 60^\circ = I_0 \times \left(\frac{1}{2}\right)^2 = I_0 / 4.$$

13. (c) $\lambda = h / p$, so $\lambda \propto 1 / p$; doubling p halves λ .

$$14. (d) \lambda_0 = \frac{hc}{\phi_0} = \frac{12400}{2.48} = 5000 \text{ Å}.$$

15. (d) Nearly all of the atom's mass and positive charge is in a tiny nucleus; most α -particles pass through the vast empty space and are not deflected.

$$16. (c) r_n = n^2 a_0; r_2 = 4 \times 0.53 = 2.12 \text{ Å}.$$

17. (a) The BE/A curve peaks (about 8.75 MeV) near mass number 56 (iron); ${}^2\text{H}$ has about 1.1 MeV, ${}^4\text{He}$ about 7.1 MeV, ${}^{235}\text{U}$ about 7.6 MeV.

18. (b) Electrons and holes diffuse across the junction and recombine, leaving behind immobile donor (+) and acceptor (-) ions in a region free of mobile carriers.

Q.2 – Fill in the blanks: answers (½ each)

No.	Ch	Answer
1	1	zero
2	3	$neAv_d$
3	4	zero
4	6	weber (Wb)
5	7	zero
6	8	heat (thermal)
7	10	twice
8	11	photoelectric effect
9	12	ultraviolet
10	14	decreases

Q.3 – Very short answers (1 mark each)

No.	Ch	Answer
1	2	$C = 4\pi\epsilon_0 R$
2	3	Current per unit area normal to the flow, $J = I / A$; unit A/m^2 .
3	5	The ratio of the intensity of magnetisation to the magnetising field, $\chi = M / H$ (a number without unit).
4	6	$U = \frac{1}{2}LI^2$
5	8	Microwaves.
6	9	Always virtual, erect and diminished.
7	10	The tangent of the polarising (Brewster) angle equals the refractive index of the medium: $\tan i_B = n$.
8	11	The minimum energy required to eject an electron from the surface of the metal.
9	12	When an electron jumps from a higher level E_i to a lower level E_f , a photon is emitted with $h\nu = E_i - E_f$.
10	13	Nuclides having the same mass number but different atomic numbers, e.g. ${}^{14}_6\text{C}$ and ${}^{14}_7\text{N}$.

Q.4–13 – Short answers (1½ marks each)**🔧 Q.4 • Ch 2 • 1½ marks**

Two charges $+4\ \mu\text{C}$ and $-4\ \mu\text{C}$ are placed 20 cm apart in air. Find the electric potential energy of the system and the electric potential at the midpoint of the line joining them.

Solution: $U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$ (½)

$$U = \frac{9 \times 10^9 \times (4 \times 10^{-6})(-4 \times 10^{-6})}{0.20} = -0.72\text{ J} \text{ (½)}$$

At the midpoint: $V = k \frac{(+4\mu\text{C})}{0.1} + k \frac{(-4\mu\text{C})}{0.1} = 0$ (½)

🔧 Q.5 • Ch 3 • 1½ marks

A copper wire of cross-section 1 mm^2 carries a current of 3.2 A. If the free-electron density of copper is $8.5 \times 10^{28}\text{ m}^{-3}$, find the drift velocity of electrons.

Solution: $I = neAv_d \Rightarrow v_d = \frac{I}{neA}$ (½)

$$v_d = \frac{3.2}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-6}} \text{ (½)}$$

$$v_d \approx 2.35 \times 10^{-4}\text{ m/s} \text{ (½)}$$

 Q.6 • Ch 5 • 1½ marks

A bar magnet of magnetic moment 2 A m^2 is placed in a uniform magnetic field of 0.5 T at 30° to the field. Find the torque on it and its potential energy.

Solution: $\tau = mB\sin\theta = 2 \times 0.5 \times \sin 30^\circ = 0.5 \text{ N m}$ (½)

$U = -mB\cos\theta$ (½)

$U = -2 \times 0.5 \times \cos 30^\circ = -0.87 \text{ J}$ (½)

 Q.7 • Ch 6 • 1½ marks

Derive the expression for the motional emf $\varepsilon = Blv$ of a rod of length l moving with speed v perpendicular to a uniform field B . Find the emf for $l = 0.4 \text{ m}$, $v = 5 \text{ m/s}$, $B = 0.25 \text{ T}$.

Solution: Rod PQ slides on a U-shaped rail; if it is at distance x from the closed end, flux through the loop $\Phi = Blx$. (½)

$\varepsilon = -\frac{d\Phi}{dt} = -Bl\frac{dx}{dt} \Rightarrow |\varepsilon| = Blv$ (½)

$\varepsilon = 0.25 \times 0.4 \times 5 = 0.5 \text{ V}$ (½)

 Q.8 • Ch 9 • 1½ marks

A convex lens of focal length 10 cm forms an image of an object placed 15 cm from it. Find the image distance, magnification and nature of the image.

Solution: $f = +10 \text{ cm}$, $u = -15 \text{ cm}$: $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30}$ (½)

$v = +30 \text{ cm}$ (on the other side) (½)

$m = \frac{v}{u} = \frac{30}{-15} = -2 \Rightarrow \text{real, inverted, magnified (twice).}$ (½)

 Q.9 • Ch 10 • 1½ marks

Write three differences between interference and diffraction of light.

Solution: (1) **Origin:** interference – superposition of waves from two separate coherent sources; diffraction – superposition of secondary wavelets from different parts of the same wavefront. (½)

(2) **Fringe width:** interference fringes are (nearly) of equal width; diffraction fringes are of unequal width (the central one is the widest). (½)

(3) **Intensity:** in interference all bright fringes have nearly the same intensity; in diffraction the intensity falls rapidly with order. (½)

 Q.10 • Ch 10 • 1½ marks

In a Young's double-slit experiment the slits are 0.5 mm apart, the screen is 1 m away and the light has wavelength $5000 \text{ \AA}$. Find the fringe width and the distance of the third bright fringe from the centre.

Solution: $\beta = \frac{\lambda D}{d} = \frac{5 \times 10^{-7} \times 1}{0.5 \times 10^{-3}} = 1 \times 10^{-3} \text{ m} = 1 \text{ mm}$ (1)

$x_3 = 3\beta = 3 \text{ mm}$ (½)

Q.11 • Ch 11 • 1½ marks

Light of wavelength $2000 \text{ \AA}$ falls on a metal of work function 4.2 eV . Find the maximum kinetic energy of the photoelectrons and the stopping potential. ($hc = 12400 \text{ eV \AA}$)

Solution: Photon energy $E = \frac{12400}{2000} = 6.2 \text{ eV}$ (½)

$$K_{\max} = E - \phi_0 = 6.2 - 4.2 = 2.0 \text{ eV} \text{ (½)}$$

$$eV_0 = K_{\max} \Rightarrow V_0 = 2.0 \text{ V} \text{ (½)}$$

Q.12 • Ch 12 • 1½ marks

Calculate the wavelength of the first line of the Lyman series of hydrogen ($n = 2 \rightarrow 1$). ($R = 1.097 \times 10^7 \text{ m}^{-1}$)

$$\text{Solution: } \frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3R}{4} \text{ (½)}$$

$$\lambda = \frac{4}{3 \times 1.097 \times 10^7} \text{ (½)}$$

$$\lambda = 1.215 \times 10^{-7} \text{ m} = 1215 \text{ \AA} \text{ (ultraviolet) (½)}$$

Q.13 • Ch 13 • 1½ marks

Write any three properties of the nuclear force.

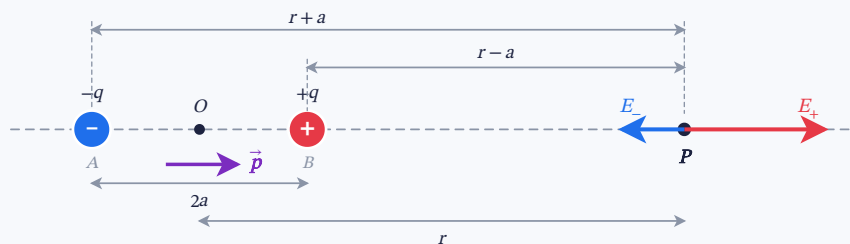
Solution: (1) It is the **strongest** fundamental force (about 100 times the electrostatic force at nuclear distances), attractive in nature. (½)

(2) It is **short-ranged** – effective only up to a few femtometres (about 2–3 fm) and negligible beyond. (½)

(3) It is **charge-independent** – the p–p, n–n and p–n forces are nearly equal – and it shows **saturation** (a nucleon interacts only with its near neighbours). (½)

Q.14–16 – Long answers (3 marks each)**Q.14 (a) • Ch 1 • 3 marks**

Derive an expression for the electric field at a point on the axial line of an electric dipole.



Solution: 1. Dipole: charges $-q$ at A and $+q$ at B, $AB = 2a$, centre O; point P on the axis at distance r from O (figure). (½)

2. Field of $+q$ at P (distance $r - a$), along $\vec{p}$: $E_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r - a)^2}$. (½)

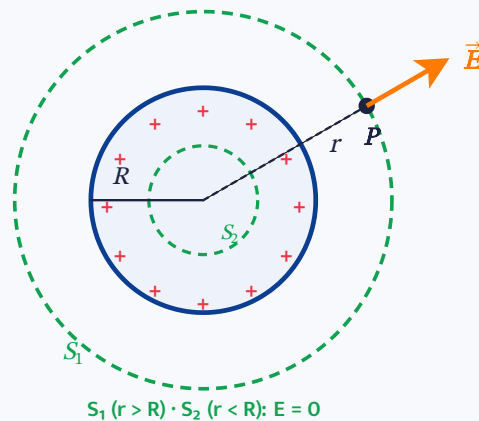
3. Field of $-q$ at P (distance $r + a$), opposite to $\vec{p}$: $E_- = \frac{1}{4\pi\epsilon_0} \frac{q}{(r + a)^2}$. (½)

4. $E = E_+ - E_- = \frac{q}{4\pi\epsilon_0} \frac{(r + a)^2 - (r - a)^2}{(r^2 - a^2)^2} = \frac{1}{4\pi\epsilon_0} \frac{4qar}{(r^2 - a^2)^2}$. (½)

5. With $p = 2qa$: $E = \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2}$ along $\vec{p}$; for $r \gg a$: $E \approx \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$. (1)

 Q.14 (b) • Ch 1 • 3 marks

Using Gauss's law, find the electric field due to a uniformly charged thin spherical shell (i) outside, (ii) inside it.



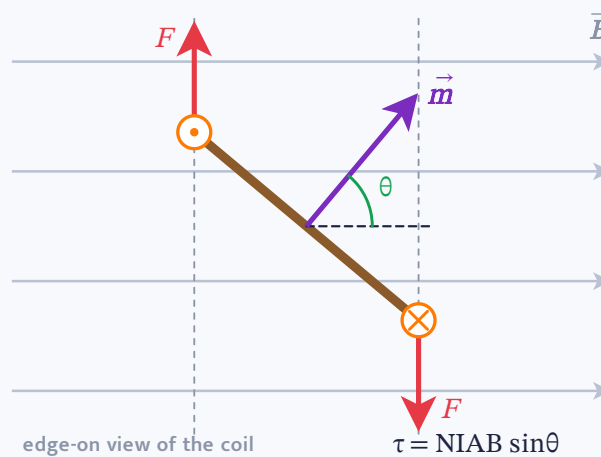
Solution: Shell of radius R and total charge Q ; by symmetry $\vec{E}$ is radial with the same magnitude on any concentric sphere (figure). (½)

(i) **Outside ($r > R$):** Gaussian sphere of radius r encloses Q : $E(4\pi r^2) = \frac{Q}{\epsilon_0}$ (½) $\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ – as if all the charge were at the centre. (1)

(ii) **Inside ($r < R$):** Gaussian sphere encloses no charge, $q_{enc} = 0 \Rightarrow E(4\pi r^2) = 0 \Rightarrow E = 0$. (1)

 Q.15 (a) • Ch 4 • 3 marks

Derive an expression for the torque on a rectangular current loop placed in a uniform magnetic field. Hence define the magnetic moment of the loop.



Solution: 1. Loop of sides a and b , current I , area $A = ab$; the normal $\hat{n}$ makes angle θ with $\vec{B}$ (figure). (½)

2. Forces on the two arms of length b are equal and opposite and act along the same line, so they cancel; likewise the net force on the loop is zero. (½)

3. The arms of length a are perpendicular to $\vec{B}$: each feels $F = IaB$, in opposite directions, forming a couple. (½)

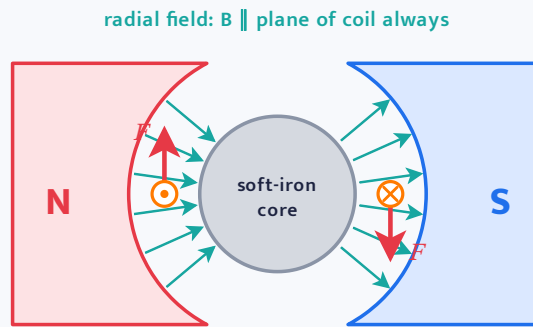
4. Perpendicular distance between the two forces $= b \sin\theta$. (½)

5. Torque $\tau = F b \sin\theta = IaB b \sin\theta \Rightarrow \tau = IAB \sin\theta$ (for N turns: $NIAB \sin\theta$). (½)

6. Defining $\vec{m} = NI\vec{A}$: $\vec{\tau} = \vec{m} \times \vec{B}$; magnetic moment $m = NIA$ (unit $A m^2$), direction along $\hat{n}$ by the right-hand rule. (½)

 Q.15 (b) • Ch 4 • 3 marks

Describe the principle, construction and working of a moving-coil galvanometer. Define its current sensitivity.



$$NIAB = k\phi \Rightarrow \phi \propto I$$

Solution: Principle: a current-carrying coil in a magnetic field experiences a torque. (½)

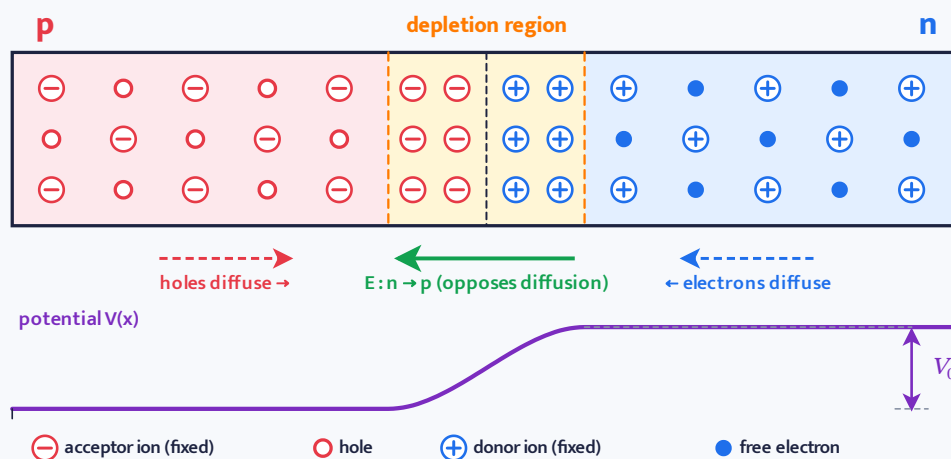
Construction: a rectangular coil of N turns on a soft-iron core, suspended between concave pole pieces of a strong magnet so that the field is **radial**; a phosphor-bronze suspension carries a restoring torque $k\phi$ and a pointer moves over the scale. (½)

Working: deflecting torque $= NIAB$ (always $\perp$ to $\vec{B}$ in a radial field); at equilibrium $NIAB = k\phi \Rightarrow \phi = \frac{NAB}{k}I$, so the deflection is proportional to the current. (1)

Current sensitivity: deflection per unit current, $\frac{\phi}{I} = \frac{NAB}{k}$; increased by more turns, larger area, stronger field and a weaker suspension. (1)

 Q.16 (a) • Ch 14 • 3 marks

Explain the formation of the depletion region and the barrier potential in a p-n junction. How do the width of the depletion region and the current change in forward and reverse bias?



Solution: 1. Diagram of the junction with depletion layer and barrier potential V_0 (figure). (1)

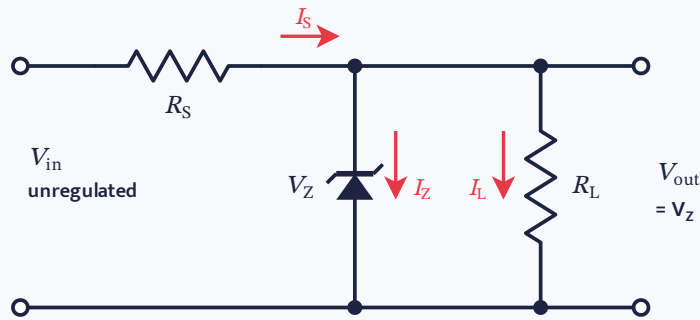
2. Formation: electrons diffuse from n to p and holes from p to n ; they recombine near the junction, leaving immobile $+$ donor ions on the n -side and $-$ acceptor ions on the p -side. This layer free of carriers is the **depletion region**. The ions set up a field from n to p which causes a drift current opposing diffusion; at equilibrium a **barrier potential** V_0 (≈ 0.7 V Si, 0.3 V Ge) exists. (1)

3. Forward bias (p to $+$): applied field opposes the barrier, depletion region **narrows**, barrier lowers, a large current (mA) flows. (½)

4. Reverse bias (p to $-$): barrier and depletion region **widen**, only a tiny minority-carrier current (μ A) flows. (½)

 Q.16 (b) • Ch 14 • 3 marks

What is a Zener diode? With a circuit diagram explain how it acts as a voltage regulator.



Zener in REVERSE bias (breakdown)

Solution: 1. Zener diode: a heavily doped p-n junction diode designed to operate in the reverse **breakdown** region, where the voltage across it stays almost constant (V_Z) over a wide range of current. (½)

2. Circuit (figure): the Zener (reverse biased) is connected in parallel with the load R_L and a series resistance R_S joins it to the unregulated input V_{in} . (½)

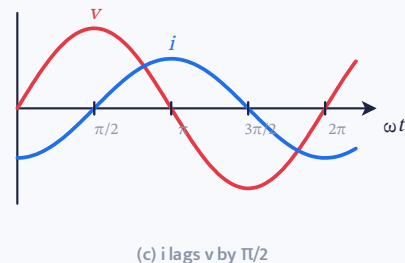
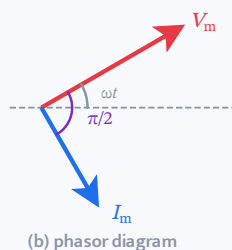
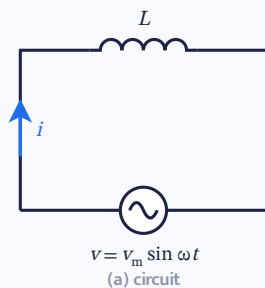
3. Working: if V_{in} rises, the Zener current rises; the extra voltage drops across R_S , so $V_{out} = V_Z$ stays constant. (1)

4. If the load current changes, the Zener current changes oppositely, so $I_S = I_Z + I_L$ and V_{out} remain constant. (1)

Q.17–18 – Essay type (4 marks each)

 Q.17 (a) • Ch 7 • 4 marks

Derive an expression for the current in a pure inductor connected to an AC source. Show that the current lags the voltage by $\pi / 2$, define inductive reactance and show that the average power is zero.



Solution: 1. Circuit (figure): $v = v_0 \sin \omega t$ across an inductor L (no resistance). (½)

2. The changing current induces an emf $-L \frac{di}{dt}$. Loop rule: $v - L \frac{di}{dt} = 0 \Rightarrow \frac{di}{dt} = \frac{v_0}{L} \sin \omega t$. (1)

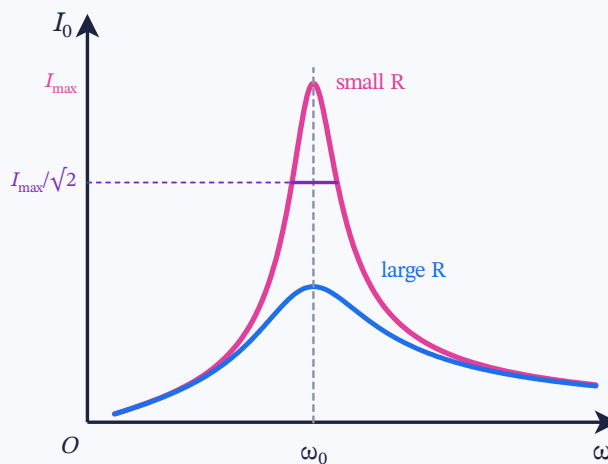
3. Integrating: $i = -\frac{v_0}{\omega L} \cos \omega t = \frac{v_0}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right) \Rightarrow i = i_0 \sin \left(\omega t - \frac{\pi}{2} \right)$, $i_0 = \frac{v_0}{\omega L}$: the current **lags** the voltage by $\pi / 2$. (1)

4. Inductive reactance $X_L = \omega L = 2\pi f L$ (unit Ω): the opposition offered by an inductor, $i_0 = v_0 / X_L$. (½)

5. Instantaneous power $p = vi = -\frac{v_0 i_0}{2} \sin 2\omega t$ – alternately positive and negative; its average over a cycle is zero, $\bar{P} = 0$ (energy is stored and returned, never dissipated). (1)

 Q.17 (b) • Ch 7 • 4 marks

A series LCR circuit has $R = 20\ \Omega$, $L = 0.1\ \text{H}$ and $C = 10\ \mu\text{F}$ and is connected to a $100\ \text{V}$ (rms) variable-frequency source. Find the resonant angular frequency and frequency, the current at resonance, the voltage across the inductor at resonance and the quality factor. State the condition for resonance.



Solution: 1. Condition: resonance occurs when $X_L = X_C$, i.e. $\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$; then $Z = R$ is minimum and the current is maximum, in phase with the voltage. (1)

2. $\omega_0 = \frac{1}{\sqrt{0.1 \times 10 \times 10^{-6}}} = \frac{1}{10^{-3}} = 1000\ \text{rad/s}$; $f_0 = \frac{\omega_0}{2\pi} = \frac{1000}{2\pi} = 159\ \text{Hz}$ (1)

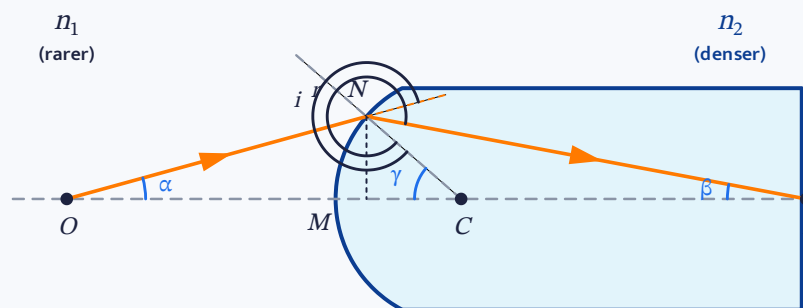
3. $I_{rms} = \frac{V}{R} = \frac{100}{20} = 5\ \text{A}$ (½)

4. $X_L = \omega_0 L = 1000 \times 0.1 = 100\ \Omega$; $V_L = IX_L = 5 \times 100 = 500\ \text{V}$ (equal to V_C) (1)

5. $Q = \frac{\omega_0 L}{R} = \frac{100}{20} = 5$ (also $V_L / V = 500 / 100$) (½)

 Q.18 (a) • Ch 9 • 4 marks

Derive the relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ for refraction at a convex spherical surface separating two media.



Solution: 1. Diagram (figure): object O in medium n_1 , spherical surface of radius R and centre C , paraxial ray ON refracting at N to image I in medium n_2 ; M is the foot of the perpendicular from N . (1)

2. For small angles: $\alpha \approx \frac{MN}{OM}$, $\gamma \approx \frac{MN}{MC}$, $\beta \approx \frac{MN}{MI}$ (angles at O, C, I). (½)

3. Exterior angles: $i = \alpha + \gamma$ ($\triangle NOC$) and $\gamma = r + \beta \Rightarrow r = \gamma - \beta$ ($\triangle NCI$). (½)

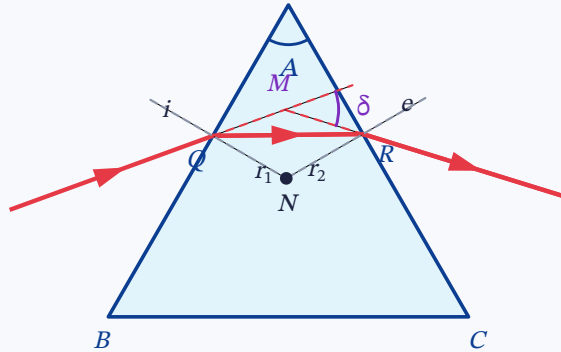
4. Snell's law for small angles: $n_1 i = n_2 r \Rightarrow n_1(\alpha + \gamma) = n_2(\gamma - \beta)$. (½)

5. Substituting: $\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}$. (½)

6. Signs: $OM = -u$, $MI = +v$, $MC = +R \Rightarrow \frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ (1)

 Q.18 (b) • Ch 9 • 4 marks

Derive the formula $n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$ for a prism. A prism of angle 60° produces a minimum deviation of 37° ; find the refractive index of its material.



Solution: 1. Ray path through prism ABC (figure): incidence i , refraction r_1 at the first face, r_2 and emergence e at the second face. (1)

2. In the quadrilateral formed with the normals: $r_1 + r_2 = A$. Total deviation $\delta = (i - r_1) + (e - r_2) = i + e - A$. (1)

3. At minimum deviation the ray passes symmetrically: $i = e, r_1 = r_2 = r \Rightarrow r = \frac{A}{2}$ and $\delta_m = 2i - A \Rightarrow i = \frac{A + \delta_m}{2}$. (½)

4. Snell's law at the first face: $n = \frac{\sin i}{\sin r} = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$ (½)

5. Numerical: $n = \frac{\sin \frac{60^\circ + 37^\circ}{2}}{\sin 30^\circ} = \frac{\sin 48.5^\circ}{0.5} = \frac{0.749}{0.5} \approx 1.5$ (1)

Score sheet

Section	Maximum	Your marks
A (Q.1–3)	24	
B (Q.4–13)	15	
C (Q.14–16)	9	
D (Q.17–18)	8	
Total	56	

Score bands

48+ excellent – polish diagrams and units • **40–47** good – recheck every numerical • **below 40** – revise the derivations and formula boxes of the chapters where you lost marks, then retry this paper after 3 days.

DAY
27

RBSE CLASS 12 PHYSICS • FULL PAPER

Model Paper 3

SS-40 • Physics

3 h 15 min

56 marks

18 questions

2025 pattern

Original paper in the 2025 BSER pattern. Write it in one sitting with a clock, then mark yourself with the marking scheme that follows.

Instructions to candidates

Paper details

- Subject: **Physics** (theory) • Class 12 • Code SS-40
- Time: **3 hours 15 minutes** • Maximum marks: **56**
- **18** questions, all compulsory; internal choice only in Q.14–18.
- Q.1: 18 MCQ (½ each) • Q.2: 10 blanks (½ each) • Q.3: 10 very short (1 each)
- Q.4–13: short answer (1½ each) • Q.14–16: 3 marks • Q.17–18: 4 marks

Constants you may use

- $e = 1.6 \times 10^{-19} \text{ C}$, $c = 3 \times 10^8 \text{ m/s}$,
 $h = 6.63 \times 10^{-34} \text{ J s}$
- $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$
- $m_e = 9.1 \times 10^{-31} \text{ kg}$, $1 \text{ u} = 931.5 \text{ MeV}/c^2$,
 $hc = 12400 \text{ eV Å}$
- Draw neat labelled diagrams wherever needed. Write the units in every numerical answer.
- If the Hindi and English versions differ, the Hindi version is valid.

Blueprint of this paper (chapter-wise marks = RBSE weightage)

Ch	Chapter	Marks	Question numbers
1	Electric Charges and Fields	4	1(1), 1(2), 2(1), 3(1), 4
2	Electrostatic Potential and Capacitance	3	1(3), 3(2), 5
3	Current Electricity	4	1(4), 2(2), 14
4	Moving Charges and Magnetism	4	1(5), 2(3), 6, 7
5	Magnetism and Matter	3	1(6), 3(3), 8
6	Electromagnetic Induction	4	1(7), 1(8), 2(4), 3(4), 9
7	Alternating Current	5	1(9), 2(5), 17
8	Electromagnetic Waves	2	1(10), 2(6), 3(5)
9	Ray Optics and Optical Instruments	7	1(11), 3(6), 10, 18
10	Wave Optics	5	1(12), 2(7), 3(7), 15
11	Dual Nature of Radiation and Matter	4	1(13), 1(14), 2(8), 3(8), 11
12	Atoms	4	1(15), 2(9), 16
13	Nuclei	3	1(16), 3(9), 12
14	Semiconductor Electronics	4	1(17), 1(18), 2(10), 3(10), 13
Total		56	

The question paper

Section A (Q.1–3 • 24 marks)



Q.1 Choose the correct option ($18 \times \frac{1}{2} = 9$)

Write the letter and the answer in your answer-book. Items are numbered 1–18.

- 1 Which of the following can be the charge on a body? ($e = 1.6 \times 10^{-19} \text{ C}$)
 (a) $3.2 \times 10^{-19} \text{ C}$ (b) $2.4 \times 10^{-19} \text{ C}$ (c) $0.8 \times 10^{-19} \text{ C}$ (d) $4.0 \times 10^{-19} \text{ C}$
- 2 The electric field due to a large uniformly charged plane sheet is
 (a) independent of the distance from the sheet
 (b) inversely proportional to the square of the distance
 (c) inversely proportional to the distance
 (d) directly proportional to the distance
- 3 The electric potential at any point on the equatorial line of an electric dipole is
 (a) $\frac{p}{4\pi\epsilon_0 r^2}$ (b) zero (c) $\frac{2p}{4\pi\epsilon_0 r^3}$ (d) $\frac{p}{4\pi\epsilon_0 r}$
- 4 Which of the following has a negative temperature coefficient of resistance?
 (a) Manganin (b) Copper (c) Silicon (d) Nichrome
- 5 A circular coil of 10 turns and radius 10 cm carries a current of 5 A. The magnetic field at its centre is ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$)
 (a) $\pi \times 10^{-5} \text{ T}$ (b) $\pi \times 10^{-4} \text{ T}$ (c) $2\pi \times 10^{-4} \text{ T}$ (d) $4\pi \times 10^{-4} \text{ T}$
- 6 Soft iron is used for the core of a transformer because it has
 (a) high retentivity and high coercivity
 (b) high hysteresis loss
 (c) low permeability
 (d) low hysteresis loss and high permeability
- 7 The emf induced in a coil is directly proportional to
 (a) the resistance of the coil
 (b) the rate of change of magnetic flux linked with it
 (c) the time for which flux changes
 (d) the magnetic flux linked with it
- 8 A long solenoid has self-inductance L . If its number of turns is doubled keeping the length and area the same, its self-inductance becomes
 (a) $2L$ (b) $4L$ (c) $L/2$ (d) $L/4$
- 9 The time period of the alternating current supplied in India (frequency 50 Hz) is
 (a) 0.05 s (b) 50 s (c) 0.2 s (d) 0.02 s

- 10 An electromagnetic wave has frequency 3×10^{10} Hz. Its wavelength in vacuum is
(a) 10 cm (b) 1 m (c) 0.1 cm (d) 1 cm
- 11 The power of a concave lens of focal length 25 cm is
(a) +0.25 D (b) +4 D (c) -4 D (d) -25 D
- 12 A path difference of $\lambda / 4$ between two waves corresponds to a phase difference of
(a) 2π (b) π (c) $\pi / 4$ (d) $\pi / 2$
- 13 In the photoelectric effect the stopping potential depends on
(a) the frequency of the incident light
(b) the distance between the source and the metal
(c) the intensity of the incident light
(d) the area of the metal plate
- 14 The momentum of a photon of wavelength 6630 Å is ($h = 6.63 \times 10^{-34}$ J s)
(a) 1×10^{-27} kg m/s (b) 1×10^{-30} kg m/s (c) 6.63×10^{-34} kg m/s (d) 6.63×10^{-27} kg m/s
- 15 The Paschen series of the hydrogen spectrum lies in the
(a) X-ray region (b) ultraviolet region
(c) infrared region (d) visible region
- 16 The energy of the Sun is produced mainly by
(a) fission of uranium
(b) burning of coal
(c) radioactive decay of carbon
(d) fusion of hydrogen nuclei into helium
- 17 In a pure (intrinsic) silicon crystal at room temperature
(a) $n_e = n_h$ (b) $n_e < n_h$ (c) $n_e > n_h$ (d) $n_e = n_h = 0$
- 18 The current in a reverse-biased p-n junction is due to
(a) donor ions (b) majority charge carriers
(c) minority charge carriers (d) acceptor ions

Q.2 Fill in the blanks ($10 \times \frac{1}{2} = 5$)

1	The SI unit of linear charge density is _____.
2	Kirchhoff's loop rule is based on the conservation of _____.
3	A magnetic force does no _____ on a moving charged particle.
4	Eddy currents produce an undesirable _____ effect in transformer cores.
5	The capacitive reactance of a capacitor C at angular frequency ω is $X_C =$ _____.
6	In an electromagnetic wave the electric and magnetic fields are _____ to each other and to the direction of propagation.
7	Two waves of intensities I and $4I$ interfere. The maximum intensity obtained is _____.
8	The photoelectric effect demonstrates the _____ nature of light.
9	The angular momentum of the electron in the n -th Bohr orbit is _____.
10	Doping silicon with a pentavalent impurity produces a _____-type semiconductor.

Q.3 Very short answer questions ($10 \times 1 = 10$)

1	Define electric dipole moment and write its SI unit.
2	Write two properties of equipotential surfaces.
3	Define Curie temperature.
4	Define self-inductance of a coil.
5	Name the electromagnetic radiation of wavelength range 400 nm to 700 nm.
6	Write the expression for the magnifying power of a simple microscope when the image is formed at the near point.
7	What is the shape of the wavefront coming from a very distant source such as the Sun?
8	Define threshold frequency.
9	Write the approximate energy released in the fission of one ^{235}U nucleus.
10	What is the function of a rectifier?

Section B (Q.4–13 • short answer • $10 \times 1\frac{1}{2} = 15$ marks)

Q.4 Two point charges of $2\ \mu\text{C}$ and $8\ \mu\text{C}$ are placed 0.3 m apart in air. Find the force between them. What would the force be if the charges were placed in a medium of dielectric constant 2? [1½]

Q.5 A parallel-plate capacitor has plates of area $100\ \text{cm}^2$ separated by 1 mm in air. Find its capacitance and the charge on it when connected to a 100 V battery. ($\epsilon_0 = 8.85 \times 10^{-12}\ \text{F/m}$) [1½]

Q.6 State the Biot–Savart law and write it in vector form. [1½]

Q.7 A galvanometer of resistance $100\ \Omega$ gives full-scale deflection for 5 mA. How can it be converted into a voltmeter of range 0–10 V? [1½]

Q.8 Write three differences between soft and hard magnetic materials. [1½]

Q.9 A coil of self-inductance 0.5 H carries a current of 4 A. Find the energy stored in it. If the current falls to zero in 0.02 s, find the average induced emf. [1½]

Q.10 A prism of refracting angle 60° is made of glass of refractive index $\sqrt{2}$. Find the angle of minimum deviation. [1½]

Q.11 An electron moves with a speed of $1 \times 10^6\ \text{m/s}$. Find its de Broglie wavelength and its kinetic energy in eV. ($m = 9.1 \times 10^{-31}\ \text{kg}$) [1½]

Q.12 Calculate the binding energy per nucleon of ${}^4_2\text{He}$. Masses: $m_p = 1.00728 \text{ u}$, $m_n = 1.00867 \text{ u}$, nuclear mass of ${}^4_2\text{He} = 4.0015 \text{ u}$; $1 \text{ u} = 931.5 \text{ MeV}/c^2$. [1½]

Q.13 Explain half-wave rectification using a p-n junction diode in three points. [1½]

Section C (Q.14–16 • long answer • $3 \times 3 = 9$ marks)

Q.14 (a) Derive the relation $I = neAv_d$ between current and drift velocity, and hence show that $\rho = \frac{m}{ne^2\tau}$. [3]

— OR —

(b) Derive the expressions for the equivalent emf and equivalent internal resistance of two cells connected in parallel. Two cells (2 V, 1 Ω) and (1 V, 1 Ω) in parallel are connected to an external resistance of 1 Ω ; find the current in it.

Q.15 (a) Using Huygens' principle, prove the law of reflection of light at a plane surface. [3]

— OR —

(b) Explain the diffraction of light at a single slit. Obtain the condition for the minima and the width of the central maximum.

Q.16 (a) Describe the Geiger–Marsden α -particle scattering experiment. State the observations and the conclusions drawn from them. [3]

— OR —

(b) Using Bohr's postulates, derive the expressions for the radius of the n-th orbit and the total energy of the electron in the n-th orbit of a hydrogen atom. What is the energy in the ground state?

Section D (Q.17–18 • essay type • $2 \times 4 = 8$ marks)

Q.17 (a) Explain the principle, construction and working of a transformer. Obtain the relation between the voltages and the number of turns. A step-down transformer converts 2200 V to 220 V; the primary has 4000 turns and the secondary delivers 5 A. Find the number of secondary turns and the primary current (ideal transformer). [4]

— OR —

(b) Derive the expression for the current in a pure capacitor connected to an AC source and show that the current leads the voltage by $\pi/2$. Show that the average power consumed is zero. A 100 μF capacitor is connected to a 220 V, 50 Hz supply; find its reactance and the rms current.

Q.18 (a) Derive the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for a concave mirror and obtain the expression for the magnification. [4]

— OR —

(b) Draw the ray diagram of an astronomical telescope in normal adjustment and derive its magnifying power and tube length. A telescope has a magnifying power of 20 and its tube length is 105 cm; find the focal lengths of the objective and the eyepiece.

— End of paper —

Solutions and marking scheme

How to use the marking scheme

- Step marks are shown as (½), (1) at the end of each step, the way an examiner awards them. A missing unit or a missing diagram costs ½ mark.
- Any other correct method and answer earns full marks. Total your score: Section A /24, B /15, C /9, D /8.

✓ Q.1 – Answer key and quick solutions

1(a) 2(a) 3(b) 4(c) 5(b) 6(d) 7(b) 8(b) 9(d) 10(d) 11(c) 12(d) 13(a) 14(a) 15(c)

16(d) 17(a) 18(c)

- (a) Charge is quantised, $q = ne$ with integer n : $3.2 \times 10^{-19} = 2e$; the others are $1.5e$, $0.5e$ and $2.5e$.
- (a) By Gauss's law $E = \sigma / 2\epsilon_0$, which does not contain the distance.
- (b) $V = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}$ and $\theta = 90^\circ$ on the equator, so $V = 0$.
- (c) In a semiconductor, heating frees more charge carriers, so resistance falls as temperature rises. Metals and alloys have a positive (alloys: very small) coefficient.
- (b) $B = \frac{\mu_0 NI}{2R} = \frac{4\pi \times 10^{-7} \times 10 \times 5}{2 \times 0.1} = \pi \times 10^{-4} \text{ T}$.
- (d) Soft iron has a narrow B-H loop (small area = small hysteresis loss per cycle) and high permeability, ideal for repeated magnetisation reversal.
- (b) Faraday's law: $\epsilon = -N d\Phi / dt$.
- (b) $L = \mu_0 \frac{N^2 A}{l} \propto N^2$, so doubling N gives $4L$.
- (d) $T = 1 / f = 1 / 50 = 0.02 \text{ s}$.
- (d) $\lambda = c / f = \frac{3 \times 10^8}{3 \times 10^{10}} = 10^{-2} \text{ m} = 1 \text{ cm}$.
- (c) $f = -0.25 \text{ m}$, $P = 1 / f = -4 \text{ D}$.
- (d) $\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$.
- (a) $eV_0 = K_{\max} = h\nu - \phi_0$: it depends on ν (and the metal), not on intensity.
- (a) $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{6.63 \times 10^{-7}} = 1 \times 10^{-27} \text{ kg m/s}$.
- (c) Lyman ($n \rightarrow 1$): ultraviolet; Balmer ($n \rightarrow 2$): visible; Paschen ($n \rightarrow 3$) and later series: infrared.
- (d) In the Sun's core four protons fuse into a helium nucleus (proton-proton cycle), releasing about 26 MeV per helium nucleus.
- (a) Each thermally broken bond creates one electron and one hole, so the two concentrations are equal.
- (c) The higher barrier blocks majority carriers; only thermally generated minority carriers are swept across, giving a tiny reverse current (μA).

Q.2 – Fill in the blanks: answers (½ each)

No.	Ch	Answer
1	1	C m^{-1}
2	3	energy
3	4	work
4	6	heating
5	7	$\frac{1}{\omega C} = \frac{1}{2\pi f C}$
6	8	perpendicular
7	10	$9I$
8	11	particle (photon)
9	12	$\frac{nh}{2\pi}$
10	14	n

Q.3 – Very short answers (1 mark each)

No.	Ch	Answer
1	1	The product of either charge and the separation between the charges, $\vec{p} = q(2\vec{a})$, directed from $-q$ to $+q$; unit C m.
2	2	(i) No work is done in moving a charge along an equipotential surface. (ii) The electric field is always perpendicular to the surface (they never intersect). Any two.
3	5	The temperature above which a ferromagnetic substance becomes paramagnetic.
4	6	The ratio of the magnetic flux linked with the coil to the current in it, $L = \Phi / I$ (equivalently, the induced emf per unit rate of change of current).
5	8	Visible light.
6	9	$M = 1 + \frac{D}{f}$
7	10	Plane wavefront.
8	11	The minimum frequency of the incident radiation below which photoelectric emission does not occur from a metal surface.
9	13	About 200 MeV.
10	14	It converts alternating current (voltage) into direct current (voltage).

Q.4–13 – Short answers (1½ marks each)**🔧 Q.4 • Ch 1 • 1½ marks**

Two point charges of $2 \mu\text{C}$ and $8 \mu\text{C}$ are placed 0.3 m apart in air. Find the force between them. What would the force be if the charges were placed in a medium of dielectric constant 2?

Solution: $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 8 \times 10^{-6}}{(0.3)^2} \text{ (½)}$

$F = 1.6 \text{ N}$ (repulsive) (½)

In the medium $F' = F / K = 1.6 / 2 = 0.8 \text{ N}$ (½)

 Q.5 • Ch 2 • 1½ marks

A parallel-plate capacitor has plates of area 100 cm^2 separated by 1 mm in air. Find its capacitance and the charge on it when connected to a 100 V battery. ($\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$)

Solution: $C = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 10^{-2}}{10^{-3}} \text{ (½)}$

$C = 8.85 \times 10^{-11} \text{ F} = 88.5 \text{ pF} \text{ (½)}$

$Q = CV = 8.85 \times 10^{-11} \times 100 = 8.85 \times 10^{-9} \text{ C} \text{ (½)}$

 Q.6 • Ch 4 • 1½ marks

State the Biot–Savart law and write it in vector form.

Solution: The magnetic field $d\vec{B}$ at a point P due to a current element $I d\vec{l}$ is directly proportional to I, dl and $\sin\theta$ and inversely proportional to the square of the distance r of P from the element ($\theta =$ angle between $d\vec{l}$ and $\vec{r}$). (1)

Vector form: $d\vec{B} = \frac{\mu_0 I d\vec{l} \times \vec{r}}{4\pi r^3}$; the direction is that of $d\vec{l} \times \vec{r}$ (right-hand rule). (½)

 Q.7 • Ch 4 • 1½ marks

A galvanometer of resistance 100Ω gives full-scale deflection for 5 mA . How can it be converted into a voltmeter of range $0\text{--}10 \text{ V}$?

Solution: Connect a high resistance R in **series** with the galvanometer. (½)

$V = I_g(G + R) \Rightarrow R = \frac{V}{I_g} - G = \frac{10}{5 \times 10^{-3}} - 100 \text{ (½)}$

$R = 2000 - 100 = 1900 \Omega \text{ (½)}$

 Q.8 • Ch 5 • 1½ marks

Write three differences between soft and hard magnetic materials.

Solution: (1) **Hysteresis loop:** soft – narrow, small area; hard – wide, large area. (½)

(2) **Retentivity and coercivity:** soft – low coercivity, easily magnetised and demagnetised; hard – high retentivity and coercivity, retain magnetism. (½)

(3) **Uses:** soft (soft iron) – cores of transformers, electromagnets, motors; hard (steel, alnico) – permanent magnets. (½)

 Q.9 • Ch 6 • 1½ marks

A coil of self-inductance 0.5 H carries a current of 4 A . Find the energy stored in it. If the current falls to zero in 0.02 s , find the average induced emf.

Solution: $U = \frac{1}{2}LI^2 = \frac{1}{2} \times 0.5 \times 4^2 = 4 \text{ J} \text{ (½)}$

$\epsilon = L \frac{\Delta I}{\Delta t} = \frac{0.5 \times (4 - 0)}{0.02} \text{ (½)}$

$\epsilon = 100 \text{ V} \text{ (½)}$

 Q.10 • Ch 9 • 1½ marks

A prism of refracting angle 60° is made of glass of refractive index $\sqrt{2}$. Find the angle of minimum deviation.

Solution: $n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}} \Rightarrow \sin \frac{60^\circ + \delta_m}{2} = \sqrt{2} \sin 30^\circ = \frac{1}{\sqrt{2}}$ (½)

$$\frac{60^\circ + \delta_m}{2} = 45^\circ \text{ (½)}$$

$$\delta_m = 30^\circ \text{ (½)}$$

 Q.11 • Ch 11 • 1½ marks

An electron moves with a speed of 1×10^6 m/s. Find its de Broglie wavelength and its kinetic energy in eV. ($m = 9.1 \times 10^{-31}$ kg)

Solution: $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^6} = 7.29 \times 10^{-10} \text{ m} \approx 7.3 \text{ Å}$ (1)

$$K = \frac{1}{2}mv^2 = \frac{1}{2} \times 9.1 \times 10^{-31} \times 10^{12} = 4.55 \times 10^{-19} \text{ J} = \frac{4.55 \times 10^{-19}}{1.6 \times 10^{-19}} \approx 2.84 \text{ eV} \text{ (½)}$$

 Q.12 • Ch 13 • 1½ marks

Calculate the binding energy per nucleon of ${}^4_2\text{He}$. Masses: $m_p = 1.00728 \text{ u}$, $m_n = 1.00867 \text{ u}$, nuclear mass of ${}^4_2\text{He} = 4.0015 \text{ u}$; $1 \text{ u} = 931.5 \text{ MeV}/c^2$.

Solution: $\Delta m = 2m_p + 2m_n - M = 2(1.00728) + 2(1.00867) - 4.0015 = 0.0304 \text{ u}$ (½)

$$E_b = 0.0304 \times 931.5 \approx 28.3 \text{ MeV} \text{ (½)}$$

$$E_b / A = \frac{28.3}{4} \approx 7.08 \text{ MeV per nucleon} \text{ (½)}$$

 Q.13 • Ch 14 • 1½ marks

Explain half-wave rectification using a p-n junction diode in three points.

Solution: (1) The AC input is applied to the diode in series with a load resistance R_L (through a transformer). (½)

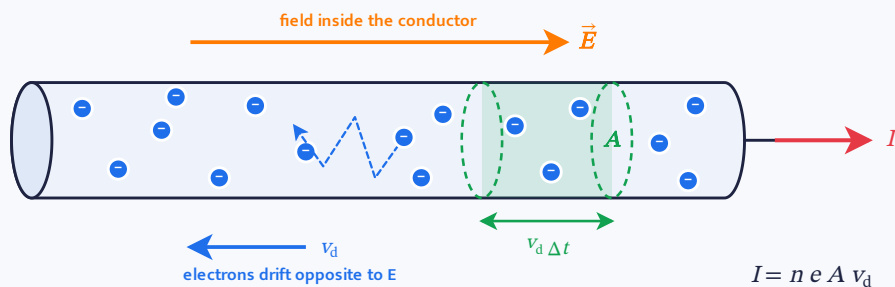
(2) In the positive half cycle the diode is forward biased and conducts; in the negative half cycle it is reverse biased and does not conduct. (½)

(3) The output across R_L is a series of unidirectional pulses (pulsating DC) of the same frequency as the input (50 Hz); only half the cycle is used. (½)

Q.14–16 – Long answers (3 marks each)

🔧 Q.14 (a) • Ch 3 • 3 marks

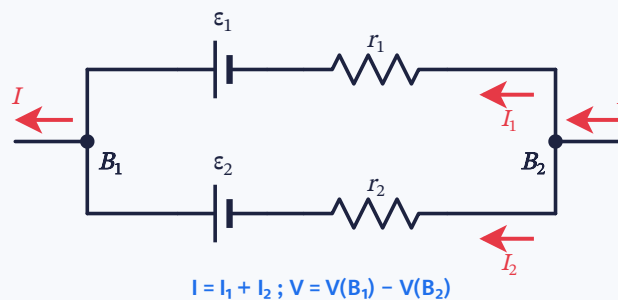
Derive the relation $I = neAv_d$ between current and drift velocity, and hence show that $\rho = \frac{m}{ne^2\tau}$.



- Solution: 1.** Free electrons (n per unit volume, charge $-e$) in a field E accelerate with $a = -\frac{eE}{m}$; they lose their extra velocity at collisions after an average relaxation time τ , so the drift velocity is $v_d = \frac{eE\tau}{m}$ (opposite to E). (1)
- 2.** In time Δt all electrons within the length $v_d \Delta t$ cross the area A (figure): charge $= n(Av_d \Delta t)e$, so $I = neAv_d$. (1/2)
- 3.** Current density $j = \frac{I}{A} = nev_d = \frac{ne^2\tau}{m}E$, i.e. $j = \sigma E$ with $\sigma = \frac{ne^2\tau}{m}$ (Ohm's law in vector form). (1/2)
- 4.** For a wire of length l , $E = \frac{V}{l}$ and $j = \frac{I}{A}$ give $V = \frac{l}{\sigma A}I = IR$, so $R = \rho \frac{l}{A}$ with $\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$ (1)

🔧 Q.14 (b) • Ch 3 • 3 marks

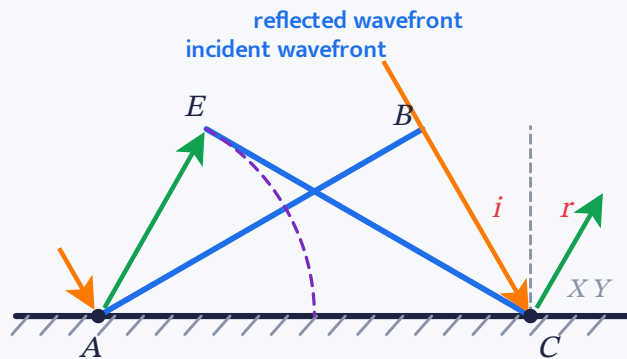
Derive the expressions for the equivalent emf and equivalent internal resistance of two cells connected in parallel. Two cells ($2\text{ V}, 1\ \Omega$) and ($1\text{ V}, 1\ \Omega$) in parallel are connected to an external resistance of $1\ \Omega$; find the current in it.



- Solution: 1.** Cells $(\varepsilon_1, r_1), (\varepsilon_2, r_2)$ are joined between B_1 and B_2 (figure); they give currents I_1, I_2 and $I = I_1 + I_2$ flows in the external circuit. If $V = V_{B_1} - V_{B_2}$: $I_1 = \frac{\varepsilon_1 - V}{r_1}, I_2 = \frac{\varepsilon_2 - V}{r_2}$. (1/2)
- 2.** $I = \frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right) \Rightarrow V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$. (1)
- 3.** Comparing with $V = \varepsilon_{eq} - Ir_{eq}$: $\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}, r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$ (1/2)
- 4.** Numerical: $\varepsilon_{eq} = \frac{2(1) + 1(1)}{1 + 1} = 1.5\text{ V}, r_{eq} = \frac{1 \times 1}{1 + 1} = 0.5\ \Omega; I = \frac{1.5}{1 + 0.5} = 1\text{ A}$ (1)

 Q.15 (a) • Ch 10 • 3 marks

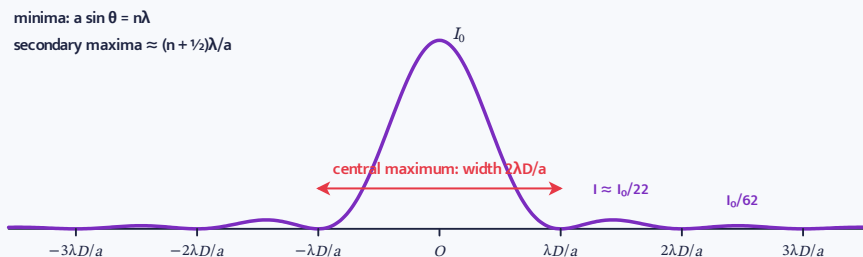
Using Huygens' principle, prove the law of reflection of light at a plane surface.



- Solution: 1.** A plane wavefront AB is incident on the reflecting surface XY; A touches the surface first, and the ray through B reaches the surface at C after a time t , so $BC = vt$ (figure). **(1)**
- 2.** In this time the secondary wavelet from A spreads in the same medium to a radius $AE = vt = BC$. **(½)**
- 3.** The tangent from C to this wavelet is the reflected wavefront CE. **(½)**
- 4.** Right-angled triangles ABC and AEC have a common hypotenuse AC and $BC = AE$, so they are congruent: $\angle BAC = \angle ECA$. **(½)**
- 5.** $\angle BAC = i$ and $\angle ECA = r$ (angles between the wavefronts and the surface equal those between the rays and the normal) $\Rightarrow i = r$; the incident ray, reflected ray and normal lie in one plane. **(½)**

 Q.15 (b) • Ch 10 • 3 marks

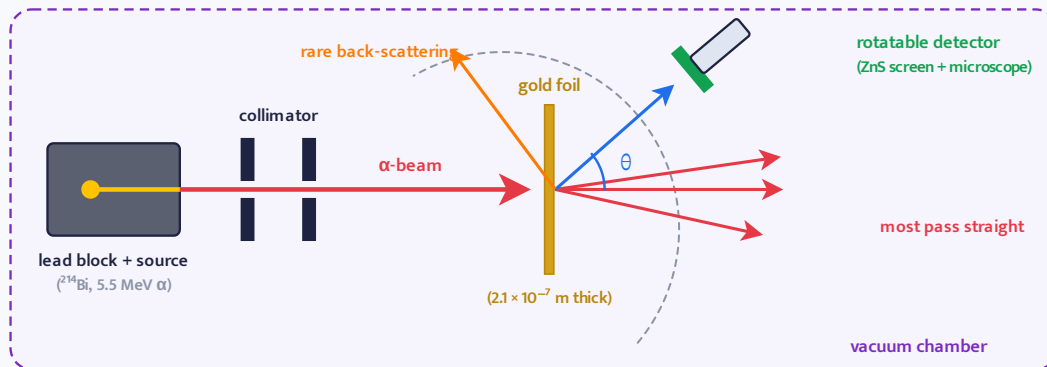
Explain the diffraction of light at a single slit. Obtain the condition for the minima and the width of the central maximum.



- Solution: 1.** Plane wave of wavelength λ falls on a slit of width a ; the pattern is observed on a screen at distance D (figure): a broad bright **central maximum** with weaker, narrower maxima on both sides. **(½)**
- 2. Minima:** at angle θ the path difference between the two edges is $a \sin \theta$. For $a \sin \theta = \lambda$ divide the slit into two halves; each point of one half has a partner in the other half with path difference $\lambda / 2$ – they cancel in pairs. In general $a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2, \dots$). **(1)**
- 3. Secondary maxima** occur near $a \sin \theta = \left(n + \frac{1}{2}\right)\lambda$; their intensity falls rapidly with n . **(½)**
- 4. Central maximum** lies between the first minima $\theta = \pm \lambda / a$; angular width $2\lambda / a$, linear width $\frac{2\lambda D}{a}$; it is twice as wide as any secondary maximum and widens when the slit is narrowed. **(1)**

 Q.16 (a) • Ch 12 • 3 marks

Describe the Geiger–Marsden α -particle scattering experiment. State the observations and the conclusions drawn from them.



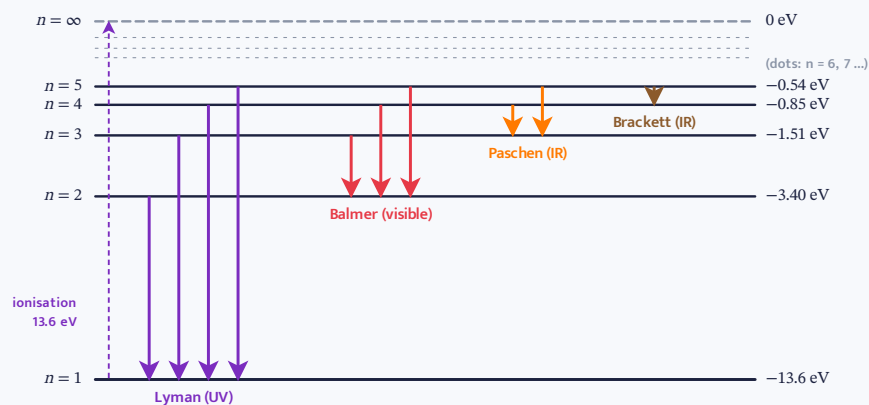
Solution: Set-up: a narrow beam of α -particles from a radioactive source in a lead box falls on a very thin gold foil (about 10^{-7} m); a rotatable detector (ZnS screen with microscope) counts the scattered α -particles at various angles, all in vacuum (figure). (½)

Observations: (i) most α -particles pass through the foil undeviated; (ii) many are deflected through small angles; (iii) only about 1 in 8000 is deflected by more than 90° , a few almost back along the path. (1)

Conclusions: (i) an atom is mostly empty space; (ii) the entire positive charge and almost all the mass are concentrated in a tiny **nucleus** (size about 10^{-15} – 10^{-14} m) at the centre, with electrons around it – the **nuclear model**; (iii) a large deflection occurs only when an α -particle comes very close to the nucleus (small impact parameter) and is strongly repelled by its Coulomb field. (1½)

 Q.16 (b) • Ch 12 • 3 marks

Using Bohr's postulates, derive the expressions for the radius of the n -th orbit and the total energy of the electron in the n -th orbit of a hydrogen atom. What is the energy in the ground state?



Solution: 1. Electron (m, v) in a circular orbit of radius r : Coulomb force provides the centripetal force,

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \dots (1); \text{ quantisation (Bohr): } mvr = \frac{nh}{2\pi} \dots (2). \quad (\frac{1}{2})$$

$$2. \text{ Eliminating } v \text{ from (1) and (2): } r_n = \frac{\epsilon_0 n^2 h^2}{\pi m e^2} = 0.53 n^2 \text{ \AA}. \quad (1)$$

$$3. \text{ From (1): } K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\epsilon_0 r} \text{ and } U = -\frac{e^2}{4\pi\epsilon_0 r}, \text{ so } E = K + U = -\frac{e^2}{8\pi\epsilon_0 r_n}. \quad (\frac{1}{2})$$

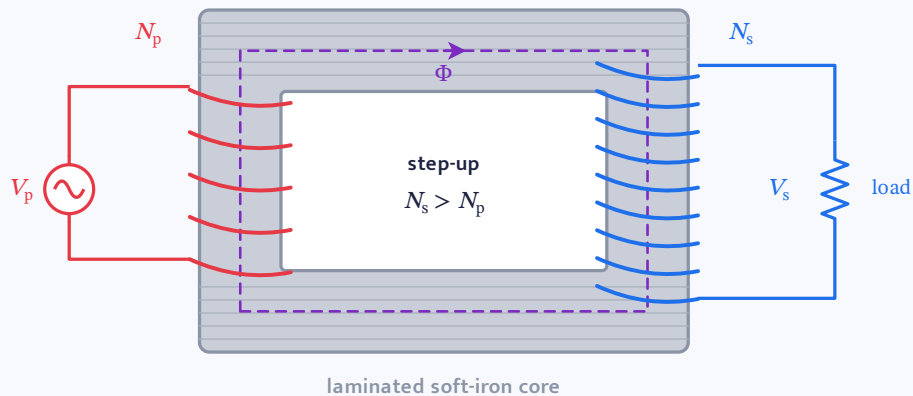
$$4. \text{ Substituting } r_n: E_n = -\frac{me^4}{8\epsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2} \text{ eV} \quad (\frac{1}{2})$$

$$5. \text{ Ground state } (n = 1): E_1 = -13.6 \text{ eV (the energy needed to ionise the atom)}. \quad (\frac{1}{2})$$

Q.17–18 – Essay type (4 marks each)

🔧 Q.17 (a) • Ch 7 • 4 marks

Explain the principle, construction and working of a transformer. Obtain the relation between the voltages and the number of turns. A step-down transformer converts 2200 V to 220 V; the primary has 4000 turns and the secondary delivers 5 A. Find the number of secondary turns and the primary current (ideal transformer).



Solution: 1. Labelled diagram: laminated soft-iron core, primary coil (N_p turns) and secondary coil (N_s turns) wound on it (figure). (1)

2. Principle: mutual induction – a changing current in the primary changes the flux in the core, inducing an emf in the secondary. (½)

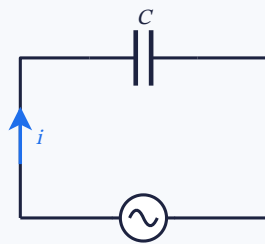
3. Relation: the same flux Φ links each turn, so $\varepsilon_p = -N_p \frac{d\Phi}{dt}$ and $\varepsilon_s = -N_s \frac{d\Phi}{dt}$. For an ideal transformer (no losses) $V_p = \varepsilon_p$, $V_s = \varepsilon_s$: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$; with $V_p I_p = V_s I_s$, $\frac{I_p}{I_s} = \frac{N_s}{N_p}$. $N_s > N_p$: step-up; $N_s < N_p$: step-down. (1)

4. Numerical: $N_s = N_p \frac{V_s}{V_p} = 4000 \times \frac{220}{2200} = 400 \text{ turns}$ (½); $I_p = I_s \frac{V_s}{V_p} = 5 \times \frac{1}{10} = 0.5 \text{ A}$ (½)

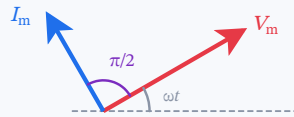
5. Losses (any one with remedy): copper loss – thick wires; eddy-current loss – laminated core; hysteresis loss – soft-iron core; flux leakage – coils wound over one another. (½)

 Q.17 (b) • Ch 7 • 4 marks

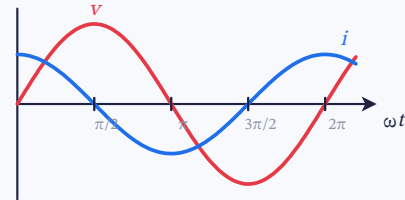
Derive the expression for the current in a pure capacitor connected to an AC source and show that the current leads the voltage by $\pi/2$. Show that the average power consumed is zero. A $100\ \mu\text{F}$ capacitor is connected to a $220\ \text{V}$, $50\ \text{Hz}$ supply; find its reactance and the rms current.



$v = v_m \sin \omega t$
(a) circuit



(b) phasor diagram



(c) i leads v by $\pi/2$

Solution: 1. Source $v = v_0 \sin \omega t$ across a capacitor C (figure): charge $q = Cv = Cv_0 \sin \omega t$. (½)

2. $i = \frac{dq}{dt} = \omega C v_0 \cos \omega t \Rightarrow i = i_0 \sin\left(\omega t + \frac{\pi}{2}\right)$, $i_0 = \omega C v_0 = \frac{v_0}{X_C}$ with $X_C = \frac{1}{\omega C}$: the current **leads** the voltage by $\pi/2$. (1)

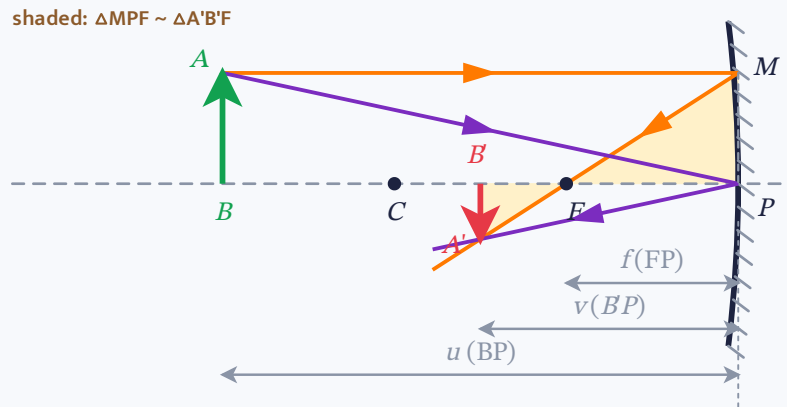
3. Power $p = vi = \frac{v_0 i_0}{2} \sin 2\omega t$: positive for a quarter cycle (capacitor charges, energy stored) and negative for the next quarter (energy returned); its mean over a cycle is zero, $\bar{P} = 0$. (1)

4. $X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} \approx 31.8\ \Omega$ (½)

5. $I_{rms} = \frac{V_{rms}}{X_C} = \frac{220}{31.8} \approx 6.9\ \text{A}$ (1)

 Q.18 (a) • Ch 9 • 4 marks

Derive the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for a concave mirror and obtain the expression for the magnification.



Solution: 1. Ray diagram (figure): object AB beyond the centre of curvature; ray AM parallel to the axis reflects through F; ray AP at the pole reflects obeying $\angle APB = \angle A'PB'$; they meet at A' to form the real image $A'B'$. For paraxial rays MP is taken perpendicular to the axis, so $MP = AB$. (1)

2. $\triangle A'B'P \sim \triangle ABP$: $\frac{A'B'}{AB} = \frac{B'P}{BP} \dots (1) \text{ (}\frac{1}{2}\text{)}$

3. $\triangle A'B'F \sim \triangle MPF$: $\frac{A'B'}{MP} = \frac{B'F}{FP}$ and $MP = AB \Rightarrow \frac{A'B'}{AB} = \frac{B'F}{FP} \dots (2) \text{ (}\frac{1}{2}\text{)}$

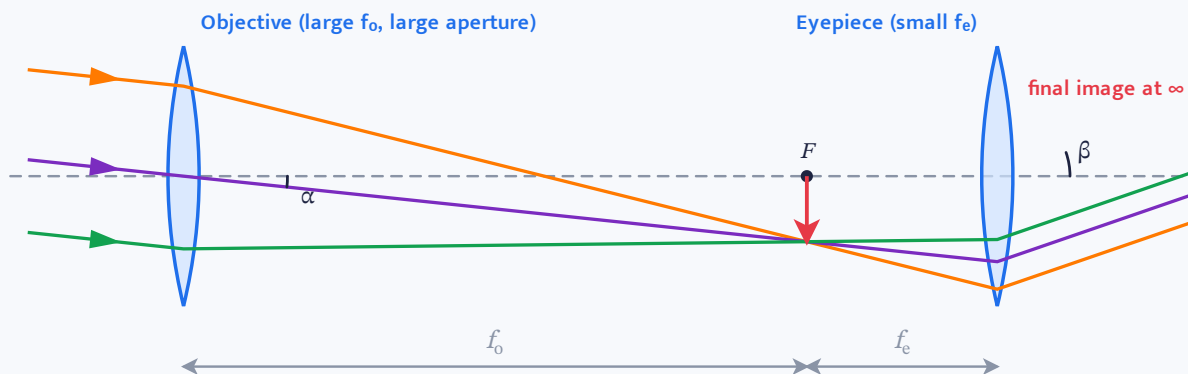
4. From (1) and (2): $\frac{B'P}{BP} = \frac{B'F}{FP} = \frac{B'P - FP}{FP} \dots (1/2)$

5. Sign convention: $B'P = -v, BP = -u, FP = -f \Rightarrow \frac{-v}{-u} = \frac{-v + f}{-f} \Rightarrow \frac{v}{u} = \frac{v - f}{f} \Rightarrow$ dividing by v: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ (1)

6. Magnification from (1): $m = \frac{\text{image height}}{\text{object height}} = -\frac{v}{u}$ (negative for a real inverted image). (1/2)

 Q.18 (b) • Ch 9 • 4 marks

Draw the ray diagram of an astronomical telescope in normal adjustment and derive its magnifying power and tube length. A telescope has a magnifying power of 20 and its tube length is 105 cm; find the focal lengths of the objective and the eyepiece.



Solution: 1. Ray diagram (figure): parallel rays from a distant object make angle α with the axis; the objective (large f_o) forms a real image $A'B'$ at its focus; the eyepiece (f_e) is placed so that $A'B'$ lies at its focus, giving the final image at infinity. **(1)**

2. $\alpha \approx \tan \alpha = \frac{A'B'}{f_o}$ and the final image subtends $\beta \approx \frac{A'B'}{f_e}$ at the eye. **(1)**

3. $M = \frac{\beta}{\alpha} = \frac{f_o}{f_e}$ **(½)**

4. Tube length $L = f_o + f_e$ **(½)**

5. Numerical: $\frac{f_o}{f_e} = 20$ and $f_o + f_e = 105 \Rightarrow 21f_e = 105 \Rightarrow f_e = 5 \text{ cm}, f_o = 100 \text{ cm}$ **(1)**

Score sheet

Section	Maximum	Your marks
A (Q.1–3)	24	
B (Q.4–13)	15	
C (Q.14–16)	9	
D (Q.17–18)	8	
Total	56	

Score bands

48+ excellent – polish diagrams and units • **40–47** good – recheck every numerical • **below 40** – revise the derivations and formula boxes of the chapters where you lost marks, then retry this paper after 3 days.

DAY

28

RBSE CLASS 12 PHYSICS • SS-40

Model Paper 4

18 questions • all compulsory

3 h 15 min

56 marks

Internal choice in Q14–18

Marking scheme with step marks

An original paper in the 2025 board pattern. Sit it like the real exam: 3 h 15 min, no book, no phone. Then mark yourself with the step-wise scheme and fill the chapter table.

Instructions to candidates



General instructions

- Subject: Physics • Code **SS-40** • Time **3 h 15 min** • Maximum marks **56**.
- All **18** questions are compulsory. Internal choice (OR) is given only in Q14 to Q18.
- Write the question number exactly as printed. Marks are shown on the right of each question.
- Draw neat, labelled diagrams wherever asked or useful. Write all steps of a numerical with units.



Constants you may use

- $e = 1.6 \times 10^{-19} \text{ C}$ • $c = 3 \times 10^8 \text{ m/s}$
- $h = 6.63 \times 10^{-34} \text{ J s}$ • $m_e = 9.1 \times 10^{-31} \text{ kg}$
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$ • $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$
- $1 \text{ u} = 931.5 \text{ MeV}/c^2$ • $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$



Time plan for this paper (195 min)

- Section A (38 items): **50 min** • Section B ($10 \times 1\frac{1}{2}$): **60 min** • Section C (3×3): **35 min** • Section D (2×4): **40 min** • revision: **10 min**.
- Do not look at the marking scheme until the 195 minutes are over.

Question paper

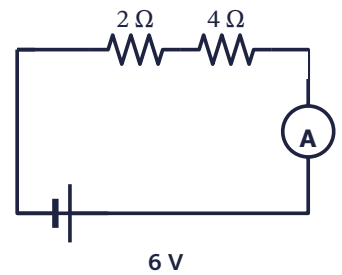
Section A – Objective and very short answer

24

Q1. Choose the correct answer ($18 \times \frac{1}{2} = 9$)

- i The SI unit of electric potential is equivalent to
 (a) C m^{-2} (b) J C (c) N C^{-1} (d) J C^{-1}
- ii A $2 \mu\text{F}$ capacitor is charged to 100 V. The energy stored in it is
 (a) 0.01 J (b) 1 J (c) 0.1 J (d) 0.02 J
- iii The resistance of a copper wire of length 10 m and cross-section 1 mm^2 is ($\rho = 1.7 \times 10^{-8} \Omega \text{ m}$)
 (a) 17 Ω (b) 0.017 Ω (c) 0.17 Ω (d) 1.7 Ω

- iv The reading of the ideal ammeter in the circuit shown is
 (a) 2 A (b) 1 A (c) 1.5 A (d) 3 A



- v A straight wire of length 0.5 m carrying 4 A is placed at 30° to a uniform field of 0.2 T. The force on it is
 (a) 0.35 N (b) 0.1 N (c) 0.4 N (d) 0.2 N

- vi The SI unit tesla is equivalent to
 (a) N A m (b) $\text{N A}^{-1} \text{ m}$ (c) $\text{N A}^{-1} \text{ m}^{-1}$ (d) N A m^{-1}

- vii An alternating voltage is $V = 100\sin(100\pi t)$ volt. Its frequency is
 (a) 314 Hz (b) 50 Hz (c) 100 Hz (d) 25 Hz

- viii The ozone layer in the atmosphere absorbs mainly
 (a) radio waves (b) ultraviolet rays
 (c) microwaves (d) infrared rays

- ix An object is placed 20 cm in front of a concave mirror of focal length 10 cm. The image is
 (a) 20 cm in front, real, inverted, same size
 (b) 40 cm in front, real, inverted, magnified
 (c) 10 cm in front, real, inverted, diminished
 (d) 20 cm behind, virtual, erect, same size

- x The speed of light in a medium of refractive index 1.5 is ($c = 3 \times 10^8 \text{ m/s}$)
 (a) $1.5 \times 10^8 \text{ m/s}$ (b) $3 \times 10^8 \text{ m/s}$ (c) $4.5 \times 10^8 \text{ m/s}$ (d) $2 \times 10^8 \text{ m/s}$

- xi In Young's experiment the distance between the two slits is halved, other things unchanged. The fringe width becomes
 (a) double (b) half (c) unchanged (d) four times

- xii The Brewster angle for light going from air to glass of refractive index $\sqrt{3}$ is
 (a) 30° (b) 45° (c) 90° (d) 60°

- xiii The energy of the electron of a hydrogen atom in the orbit $n = 2$ is
 (a) -6.8 eV (b) -3.4 eV (c) -13.6 eV (d) -1.51 eV

- xiv Hydrogen atoms are excited to the level $n = 4$. The maximum number of different spectral lines they can emit is
 (a) 3 (b) 6 (c) 10 (d) 4

- xv The number of neutrons in the nucleus of ${}_{92}^{238}\text{U}$ is
 (a) 54 (b) 92 (c) 238 (d) 146

- xvi** The ratio of the radii of two nuclei of mass numbers 64 and 8 is
 (a) 1 : 2 (b) 4 : 1 (c) 8 : 1 (d) 2 : 1
- xvii** A p-type semiconductor is obtained by doping pure silicon with
 (a) indium (b) phosphorus (c) antimony (d) arsenic
- xviii** When a p-n junction diode is forward biased, the width of its depletion region
 (a) remains the same (b) decreases
 (c) increases (d) becomes infinite

- 2 Fill in the blanks ($10 \times \frac{1}{2} = 5$)** **[10 × ½]**
- (i) The equipotential surfaces of a uniform electric field are planes ____ to the field lines.
 (ii) The resistivity of a metallic conductor ____ with rise in temperature.
 (iii) The magnetic moment of a coil of N turns, area A and current I is ____.
 (iv) The magnetic susceptibility of a ferromagnetic substance is very large and ____.
 (v) The SI unit of inductive reactance is ____.
 (vi) The current produced by a time-varying electric flux is called ____ current.
 (vii) The power of a lens of focal length 50 cm is ____ dioptre.
 (viii) In Rutherford's α -scattering experiment, the impact parameter for a head-on collision is ____.
 (ix) Nuclei having the same mass number but different atomic number are called ____.
 (x) A Zener diode is used as a voltage regulator in ____ bias.

- 3 Answer in one or two sentences ($10 \times 1 = 10$)** **[10 × 1]**
- (i) Define electric dipole moment. Write its SI unit.
 (ii) State Kirchhoff's junction rule.
 (iii) Write the expression for the Lorentz force on a charge q moving with velocity $\vec{v}$ in electric field $\vec{E}$ and magnetic field $\vec{B}$.
 (iv) Define the angle of dip.
 (v) Define 1 henry.
 (vi) Write the relation between the speed of electromagnetic waves in vacuum and the constants ϵ_0 and μ_0 .
 (vii) State Malus' law.
 (viii) Write the expression for the de Broglie wavelength of a particle of mass m moving with velocity v .
 (ix) Which series of the hydrogen spectrum lies in the visible region?
 (x) What is meant by doping of a semiconductor?

Section B – Short answer

15

- 4** Three capacitors of $4 \mu\text{F}$, $12 \mu\text{F}$ and $6 \mu\text{F}$ are connected in series across a 60 V battery. Find (i) the equivalent capacitance and (ii) the potential difference across the $4 \mu\text{F}$ capacitor. **[1½]**

- 5 A cell of emf 12 V and internal resistance $2\ \Omega$ is connected to a $10\ \Omega$ resistor. Find the current and the terminal voltage. [1½]
- 6 A galvanometer of resistance $100\ \Omega$ shows full-scale deflection for 5 mA. How can it be converted into a voltmeter of range 0–10 V? [1½]
- 7 Write any three differences between diamagnetic and paramagnetic substances. [1½]
- 8 A rod 40 cm long moves with a speed of 5 m/s perpendicular to a uniform magnetic field of 0.5 T. It forms part of a closed circuit of total resistance $2\ \Omega$. Find the induced emf, the current and the force needed to keep the rod moving uniformly. [1½]
- 9 The mutual inductance of two coils is 1.2 H. The current in the primary changes from 0 to 5 A in 0.2 s. Find the emf induced in the secondary and the change in flux linked with it. [1½]
- 10 An object is placed 30 cm from a convex lens of focal length 20 cm. Find the position of the image and the magnification. State the nature of the image. [1½]
- 11 Calculate the wavelength of the first line ($H\alpha$) of the Balmer series of hydrogen. ($R = 1.097 \times 10^7\ \text{m}^{-1}$) [1½]
- 12 Define mass defect. Calculate the binding energy of a nucleus whose mass defect is 0.0304 u. ($1\ \text{u} = 931.5\ \text{MeV}/c^2$) [1½]
- 13 Write any three differences between n-type and p-type semiconductors. [1½]

Section C – Long answer

9

- 14 Using Gauss's law, derive the expression for the electric field at a distance r from an infinitely long straight wire of uniform linear charge density λ . Draw a diagram. [3]
- OR -----
- State Gauss's law. Use it to find the electric field due to a uniformly charged thin spherical shell (i) outside it and (ii) inside it.
- 15 In Young's double-slit experiment derive the expression for the fringe width. [3]
- OR -----
- Using Huygens' principle, prove the law of reflection of light.
- 16 State Einstein's photoelectric equation. Using it explain (i) the existence of a threshold frequency, (ii) the independence of the maximum kinetic energy from the intensity of light, (iii) the absence of time lag. [3]
- OR -----
- State the de Broglie hypothesis. Derive the expression for the de Broglie wavelength of an electron accelerated from rest through a potential difference V and hence calculate it for $V = 100\ \text{V}$.

Section D – Essay type

8

- 17 (a) A source $v = v_m \sin \omega t$ drives a series LCR circuit. Using a phasor diagram, derive the expressions for the impedance and the phase angle. (3) [4]
(b) Find the impedance and the rms current if $R = 30 \, \Omega$, $X_L = 70 \, \Omega$, $X_C = 30 \, \Omega$ and the rms source voltage is 100 V. (1)

----- OR -----

- (a) What is resonance in a series LCR circuit? Obtain the resonant frequency and draw the graph of current against angular frequency. (3)
(b) Find ω_0 and the quality factor Q for $L = 2 \, \text{H}$, $C = 8 \, \mu\text{F}$, $R = 10 \, \Omega$. (1)

- 18 (a) Derive the relation $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$ for refraction at a spherical surface. (3) [4]
(b) A point object is 60 cm from a convex spherical surface of radius 20 cm separating air from glass ($n = 1.5$). Find the position of the image. (1)

----- OR -----

- (a) Draw the ray diagram of an astronomical telescope in normal adjustment and derive its magnifying power. (3)
(b) The focal lengths of the objective and the eyepiece are 100 cm and 5 cm. Find the magnifying power and the length of the tube. (1)

— End of question paper —

Marking scheme and model answers



How to mark yourself

- Marks in **bold brackets** are awarded for that step. A correct alternative method gets full marks.
- Numerical: formula (½), substitution (½), answer with unit (½). Missing or wrong unit: deduct ½ once.
- Be strict about the labelled diagram – it carries marks in every derivation.



Q1 – answer key with reasons (½ each)

i(d) ii(a) iii(c) iv(b) v(d) vi(c) vii(b) viii(b) ix(a) x(d) xi(a) xii(d) xiii(b) xiv(b)

xv(d) xvi(d) xvii(a) xviii(b)

(i) (d) Potential is work per unit charge, $V = W / q$, so 1 volt = 1 joule per coulomb.

(ii) (a) $U = \frac{1}{2}CV^2 = \frac{1}{2}(2 \times 10^{-6})(100)^2 = 0.01 \text{ J}$.

(iii) (c) $R = \rho l / A = \frac{1.7 \times 10^{-8} \times 10}{10^{-6}} = 0.17 \Omega$.

(iv) (b) Resistors in series: $R = 2 + 4 = 6 \Omega$; $I = 6 / 6 = 1 \text{ A}$ (ideal ammeter has zero resistance).

(v) (d) $F = BIl \sin \theta = 0.2 \times 4 \times 0.5 \times 0.5 = 0.2 \text{ N}$.

(vi) (c) From $F = BIl$: $B = F / (Il)$, so $1 \text{ T} = 1 \text{ N A}^{-1} \text{ m}^{-1}$.

(vii) (b) $\omega = 100\pi \text{ rad/s}$, so $\nu = \omega / 2\pi = 50 \text{ Hz}$.

(viii) (b) Ozone absorbs the harmful ultraviolet radiation of the Sun.

(ix) (a) Object at the centre of curvature ($u = -2f = -20 \text{ cm}$): $1 / v = 1 / f - 1 / u$ gives $v = -20 \text{ cm}$, $m = -1$.

(x) (d) $v = c / n = 3 \times 10^8 / 1.5 = 2 \times 10^8 \text{ m/s}$.

(xi) (a) $\beta = \lambda D / d$, so halving d doubles β .

(xii) (d) $\tan i_B = n = \sqrt{3} \Rightarrow i_B = 60^\circ$.

(xiii) (b) $E_n = -13.6 / n^2 \text{ eV} = -13.6 / 4 = -3.4 \text{ eV}$.

(xiv) (b) Number of lines = $n(n-1) / 2 = 4 \times 3 / 2 = 6$.

(xv) (d) Neutrons = $A - Z = 238 - 92 = 146$.

(xvi) (d) $R = R_0 A^{1/3}$: ratio = $(64 / 8)^{1/3} = 2$.

(xvii) (a) Trivalent impurities (B, Al, In, Ga) give p-type; pentavalent ones (P, As, Sb) give n-type.

(xviii) (b) The external field opposes the barrier field, so the depletion layer narrows.

Q2 – Fill in the blanks (½ each)

	Ch	Answer
(i)	2	perpendicular
(ii)	3	increases
(iii)	4	N/A
(iv)	5	positive
(v)	7	ohm (Ω)
(vi)	8	displacement
(vii)	9	+2
(viii)	12	zero
(ix)	13	isobars
(x)	14	reverse

Q3 – Very short answer (1 mark each)

	Ch	Model answer (marking points)
(i)	1	Product of either charge and the separation between them: $\vec{p} = q(2\vec{a})$, directed from -q to +q (½); unit C m (½).
(ii)	3	The algebraic sum of the currents meeting at a junction is zero, $\sum I = 0$ (1); it expresses conservation of charge.
(iii)	4	$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ (1)
(iv)	5	The angle between the earth's total magnetic field and the horizontal at that place (1).
(v)	6	The self-inductance of a coil in which an emf of 1 V is induced when the current changes at the rate of 1 A per second (1).
(vi)	8	$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (1)
(vii)	10	When plane-polarised light of intensity I_0 falls on an analyser, the transmitted intensity is $I = I_0 \cos^2 \theta$, θ = angle between the pass axes (1).
(viii)	11	$\lambda = \frac{h}{mv} = \frac{h}{p}$ (1)
(ix)	12	Balmer series (transitions to $n = 2$) (1).
(x)	14	Deliberately adding a small amount of a suitable impurity to a pure semiconductor to increase its conductivity (1).

Section B – Q4 to Q13 (1½ marks each)

🔧 Q4 • Ch 2 • Electrostatic Potential and Capacitance [1½]

Series: $\frac{1}{C} = \frac{1}{4} + \frac{1}{12} + \frac{1}{6} = \frac{3+1+2}{12} = \frac{1}{2}$ (½)

(i) $C = 2 \mu\text{F}$ (½)

(ii) $Q = CV = 2 \times 60 = 120 \mu\text{C}$ (same on each); $V_4 = Q / C_1 = 120 / 4 = 30 \text{ V}$ (½)

 Q5 • Ch 3 • Current Electricity [1½]

$$I = \frac{\varepsilon}{R + r} \quad (1/2)$$

$$I = \frac{12}{10 + 2} = 1 \text{ A} \quad (1/2)$$

$$V = IR = 1 \times 10 = 10 \text{ V (also } \varepsilon - Ir = 12 - 2) \quad (1/2)$$

 Q6 • Ch 4 • Moving Charges and Magnetism [1½]

A high resistance R is joined in **series**: $V = I_g(G + R)$ (1/2)

$$R = \frac{V}{I_g} - G = \frac{10}{5 \times 10^{-3}} - 100 \quad (1/2)$$

$R = 1900 \, \Omega$ in series with the galvanometer (1/2)

 Q7 • Ch 5 • Magnetism and Matter [1½]

Susceptibility: diamagnetic – small and negative (μ_r slightly < 1); paramagnetic – small and positive (μ_r slightly > 1). (1/2)

In a non-uniform field: diamagnetic – weakly repelled (moves to the weaker part); paramagnetic – weakly attracted (moves to the stronger part). (1/2)

Examples: Bi, Cu, Au, water – diamagnetic; Al, Na, O_2 – paramagnetic (susceptibility of paramagnetics depends on temperature, $\chi \propto 1/T$). (1/2)

 Q8 • Ch 6 • Electromagnetic Induction [1½]

$$\varepsilon = Blv = 0.5 \times 0.4 \times 5 = 1 \text{ V} \quad (1/2)$$

$$I = \varepsilon / R = 1 / 2 = 0.5 \text{ A} \quad (1/2)$$

$$F = BIl = 0.5 \times 0.5 \times 0.4 = 0.1 \text{ N (equal and opposite to the magnetic force, by Lenz's law)} \quad (1/2)$$

 Q9 • Ch 6 • Electromagnetic Induction [1½]

$$|\varepsilon| = M \frac{\Delta I}{\Delta t} \quad (1/2)$$

$$= 1.2 \times \frac{5}{0.2} = 30 \text{ V} \quad (1/2)$$

$$\Delta \Phi = M \Delta I = 1.2 \times 5 = 6 \text{ Wb} \quad (1/2)$$

 Q10 • Ch 9 • Ray Optics and Optical Instruments [1½]

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \text{ with } u = -30, f = +20 \text{ cm} \quad (1/2)$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60} \Rightarrow v = +60 \text{ cm (other side of the lens)} \quad (1/2)$$

$$m = v / u = 60 / (-30) = -2: \text{ real, inverted, twice the size} \quad (1/2)$$

 Q11 • Ch 12 • Atoms [1½]

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \text{ (H}\alpha: 3 \rightarrow 2) \quad (1/2)$$

$$\frac{1}{\lambda} = R \cdot \frac{5}{36} \Rightarrow \lambda = \frac{36}{5R} = \frac{36}{5 \times 1.097 \times 10^7} \quad (1/2)$$

$$\lambda \approx 6.56 \times 10^{-7} \text{ m} = 656 \text{ nm (red)} \quad (1/2)$$

🔧 Q12 • Ch 13 • Nuclei [1½]

Mass defect: the difference between the total mass of the separate nucleons and the mass of the nucleus,

$$\Delta m = [Zm_p + (A - Z)m_n] - M. \text{ (½)}$$

$$E_b = \Delta m c^2 = \Delta m \times 931.5 \text{ MeV} \text{ (½)}$$

$$E_b = 0.0304 \times 931.5 = \underline{28.3 \text{ MeV}} \text{ (½)}$$

🔧 Q13 • Ch 14 • Semiconductor Electronics [1½]

Majority carriers: n-type – electrons (holes are minority); p-type – holes (electrons are minority). (½)

Dopant: n-type – pentavalent donor (P, As, Sb); p-type – trivalent acceptor (B, Al, In). (½)

Energy band: n-type – donor level just below the conduction band; p-type – acceptor level just above the valence band. (½)

Section C – Q14 to Q16 (3 marks each)

🔧 Q14 alternative 1 • Ch 1 • Field of an infinite line charge [3] ★

Labelled diagram: coaxial cylinder of radius r and length l as the Gaussian surface. (½)

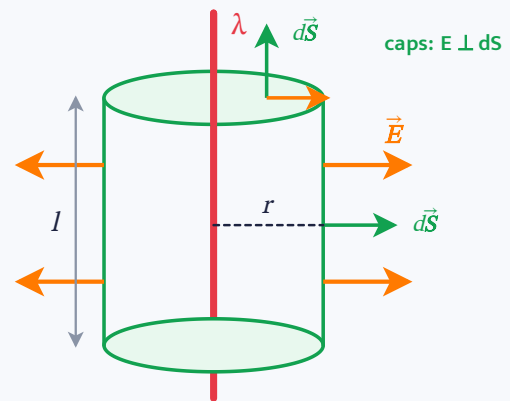
By symmetry $\vec{E}$ is radial and of constant magnitude on the curved surface; on the two flat ends $\vec{E} \perp d\vec{S}$, so their flux is zero. (½)

$$\text{Flux } \Phi = E(2\pi r l). \text{ (½)}$$

$$\text{Enclosed charge } q = \lambda l. \text{ (½)}$$

$$\text{Gauss's law } \Phi = q / \epsilon_0; E(2\pi r l) = \lambda l / \epsilon_0. \text{ (½)}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \text{ (radially outward for } \lambda > 0; E \propto 1/r). \text{ (½)}$$



cylindrical Gaussian surface

Cylindrical Gaussian surface around a line charge

🔧 Q14 alternative 2 (OR) • Ch 1 • Field of a charged spherical shell [3] ★

Gauss's law: the net flux through a closed surface equals

$$q_{in} / \epsilon_0. \text{ (½)}$$

Labelled diagram: shell of radius R , charge Q ; concentric

Gaussian spheres of radius $r > R$ and $r < R$. (½)

(i) $r > R$: $\vec{E}$ is radial and constant on the sphere, so

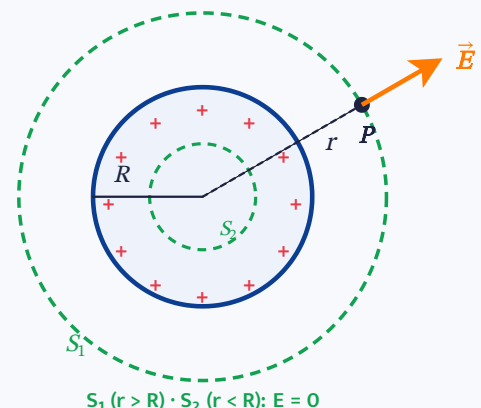
$$E(4\pi r^2) = Q / \epsilon_0. \text{ (½)}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} \text{ - as if all the charge were at the centre. (½)}$$

(ii) $r < R$: the Gaussian sphere encloses no charge, so

$$E(4\pi r^2) = 0. \text{ (½)}$$

$$E = 0 \text{ everywhere inside the shell. (½)}$$



$S_1 (r > R) \cdot S_2 (r < R): E = 0$

Gaussian spheres inside and outside the shell

🔧 Q15 alternative 1 • Ch 10 • Fringe width in Young's experiment [3] ★

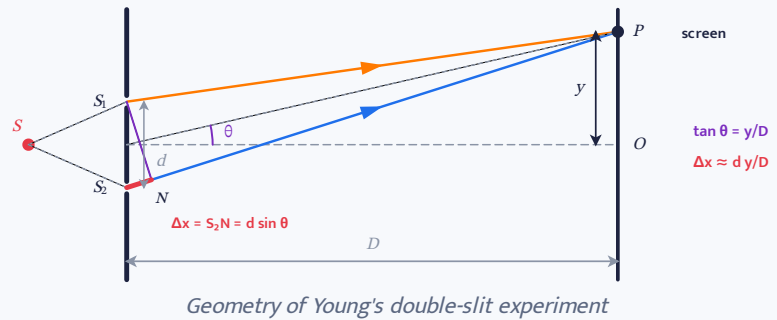
Diagram: slits S_1, S_2 separated by d ,
screen at distance D , point P at height y
from the central point O . (½)
 $(S_2P)^2 - (S_1P)^2 = 2yd \Rightarrow$ path
difference

$$\Delta = S_2P - S_1P = \frac{2yd}{S_2P + S_1P} \approx \frac{yd}{D} \text{ (for } y, d \ll D\text{)}. \text{ (1)}$$

Bright fringe: $\Delta = n\lambda \Rightarrow y_n = \frac{n\lambda D}{d}$. (½)

Dark fringe: $\Delta = (n + \frac{1}{2})\lambda \Rightarrow y_n = \frac{(n + \frac{1}{2})\lambda D}{d}$. (½)

Fringe width $\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$, the same for all fringes. (½)



🔧 Q15 alternative 2 (OR) • Ch 10 • Law of reflection by Huygens' principle [3] ★

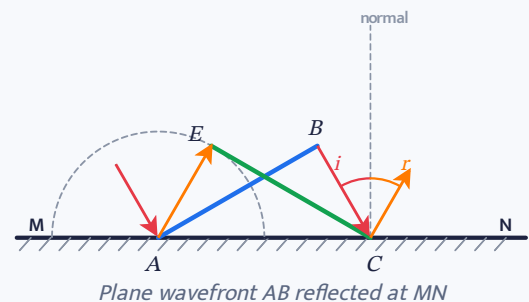
Huygens' principle: every point of a wavefront is a source of secondary wavelets; the new wavefront is the forward envelope of the wavelets after time t . (½)

Diagram: plane wavefront AB meets the mirror MN at A . While B travels $BC = vt$ to reach C , the wavelet from A spreads to radius $AE = vt$; the tangent CE from C is the reflected wavefront. (1)

In right triangles ABC and AEC : AC is common and $BC = AE = vt \Rightarrow$ the triangles are congruent. (½)

Hence $\angle BAC = \angle ECA$, i.e. **angle of incidence i = angle of reflection r** . (½)

The incident ray, the reflected ray and the normal lie in one plane. (½)



🔧 Q16 alternative 1 • Ch 11 • Einstein's photoelectric equation [3]

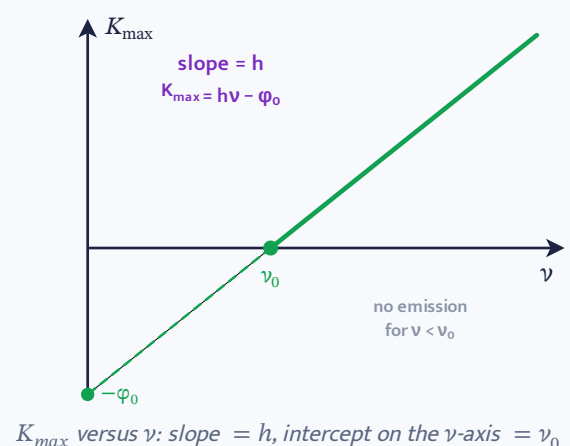
Each photon of energy $h\nu$ gives all its energy to one electron:

$$h\nu = \phi_0 + K_{max}, \text{ i.e. } K_{max} = h\nu - \phi_0 = h(\nu - \nu_0). \text{ (1)}$$

(i) K_{max} cannot be negative, so emission needs $\nu \geq \nu_0 = \phi_0 / h$; below ν_0 there is no emission however intense the light. (½)

(ii) K_{max} depends only on ν and ϕ_0 . Greater intensity means more photons per second, hence more photoelectrons (larger current), but the energy of each photon is unchanged. (1)

(iii) A photon-electron interaction is a single one-to-one event, so the electron is ejected at once (about 10^{-9} s) – no time is needed to build up energy. (½)



🔧 Q16 alternative 2 (OR) • Ch 11 • de Broglie wavelength of an accelerated electron [3] ★

de Broglie hypothesis: a moving particle has a wave associated with it, of wavelength $\lambda = h / p$. (½)

Work done by the field = kinetic energy gained: $eV = \frac{1}{2}mv^2 = \frac{p^2}{2m}$. (1)

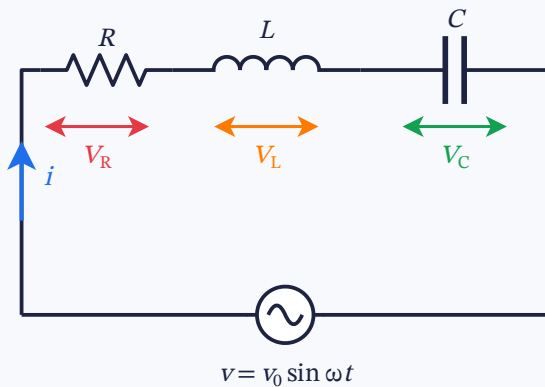
$$p = \sqrt{2meV} \text{ . (½)}$$

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{1.227}{\sqrt{V}} \text{ nm (½)}$$

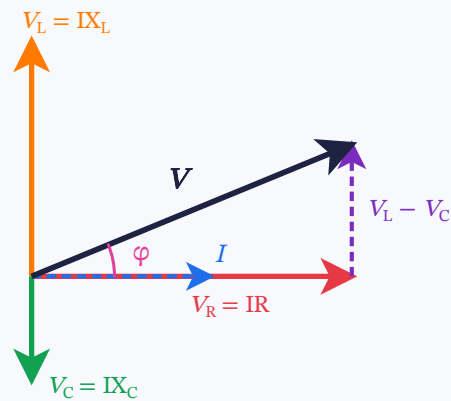
$$\text{For } V = 100 \text{ V: } \lambda = 1.227 / 10 = 0.1227 \text{ nm } (\approx 1.23 \text{ Å}). \text{ (½)}$$

Section D – Q17 and Q18 (4 marks each)

🔧 Q17 alternative 1 • Ch 7 • Series LCR: impedance [4] ★



Series LCR circuit



Phasor diagram ($X_L > X_C$)

Labelled circuit and phasor diagram. In series the same current $i = i_m \sin(\omega t - \phi)$ flows everywhere. (½)

Phasors: $V_R = i_m R$ along $\vec{I}$; $V_L = i_m X_L$ leads by 90° ; $V_C = i_m X_C$ lags by 90° . (½)

For $X_L > X_C$ the net reactive phasor is $V_L - V_C$, perpendicular to V_R . (½)

Pythagoras: $v_m^2 = V_R^2 + (V_L - V_C)^2 = i_m^2 [R^2 + (X_L - X_C)^2]$. (½)

$$Z = \frac{v_m}{i_m} = \sqrt{R^2 + (X_L - X_C)^2} \text{ (½)}$$

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R} \text{ (½)}$$

$$(b) Z = \sqrt{30^2 + (70 - 30)^2} = \sqrt{900 + 1600} = 50 \Omega \text{ (½)}$$

$$I_{rms} = V_{rms} / Z = 100 / 50 = 2 \text{ A (½)}$$

🔧 Q17 alternative 2 (OR) • Ch 7 • Series resonance [4] ★

Resonance: at a particular frequency $X_L = X_C$, so $Z = R$ is minimum and the current $I = V / R$ is maximum; V and I are in phase. (½)

$$\text{Condition } \omega_0 L = \frac{1}{\omega_0 C}. \quad (½)$$

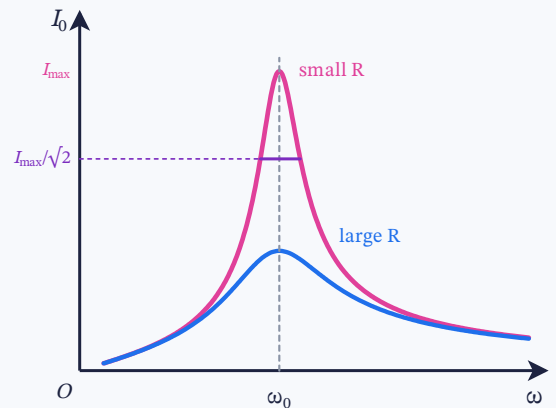
$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad \nu_0 = \frac{1}{2\pi\sqrt{LC}} \quad (½)$$

At resonance $\phi = 0$, power factor $\cos\phi = 1$ and $i_m = v_m / R$. (½)

Graph: bell-shaped $I-\omega$ curve with the peak V / R at ω_0 ; smaller R gives a taller, sharper peak. (1)

$$(b) \omega_0 = \frac{1}{\sqrt{2 \times 8 \times 10^{-6}}} = \frac{1}{4 \times 10^{-3}} = 250 \text{ rad/s} \quad (½)$$

$$Q = \frac{\omega_0 L}{R} = \frac{250 \times 2}{10} = 50 \quad (½)$$



Current versus ω : peak at ω_0

🔧 Q18 alternative 1 • Ch 9 • Refraction at a spherical surface [4] ★

Labelled diagram: object O in medium n_1 , image I in n_2 , pole P , centre C , ray hitting at N with angles $\alpha, \beta, \gamma, i, r$ (paraxial rays). (½)

For small angles: $\alpha \approx \frac{NM}{MO}, \beta \approx \frac{NM}{MC}$,

$$\gamma \approx \frac{NM}{MI}. \quad (½)$$

Exterior angles: $i = \alpha + \beta$ (in $\triangle NOC$),

$r = \beta - \gamma$ (in $\triangle NCI$). (½)

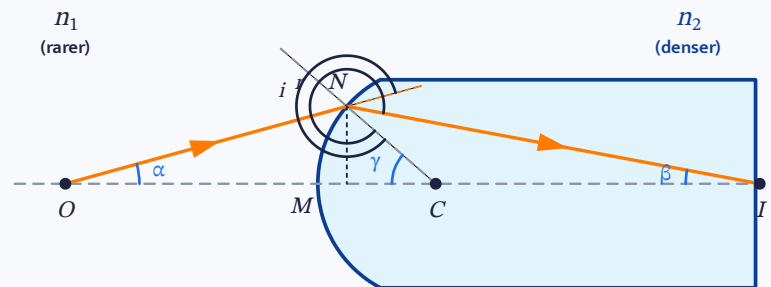
Snell's law for small angles: $n_1 i = n_2 r$. (½)

$$\text{Substituting: } \frac{n_1}{MO} + \frac{n_2}{MI} = \frac{n_2 - n_1}{MC}. \quad (½)$$

$$\text{Sign convention: } MO = -u, MI = +v, MC = +R \Rightarrow \frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad (½)$$

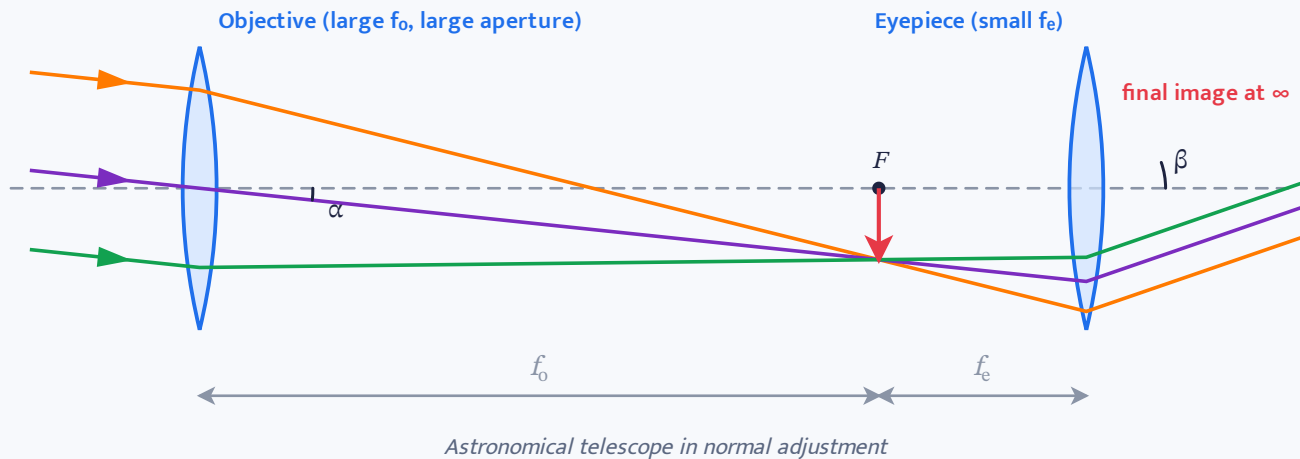
$$(b) \frac{1.5}{v} - \frac{1}{-60} = \frac{1.5 - 1}{20} \Rightarrow \frac{1.5}{v} = \frac{1}{40} - \frac{1}{60} = \frac{1}{120} \quad (½)$$

$$v = +180 \text{ cm} \quad (\text{real image inside the glass}). \quad (½)$$



Refraction at a convex spherical surface

 Q18 alternative 2 (OR) • Ch 9 • Astronomical telescope [4] ★



Ray diagram: parallel rays from a distant object form a real, inverted image at the focus F_o of the objective, which coincides with F_e of the eyepiece; the final image is at infinity. (1)

Angle subtended by the object ($\approx$ at the objective): $\alpha = \frac{h}{f_o}$. (1/2)

Angle subtended by the final image at the eye: $\beta = \frac{h}{f_e}$. (1/2)

$$m = \frac{\beta}{\alpha} = \frac{f_o}{f_e} \text{ (1/2)}$$

Tube length $L = f_o + f_e$. (1/2)

(b) $m = 100 / 5 = 20$ (1/2)

$L = 100 + 5 = 105 \text{ cm}$ (1/2)

Score card and chapter analysis

Section	Questions	Max	My marks
A – objective + VSA	Q1–Q3	24	
B – short answer	Q4–Q13	15	
C – long answer	Q14–Q16	9	
D – essay type	Q17–Q18	8	
Total		56	

Now fill the chapter table. Any chapter where you scored **less than 60 %** goes on your revision list for Day 30 and the exam-week plan.

Ch	Chapter	Questions in this paper	Max	Mine
1	Electric Charges and Fields	Q3(i) · Q14	4	
2	Electrostatic Potential and Capacitance	Q1(i,ii) · Q2(i) · Q4	3	
3	Current Electricity	Q1(iii,iv) · Q2(ii) · Q3(ii) · Q5	4	
4	Moving Charges and Magnetism	Q1(v,vi) · Q2(iii) · Q3(iii) · Q6	4	
5	Magnetism and Matter	Q2(iv) · Q3(iv) · Q7	3	
6	Electromagnetic Induction	Q3(v) · Q8 · Q9	4	
7	Alternating Current	Q1(vii) · Q2(v) · Q17	5	
8	Electromagnetic Waves	Q1(viii) · Q2(vi) · Q3(vi)	2	
9	Ray Optics and Optical Instruments	Q1(ix,x) · Q2(vii) · Q10 · Q18	7	
10	Wave Optics	Q1(xi,xii) · Q3(vii) · Q15	5	
11	Dual Nature of Radiation and Matter	Q3(viii) · Q16	4	
12	Atoms	Q1(xiii,xiv) · Q2(viii) · Q3(ix) · Q11	4	
13	Nuclei	Q1(xv,xvi) · Q2(ix) · Q12	3	
14	Semiconductor Electronics	Q1(xvii,xviii) · Q2(x) · Q3(x) · Q13	4	
Total			56	



Score bands

- **48–56:** board-ready – keep writing derivations once a week. **40–47:** fix the lost derivation steps and unit errors. **32–39:** revisit chapters below 60 % using their Board question bank. **below 32:** re-read notes of the weak chapters before the next paper.

DAY

29

RBSE CLASS 12 PHYSICS • SS-40

Model Paper 5

18 questions • all compulsory

3 h 15 min

56 marks

Internal choice in Q14–18

Marking scheme with step marks

An original paper in the 2025 board pattern. Sit it like the real exam: 3 h 15 min, no book, no phone. Then mark yourself with the step-wise scheme and fill the chapter table.

Instructions to candidates



General instructions

- Subject: Physics • Code **SS-40** • Time **3 h 15 min** • Maximum marks **56**.
- All **18** questions are compulsory. Internal choice (OR) is given only in Q14 to Q18.
- Write the question number exactly as printed. Marks are shown on the right of each question.
- Draw neat, labelled diagrams wherever asked or useful. Write all steps of a numerical with units.



Constants you may use

- $e = 1.6 \times 10^{-19} \text{ C}$ • $c = 3 \times 10^8 \text{ m/s}$
- $h = 6.63 \times 10^{-34} \text{ J s}$ • $m_e = 9.1 \times 10^{-31} \text{ kg}$
- $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$ • $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2$
- $1 \text{ u} = 931.5 \text{ MeV}/c^2$ • $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$



Time plan for this paper (195 min)

- Section A (38 items): **50 min** • Section B ($10 \times 1\frac{1}{2}$): **60 min** • Section C (3×3): **35 min** • Section D (2×4): **40 min** • revision: **10 min**.
- Do not look at the marking scheme until the 195 minutes are over.

Question paper

Section A – Objective and very short answer

24

Q1. Choose the correct answer ($18 \times \frac{1}{2} = 9$)

- i Which of the following can be the charge on a body?
- (a) $2.4 \times 10^{-19} \text{ C}$ (b) $4.8 \times 10^{-19} \text{ C}$ (c) $0.8 \times 10^{-19} \text{ C}$ (d) $1.0 \times 10^{-19} \text{ C}$

- ii The electric field at a distance of 3 cm from a point charge of 2 nC in vacuum is
- (a) $6 \times 10^3 \text{ N/C}$ (b) $2 \times 10^3 \text{ N/C}$ (c) $2 \times 10^4 \text{ N/C}$ (d) $6 \times 10^4 \text{ N/C}$

- iii A current of 0.8 A flows in a wire for 4 s. The number of electrons crossing a cross-section is
- (a) 8×10^{18} (b) 5×10^{19} (c) 2×10^{20} (d) 2×10^{19}

- iv The power consumed by a $100\ \Omega$ resistor connected across a $220\ \text{V}$ supply is
 (a) $484\ \text{W}$ (b) $22\ 000\ \text{W}$ (c) $2.2\ \text{W}$ (d) $48.4\ \text{W}$
- v A material has magnetic susceptibility $\chi = +2.5 \times 10^{-5}$ at room temperature. It is
 (a) paramagnetic (b) diamagnetic
 (c) a perfect diamagnet (d) ferromagnetic
- vi The reactance of a $50\ \mu\text{F}$ capacitor at an angular frequency of $400\ \text{rad/s}$ is
 (a) $50\ \Omega$ (b) $0.02\ \Omega$ (c) $20\ \Omega$ (d) $200\ \Omega$
- vii The frequency of an electromagnetic wave of wavelength $3\ \text{m}$ in vacuum is
 (a) $3 \times 10^8\ \text{Hz}$ (b) $9 \times 10^8\ \text{Hz}$ (c) $10^8\ \text{Hz}$ (d) $10^9\ \text{Hz}$
- viii Two thin lenses of powers $+5\ \text{D}$ and $-2\ \text{D}$ are placed in contact. The focal length of the combination is
 (a) $50\ \text{cm}$ (b) $3.3\ \text{cm}$ (c) $33.3\ \text{cm}$ (d) $14.3\ \text{cm}$
- ix The critical angle for light going from a medium of refractive index 2 to air is
 (a) 60° (b) 30° (c) 45° (d) 90°
- x Two coherent sources of intensities I and $4I$ interfere. The maximum intensity of the resultant is
 (a) $9I$ (b) $3I$ (c) I (d) $5I$
- xi In single-slit diffraction (slit width a), the first minimum occurs at the angle θ given by
 (a) $a \sin \theta = 3\lambda / 2$ (b) $a \sin \theta = 2\lambda$ (c) $a \sin \theta = \lambda$ (d) $a \sin \theta = \lambda / 2$
- xii Light of wavelength $200\ \text{nm}$ falls on a metal of work function $4.2\ \text{eV}$. The stopping potential is ($hc = 1240\ \text{eV nm}$)
 (a) $6.2\ \text{V}$ (b) $4.2\ \text{V}$ (c) $1.0\ \text{V}$ (d) $2.0\ \text{V}$
- xiii The de Broglie wavelength of an electron accelerated through $150\ \text{V}$ is about
 (a) $0.01\ \text{nm}$ (b) $10\ \text{nm}$ (c) $1\ \text{nm}$ (d) $0.1\ \text{nm}$
- xiv The total energy of the electron in the third orbit ($n = 3$) of hydrogen is
 (a) $-0.85\ \text{eV}$ (b) $-1.51\ \text{eV}$ (c) $-3.4\ \text{eV}$ (d) $-4.53\ \text{eV}$
- xv The shortest wavelength (series limit) of the Balmer series of hydrogen is nearly ($R = 1.097 \times 10^7\ \text{m}^{-1}$)
 (a) $486\ \text{nm}$ (b) $365\ \text{nm}$ (c) $91\ \text{nm}$ (d) $656\ \text{nm}$
- xvi The density of nuclear matter
 (a) is zero at the centre of the nucleus
 (b) decreases with mass number
 (c) increases with mass number
 (d) is nearly the same for all nuclei
- xvii The binding energy per nucleon is the greatest for nuclei near mass number
 (a) $4\ (\text{He})$ (b) $238\ (\text{U})$ (c) $56\ (\text{Fe})$ (d) $2\ (\text{H})$

xviii

A full-wave rectifier is fed with a 50 Hz AC. The frequency of the ripple in its output is

- (a) 50 Hz (b) 100 Hz (c) 25 Hz (d) 200 Hz

2 Fill in the blanks ($10 \times \frac{1}{2} = 5$)

[10 × ½]

- (i) Two electric field lines never _____ each other.
- (ii) The reciprocal of resistivity is called _____.
- (iii) A transformer that increases the AC voltage is called a _____ transformer.
- (iv) Electromagnetic waves are produced by _____ charges.
- (v) If the critical angle of a medium is C, its refractive index is $n =$ _____.
- (vi) The locus of all points having the same phase of disturbance is called a _____.
- (vii) The minimum frequency of light that can cause photoelectric emission from a metal is its _____ frequency.
- (viii) The lowest energy state of an atom is called its _____ state.
- (ix) One atomic mass unit (1 u) is _____ of the mass of a carbon-12 atom.
- (x) The impurity atom added to make a p-type semiconductor is called an _____.

3 Answer in one or two sentences ($10 \times 1 = 10$)

[10 × 1]

- (i) State Gauss's law in electrostatics.
- (ii) Why are manganin and constantan used for making standard resistances?
- (iii) Define the magnetic declination at a place.
- (iv) What are eddy currents? Give one use.
- (v) Why is a choke coil preferred to a resistor for controlling the current in an AC circuit such as a tube light?
- (vi) Name the electromagnetic waves used (i) in water purifiers to kill germs, (ii) in the remote control of a TV.
- (vii) State Brewster's law.
- (viii) What is the shape of the wavefront that emerges from a convex lens when a point source is placed at its focus?
- (ix) What is the effect of increasing the intensity of the incident light on the stopping potential?
- (x) State Bohr's frequency condition.

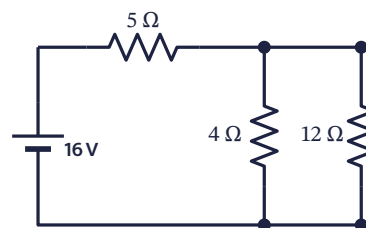
Section B – Short answer

15

- 4** Two charges of $+2 \mu\text{C}$ and $-2 \mu\text{C}$ are 4 cm apart, forming a dipole. Find its dipole moment and the torque on it in a uniform field of $5 \times 10^4 \text{ N/C}$ when the dipole axis makes 30° with the field.

[1½]

- 5 In the circuit shown, find (i) the total current drawn from the 16 V battery (negligible internal resistance) and (ii) the current through the $12\ \Omega$ resistor.



[1½]

- 6 A bar magnet of magnetic moment $2.5\ \text{A m}^2$ is placed in a uniform field of $0.4\ \text{T}$ at 30° to the field. Find the torque on it and its potential energy. [1½]
- 7 A coil of resistance $6\ \Omega$ and inductive reactance $8\ \Omega$ is connected to a $100\ \text{V}$ (rms) AC source. Find the impedance, the rms current and the average power consumed. [1½]
- 8 A step-down transformer has 2000 turns in the primary and 100 turns in the secondary. The primary is connected to $2200\ \text{V}$ AC. Find the secondary voltage and, for an ideal transformer, the primary current when the secondary delivers $10\ \text{A}$. [1½]
- 9 A prism of angle 60° gives a minimum deviation of 30° . Find the refractive index of its material. [1½]
- 10 In Young's experiment the slit separation is $0.4\ \text{mm}$, the screen is $1.2\ \text{m}$ away and the fringe width is $1.8\ \text{mm}$. Find the wavelength of the light and the distance of the third bright fringe from the centre. [1½]
- 11 Light of wavelength $400\ \text{nm}$ is incident on a metal of work function $2.3\ \text{eV}$. Find (i) the maximum kinetic energy of the photoelectrons, (ii) the stopping potential, (iii) the threshold wavelength. ($hc = 1240\ \text{eV nm}$) [1½]
- 12 The radius of the first Bohr orbit is $0.53\ \text{\AA}$ and the speed of the electron in it is $2.18 \times 10^6\ \text{m/s}$. Find the radius of the second orbit and the speed of the electron in it. [1½]
- 13 State Einstein's mass–energy relation. Calculate the energy released when $0.5\ \text{mg}$ of mass is completely converted into energy, in joule and in kWh. ($c = 3 \times 10^8\ \text{m/s}$) [1½]

Section C – Long answer

9

- 14 Derive the expression for the capacitance of a parallel-plate capacitor in vacuum. How does it change when a dielectric of constant K fills the space between the plates? [3]

----- OR -----

Derive the expression for the energy stored in a charged capacitor. Hence obtain the energy density of the electric field.

- 15 Derive the expression for the motional emf induced in a conducting rod of length l moving with speed v perpendicular to a uniform magnetic field B . Show that the mechanical work done equals the electrical energy dissipated. [3]

----- OR -----

Obtain the expression for the self-inductance of a long air-cored solenoid of length l , area of cross-section A and N turns.

- 16 Explain the formation of the depletion region and the barrier potential in a p-n junction. [3]

----- OR -----

Draw the circuit diagram of a full-wave rectifier and explain its working with the input and output waveforms.

Section D – Essay type

8

- 17 (a) Using the Biot–Savart law, derive the magnetic field at a point on the axis of a circular current loop. Hence write the field at its centre. (3) [4]

(b) Find the field at the centre of a coil of 100 turns, radius 10 cm, carrying 2 A. ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$) (1)

----- OR -----

(a) State Ampere's circuital law and use it to derive the magnetic field inside a long solenoid. (3)

(b) Find the field inside a solenoid having 1000 turns per metre and carrying 2 A. ($\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$) (1)

- 18 (a) Draw the ray diagram for refraction through a prism and derive the formula $n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$. (3) [4]

(b) For a prism of angle 60° and refractive index 1.5 find the angle of minimum deviation. (1)

----- OR -----

(a) Derive the lens maker's formula for a thin lens of refractive index n placed in air. (3)

(b) A biconvex lens of $n = 1.5$ has both faces of radius 20 cm. Find its focal length. (1)

— End of question paper —

Marking scheme and model answers



How to mark yourself

- Marks in **bold brackets** are awarded for that step. A correct alternative method gets full marks.
- Numerical: formula (½), substitution (½), answer with unit (½). Missing or wrong unit: deduct ½ once.
- Be strict about the labelled diagram – it carries marks in every derivation.



Q1 – answer key with reasons (½ each)

i(b) ii(c) iii(d) iv(a) v(a) vi(a) vii(c) viii(c) ix(b) x(a) xi(c) xii(d) xiii(d) xiv(b)

xv(b) xvi(d) xvii(c) xviii(b)

(i) (b) Charge is quantised, $q = ne$. $4.8 \times 10^{-19} / 1.6 \times 10^{-19} = 3$ is an integer; the others give 1.5, 0.5 and 0.625.

(ii) (c) $E = \frac{kq}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-9}}{(0.03)^2} = 2 \times 10^4 \text{ N/C}$.

(iii) (d) $q = It = 3.2 \text{ C}$; $n = q / e = 3.2 / 1.6 \times 10^{-19} = 2 \times 10^{19}$.

(iv) (a) $P = V^2 / R = 220^2 / 100 = 484 \text{ W}$.

(v) (a) A small positive susceptibility means paramagnetic; diamagnetics have small negative χ , ferromagnetics very large positive χ .

(vi) (a) $X_C = \frac{1}{\omega C} = \frac{1}{400 \times 50 \times 10^{-6}} = 50 \Omega$.

(vii) (c) $\nu = c / \lambda = 3 \times 10^8 / 3 = 10^8 \text{ Hz}$ (100 MHz, FM/TV range).

(viii) (c) $P = 5 - 2 = 3 \text{ D}$, so $f = 1 / 3 \text{ m} = 33.3 \text{ cm}$.

(ix) (b) $\sin C = 1 / n = 1 / 2 \Rightarrow C = 30^\circ$.

(x) (a) $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = (\sqrt{I} + 2\sqrt{I})^2 = 9I$.

(xi) (c) Minima: $a \sin \theta = n\lambda$ ($n = 1, 2, \dots$); the first is $a \sin \theta = \lambda$.

(xii) (d) Photon energy = $1240 / 200 = 6.2 \text{ eV}$; $eV_0 = 6.2 - 4.2 = 2.0 \text{ eV}$, so $V_0 = 2.0 \text{ V}$.

(xiii) (d) $\lambda = 1.227 / \sqrt{V} \text{ nm} = 1.227 / \sqrt{150} = 0.100 \text{ nm}$.

(xiv) (b) $E_3 = -13.6 / 9 = -1.51 \text{ eV}$.

(xv) (b) Limit: $n \rightarrow \infty$ to 2, $\lambda = 4 / R = 4 / (1.097 \times 10^7) = 365 \text{ nm}$.

(xvi) (d) $R \propto A^{1/3}$ so volume $\propto A$, and mass $\propto A$: the density ($\approx 2.3 \times 10^{17} \text{ kg/m}^3$) is independent of A .

(xvii) (c) The BE/A curve peaks (about 8.75 MeV) near $A = 56$ (iron); light and heavy nuclei have lower values.

(xviii) (b) Both half-cycles are rectified, so the output pulses repeat twice per input cycle: $2 \times 50 = 100 \text{ Hz}$.

Q2 – Fill in the blanks (½ each)

	Ch	Answer
(i)	1	intersect (cross)
(ii)	3	conductivity
(iii)	7	step-up
(iv)	8	accelerated
(v)	9	$\frac{1}{\sin C}$
(vi)	10	wavefront
(vii)	11	threshold
(viii)	12	ground
(ix)	13	$\frac{1}{12}$
(x)	14	acceptor

Q3 – Very short answer (1 mark each)

	Ch	Model answer (marking points)
(i)	1	The total electric flux through any closed surface equals $1 / \epsilon_0$ times the net charge enclosed by it: $\oint \vec{E} \cdot d\vec{S} = q_{in} / \epsilon_0$ (1) .
(ii)	3	Their resistivity is high and its temperature coefficient is very small, so the resistance hardly changes with temperature (1) .
(iii)	5	The angle between the magnetic meridian and the geographic meridian at that place (1) .
(iv)	6	Circulating currents induced in a bulk conductor when the magnetic flux through it changes (½) ; used in induction furnaces / electromagnetic braking / energy meters (½) .
(v)	7	A choke has large inductive reactance and very small resistance, so it reduces the current with almost no power loss ($P = VI \cos \phi \approx 0$), whereas a resistor wastes power as heat (1) .
(vi)	8	(i) ultraviolet rays (½) ; (ii) infrared rays (½) .
(vii)	10	The reflected light is completely plane-polarised when the tangent of the angle of incidence equals the refractive index of the medium: $\tan i_B = n$ (1) .
(viii)	10	Plane wavefront (the refracted rays are parallel) (1) .
(ix)	11	No effect – the stopping potential depends only on the frequency of light and the metal, not on intensity (1) .
(x)	12	When an electron jumps from an orbit of energy E_i to a lower one E_f , a photon is emitted: $h\nu = E_i - E_f$ (1) .

Section B – Q4 to Q13 (1½ marks each)

 Q4 • Ch 1 • Electric Charges and Fields [1½]

$$p = q(2a) = 2 \times 10^{-6} \times 0.04 = 8 \times 10^{-8} \text{ C m } (1/2)$$

$$\tau = pE \sin \theta (1/2)$$

$$\tau = 8 \times 10^{-8} \times 5 \times 10^4 \times \sin 30^\circ = 2 \times 10^{-3} \text{ N m } (1/2)$$

 Q5 • Ch 3 • Current Electricity [1½]

$$4 \Omega \text{ and } 12 \Omega \text{ are in parallel: } R_p = \frac{4 \times 12}{4 + 12} = 3 \Omega; \text{ in series with } 5 \Omega: R = 8 \Omega. (1/2)$$

$$(i) I = V / R = 16 / 8 = 2 \text{ A } (1/2)$$

$$(ii) \text{ p.d. across the parallel part } = 2 \times 3 = 6 \text{ V, so } I_{12} = 6 / 12 = 0.5 \text{ A } (1/2)$$

 Q6 • Ch 5 • Magnetism and Matter [1½]

$$\tau = mB \sin \theta \text{ and } U = -mB \cos \theta (1/2)$$

$$\tau = 2.5 \times 0.4 \times \sin 30^\circ = 0.5 \text{ N m } (1/2)$$

$$U = -2.5 \times 0.4 \times \cos 30^\circ = -0.87 \text{ J } (\approx -0.866 \text{ J}) (1/2)$$

 Q7 • Ch 7 • Alternating Current [1½]

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{36 + 64} = 10 \Omega (1/2)$$

$$I_{rms} = V / Z = 100 / 10 = 10 \text{ A } (1/2)$$

$$P = I^2 R = 100 \times 6 = 600 \text{ W } (\text{check: } VI \cos \phi = 100 \times 10 \times 0.6) (1/2)$$

 Q8 • Ch 7 • Alternating Current [1½]

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} (1/2)$$

$$V_s = 2200 \times \frac{100}{2000} = 110 \text{ V } (1/2)$$

$$\text{Ideal: } V_p I_p = V_s I_s \Rightarrow I_p = \frac{110 \times 10}{2200} = 0.5 \text{ A } (1/2)$$

 Q9 • Ch 9 • Ray Optics and Optical Instruments [1½]

$$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}} (1/2)$$

$$n = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{0.7071}{0.5} (1/2)$$

$$n = \sqrt{2} \approx 1.41 (1/2)$$

 Q10 • Ch 10 • Wave Optics [1½]

$$\beta = \frac{\lambda D}{d} \Rightarrow \lambda = \frac{\beta d}{D} (1/2)$$

$$\lambda = \frac{1.8 \times 10^{-3} \times 0.4 \times 10^{-3}}{1.2} = 6 \times 10^{-7} \text{ m} = 600 \text{ nm } (1/2)$$

$$y_3 = 3\beta = 3 \times 1.8 = 5.4 \text{ mm } (1/2)$$

 Q11 • Ch 11 • Dual Nature of Radiation and Matter [1½]

$$(i) E = \frac{1240}{400} = 3.1 \text{ eV}; K_{max} = 3.1 - 2.3 = 0.8 \text{ eV} \text{ (½)}$$

$$(ii) eV_0 = K_{max} \Rightarrow V_0 = 0.8 \text{ V} \text{ (½)}$$

$$(iii) \lambda_0 = \frac{hc}{\phi_0} = \frac{1240}{2.3} = 539 \text{ nm} \text{ (½)}$$

 Q12 • Ch 12 • Atoms [1½]

$$r_n \propto n^2 \text{ and } v_n \propto \frac{1}{n} \text{ (½)}$$

$$r_2 = 4r_1 = 4 \times 0.53 = 2.12 \text{ Å} \text{ (½)}$$

$$v_2 = \frac{v_1}{2} = 1.09 \times 10^6 \text{ m/s} \text{ (½)}$$

 Q13 • Ch 13 • Nuclei [1½]

$$E = mc^2 - \text{mass and energy are equivalent. (½)}$$

$$E = 0.5 \times 10^{-6} \times (3 \times 10^8)^2 = 4.5 \times 10^{10} \text{ J} \text{ (½)}$$

$$E = \frac{4.5 \times 10^{10}}{3.6 \times 10^6} = 12\,500 \text{ kWh} \text{ (½)}$$

Section C – Q14 to Q16 (3 marks each)

 Q14 alternative 1 • Ch 2 • Parallel-plate capacitor [3] ★

Labelled diagram: plates of area A, separation d, charges +Q and -Q. (½)

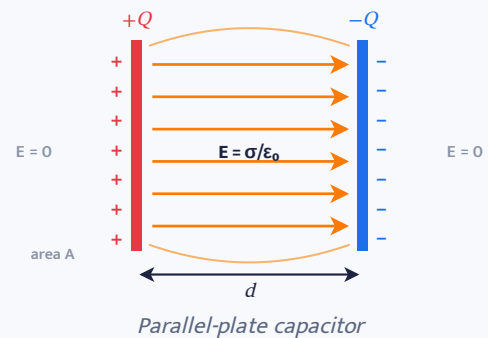
Each plate gives a field $\sigma / 2\epsilon_0$; they add between the plates and

$$\text{cancel outside: } E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}. \text{ (1)}$$

$$\text{Potential difference } V = Ed = \frac{Qd}{\epsilon_0 A}. \text{ (½)}$$

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d} \text{ (½)}$$

With a dielectric the field is reduced to E / K , so $C = \frac{K\epsilon_0 A}{d}$ (K times larger). (½)



 Q14 alternative 2 (OR) • Ch 2 • Energy stored in a capacitor [3] ★

While charging, when the charge on the plates is q the p.d. is $v = q / C$. (½)

$$\text{Work to add a further charge } dq: dW = v dq = \frac{q}{C} dq. \text{ (½)}$$

$$\text{Total work } W = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C}. \text{ (1)}$$

$$U = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV \text{ (stored in the field). (½)}$$

$$\text{Using } C = \epsilon_0 A / d \text{ and } V = Ed: U = \frac{1}{2} \epsilon_0 E^2 (Ad), \text{ so energy density } u = \frac{U}{V} = \frac{1}{2} \epsilon_0 E^2. \text{ (½)}$$

🔧 Q15 alternative 1 • Ch 6 • Motional emf [3] ★

Labelled diagram: rod PQ on U-shaped rails in a field $B \perp$ plane; the closed circuit has resistance R. (½)

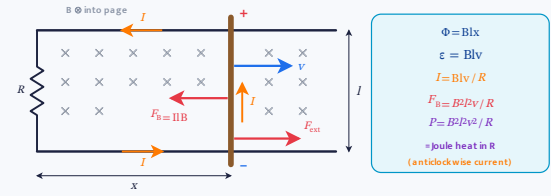
At a distance x the flux $\Phi = Blx$. (½)

Faraday's law: $|\varepsilon| = \frac{d\Phi}{dt} = Bl \frac{dx}{dt} \Rightarrow \varepsilon = Blv$ (direction by Lenz's law). (½)

Current $I = \frac{Blv}{R}$; magnetic force on the rod $F = BIl = \frac{B^2 l^2 v}{R}$, opposing the motion. (½)

To keep v constant an equal external force acts: mechanical power $P = Fv = \frac{B^2 l^2 v^2}{R}$. (½)

Electrical power dissipated $I^2 R = \frac{B^2 l^2 v^2}{R} = Fv$ – energy is conserved. (½)



Rod PQ sliding on rails in a magnetic field

🔧 Q15 alternative 2 (OR) • Ch 6 • Self-inductance of a long solenoid [3] ★

Labelled diagram of the solenoid carrying current I ; turns per unit length

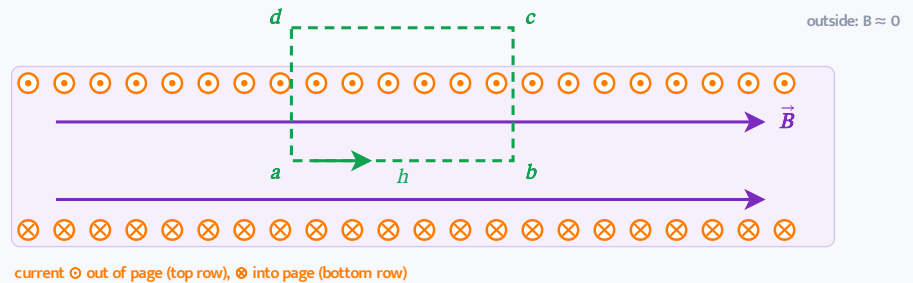
$n = N / l$. (½)

Field inside (far from the ends): $B = \mu_0 nI = \mu_0 \frac{N}{l} I$. (½)

Flux linked with the whole coil:

$$N\Phi = N(BA) = \mu_0 \frac{N^2}{l} AI = \mu_0 n^2 AlI. (1)$$

By definition $N\Phi = LI$, so $L = \mu_0 n^2 Al = \frac{\mu_0 N^2 A}{l}$ (with a core of relative permeability μ_r : $\mu_r \mu_0 n^2 Al$). (1)



$$Bh = \mu_0 (nh)I \Rightarrow B = \mu_0 nI$$

Long solenoid

🔧 Q16 alternative 1 • Ch 14 • Formation of a p-n junction [3] ★

p-side has many holes and n-side many free electrons; this concentration difference makes electrons diffuse $n \rightarrow p$ and holes diffuse $p \rightarrow n$ across the junction. (½)

Diffusing carriers recombine, leaving immobile positive donor ions on the n-side and negative acceptor ions on the p-side near the junction.

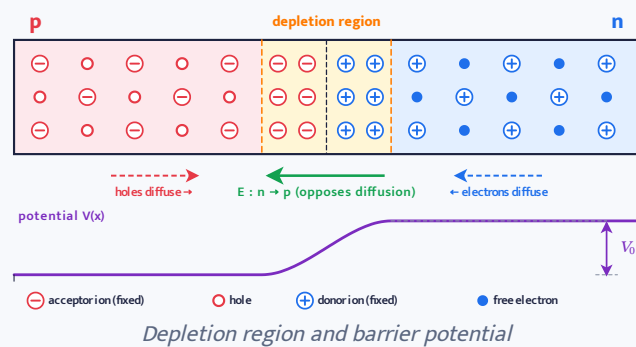
(½)

This layer, free of mobile carriers, is the **depletion region** (thickness $\sim 1 \mu\text{m}$). (½)

The ions set up an electric field directed $n \rightarrow p$; it drives a **drift** current (minority carriers) opposite to the diffusion current. (½)

In equilibrium diffusion current = drift current; the potential difference across the layer is the **barrier potential** V_0 ($\approx 0.7 \text{ V}$ for Si, 0.3 V for Ge). (½)

Labelled diagram showing the ions, the depletion width and the barrier potential. (½)



Depletion region and barrier potential

🔧 Q16 alternative 2 (OR) • Ch 14 • Full-wave rectifier [3]

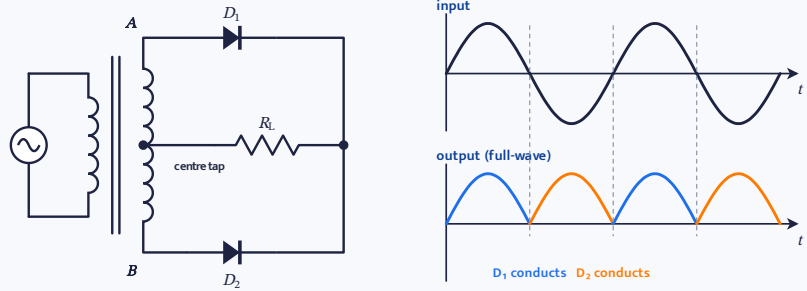
Circuit: centre-tapped transformer secondary, two diodes D_1, D_2 and load resistance R_L (labelled). (1)

Positive half-cycle: the upper end is positive, D_1 is forward biased and conducts, D_2 is reverse biased. (½)

Negative half-cycle: the lower end becomes positive, D_2 conducts, D_1 is off.

In both cases the current through R_L flows in the **same direction**. (½)

Waveforms: input sine wave; output a train of unidirectional pulses (both half-cycles). Output ripple frequency = $2 \times$ input frequency (100 Hz for 50 Hz mains). (1)



Centre-tap full-wave rectifier and its waveforms

Section D – Q17 and Q18 (4 marks each)

🔧 Q17 alternative 1 • Ch 4 • Field on the axis of a circular loop [4] ★

Biot–Savart law: $dB = \frac{\mu_0 I dl \sin \theta}{4\pi r^2}$. (½)

Diagram: loop of radius R , axial point P at distance x ; for every element $d\vec{l} \perp \vec{r}$ (so $\sin \theta = 1$) and $r^2 = R^2 + x^2$. (½)

Components of dB perpendicular to the axis cancel in pairs; the axial component is

$$dB \cos \phi = dB \frac{R}{r}. \quad (\text{½})$$

$$B = \frac{\mu_0 I R}{4\pi r^3} \oint dl = \frac{\mu_0 I R}{4\pi r^3} (2\pi R) \quad (\text{½})$$

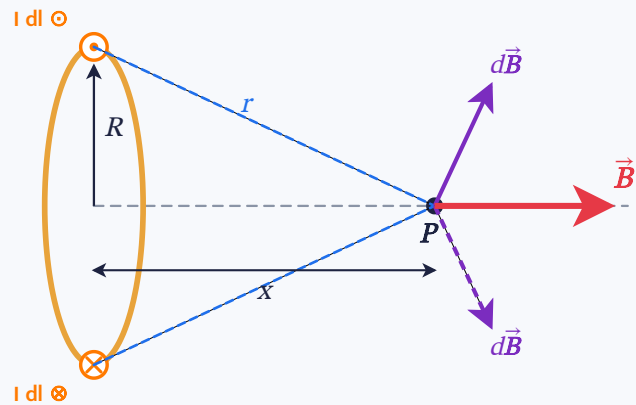
$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}} \quad (\text{for } N \text{ turns multiply by } N).$$

(½)

At the centre $x = 0$: $B = \frac{\mu_0 NI}{2R}$. (½)

(b) $B = \frac{4\pi \times 10^{-7} \times 100 \times 2}{2 \times 0.1}$ (½)

$$B = 1.26 \times 10^{-3} \text{ T} \quad (\text{½})$$



$\perp$ components cancel; axial ones add

Circular loop, point P on the axis

🔧 Q17 alternative 2 (OR) • Ch 4 • Ampere's law: solenoid [4] ★

Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$. (½)

Diagram: long solenoid with n turns per unit length and a rectangular Amperian loop $abcd$ of length h (side ab inside, cd outside). (½)

Inside the field is uniform and axial; outside it is negligible; on bc and da the field is perpendicular to $d\vec{l}$ (or zero), contributing nothing. (½)

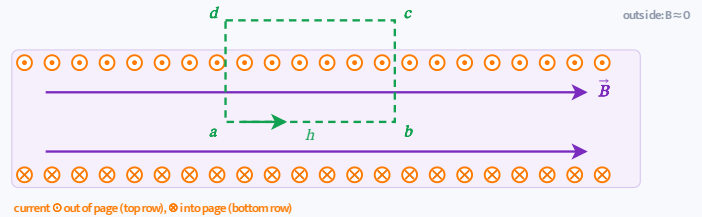
$$\oint \vec{B} \cdot d\vec{l} = Bh. \quad (½)$$

Enclosed current $I_{enc} = (nh)I$. (½)

$$Bh = \mu_0 nhI \Rightarrow B = \mu_0 nI \quad (½)$$

$$(b) B = 4\pi \times 10^{-7} \times 1000 \times 2 \quad (½)$$

$$B = 2.5 \times 10^{-3} \text{ T} \quad (½)$$



$$Bh = \mu_0 (nh)I \Rightarrow B = \mu_0 nI$$

Amperian loop $abcd$ for a solenoid

🔧 Q18 alternative 1 • Ch 9 • Prism: refractive index and minimum deviation [4] ★

Labelled ray diagram: angles i, r_1, r_2, e , prism angle A , deviation δ . (½)

Geometry: $r_1 + r_2 = A$ and

$$\delta = (i - r_1) + (e - r_2) = i + e - A. \quad (½)$$

At minimum deviation the ray passes symmetrically: $i = e, r_1 = r_2 = r = \frac{A}{2}$, and

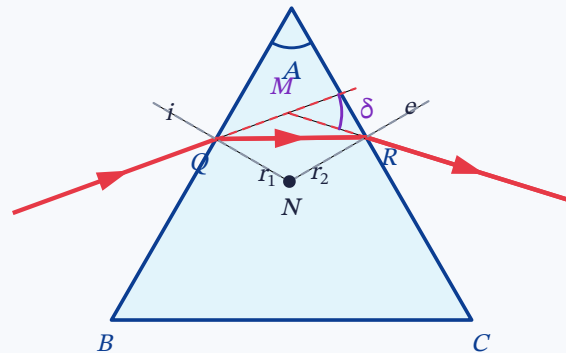
$$\delta_m = 2i - A \Rightarrow i = \frac{A + \delta_m}{2}. \quad (1)$$

$$\text{Snell's law: } n = \frac{\sin i}{\sin r} = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}} \quad (1)$$

$$(b) \sin \frac{A + \delta_m}{2} = 1.5 \sin 30^\circ = 0.75 \Rightarrow$$

$$\frac{A + \delta_m}{2} = 48.6^\circ \quad (½)$$

$$\delta_m = 2(48.6^\circ) - 60^\circ = 37.2^\circ \quad (½)$$



Refraction through a prism, minimum deviation

🔧 Q18 alternative 2 (OR) • Ch 9 • Lens maker's formula [4] ★

Labelled diagram; assumptions: thin lens, paraxial rays, same medium (air) on both sides. (½)

$$\text{First surface (radius } R_1): \frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1}$$

it forms an image at v_1 (inside the glass). (½)

Second surface (radius R_2), with the image

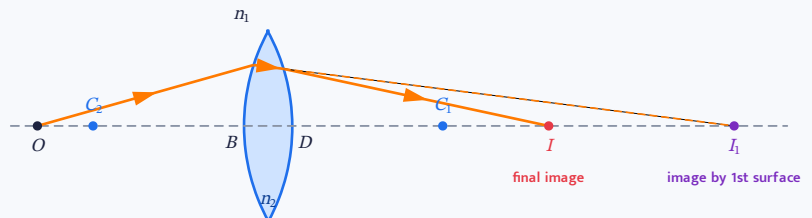
$$I_1 \text{ as virtual object: } \frac{1}{v} - \frac{n}{v_1} = \frac{1-n}{R_2}. \quad (½)$$

$$\text{Adding: } \frac{1}{v} - \frac{1}{u} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right). \quad (1)$$

$$\text{For } u \rightarrow -\infty, v = f: \frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (½)$$

$$(b) R_1 = +20, R_2 = -20: \frac{1}{f} = 0.5 \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \quad (½)$$

$$f = 20 \text{ cm} \quad (½)$$



Refraction at the two surfaces of a thin lens

Score card and chapter analysis

Section	Questions	Max	My marks
A – objective + VSA	Q1–Q3	24	
B – short answer	Q4–Q13	15	
C – long answer	Q14–Q16	9	
D – essay type	Q17–Q18	8	
Total		56	

Now fill the chapter table. Any chapter where you scored **less than 60 %** goes on your revision list for Day 30 and the exam-week plan.

Ch	Chapter	Questions in this paper	Max	Mine
1	Electric Charges and Fields	Q1(i,ii) · Q2(i) · Q3(i) · Q4	4	
2	Electrostatic Potential and Capacitance	Q14	3	
3	Current Electricity	Q1(iii,iv) · Q2(ii) · Q3(ii) · Q5	4	
4	Moving Charges and Magnetism	Q17	4	
5	Magnetism and Matter	Q1(v) · Q3(iii) · Q6	3	
6	Electromagnetic Induction	Q3(iv) · Q15	4	
7	Alternating Current	Q1(vi) · Q2(iii) · Q3(v) · Q7 · Q8	5	
8	Electromagnetic Waves	Q1(vii) · Q2(iv) · Q3(vi)	2	
9	Ray Optics and Optical Instruments	Q1(viii,ix) · Q2(v) · Q9 · Q18	7	
10	Wave Optics	Q1(x,xi) · Q2(vi) · Q3(vii,viii) · Q10	5	
11	Dual Nature of Radiation and Matter	Q1(xii,xiii) · Q2(vii) · Q3(ix) · Q11	4	
12	Atoms	Q1(xiv,xv) · Q2(viii) · Q3(x) · Q12	4	
13	Nuclei	Q1(xvi,xvii) · Q2(ix) · Q13	3	
14	Semiconductor Electronics	Q1(xviii) · Q2(x) · Q16	4	
	Total		56	



Score bands

- **48–56:** board-ready – keep writing derivations once a week. **40–47:** fix the lost derivation steps and unit errors. **32–39:** revisit chapters below 60 % using their Board question bank. **below 32:** re-read notes of the weak chapters before the next paper.

DAY

30

THE FINAL SPRINT

Final Toolkit

14 formula sheets with units

61 starred derivations

Errors log

Exam-day plan

Hindi + English

The last day: every formula with its unit on a few pages, every derivation the board asks (tick it three times), the mistakes that cost marks, and a minute-by-minute plan for the exam hall.

A • Master formula sheet (all 14 chapters, with units)

How to use: cover the middle and right columns, say the meaning and unit of each formula, then check. Tick the chapter when you can do a full sheet without a mistake. Each sheet is written in the order in which the board questions use it.

 Constants and conversions to know

- $e = 1.6 \times 10^{-19} \text{ C}$ • $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ • $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$ • $\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
- $c = 3 \times 10^8 \text{ m/s}$ • $h = 6.63 \times 10^{-34} \text{ J s}$ • $hc = 1240 \text{ eV nm}$ • $m_e = 9.1 \times 10^{-31} \text{ kg}$ • $m_p = 1.67 \times 10^{-27} \text{ kg}$ • $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$
- $1 \text{ u} = 931.5 \text{ MeV/c}^2$ • $1 \text{ Å} = 10^{-10} \text{ m}$ • $1 \text{ nm} = 10^{-9} \text{ m}$ • $1 \text{ fm} = 10^{-15} \text{ m}$ • $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ • $D = 25 \text{ cm}$

Ch 1 • Electric Charges and Fields (4 marks)

Formula	Meaning / condition	SI unit
$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$	Coulomb's law (vacuum); $k = 9 \times 10^9$	N
$F_m = \frac{F}{K}$	Force in a medium of dielectric constant K	N
$q = ne$	Quantisation of charge, $e = 1.6 \times 10^{-19} \text{ C}$	C
$E = \frac{F}{q} = \frac{kq}{r^2}$	Field of a point charge	$\text{N C}^{-1} = \text{V m}^{-1}$
$p = q(2a)$	Dipole moment (from – to +)	C m
$E_{ax} = \frac{2kp}{r^3}$, $E_{eq} = \frac{kp}{r^3}$	Short dipole, axial and equatorial (E_{eq} opposite to $\vec{p}$)	N C^{-1}
$\tau = pE \sin\theta$, $U = -pE \cos\theta$	Torque and PE of a dipole in a uniform field	N m, J
$\Phi = \oint \vec{E} \cdot d\vec{S} = \frac{q_{in}}{\epsilon_0}$	Gauss's law	$\text{N m}^2 \text{ C}^{-1}$
$E = \frac{\lambda}{2\pi\epsilon_0 r}$	Infinite line charge	N C^{-1}
$E = \frac{\sigma}{2\epsilon_0}$	Infinite plane sheet (just outside a conductor: σ / ϵ_0)	N C^{-1}

$$E = \frac{kQ}{r^2} \quad (r \geq R), \quad E = 0 \quad (r < R)$$

Thin spherical shell

N C⁻¹

$$E = \frac{kQr}{R^3} \quad (r < R)$$

Inside a uniformly charged solid sphere

N C⁻¹

Ch 2 • Electrostatic Potential and Capacitance (3 marks)

Formula	Meaning / condition	SI unit
$V = \frac{W}{q} = \frac{kq}{r}$	Potential of a point charge (scalar)	V = J C ⁻¹
$V = \frac{kpcos\theta}{r^2}$	Dipole potential (zero on the equatorial line)	V
$E = -\frac{dV}{dr}$	Field from potential	V m ⁻¹
$U = \frac{kq_1q_2}{r}$	PE of a pair; add over all pairs	J
$C = \frac{Q}{V}, \quad C = 4\pi\epsilon_0 R$	Capacitance; isolated sphere	F = C V ⁻¹

$C = \frac{K\epsilon_0 A}{d}$	Parallel plates (K = 1 in vacuum)	F
$C = \frac{\epsilon_0 A}{d - t + t/K}$	Plates with a dielectric slab of thickness t	F
$\frac{1}{C} = \sum \frac{1}{C_i}, \quad C = \sum C_i$	Series (same Q); parallel (same V)	F
$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$	Energy stored in a capacitor	J
$u = \frac{1}{2}\epsilon_0 E^2$	Energy density of the field	J m ⁻³

$V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$	Common potential on joining; heat lost $= \frac{C_1C_2(V_1 - V_2)^2}{2(C_1 + C_2)}$	V
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Ch 3 • Current Electricity (4 marks)

Formula	Meaning / condition	SI unit
$I = \frac{Q}{t} = neAv_d$	Current and drift velocity	A = C s ⁻¹
$v_d = \frac{eE\tau}{m}, \quad \mu = \frac{v_d}{E}$	Drift velocity; mobility	m s ⁻¹ ; m ² V ⁻¹ s ⁻¹
$R = \frac{V}{I} = \frac{\rho l}{A}$	Ohm's law; resistance	Ω
$\rho = \frac{m}{ne^2\tau}, \quad \sigma = \frac{1}{\rho}$	Resistivity; conductivity	$\Omega \text{ m}$; S m ⁻¹
$R_T = R_0[1 + \alpha(T - T_0)]$	Temperature dependence	Ω ; α in K ⁻¹

$R_s = \sum R_i, \frac{1}{R_p} = \sum \frac{1}{R_i}$	Series; parallel	Ω
$P = VI = I^2R = \frac{V^2}{R}$	Power; 1 kWh = 3.6×10^6 J	W
$V = \varepsilon - Ir, I = \frac{\varepsilon}{R + r}$	Terminal voltage (discharging)	V; A
$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2, r_{eq} = r_1 + r_2$	Cells in series (same polarity)	V, Ω
$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}, r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$	Two cells in parallel	V, Ω

$\sum I = 0, \sum \varepsilon = \sum IR$	Kirchhoff: junction rule, loop rule	
$\frac{P}{Q} = \frac{R}{S}, X = R \frac{100 - l}{l}$	Wheatstone balance; metre bridge (R in left gap, l from left)	Ω
$P_{max} = \frac{\varepsilon^2}{4r} \quad (R = r)$	Maximum power transfer	W

Ch 4 • Moving Charges and Magnetism (4 marks)

Formula	Meaning / condition	SI unit
$\vec{F} = q(\vec{v} \times \vec{B}), F = qvB\sin\theta$	Magnetic force (Lorentz force adds $q\vec{E}$)	N
$r = \frac{mv}{qB}, T = \frac{2\pi m}{qB}$	Circular path $\perp$ B (T independent of v)	m; s
$\vec{F} = I \vec{l} \times \vec{B}, F = BIl\sin\theta$	Force on a current-carrying wire	N
$dB = \frac{\mu_0 I dl \sin\theta}{4\pi r^2}$	Biot-Savart law; $\mu_0 = 4\pi \times 10^{-7}$	T = Wb m ⁻²
$B = \frac{\mu_0 I}{2\pi r}$	Long straight wire	T

$B = \frac{\mu_0 NI}{2R}; B = \frac{\mu_0 NIR^2}{2(R^2 + x^2)^{3/2}}$	Loop: centre; on the axis	T
$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$	Ampere's circuital law	T m
$B = \mu_0 nI; B = \frac{\mu_0 NI}{2\pi r}$	Solenoid (n = turns per metre); toroid	T
$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$	Parallel wires: attract if currents are parallel	N m ⁻¹
$\vec{\tau} = \vec{m} \times \vec{B}, m = NIA$	Torque on a loop; magnetic moment	N m; A m ²

$S = \frac{I_g G}{I - I_g}, R = \frac{V}{I_g} - G$	Ammeter shunt (parallel); voltmeter resistor (series)	Ω
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Ch 5 • Magnetism and Matter (3 marks)

Formula	Meaning / condition	SI unit
$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}, B_{eq} = \frac{\mu_0 m}{4\pi r^3}$	Short bar magnet (axial, equatorial)	T
$\tau = mB\sin\theta, U = -mB\cos\theta$	Torque and PE of a magnetic dipole	N m; J
$\oint \vec{B} \cdot d\vec{S} = 0$	Gauss law for magnetism (no monopoles)	Wb
$B = \mu_0(H + M), M = \chi H$	Magnetisation, susceptibility	A m ⁻¹ ; χ no unit
$\mu_r = 1 + \chi, \mu = \mu_0\mu_r$	Relative permeability, permeability	T m A ⁻¹
$\chi = \frac{C}{T}$	Curie's law (paramagnetic)	no unit
$B_H = B_E\cos\delta, B_V = B_E\sin\delta, \tan\delta = \frac{B_V}{B_H}$	Earth's field and dip	T

Ch 6 • Electromagnetic Induction (4 marks)

Formula	Meaning / condition	SI unit
$\Phi = BA\cos\theta$	Magnetic flux	Wb = T m ²
$\varepsilon = -N\frac{d\Phi}{dt}$	Faraday–Lenz law; charge flown $q = \Delta\Phi / R$	V; C
$\varepsilon = Blv, \varepsilon = \frac{1}{2}B\omega l^2$	Moving rod; rod rotating about one end	V
$L = \frac{N\Phi}{I}, \varepsilon = -L\frac{dI}{dt}$	Self-inductance	H = Wb A ⁻¹
$L = \mu_0 n^2 Al$	Long solenoid ($\times \mu_r$ with a core)	H
$U = \frac{1}{2}LI^2$	Energy in an inductor	J
$\varepsilon_2 = -M\frac{dI_1}{dt}, M = \frac{N_2\Phi_2}{I_1}$	Mutual inductance	H
$M = \frac{\mu_0 \pi r_1^2}{2r_2}$	Two concentric coplanar single-turn loops, $r_1 \ll r_2$	H
$M = \mu_0 n_1 n_2 \pi r_1^2 l$	Two coaxial solenoids (inner radius r_1 , length l)	H

Ch 7 • Alternating Current (5 marks)

Formula	Meaning / condition	SI unit
$\varepsilon = NBA\omega\sin\omega t$	AC generator (peak $\varepsilon_0 = NBA\omega$)	V
$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707I_0$	rms value; mean over a full cycle = 0, over a half cycle = $0.637I_0$	A
$X_L = \omega L, X_C = \frac{1}{\omega C}$	Reactances (I lags V in L, leads in C by $\pi / 2$)	Ω
$Z = \sqrt{R^2 + (X_L - X_C)^2}, \tan\phi = \frac{X_L - X_C}{R}$	Series LCR impedance and phase	Ω
$\omega_0 = \frac{1}{\sqrt{LC}}, \nu_0 = \frac{1}{2\pi\sqrt{LC}}$	Resonance ($Z = R, I$ maximum)	rad s ⁻¹ ; Hz

$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$	Quality factor; $2\Delta\omega = R / L$	no unit
$\bar{P} = V_{rms} I_{rms} \cos\phi$, $\cos\phi = \frac{R}{Z}$	Average power and power factor (0 for pure L or C)	W
$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$	Ideal transformer; $\eta = P_{out} / P_{in}$	no unit

Ch 8 • Electromagnetic Waves (2 marks)

Formula	Meaning / condition	SI unit
$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$	Displacement current	A
$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d)$	Ampere–Maxwell law	T m
$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$, $\frac{E_0}{B_0} = c$	Speed of EM waves in vacuum	m s ⁻¹
$v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{n}$	Speed in a medium	m s ⁻¹
$u = \frac{1}{2} \epsilon_0 E^2 + \frac{B^2}{2\mu_0}$	Energy density (electric and magnetic parts are equal)	J m ⁻³

$p = \frac{U}{c}$, pressure = $\frac{I}{c}$	Momentum; radiation pressure on a perfect absorber	kg m s ⁻¹ ; Pa
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Ch 9 • Ray Optics and Optical Instruments (7 marks)

Formula	Meaning / condition	SI unit
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, $f = \frac{R}{2}$, $m = -\frac{v}{u}$	Mirror formula (Cartesian signs)	m or cm
$n = \frac{c}{v} = \frac{\sin i}{\sin r}$, $n_{12} = \frac{n_2}{n_1}$	Refractive index; Snell's law	no unit
$n = \frac{\text{real depth}}{\text{apparent depth}}$	Apparent depth; normal shift = $t(1 - 1/n)$	m
$\sin C = \frac{1}{n}$	Critical angle (denser to rarer; TIR if $i > C$)	°
$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	Refraction at a spherical surface	m ⁻¹

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, $m = \frac{v}{u}$	Thin lens formula	m
$\frac{1}{f} = \left(\frac{n_l}{n_m} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$	Lens maker's formula	m ⁻¹
$P = \frac{1}{f}$, $P = P_1 + P_2$	Power of a lens (f in metres); contact combination	diopetre (D)
$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$, $\delta = i + e - A$	Prism; thin prism $\delta = (n - 1)A$	no unit
$m = 1 + \frac{D}{f}$ (or $\frac{D}{f}$)	Simple microscope (image at $D = 25$ cm; at infinity)	no unit

$$m = \frac{v_o}{|u_o|} \left(1 + \frac{D}{f_e} \right) \approx \frac{L}{f_o f_e} \frac{D}{f_e}$$

Compound microscope

no unit

$$m = \frac{f_o}{f_e}, \quad L = f_o + f_e$$

Astronomical telescope, normal adjustment

no unit; m

Ch 10 • Wave Optics (5 marks)

Formula	Meaning / condition	SI unit
$\Delta x = d \sin \theta \approx \frac{yd}{D}$	Path difference in Young's experiment	m
$y_n = \frac{n\lambda D}{d}$ (bright), $\frac{(n + \frac{1}{2})\lambda D}{d}$ (dark)	Fringe positions	m
$\beta = \frac{\lambda D}{d}$	Fringe width	m
$\phi = \frac{2\pi}{\lambda} \Delta x$	Phase difference from path difference	rad
$I_{max/min} = (\sqrt{I_1} \pm \sqrt{I_2})^2$	Resultant intensity; $I \propto a^2$	W m^{-2}

$a \sin \theta = n\lambda$ ($n = 1, 2, \dots$)	Single slit: minima; central maximum width $= 2\lambda D / a$	m
$z_F = \frac{a^2}{\lambda}$	Fresnel distance (ray optics valid for $z < z_F$)	m
$I = I_0 \cos^2 \theta$, $I = \frac{I_0}{2}$	Malus' law; unpolarised light through a polaroid	W m^{-2}
$\tan i_B = n$, $i_B + r = 90^\circ$	Brewster's law	°

Ch 11 • Dual Nature of Radiation and Matter (4 marks)

Formula	Meaning / condition	SI unit
$E = h\nu = \frac{hc}{\lambda}$, $E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$	Photon energy; $h = 6.63 \times 10^{-34} \text{ J s}$	J or eV
$K_{max} = h\nu - \phi_0 = eV_0$	Einstein's equation, $\phi_0 = h\nu_0$	J or eV
$V_0 = \frac{h}{e}(\nu - \nu_0)$	V_0 - ν graph: slope $= h / e$	V
$p = \frac{h}{\lambda} = \frac{E}{c}$	Photon momentum	kg m s^{-1}
$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}$	de Broglie wavelength	m

$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$	Electron accelerated through V volts	nm
$n = \frac{P}{h\nu}$	Photons emitted per second by a source of power P	s^{-1}

Ch 12 • Atoms (4 marks)

Formula	Meaning / condition	SI unit
$r_n = n^2 a_0, a_0 = 0.529 \text{ \AA}$	Bohr radius of the nth orbit	m
$v_n = \frac{2.18 \times 10^6}{n}$	Speed of the electron	m s^{-1}
$mvr = \frac{nh}{2\pi}$	Quantisation of angular momentum	J s
$E_n = -\frac{13.6}{n^2} \text{ eV}$	Energy; KE = $+13.6 / n^2$, PE = $-27.2 / n^2$	eV
$h\nu = E_i - E_f$	Bohr's frequency condition	J

$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$	Spectral lines; $R = 1.097 \times 10^7 \text{ m}^{-1}$	m^{-1}
$\frac{n(n-1)}{2}$	Maximum number of lines from level n	no unit
$r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K}$	Distance of closest approach (head-on α)	m

Ch 13 • Nuclei (3 marks)

Formula	Meaning / condition	SI unit
$R = R_0 A^{1/3}, R_0 = 1.2 \text{ fm}$	Nuclear radius; density $\approx 2.3 \times 10^{17} \text{ kg m}^{-3}$ (same for all nuclei)	m
$1 \text{ u} = 931.5 \text{ MeV} / c^2 = 1.66 \times 10^{-27} \text{ kg}$	Atomic mass unit	MeV/c^2 ; kg
$\Delta m = Zm_p + (A - Z)m_n - M$	Mass defect	u
$E_b = \Delta m c^2, \frac{E_b}{A}$	Binding energy; per nucleon (max $\approx 8.75 \text{ MeV}$ near $A = 56$)	MeV
$Q = (m_{in} - m_{out})c^2$	Energy released in a reaction (fission $\approx 200 \text{ MeV}$ per U-235)	MeV

$N = N_0 e^{-\lambda t}, T_{1/2} = \frac{0.693}{\lambda}$	Decay law, half-life (outside current NCERT; asked by RBSE in 2023)	s
${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\text{He}$	α decay (β^- : $Z + 1, A$ same; γ : no change)	

Ch 14 • Semiconductor Electronics (4 marks)

Formula	Meaning / condition	SI unit
$n_e n_h = n_i^2$	Intrinsic semiconductor; $n_e = n_h = n_i$	m^{-6}
$\sigma = e(n_e \mu_e + n_h \mu_h)$	Conductivity	S m^{-1}
E_g : Si 1.1, Ge 0.67 eV; $E_g > 3 \text{ eV}$ insulator	Forbidden energy gap	eV
$V_0 \approx 0.7 \text{ V (Si)}, 0.3 \text{ V (Ge)}$	Barrier potential / knee voltage	V
$r_d = \frac{\Delta V}{\Delta I}$	Dynamic resistance of a diode	Ω

$$\nu_{out} = \nu_{in} \text{ (half-wave), } 2\nu_{in} \text{ (full-wave)}$$

Ripple frequency of a rectifier

Hz

$$R_s = \frac{V_{in} - V_Z}{I_Z + I_L}$$

Series resistor of a Zener regulator

 Ω

B • Derivation and diagram checklist ★

Every derivation the board has asked, or is likely to ask, in one table. Write each one on paper with its labelled diagram and tick a box: ☐ **first round (Phase 2)** • ☐ **second round (days 31–35)** • ☐ **last week**. A derivation is "done" only when the diagram, the steps and the boxed result are all on the page. "Marks" shows the usual value; the board history is from the official papers of 2020 and 2023–2026* (* = CWSN paper).

Ch 1 • Electric Charges and Fields

Derivation ★	Marks	Final result (box it)	Asked
☐ ☐ ☐ Field of a dipole on the axis	3	$E = \frac{2kp}{r^3}$	2023 · 2024 · 2026*
☐ ☐ ☐ Field of a dipole on the equatorial line	3	$E = \frac{kp}{r^3}$	2024 · 2026*
☐ ☐ ☐ Torque on a dipole in a uniform field	3	$\tau = pE \sin \theta$	—
☐ ☐ ☐ Gauss: infinite plane sheet	3	$E = \frac{\sigma}{2\epsilon_0}$	2024 · 2026*
☐ ☐ ☐ Gauss: infinite line charge	3	$E = \frac{\lambda}{2\pi\epsilon_0 r}$	2025 · 2020 (numerical)
☐ ☐ ☐ Gauss: spherical shell (in and out)	3	$E = \frac{kQ}{r^2}, 0$	2023 (concept)

Ch 2 • Electrostatic Potential and Capacitance

Derivation ★	Marks	Final result (box it)	Asked
☐ ☐ ☐ Potential due to a point charge	1½	$V = \frac{kq}{r}$	2025
☐ ☐ ☐ Potential of a dipole at (r, θ)	3	$V = \frac{kpcos\theta}{r^2}$	2020
☐ ☐ ☐ PE of a system of three charges	1½	$U = k \sum \frac{q_i q_j}{r_{ij}}$	2023
☐ ☐ ☐ Capacitance of a parallel-plate capacitor (with slab)	3	$C = \frac{K\epsilon_0 A}{d}$	—
☐ ☐ ☐ Energy stored in a capacitor; energy density	3	$U = \frac{1}{2}CV^2, u = \frac{1}{2}\epsilon_0 E^2$	2025 (MCQ)
☐ ☐ ☐ Spherical capacitor; series and parallel	3	$C = \frac{4\pi\epsilon_0 ab}{b-a}$	2020 · 2024

Ch 3 • Current Electricity

Derivation ★	Marks	Final result (box it)	Asked
□□□ Drift velocity, $I = neAv_d$, $\rho = m / ne^2\tau$	3	$\rho = \frac{m}{ne^2\tau}$	2024 • 2025 (mobility)
□□□ Wheatstone bridge balance by Kirchhoff rules	3	$\frac{P}{Q} = \frac{R}{S}$	2023 • 2024 • 2025
□□□ Cells in series and parallel (ε_{eq} , r_{eq})	3	$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$	2025

Ch 4 • Moving Charges and Magnetism

Derivation ★	Marks	Final result (box it)	Asked
□□□ Biot–Savart: field on the axis of a loop	3 / 4	$B = \frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}}$	2024 • 2025 • 2026*
□□□ Ampere's law: infinite straight wire	3 / 4	$B = \frac{\mu_0 I}{2\pi r}$	2025 • 2026*
□□□ Ampere's law: long solenoid	3	$B = \mu_0 nI$	2023
□□□ Force per unit length between parallel wires	3	$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$	2024 • 2023 (blank)
□□□ Torque on a current loop; ammeter / voltmeter conversion	3 / 1½	$\tau = NIAB\sin\theta$	2024 • 2025 (blanks)

Ch 5 • Magnetism and Matter

Derivation ★	Marks	Final result (box it)	Asked
□□□ Bar magnet as a solenoid: axial and equatorial field	3	$B_{ax} = \frac{\mu_0 2m}{4\pi r^3}$	—
□□□ Torque and PE of a magnetic dipole	1½	$U = -mB\cos\theta$	2025 (MCQ)

Ch 6 • Electromagnetic Induction

Derivation ★	Marks	Final result (box it)	Asked
□□□ Motional emf (with energy conservation)	3	$\varepsilon = Blv$	2024 (numerical)
□□□ Mutual inductance of concentric coplanar coils	3	$M = \frac{\mu_0 \pi r_1^2}{2r_2}$	2025 • 2026*
□□□ Self-inductance of a long solenoid	3	$L = \mu_0 n^2 Al$	—
□□□ Energy stored in an inductor	3	$U = \frac{1}{2}LI^2$	2023 (L numerical)

Ch 7 • Alternating Current

Derivation ★	Marks	Final result (box it)	Asked
□□□ AC generator (labelled diagram)	3	$\varepsilon = NBA\omega \sin \omega t$	2024 · 2025 (MCQ)
□□□ rms value of AC	3 / 4	$I_{rms} = \frac{I_0}{\sqrt{2}}$	2024 · 2025
□□□ AC through a capacitor (phasor, reactance, power)	4	$X_C = \frac{1}{\omega C}$	2025
□□□ AC through an inductor (average power = 0)	3 / 4	$X_L = \omega L$	2024 · 2025 (blank)
□□□ Series LCR: impedance by phasors	4	$Z = \sqrt{R^2 + (X_L - X_C)^2}$	2024 · 2026*
□□□ Resonance and the I - ω graph	4	$\omega_0 = \frac{1}{\sqrt{LC}}$	2025
□□□ Average power; power factor; wattless current	3 / 4	$\bar{P} = VI \cos \phi$	2023 · 2024 · 2026*
□□□ Transformer: theory, losses and remedies	3	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$	2023 · 2026*

Ch 8 • Electromagnetic Waves

Derivation ★	Marks	Final result (box it)	Asked
□□□ Displacement current (Ampere–Maxwell)	1½	$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$	2024 · 2026*
□□□ Properties of EM waves; $c = 1 / \sqrt{\mu_0 \varepsilon_0}$	1½	$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$	2023 · 2024 · 2025

Ch 9 • Ray Optics and Optical Instruments

Derivation ★	Marks	Final result (box it)	Asked
□□□ Mirror formula	4	$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	2023 · 2024
□□□ Refraction at a spherical surface	4	$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	NCERT standard
□□□ Lens maker's formula and lens formula	4	$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$	2024
□□□ Prism: refractive index by minimum deviation	4	$n = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$	2023 · 2025
□□□ Total internal reflection; critical angle	4	$\sin C = \frac{1}{n}$	2024 · 2026*
□□□ Compound microscope: magnifying power	4	$m \approx \frac{L}{f_o} \frac{D}{f_e}$	2025 · 2024 (MCQ)
□□□ Astronomical telescope	3 / 4	$m = \frac{f_o}{f_e}$	2024 (numerical) · 2020

Ch 10 • Wave Optics

Derivation ★	Marks	Final result (box it)	Asked
□□□ Huygens' principle: Snell's law (refraction)	3	$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$	2024 · 2025 · 2026*
□□□ Huygens' principle: law of reflection	3	$i = r$	2020 (wavefront)
□□□ Young's experiment: fringe width	3	$\beta = \frac{\lambda D}{d}$	2023
□□□ Single-slit diffraction: first minimum	3	$a \sin \theta = \lambda$	2020 (compare)

Ch 11 • Dual Nature of Radiation and Matter

Derivation ★	Marks	Final result (box it)	Asked
□□□ Einstein's equation; V_0 - ν and I - V graphs	3	$K_{max} = h\nu - \phi_0$	2025
□□□ de Broglie wavelength of an accelerated electron	3	$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$	2023 · 2024 · 2026*
□□□ Davisson–Germer experiment	3	$\lambda_{exp} = \lambda_{dB}$	2020 (aim)

Ch 12 • Atoms

Derivation ★	Marks	Final result (box it)	Asked
□□□ Bohr model: radius and total energy of the orbit	3	$r_n = n^2 a_0, E_n = -\frac{13.6}{n^2}$	2025 · 2023
□□□ Bohr's 2nd postulate from de Broglie	1½	$mvr = \frac{nh}{2\pi}$	2024 · 2026* · 2020
□□□ Rutherford's α -scattering: drawbacks of the model	1½	$b \rightarrow 0$ head-on	2023 · 2024 (MCQ)

Ch 13 • Nuclei

Derivation ★	Marks	Final result (box it)	Asked
□□□ BE per nucleon curve; nuclear force	3	$E_b = \Delta m c^2$	2025
□□□ Fission and fusion	1½	$Q = (m_{in} - m_{out})c^2$	2023 · 2024
□□□ Radioactive decay law (outside current NCERT)	1½	$N = N_0 e^{-\lambda t}$	2023

Ch 14 • Semiconductor Electronics

Derivation ★	Marks	Final result (box it)	Asked
□□□ Energy bands: conductor, insulator, semiconductor	3	E_g gap	2024 · 2025 · 2026*
□□□ n-type and p-type band diagrams	3	donor / acceptor level	2024
□□□ p-n junction: depletion layer and barrier potential	3 / 4	$V_0 \approx 0.7 \text{ V (Si)}$	2023 · 2026*
□□□ Full-wave rectifier with waveforms	4	$v_{out} = 2v_{in}$	2023
□□□ Zener diode as a voltage regulator	3	$V_{out} = V_Z$	2020

 How a derivation earns its marks

- **Diagram ½–1:** labelled, with the Gaussian surface / loop / ray path drawn. No diagram, no mark.
- **Law or starting formula ½:** write the law by name before using it (Gauss, Biot–Savart, Snell, Faraday ...).
- **Steps ½ each:** one idea per line – symmetry, small-angle approximation, integration limits, sign convention.
- **Result ½:** boxed, with the condition (e.g. $r \gg a$) and the unit.

Diagrams to draw in 60 seconds each

Tick when you can draw it neatly with all labels, from memory

- ☐ Ch 1 · field lines of a dipole; cylindrical and spherical Gaussian surfaces
- ☐ Ch 2 · equipotential surfaces; capacitor with dielectric slab; series and parallel circuits
- ☐ Ch 3 · Wheatstone bridge; metre bridge; potentiometer / cells in series and parallel
- ☐ Ch 4 · Amperian loop for a solenoid; loop on the axis; moving-coil galvanometer; cyclotron
- ☐ Ch 5 · bar-magnet field lines; hysteresis loop; B–H for soft iron and steel
- ☐ Ch 6 · Lenz-law pictures; rod on rails; AC generator
- ☐ Ch 7 · phasor diagrams for L, C and LCR; $I - \omega$ resonance curve; transformer
- ☐ Ch 8 · propagation of an EM wave ($\vec{E}$, $\vec{B}$, direction); EM spectrum
- ☐ Ch 9 · ray diagrams of mirrors and lenses; prism and $\delta - i$ graph; microscope; telescope; TIR
- ☐ Ch 10 · Huygens' constructions (reflection, refraction); Young's set-up; intensity pattern; polaroids
- ☐ Ch 11 · photocell set-up; $I - V$ for two intensities / frequencies; $V_0 - \nu$ graph
- ☐ Ch 12 · α -scattering tracks; energy-level diagram with H series; standing wave in orbit
- ☐ Ch 13 · BE per nucleon versus A; fission chain; fusion in the Sun
- ☐ Ch 14 · band diagrams; p-n junction; I–V curves (diode, Zener); half- and full-wave rectifier

C • Common-errors log

41 common mistakes that cost marks, with the correct version. Read a chapter's rows just before you revise that chapter. Then keep **your own log** below: every wrong answer from the model papers goes in with its cause.

Ch	Wrong	Right
1	✗ Field inside a charged conducting shell is kQ / r^2 .	✓ Zero inside ($r < R$); kQ / r^2 only for $r \geq R$.
1	✗ Using $2kp / r^3$ on the equatorial line.	✓ Axial $2kp / r^3$; equatorial kp / r^3 , opposite to $\vec{p}$.
1	✗ Flux through one face of a cube with q at the centre is q / ϵ_0 .	✓ It is $q / 6\epsilon_0$; the whole cube gives q / ϵ_0 .
2	✗ Capacitors in series have the same voltage.	✓ Series: same charge. Parallel: same voltage.
2	✗ Adding potentials of charges as vectors, or forgetting the signs of q .	✓ Potential is a scalar: add algebraically with signs.
2	✗ Dielectric inserted: energy always rises.	✓ Battery off (Q fixed): $U \rightarrow U / K$. Battery on (V fixed): $U \rightarrow KU$.
3	✗ Drift speed is how fast the current travels.	✓ Drift speed $\approx 10^{-4}$ m/s; the signal travels near c .
3	✗ Stretching a wire to n times length: $R' = nR$.	✓ Volume is constant, so $R' = n^2R$.
3	✗ Terminal voltage equals emf.	✓ $V = \epsilon - Ir$ on discharge, $\epsilon + Ir$ while charging; equal only on open circuit.
4	✗ The magnetic force changes the speed of a charge.	✓ $\vec{F} \perp \vec{v}$: no work, speed and KE are constant; only direction changes.
4	✗ Solenoid field $\mu_0 NI$.	✓ $B = \mu_0 nI$ with n = turns per metre.
4	✗ Shunt in series, high resistance in parallel.	✓ Ammeter: low-resistance shunt in parallel with G. Voltmeter: high resistance in series with G.
4	✗ Parallel currents repel.	✓ Parallel currents attract; antiparallel repel.
5	✗ Field lines of a magnet start at N and end at S.	✓ They are closed loops: outside $N \rightarrow S$, inside $S \rightarrow N$.
5	✗ Diamagnetic susceptibility is positive.	✓ Dia: small negative. Para: small positive. Ferro: large positive.
6	✗ Lenz's law gives the magnitude of the emf.	✓ Faraday gives the magnitude; Lenz (the minus sign) gives the direction – conservation of energy.
6	✗ A rod moving along $\vec{B}$ gets an emf.	✓ No flux is cut, so $\epsilon = 0$; need $\vec{v}$ and l perpendicular to $\vec{B}$.
6	✗ Self-inductance depends on the current.	✓ L depends only on geometry and the core: $\mu_0 n^2 Al$.

7	✗ The AC ammeter reads the peak value; mean AC is $0.707I_0$.	✓ It reads rms. Mean over a full cycle is 0; over a half cycle $0.637I_0$.
7	✗ $X_C = \omega C$ and I lags V in a capacitor.	✓ $X_C = 1 / \omega C$; in C the current leads by $\pi / 2$, in L it lags by $\pi / 2$.
7	✗ Power is always VI .	✓ $\bar{P} = V_{rms}I_{rms}\cos\phi$; zero in a pure L or C.
7	✗ Transformer works with DC if the voltage is large.	✓ DC gives constant flux: no induced emf. It needs changing flux (AC).
8	✗ Displacement current is a flow of charge across the gap.	✓ It comes from a changing electric flux; no charge crosses the gap.
8	✗ EM waves are deflected by electric and magnetic fields.	✓ They carry no charge; $\vec{E} \perp \vec{B} \perp$ direction of travel.

9	✗ Mixing sign conventions: positive u , wrong sign of R and f .	✓ Cartesian: distances against the incident light are negative; concave mirror $f < 0$.
9	✗ Lens formula written as $1 / v + 1 / u = 1 / f$.	✓ Lens: $1 / v - 1 / u = 1 / f$. Mirror: $1 / v + 1 / u = 1 / f$.
9	✗ TIR happens whenever light enters a rarer medium.	✓ Only if $i > C$ and light goes from denser to rarer.
9	✗ A lens has the same focal length in water.	✓ It changes with the surrounding medium: $\frac{f_w}{f_a} = \frac{n_g - 1}{n_g / n_w - 1}$ (longer).
10	✗ Fringe width depends on the intensity of light.	✓ $\beta = \lambda D / d$ only; intensity changes brightness.
10	✗ Unpolarised light through a polaroid: $I_0 \cos^2 \theta$.	✓ First polaroid gives $I_0 / 2$; Malus' law applies only to the next one.

10	✗ Phase difference = path difference.	✓ $\phi = 2\pi \Delta x / \lambda$.
11	✗ Stopping potential increases with intensity.	✓ V_0 depends on ν and the metal only; intensity raises the saturation current.
11	✗ Mixing joule and eV in $h\nu - \phi_0$.	✓ Use $hc = 1240 \text{ eV nm}$ and keep everything in eV.
12	✗ Energy of the electron in an orbit is positive.	✓ Bound state: $E = -13.6 / n^2 \text{ eV}$; KE = $-E$, PE = $2E$.
12	✗ Number of lines from level n is n^2 .	✓ $n(n - 1) / 2$.
12	✗ Lyman is visible, Balmer is ultraviolet.	✓ Lyman: UV ($n_f = 1$). Balmer: visible ($n_f = 2$). Paschen onwards: IR.

Morning of the exam

- ☐ Light breakfast, water. Reach the centre 30 minutes early.
- ☐ Read only the one-page formula list of the "memory" boxes; do not discuss answers with friends outside the hall.
- ☐ Write roll number and subject code (SS-40) correctly on the answer-book and every supplement.

Inside the hall: the 195-minute plan

5'	Section A 50'	Section B 60'	Section C 35'	Section D 40'	5'
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Total 195 min: the first 5 min (purple) to read the whole paper and mark the OR choices, the last 5 min (grey) to check numbering and units.

Section	How to write	Marks-saving rule
A • Q1 MCQ (9)	Write the option letter and its content. One minute per question; no negative marking, so never leave one blank.	Eliminate two options first; check units in numerical MCQs.
A • Q2 blanks (5)	Write the missing word/value only, in the same order, with the unit if it is a number.	Use the exact textbook word (e.g. "displacement" current).
A • Q3 VSA (10)	One or two sentences with the keyword or formula; add the unit. No story.	Definition = keyword + unit, e.g. dipole moment: $q \times 2a$, C m.
B • Q4–13 (15)	Short answers: 3 points of $\frac{1}{2}$ each, or formula – substitution – answer with unit. 6 minutes per question.	A numerical without unit loses $\frac{1}{2}$; a wrong unit loses $\frac{1}{2}$.
C • Q14–16 (9)	Labelled diagram ($\frac{1}{2}$ –1) → law → steps → boxed result. Choose the OR option you know best in the first 30 seconds.	Cross out any rough work; never answer both alternatives.
D • Q17–18 (8)	Part (a) derivation with figure (3), part (b) numerical (1). Do the numerical even if the derivation is incomplete.	Start Section D with 40 min left on the clock; it is the highest marks per question.

If you get stuck

- ☐ Write what you know: the law, the formula, the diagram – partial marks are real.
- ☐ Leave space (3–4 lines) and move on; return after Section D.
- ☐ Do not spend more than 8 minutes on any $1\frac{1}{2}$ -mark question.

The last 10 minutes

Final check

- ☐ All 18 question numbers attempted and written correctly; the OR choice is clear.
- ☐ Units on every numerical; boxed final answers; sub-parts (a)/(b) labelled.
- ☐ Diagrams labelled (names of rays, angles, plates, terminals).
- ☐ Blanks (Q2) and MCQs (Q1) all answered.

**Five lines to remember on the way in**

- Diagram + law + steps + boxed answer = full marks.
- Unit on every number. Sign on every vector.
- Rms for AC meters; peak only when asked.
- Lens: $1/v - 1/u$. Mirror: $1/v + 1/u$. Cartesian signs always.
- You have solved five full papers. Trust your preparation.

$$1/f = 1/v - 1/u$$

All the best for your board exam!

Diagram, formula, boxed answer – every time.

Also by VS Sir

NEET + JEE Physics 30-Day Rapid Revision – Class 11 & Class 12 (English & Hindi)

All books: vcakota20-web.github.io/ai-mastery-with-vs-sir/books/

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$$E = h\nu$$

$$V = IR$$

$$\varepsilon = -d\Phi/dt$$

$$\lambda = h/p$$

